

Regulations and Curriculum for
Bachelor of Technology (B.Tech.)
in
Computer Science and Engineering
(Cyber Security)

Version 2024.02



(Established under Section 3 of UGC Act, 1956)
Placed under Category 'A' by MHRD, GoI | Accredited with 'A+' Grade by NAAC

Regulations and Curriculum for

Bachelor of Technology (B. Tech.)
in
Computer Science and Engineering
(Cyber Security)
Choice Based Credit System (CBCS)
Effective for the Batch admitted in
AY 2024-25



(Deemed to be University under Section 3 of UGC Act, 1956)
(Placed under Category 'A' by MHRD, Govt. of India, Accredited with 'A+' Grade
by NAAC)

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VISION

To build a humane society through excellence in the education and healthcare

MISSION

*To develop
Nitte (Deemed to be University)
As a center of excellence imparting quality education,
Generating competent, skilled manpower to face the scientific and social
challenges with a high degree of credibility, integrity,
ethical standards and social concern*

Regulations and Curriculum

B.Tech. Degree Programs
Choice based Credit System
(CBCS)

Effective for the Batch admitted in
Academic Year
2024 – 2025

Curriculum for Acquiring Professional Skills (CAPS)

With Scheme of Teaching & Examination

REGULATIONS: 2024

**COMMON TO ALL
B.Tech. DEGREE PROGRAMS
CHOICE BASED CREDIT SYSTEM
(CBCS)**

Version 2024.02

Choice Based Credit System (CBCS)

1. Choice for the selection of courses during each semester.
2. Choice in planning the academic activities by selecting desired number of courses per semester.
3. Balanced curriculum with engineering, science, humanities, and management courses.
4. Project based learning (PBL) which focusses on experiential learning.
5. Opportunities to study inter-disciplinary courses.
6. Enabling slow learners by offering important courses in all semesters.
7. Optional Summer semester.
8. Opportunity to get associated in research projects to acquire research experience.
9. Value addition with Honors / Minor credentials.

Curriculum for Acquiring Professional Skills (CAPS)

1. Practicing outcome-based education (OBE) where Courses made student-centric rather than teacher-centric.
2. Provisions for courses integrated with Lab/ PBL component.
3. Focus on experiential learning.
4. Ability enhancement and skill development courses as per National Education Policy (NEP) 2020.
5. Focus on Industry Internship and Research Internship.
6. Students to work on real world/interdisciplinary problems in major project.
7. Importance is given to creativity, innovation, and development of entrepreneurship skills.

Key Information

Program Title	Bachelor of Technology Abbreviated as B.Tech. Computer Science and Engineering (Cyber Security)
Short description	Four-year, eight semester Choice Based Credit System (CBCS) type of Undergraduate Engineering Degree Program with English as medium of instruction.
Program Code	14ENGR21D2
Revision version	2024.02 These regulations may be modified from time to time as mandated by the policies of the University. Revisions are to be recommended by the Board of Studies for • Computer Science and Engineering (Cyber Security) • Humanities • Mathematics • Science and approved by the Academic Council.
Effective from	01-08-2024
Approvals	Approved in the 57th Academic Council meeting of NITTE (Deemed to be University), held on 26.06.2024 and vide Notification of Registrar • Ref: N(DU)/REG/AC-NMAMIT/2023-24/1413 dated 09.07.2024 • Ref: N(DU)/REG/AC-NMAMAIT/2023-24/1419 dated 09.07.2024 • Ref: N(DU)/REG/AC-NMAMIT/2023-24/1420 dated 09.07.2024 • Ref: N(DU)REG/AC-NMAMIT/2023-24/1421 dated 09.07.2024. Approved in the 60th Academic Council meeting of NITTE (Deemed to be University), held on 05.02.2025 and vide Notification of Registrar Ref: N(DU)/REG/AC-NMAMIT/2024-25/748-A dated 15.02.2025.
Program offered at	NMAM Institute of Technology, Off -Campus Centre, Nitte, 574110, Karkala Taluk
Grievance and dispute resolution	All disputes arising from this set of regulations shall be addressed to the Executive Council. The decision of the Executive Council is final and binding on all parties concerned. Further, any legal disputes arising out of this set of regulations shall be limited to jurisdiction of Courts of Mangalore only

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PREAMBLE

NMAM Institute of Technology (NMAMIT) was established in 1986 and is located at Nitte and off-campus center of NITTE (Deemed to be University), accredited by National Assessment & Accreditation Council (NAAC) with 'A+' grade. NMAMIT is recognized by the All-India Council for Technical Education (AICTE), New Delhi.

The Bachelor of Technology (B. Tech.) Programs focus on Pursuing Excellence, Empowering people, and Partnering in Community Development. Out of fourteen UG Programs i.e., Artificial Intelligence & Machine Learning (AM), Artificial Intelligence & Data Science (AD), Biotechnology (BT), Computer & Communication Engineering (CC), Computer Science & Engineering (CS), Civil Engineering (CV), Electronics & Communication Engineering (EC), Electrical & Electronics Engineering (EE), Information Science & Engineering (IS), Mechanical Engineering (ME), Robotics & Artificial Intelligence (RI), Computer Science & Engineering - Cyber Security (CB), Electronics Engineering - VLSI Design and Technology (VT), and Electronics & Communication - Advanced Communication Technology (AC), all seven eligible UG Programs i.e., BT, CS, CV, EC, EE, IS and ME are accredited by NBA, New Delhi under Tier - I category till 30th June 2025.

The curriculum is jointly approved by members of the Board of Studies (BoS) and Academic Council drawn from academia, Industry, Alumni, and working professionals from Industry, and has been designed to integrate hands-on practical training with the concepts of theory courses to enhance the learning experience.

The Curriculum focuses on students Acquiring Professional Skills (CAPS) through rigorous theoretical training using innovations in pedagogy, experiential learning, active learning, collaborative learning, critical thinking, project planning, Project Based Learning (PBL), Ability enhancement courses for skill-building, effective communication, professional practice, creativity & innovation and developing entrepreneurial skills.

The focus of the Institution is to impart Quality Education to generate competent, Skilled, and Humane Manpower to face emerging Scientific, Technological, Managerial and Social Challenges with Credibility, Integrity, Ethics, and Social Concern.

In the present scenario, students wish to make plans for a bright future. However, student aspirations and industry demands are highly diverse. Employers expect the graduates possess multi-disciplinary competency, Information and Communication Technology (ICT), and leadership skills. In this context, NMAMIT offers the opportunity to the students to select the courses of their choice and helps them in grooming to have well-rounded personalities and become industry ready.

Efforts have been made to make the syllabus compliant with international professional societies. As part of providing quality engineering education, at NMAMIT, Nitte, it has initiated the Choice Based Credit System (CBCS) into its academic curriculum. By this, the students can register for courses of their choice and alter the pace of learning within the broad framework of academic courses and credit requirements. CBCS allows students to plan for their academic load and alter it as they progress in learning. Students also have the option of choosing courses from a pool of courses within each classification. Ample options are given to choose interdisciplinary courses

from other programs which will help the student to develop additional skills. Slow learners will also be benefitted since important courses are offered in all semesters. This arrangement helps the students to re-register and clear the backlog courses in the subsequent semester. Suitable provisions are made for fast learners to associate them with research activities of faculty members and contribute to research beyond the working hours.

A faculty advisor helps the student in identifying the courses to be studied in each semester based on program requirements, course prerequisites, student's interest in various disciplines, past academic performance, and courses offered by the departments.

Learning becomes more 'experiential' by carrying out labs associated with theory, mini-projects, and Project Based Learning (PBL) as a part of many courses which enhances the capability of students in understanding and apply Engineering /Technology concepts to solve real life-problems. Hence students will develop the ability to apply the gained knowledge in multi-disciplinary projects and be able to take up major projects based on real-world problems and come up with better solutions while addressing social concerns.

REGULATIONS

COMMON TO ALL B.Tech. (CBCS) DEGREE PROGRAMS OF

NITTE (Deemed to be University)

1. INTRODUCTION

- 1.1** The general regulations are common to all B.Tech.(CBCS) Degree Programs conducted at the NMAM Institute of Technology (NMAMIT), off-campus center of NITTE (Deemed to be University) and shall be called “B.Tech. Regulations”.
- 1.2** The provisions contained in this set of regulations govern the policies and procedures on the Registration of students, imparting instructions of courses, the conduct of the examination & evaluation, certification of student performance, and all amendments related to the said Degree program(s).
- 1.3** This set of Regulations, on approval by the Academic Council and Governing Council, shall supersede all the corresponding earlier sets of regulations of the B. Tech Degree program of NITTE (Deemed to be University) along with all the amendments thereto, and shall be binding on all students undergoing the Graduate Degree Program(s) (Choice Based Credit System) conducted at the NMAMIT, Nitte with effect from its date of approval. This set of Regulations may evolve and get modified or changed through appropriate approvals from the Academic Council / Governing Council from time to time and shall be binding on all stakeholders (The Students, Faculty, Staff of Departments of NMAMIT, Nitte). The decisions of the Academic Council / Governing Council shall be final and binding.
- 1.4** To guarantee fairness and justice to the parties concerned given the periodic evolutionary refinements, any specific issues or matters of concern shall be addressed separately, by the appropriate authorities, as and when found necessary.
- 1.5** The Academic Council may consider any issues or matters of Concern relating to any or all the academic activities of Engineering courses for appropriate action, irrespective of whether a reference is made here in this set of Regulations or otherwise.
- 1.6** The program shall be called **Bachelor of Technology**, abbreviated as B.Tech. (Program Specialization).

2. ELIGIBILITY FOR ADMISSION

Sl. No	Program	Duration	Eligibility
1	B. Tech.	4 years	Passed 10+2 examination with Physics/ Mathematics / Chemistry/ Computer Science/ Electronics/ Information Technology/ Biology/ Informatics Practices/ Biotechnology/Technical Vocational subject as per Table-1 Obtained at least 45% marks (40% marks in case of candidates belonging to reserved category) in the above subjects taken together.
2	B.Tech. (Lateral Entry to Second year)	3 years	Passed Minimum THREE years / TWO years (Lateral Entry) Diploma examination with at least 45% marks (40% marks in case of candidates belonging to reserved category) in relevant branch of Engineering and Technology. (The University will offer suitable bridge courses such as Mathematics, Physics, Engineering drawing, etc., for the students coming from diverse backgrounds to prepare Level playing field and desired learning outcomes of the program)

Table-1 Academic Level and Credit Framework for admission to Bachelor of Technology (B.Tech.) degree program				
Sl. No.	Academic Level	Desired Entry Qualifications at different levels.	NHEQF / NSQF Level at Exit	Unified Credit Level (UCF) at Exit
1	12 th Std.	-	4	4
2	First Year B.Tech. Degree	12 th Completed (NHEQF /UCF level 4 completed)	5	4.5
3	Second Year B.Tech. Degree	A candidate with a Diploma in the appropriate branch of Engineering/Equivalent Vocational or Technical Program with NHEQF level 5/UCF level 4.5 completed.	6	5

2.1 Qualifications from foreign countries

Candidates with qualifications from educational institutions outside of India may be admitted to the program(s) subject to the establishment of equivalence by the university. The Program Committee will evaluate and establish the eligibility of such candidates.

3. PROGRAM PATHS, EXIT OPTIONS, AND DURATION OF THE B. TECH. PROGRAM

3.1 Program paths, exit options

Sr. No	Academic Level	Entry Level Qualifications	Qualifications at Exit	NCrF Level
1	1 st yr. of UG Degree	A candidate completing 10+2 years with Diploma of Vocation or passed 12 th std. or equivalent vocational training with NCrF level 4	UG Certificate	4.5
2	2 nd yr. of UG Degree	A candidate with Diploma in appropriate branch of Engineering/ UG Certificate/ Equivalent Vocational or Technical Program NCrF level 4.5	UG Diploma (Engg.)	5.0
3	3 rd yr. of UG Degree	A candidate with 10+3+1/12+2/ UG Diploma (Engg.) in appropriate domain with NCrF level 5	B. Sc (Engg.)	5.5
4	Final yr. of UG Degree	A candidate with 3 yrs. Bachelor degree in Vocation / B.Sc. (Engg.) with NCrF level 5.5	B. Tech (on completion of 164 credits with a minimum CGPA of 5)	6
	Final yr. of UG Degree with Honours	A candidate with 3 yrs. Bachelor degree in Vocation / B.Sc. (Engg.) with NCrF level 5.5	B. Tech (Honors) 182 credits (Additional 18 credits over and above 164 credits in the same discipline	6
	Final yr. of UG Degree with a minor in (Other Discipline).	A candidate with 3 yrs. Bachelor degree in Vocation / B.Sc. (Engg.) with NCrF level 5.5	B. Tech with Minor 182 credits. Additional 18 credits over and above 164 credits in other disciplines	6

*** It is mandatory to earn 10 credits through Internship/ Training/ Specialized courses before the award of Qualifications at Exit.**

3.2 Duration of the B. Tech. program

- The B. Tech Program shall extend over a period of a total duration of 4 years for students admitted during the first year of the program.
- The total duration shall be 3 years for students admitted to the second year under the lateral entry scheme.
- The maximum period which a student can take to complete a full-time academic program is eight years / Six years for Lateral entry diploma students for B.Tech.
- Each year shall have the following schedule with 5½ days a week. Suggested break down of Academic Year into Semesters

1.	No. of Semesters / Year	There are three semesters in an academic year. Two Main semesters (Odd, Even) followed by a summer semester. Normally the Odd Semester will be from August to December and Even Semester from January to May during a calendar year. The optional summer semester is offered during the vacation period of the even semester. The summer semester is offered considering the demand for such courses of needy students, subject to the availability of time, faculty, and other resources under a fast-track mode as the available instructional days during even semester vacation periods are less. However, the number of instructional hours needed to cover the syllabi shall be maintained (equivalent to that in the regular semester) with a greater number of instruction hours per week. (Note: The summer semester is primarily to assist slow learners and/or failed students in the main semesters. The summer semester may be used to arrange Add-On courses for other students and/or for deputing them for practical training elsewhere)												
2.	Semester Duration	Main semester (Odd, Even) each 20 Weeks; Summer Semester 8 Weeks												
3.	Academic Activities (Weeks)	ODD / EVEN Semester <table><tr><td>Registration of Courses & Course Work</td><td>(16)</td></tr><tr><td>Examination Preparation and Examination</td><td>(04)</td></tr><tr><td>Total</td><td>(20)</td></tr></table> Summer Semester <table><tr><td>Registration of Courses & Course Work</td><td>(05)</td></tr><tr><td>Examination Preparation and Examination</td><td>(03)</td></tr><tr><td>Total</td><td>(08)</td></tr></table> Declaration of results: 02 weeks from the date of the last examination Inter-Semester Recess: After each Main Semester (02) Total Vacation: 10 weeks (for those who do not register for the summer semester) and 4 weeks (for those who register for the summer semester)	Registration of Courses & Course Work	(16)	Examination Preparation and Examination	(04)	Total	(20)	Registration of Courses & Course Work	(05)	Examination Preparation and Examination	(03)	Total	(08)
Registration of Courses & Course Work	(16)													
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Registration of Courses & Course Work	(05)													
Examination Preparation and Examination	(03)													
Total	(08)													

(Note: In each semester, there will be provision for students to register for courses at the beginning, dropping of courses in the middle, and withdraw from courses towards the end, under the advice of a faculty member. These facilities are expected to enhance the learning capabilities of students, minimizing their chances of failure in courses registered and ensuring their better monitoring by Faculty Advisors).

A candidate shall be allowed a maximum duration of eight years from the first semester of admission to become eligible for the award of a Bachelor's degree.

The calendar of events in respect of the program shall be fixed by the Institution from time to time, but preferably in line with the suggested academic calendar of the NITTE (Deemed to be University).

4. DEGREE PROGRAMS

4.1 Undergraduate B. Tech. Degree Programs are offered in the following disciplines by the respective program hosting departments listed below:

i)	Biotechnology Engineering	(BT)
ii)	Computer Science & Engineering	(CS)
iii)	Computer Science and Engineering (Cyber Security)	(CB)
iv)	Civil Engineering	(CV)
v)	Electronics & Communication Engineering	(EC)
vi)	Electronics Engineering (VLSI Design and Technology)	(VT)
vii)	Electronics & Communication (Advanced Communication Technology)	(AC)
viii)	Electrical & Electronics Engineering	(EE)
ix)	Information Science & Engineering	(IS)
x)	Mechanical Engineering	(ME)
xi)	Artificial Intelligence and Machine Learning Engineering	(AM)
xii)	Computer and Communication Engineering	(CC)
xiii)	Robotics and Artificial Intelligence Engineering	(RI)
xiv)	Artificial Intelligence and Data Science	(AD)
Other teaching departments are –		
i)	Chemistry	(CY)
ii)	Humanities	(HU)
iii)	Management and Social Sciences	(MG)
iv)	Mathematics	(MA)
v)	Physics	(PH)

4.2 The provisions of these regulations shall apply to any new discipline that may be introduced from time to time and appended to the above list.

5. CREDIT SYSTEM

In the Credit System, the course work of students is unitized, and each unit is assigned one credit after a student completes the teaching-learning process as prescribed for that unit and is successful in its assessment.

5.1 Credit Definition: The following widely accepted definition for credit can provide good flexibility to the students and strengthens CBCS under the University. Here, one unit of course work and its corresponding one credit (while referring to the main semester) shall be equal to:

- Four-credit theory courses shall be designed for 50 hours of the Teaching-Learning process.
- Three-credit theory courses shall be designed for 40 hours of the Teaching-Learning process.
- Two-credit theory courses shall be designed for 25 hours of the Teaching-

Learning process.

- One credit theory course shall be designed for 15 hours of the Teaching-Learning process.

The above figures shall also be applicable in the case of the summer semester. Other student activities which are not demanding intellectually, or which do not lend to effective assessment, like practical training, study tours, and attending guest lectures shall not carry any credit.

5.2 Credit Assignment and Lower & Upper Limits for Course Credits Registration in a Semester

All courses comprise of specific Lecture/Tutorial/Practical/Project (L-T-P-J) schedule. The course credits are fixed based on the following norms.

Lecture / Tutorials / Practical:

- 1-hour Lecture per week is assigned 1.0 Credit.
- 2-hour Tutorial session per week is assigned 1.0 Credit.
- 2-hour Lab Session/project work per week is assigned 1.0 credit.

For example,

- A theory course with L-T-P schedule of 3-2-0 hours will be assigned 4.0 credits.
- A laboratory practical course with L-T-P schedule of 0-0-2 hours will be assigned 1.0 credit.
- Calculation of Contact Hours / Week – A Typical Example

Example:

An LTP-C of 2-2-2-4 means 2 instructional units based on classroom lecture (L), one instructional unit of the tutorial (T), and one laboratory (P) based instructional unit all delivered during a calendar week and repeated for the entire duration of the semester to earn 4 credits (C) after passing the course.

- As advised by the faculty advisor, a student may register, between a minimum of **16 credits and up to a maximum of 28 credits.**

The maximum number of credits a student can register during a summer semester shall be 16. However, in special cases, the student may be permitted to register additional credits with the approval of the Department Undergraduate Committee (DUGC). There is no minimum number of credits fixed for course registration during the summer semester.

6. REGISTRATION

- 6.1** Every student after consulting his/ her Faculty Advisor in the parent department shall register for the approved courses (core and elective) to earn credits for meeting the requirements of a degree program at the commencement of each Semester on the days fixed for such registration and notified in the academic calendar. Students who fail to register on or before the specified date will be allowed to register within one week of the last date by paying a late fee. Such courses together with their grade and credits earned will be included in the grade card issued by the University at the end of each semester, like ODD, EVEN, and summer and it forms the basis for determining the student's performance in that semester.

6.1.1 Each course will be identified by a unique Course Code of seven alpha-numerals (two alphabets followed by 5 digits). The alphabets reflect the discipline to which the course belongs. The first numeral (after the alphabet) indicates the learning level (based on prerequisites) of the course, and the rest of the three numerals indicate a running serial number. Each course also has its version to track the revisions carried out in its syllabus over time as represented by the last numerical separated by a hyphen (-). Example: EE1001-1 represents the course offered by EE Dept., Level-1, course serial number is 001 and the version is 1.

6.2 Mandatory Pre-Registration for higher semester

To facilitate proper planning of the academic activities of the Semester, the students must declare their intention to register for courses of higher semesters (3rd and above) at least two weeks before the end of the current semester choosing the courses offered by each department in the next higher semester which is displayed on the Department Notice Board at least 4 weeks before the last working day of the semester.

Registration to a higher semester is allowed only if the student fulfills the following conditions.

- Satisfied all the academic requirements to continue with the program of studies.
- Cleared all Institute, hostel, and library dues and fines, if any, of the previous semester.
- Paid all required fees of the Institute and the hostel for the current semester.
- Has not been debarred from registering on any specific grounds by the Institute.

6.3 Registering for Backlog Courses

- i) Students who have not cleared a course (Theory/ Lab/ project) are shown with “F” grade. A course having an ‘F’ grade will be considered as a backlog and it has to be re-registered in the subsequent semesters. F-graded courses are eligible to register for the next level course (pre-requisite is met).
- ii) Re-registration fee will be as per the university norms existing at the time of re-registration. When a course is re-registered, the evaluation marks of that course shall be treated as canceled/ reset.
- iii) To provide an early opportunity for students to clear their backlog of courses, efforts will be made to offer as many courses as possible during Odd, Even and summer semesters.

7. ADD/DROP/AUDIT OPTIONS

7.1 Registration of courses

Each student shall have to register for course work at the beginning of a semester within 2 to 3 days of commencement after discussing with the course teacher and under faculty advice. The permissible course load is to be either average credits (20) or to be within the limits of minimum (16) and maximum (28) credits.

7.2 DROP-option

During a specified period in the middle of a semester student’s performance in CIE is reviewed by the faculty advisor. Following a poor performance by a student, he/she can be facilitated to drop identified course(s) (up to the minimum credits

specified for these semester). Such course(s) will not be mentioned in the Grade card. Such courses are to be re-registered by these students and taken up for study at a later point in time.

7.3 **Withdrawal from courses (Letter Grade “W”)**

During a specific period specified towards the end of the semester, a student's performance in CIE is reviewed by the faculty advisors. Following a poor performance by a student in the identified course (s) he/she is advised to withdraw from such course(s) (up to the minimum credits specified for the semester) with a mention in the Grade card (Grade “W”). Such courses to be re-registered by these students and taken up for study at a later point in time.

7.4 **AUDIT-option (Letter Grade “U”)**

A student can register for courses for audit only, to supplement his/her knowledge and/or skills. The audit courses shall not be considered in determining the student's academic performance (SGPA and CGPA) in the semester. “U” grade is awarded to such courses and will be reflected in the grade card on satisfying the attendance requirements and CIE requirements. The candidate need not appear for SEE in such courses. However, CORE courses shall not be made available for audit.

8. **COURSE STRUCTURE:**

8.1 **Types of courses**

A “Course” is defined as a unit of learning that typically lasts one semester, led by one or more teachers, for a fixed roster of students. A course has identified course outcomes, modules/units of study, specified teaching-learning methods, and assessment schemes. A course may be designed to include lectures, tutorials, practical, laboratory work, field work, project work, internship experiences, seminars, self-study components, online learning modules, etc. in any combination. The following types of courses are included in the B. Tech. program:

- (a) **Humanities, Social Sciences, and Management Courses (HSMC):** These are common courses for all disciplines.
- (b) **Basic Science Courses (BSC):** Physics, Chemistry, and Mathematics: These are mandatory for all disciplines.
- (e) **Engineering Science Courses (ESC):** Basics of Electrical/ Electronics/ Civil/ Mechanical/ Computer Engineering, etc. These are mandatory for all disciplines.
- (f) **Professional Core Courses (PCC):** These are the professional Core Courses, relevant to the chosen specialization/ branch. The core courses shall be compulsorily studied by students, and it is mandatory to complete them to fulfill the requirements of a Program.
- (g) **Professional Elective Courses (PEC):** These are professional Electives, relevant to the chosen specialization/branch and can be chosen from the pool of courses. It shall be supportive to the discipline providing extended scope/enabling exposure to some other discipline /domain and nurturing student proficiency skills.
- (h) **Open Elective Courses (OEC):** These are the Elective Courses from other technical areas and/ or emerging fields. Students of other departments shall opt

for these courses for fulfilling the eligibility and prerequisites mentioned in the syllabus.

- (i) **Integrated Professional Core Courses (IPCC):** It refers to Professional Theory Core Course Integrated with Practical of the same course. Credit for IPCC shall be 04 considering L: T: P as 3:0:1 or L: T:P as 2:1:1, (where L, T, and P represent credits not hours per week)
- (j) **Holistic Education Courses (HEC):** These courses are designed to look into the emotional, social, ethical and academic needs of students in an integrated learning format. It helps in the engagement of all aspects of the learner including body, mind and spirit.
- (k) **Vocational Education Courses (VEC):** These courses are designed to prepare students for jobs that are based on manual or practical activities, traditionally non-academic related to a specific trade, occupation or vocation.
- (l) **Emerging Technology Courses (ETC):** These courses are designed to teach students about developing technologies that will be available within the next five to ten years and are expected to create significant social or economic effects.
- (m) **Programming Language Courses (PLC):** These courses are designed to teach students languages that can be used to communicate with computers for developing and working on different applications.
- (n) **University Core Courses (UCC): These are compulsory core courses with common course codes across all the disciplines.**
 - i. **Project Work (PROJ):** Provide experiential learning opportunities for students. Students are required individually, or in a small group, to select and complete a project that may include review, design, development, curation, analysis, etc. with the application of skills and knowledge relevant to the area of study. Mini-project and Project work carried out at the parent Institution, or any university / Government recognized organization without affecting the regular class work.
 - ii. **Internship (INT):** The internship (a form of experimental learning) program is a workplace-based professional learning experience that offers supervised exposure to real-life work experience in an area related to the field of study or career interest. An internship may be undertaken at a workplace such as an industry/R&D organization/Government organization, or any other reputed organization/ institution recognized for the purpose by the University. The internship program not only helps fresh pass-outs in gaining professional know-how but also benefits corporate sectors. The internship also enhances the employability skills of the student passing out from Technical Institutions.
- (o) **Mandatory Non-Credit Courses (MNC):** These courses are mandatory, without the benefit of a grade or credit, passing each mandatory course is required to qualify for the award of a degree.
 - Assessment of these courses is conducted in the college and will include Continuous Internal Evaluation (CIE). University Semester End Evaluation (SEE) may not be necessary for these courses.

- A minimum of 40% of the prescribed marks of CIE and SEE (If any) are required to secure a passing grade in these courses.
 - The ‘PP’ grade is awarded for a Pass in the course and the ‘NP’ grade is awarded for a Fail in the course. In case an ‘NP’ grade is awarded, the student has to re-register for the same course wherein he has no alternative options.
 - The “PP” and “NP” grades do not carry grade points and are hence not included in the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA) computations. However, such non-credit mandatory courses are required to be included in the students’ performance records (transcript) with Pass or Fail (PP or NP).
 - Courses that come under this category are the following.
 - Engineering Visualization, Employability Skill Development, Environmental Science, Kannada etc.
- (p) **Ability Enhancement Courses (AEC)** These courses are designed to help students to enhance their skills in language, communication, personality development, etc. They also promote a deeper understanding of courses like social sciences, ethics, culture, human behavior human rights, and the law. Ability Enhancement Courses are based upon the content that leads to Knowledge enhancement.

8.2 Typical Breakdown for the B.Tech. Degree Curriculum:

Sl. No.	Course Category	Credit Range	Suggested Credits
1.	Basic Science Courses (BSC)	18-23	22
2.	Engineering Science Courses (ESC)	10-15	13
3.	Emerging Technology Courses (ETC)	03-05	03
4.	Programming Language Courses (PLC)	03-05	03
5.	Professional Core Courses (PCC)	52 – 58	55
6.	Professional Elective Courses (PEC)	12-18	15
7.	Open Elective Courses (OEC)	6	6
8.	Humanities, Social Sciences and Management courses (HSMC)	09-15	15
9.	Ability Enhancement Courses (AEC)	8	8
10.	Mandatory Non-credit Courses (MNC)	Non-Credit	0
11.	Holistic Education Courses (HEC)	2	2
12.	Vocational Education Courses (VEC)	1	1
13.	Project Work (PROJ) (UCC)	10-12	10
14.	Internship (INT) (UCC)	8-12	11
15.	Note: Student can register between 16 to 28 credits per semester		164
	Total minimum Credits to be earned: 164		

- i) The Department Undergraduate Committee (DUGC) will discuss and recommend the exact credits offered for the program for the above components, the semester-wise distribution among them, as well as the syllabi of all undergraduate courses offered by the department from time to time before sending the same to the Board of Studies (BOS). The BOS will consider

the proposals from the departments and make recommendations to the Academic Council for consideration and approval.

8.3 The earned Credit Requirements for the B.Tech. Degree is 164.

Degree is awarded by prescribing the total number of credits to be earned, rather than by using the program duration, giving flexibility to a student to plan their career.

8.4 Program structure and suggested Course offerings

I SEMESTER (AI&DS, AI&ML, CC, CS, IS, RI)

SI No.	Course and Course code		Course Title	Teaching Department	Teaching hours/Week			SL #	Examination				Credits
					Theory Lecture	Tutorial	Practical/ Drawing		Duration in hours	CIE	SEE	Total Marks	
					L	T	P						
1	BSC	MA1002 – 1	Calculus and Differential Equations	MA	3	0	0	2	3	50	50	100	3
2	BSC	PH1004-1	Quantum Computing and Modern Physics	PH	3	0	2	2	3	50	50	100	4
3	ESC	CS1001-2	Problem Solving through Programming	CS	3	0	2	2	3	50	50	100	4
4	ESC	EC1001-1	Basic Electronics	EC	3	0	0	2	3	50	50	100	3
5	ETC	IS1101-1	Fundamentals of Cyber Security	CS	3	0	0	2	3	50	50	100	3
6	HSMC	HU1001-1	Technical English	HU	1	0	2	-	3	50	50	100	2
7	UCC	HU1002-1	Constitution of India	HU	1	0	0	-	0	50	0	50	1
8	BSC	MA1006 – 1	Mathematics with MATLAB	MAT	0	0	2	-	0	50	0	50	1
TOTAL					17	0	8	10	18	400	300	700	21
Additional Academic activities beyond class hours					-	-	-	5					
Learning hours per week					17	-	8	15					
Total notional learning hours (L+T+P+J+SL)			Per week		40								
			Per Semester		40x16=640								

II SEMESTER (AI&DS, AI&ML, CC, CS, IS, RI)

Sl. No.	Course and Course code		Course Title	Teaching Department	Teaching hours/Week			SL #	Examination				Credits
					Theory Lecture	Tutorial	Practical/ Drawing		Duration in hours	CIE	SEE	Total Marks	
					L	T	P						
1	BSC	MA1007 – 1	Discrete Mathematics and Transform Techniques	MA	4	0	0	3	3	50	50	100	4
2	BSC	CY1003-1	Materials Chemistry for Computer Systems	CY	3	0	2	2	3	50	50	100	4
3	ESC	EC1002-2	Applied Digital Logic Design	EC	2	0	2	1	3	50	50	100	3
4	PLC	CS1005-2	Introduction to Python Programming	CS	2	0	2	1	3	50	50	100	3
5	ESC	EE1001-2	Basic Electrical Engineering	EE	2	0	2	1	3	50	50	100	3
6	AEC	BT1651-1	Biology for Engineers	BT	1	0	0	-	1	50	50	100	1
7	ESC	ME1004-1	Engineering Visualization	ME	0	0	2	-	0	50	0	50	1
8	MNC	CV1005-1	Environmental Studies and Sustainability	CV	1	0	0	-	0	50	0	50	0
TOTAL					15	0	10	8	16	400	300	700	19
Additional Academic activities beyond class hours					-	-	-	3					
Learning hours per week					15	-	10	11					
Total notional learning hours (L+T+P+J+SL)			Per week		36								
			Per Semester		36x16=576								

Mandatory Internship-I*								
1.	INT	UC1001-1	Internship – I	Mandatory Intra Institutional Internship of duration (80 - 90 Hours) to be completed during I & II Semesters. *The grades will be included in the IV semester grade card (Refer 11.5.2 for details)	100	--	100	2

III SEMESTER											
Sl. No.	Course and Course code		Course Title	Teachin g Dept.	Contact Hours/week				SL*	Credits	
					L	T	P	J			
1.	BSC	MA2001-1	Statistics and Probability Theory	MA	3	0	0	0	1	3	
2.	IPCC	CS2001-1	Data Structures	CB	3	0	2	0	2	4	
3.	IPCC	CS2002-1	Object Oriented Programming	CB	3	0	2	0	2	4	
4.	PCC	CS2101-1	Computer Organization and Architecture	CB	3	0	0	0	1	3	
5.	PCC	CB1102-1	Fundamentals of Data Communications	CB	3	0	0	0	1	3	
6.	PCC	CB1602-1	UNIX and Shell Programming	CB	0	0	2	0	2	1	
7.	HSM C	HU2002-1	Developing Professional Skills	HU	1	0	0	0	-	1	
8.	HEC	HU1005-1	Essence of Indian Culture	HU	1	0	0	0	-	0	
9.	AEC	ME1654-1	Innovation and Design Thinking	Any Dept.	1	0	0	0	1	1	
10.	MNC	HU1003-1	Kannada (Balake / Samskrithika)	Any Dept.	1	0	0	0	-	0	
TOTAL					19	0	6	0	10	20	
Additional Academic activities beyond class hours					-	-	-	-	3		
Learning hours per week					19	0	6	-	13		
Total notional learning hours (L+T+P+J+SL)				Per week	38						
				Per Semester	38						
					X						
					16						
					=						
					608						

*SL: Supervised Learning

Course prescribed to lateral entry Diploma holders admitted to III semester of Engineering programs													
10	MNC	MA1007-1	Bridge Course – Probability and Differential Equations	MA	3	0	0	0	3	100	0	100	0

IV SEMESTER										
Sl. No.	Course and Course code		Course Title	Teach ing Dep t.	Contact Hours/week				SL*	Credits
					L	T	P	J		
1.	BSC	MA2005-1	Linear Algebra and its Applications	MA	3	0	0	0	2	3
2.	IPCC	CS3004-1	Design and Analysis of Algorithms	CB	3	0	2	0	2	4
3.	IPCC	CS3005-1	Microprocessor and Embedded Systems	CB	3	0	2	0	2	4
4.	PCC	CS2103-1	Software Engineering and Project Management	CB	3	0	0	0	1	3
5.	PCC	CB2102-1	Fundamentals of Computer Networks	CB	3	0	0	0	1	3
6.	PCC	CB2601-1	UNIX System Programming and Networking Lab	CB	0	0	2	√	2	1
7.	HSM C	HU1004-1	Universal Human Values	Any Dept.	1	0	0	0	-	1
8.	HSM C	HU2001-1	Enhancing Self Competence	HU	2	0	0	0	-	2
9.	VEC	CB1551-1	Department specific Vocational Education Course - NPTEL course 4 weeks	-	-	-	-	-	1	1
10.	UCC	UC1001-1	Internship – I (Activity based Internship)	-	-	-	-	-	-	2
TOTAL					18	0	6	0	11	24
Additional Academic activities beyond class hours					-	-	-	-	10	
Learning hours per week					18	0	6	-	21	
Total notional learning hours (L+T+P+J+SL)				Per week	45					
				Per Semester	45 X 16 = 720					

V SEMESTER

Sl. No.	Course and Course code		Course Title	Teaching Dept.	Contact Hours/week				SL *	Credits
					L	T	P	J		
1.	IPCC	CB3001-1	Information and Network Security	CB	3	0	2	0	2	4
2.	IPCC	CB2004-1	Principles of Database Management Systems	CB	3	0	2	0	2	4
3.	PCC	CB3003-1	Essentials of Operating Systems	CB	3	0	0	0	2	3
4.	PCC	CB3603-1	Secure Full Stack Web Development	CB	0	0	2	√	2	1
5.	PEC	CSxxxx-x	Professional Elective -1 [Group 1]	CB	3	0	0	0	1	3
6.	HSM C	HU1006-1	Introduction to IPR	Any Dept.	1	0	0	0	1	1
7.	AEC	CS3601-1	Program Specific Ability Enhancement Course Back End Development	CB	1	0	2	0	-	2
		HU1010-1	Research Methodology	Any Dept.	2	0	0	0		
8.	AEC	HU1007-1	Social Connect & Responsibility	Any Dept.	1	0	0	0	-	1
9.	AEC	UM1003-1	Employability Skill Development	Any Dept.	1	0	0	0	1	1
TOTAL					16	0	8	0	11	20
Additional Academic activities beyond class hours					-	-	-	-	3	
Learning hours per week					16	0	8	-	14	
Total notional learning hours (L+T+P+J+SL)				Per week	38					
				Per Semester	38 X 16 = 608					

Value added Course

9	Value added Course	Introduction to DevOps		0	0	2	0	3	0	PP/NP
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*SL: Supervised Learning

VI SEMESTER										
Sl. No.	Course and Course code		Course Title	Teaching Dept.	Contact Hours/week				SL*	Credits
					L	T	P	J		
1.	IPCC	CB2003-1	Machine Learning for Cyber Security	CB	3	0	2	0	2	4
2.	PCC	CB3101-1	Firewall and UTM Architecture	CB	3	0	0	0	2	3
3.	PCC	CB3602-1	Ethical Hacking and Network Defense	CB	0	0	2	0	2	1
4.	PEC	CBxxxx-1	Professional Elective -II [Group 1]	CB	3	0	0	0	1	3
5.	PEC	CBxxxx-1	Professional Elective -III [Group 2]	CB	3	0	0	0	1	3
6.	OEC	XXX5X X-1	Open Elective –I	Any Dept.	3	0	0	0	2	3
7.	HSM C	MG1003 -1	Management & Entrepreneurship	Any Dept.	3	0	0	0	1	3
8.	AEC	HU1008 -1	Life Skills for Engineers	Any Dept.	1	0	0	0	1	1
TOTAL					19	0	4	0	12	21
Additional Academic activities beyond class hours					-	-	-	-	4	
Learning hours per week					19	0	4	-	16	
Total notional learning hours (L+T+P+J+SL)				Per week	39					
				Per Semester	39 X 16 = 624					

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VII SEMESTER

Sl. No.	Course and Course code		Course Title	Teaching Dept.	Contact Hours/week				SL *	Credits
					L	T	P	J		
1.	IPCC	CB3002-1	Cyber Forensics	CB	3	0	2	0	2	4
2.	PCC	CB3603-1	Secured Mobile Application Development	CB	0	0	2	0	2	1
3.	PEC	CBxxxx-x	Professional Elective -IV [Group 1]	CB	3	0	0	0	1	3
4.	PEC	CBxxxx-x	Professional Elective -V [Group 2]	CB	3	0	0	0	1	3
5.	OEC	XXX5X X-1	Open Elective –II	Any Dept.	3	0	0	0	1	3
6.	HSM C	MG1002-1	Financial Management	Any Dept.	3	0	0	0	1	3
7.	HEC	HU1009-1	Indian Knowledge Systems	Any Dept.	1	0	0	0	-	1
8.	UCC	UC3001-1	Major Project Phase I	CB	-	-	4	-	3	2
TOTAL					16	0	8	0	11	20
Additional Academic activities beyond class hours					-	-	-	-	3	
Learning hours per week					16	0	8	-	14	
Total notional learning hours (L+T+P+J+SL)				Per week	38					
				Per Semester	38 X 16 = 608					

VIII SEMESTER

Sl. No.	Course and Course code		Course Title	Teaching Dept.	Contact Hours/week				SL*	Credits
					L	T	P	J		
1.	U C C	UC2001-1	Internship- II (Societal internship and Research/Industry Internship)		Mandatory Societal internship for 2 weeks (80 – 90 h) and Research Internship / Industry Internship of 6 weeks (240 – 270 h) or Research Internship / Industry internship for a total of 8 weeks (320 – 360 h) to be completed in one/two stretches during the vacation periods between IV to VII semesters					8
2.	U C C	UC3002-1	Major Project Phase II	CB	Students should carry out projects in research institute/industry/intra institute Center of Excellences. Two contact hours /week for interaction between the project guide and students.					8
TOTAL					-	-	-	-		16
Additional Academic activities beyond class hours					-	-	-	-		
Learning hours per week					-	-	-	-		
Total notional learning hours (L+T+P+J+SL)				Per week	-					
				Per Semester	680					

8.5 Eligibility for submission of Project Work Report

- Project work during the 8th semester shall be taken up batch-wise and report can be submitted for evaluation only on completion of a minimum of **122 credits** and for Diploma lateral entry students (those who have joined the second year B.Tech.) the same is **88 credits**.
- Project work can be carried out as domain-specific /interdisciplinary under the guidance of faculty/ faculty members. They can also opt for an advanced Internship or research Internship in an Industry / Research Institution/Center of excellence.
- Project viva-voce examination shall be conducted individually.

8.6 ELECTIVES

- A candidate shall take electives in each semester from groups of electives, commencing from the 5th semester.
- The minimum number of students to be registered for any Elective offered shall not be less than fifteen (15) and should not exceed forty (40).
- A candidate shall opt for his/her choice of electives and register for the same at the beginning of each of the 5th to 7th semesters if pre-registration is not done. The candidate is permitted to opt for a change of elective within 15 days from the date of commencement of the semester as per the academic calendar of the college.

9. ATTENDANCE REQUIREMENT:

- 9.1** Each semester is considered as a unit and the candidate has to put in a minimum attendance of 85% in each subject with a provision of condoning 10% of the attendance by the Principal for reasons such as medical grounds, participation in University level sports, cultural activities, seminars, workshops, and paper presentation.
- 9.2** The basis for the calculation of the attendance shall be the term prescribed by the institution by its calendar of events. For the first semester students, the same is reckoned from the date of admission to the course.
- 9.3** The students shall be informed about their attendance position in the first week of every month by the College so that the students shall be cautioned to make up for the shortage.
- 9.4** A candidate having a shortage of attendance (<75%) in any course(s) registered shall not be allowed to appear for SEE of such course(s). Such students will be awarded an ‘N’ grade in these courses.
- 9.5** He/she shall have to repeat those course(s) with an ‘N’ grade and shall re-register for the same course(s) core or elective, as the case may be when the particular course is offered next either in a main (odd/even) or summer semester.
- 9.6 Attendance in CIE and SEE:**
Attendance in all examinations both CIE and SEE of each course registered shall be compulsory and there shall not be any provision for re-examinations. Any student against whom any disciplinary action is pending shall not be permitted to attend any SEE in that semester.

10. WITHDRAWAL FROM THE PROGRAM

10.1 Temporary Withdrawal

- a) A student who has been admitted to a degree program of the college may be permitted once during the course to withdraw temporarily, for one semester, on the grounds of prolonged illness or grave calamity in the family, etc., provided—
- The student applies to the College within 6 weeks of the commencement of the college stating fully the reasons for withdrawal together with supporting documents and endorsement from his parent/guardian.
 - The College is satisfied with the genuineness of the case and that even by considering the expected period of withdrawal, the student can complete the program requirements (credits) within the time limits specified by the university.
 - The student does not have any dues or demands at the College / University including tuition and other fees as well as library material.
 - A student availing of temporary withdrawal shall be required to pay such fees and/or charges as may be fixed by the college until his/her name appears on the student’s roll list. The fees/charges once paid shall not be refunded.
 - A student will be entitled to avail of the temporary withdrawal facility only once during his/her studentship. However, any other concession for the concerned student shall have to be approved by the academic council.

10.2 Permanent Withdrawal

Any student who withdraws the admission before the closing date of admission for the Academic Session is eligible for the refund of the deposits only. Fees once paid will not be refunded on any account.

Once the admission for the year is closed, the following conditions govern withdrawal of admissions.

- i) A student who wants to leave the College for good will be permitted to do so (and take a Transfer Certificate from the College, if needed), only after clearing all other dues if any.
- ii) Those students who have received any scholarship, stipend, or other forms of assistance from the College shall repay all such amounts.
- iii) The decision of the Principal of the College regarding the withdrawal of a student is final and binding.

11. EVALUATION SYSTEM

11.1 The Academic Performance Evaluation of a student shall be according to a Letter Grading System, based on the Class Performance Distribution.

11.2 The Letter grades O, A+, A, B+, B, C, P, and F indicate the level of academic achievement, assessed on a decimal (0-10) scale.

11.3 The Letter grade awarded to a student in a course, for which he has registered shall be based on his performance in quizzes, tutorials, assignments, etc., as applicable, in addition to two mid-semester examinations and one semester-end examination. The distribution of weightage among these components may be as follows.

Semester End Examination (SEE)	:	50% (50 marks)
Continuous Internal Evaluation (CIE)	:	50% (50marks)
CIE for Non-PBL Courses		
i) Quizzes, Tutorials, Assignments, Seminars, etc.	:	10 marks
ii) Mid-semesterExaminations	:	40marks
CIE for PBL/IPCC Courses		
i) Project Based Learning (PBL)	:	50 marks
ii) Mid-semesterExaminations	:	40 marks
iii) Quizzes, Tutorials, Assignments, Seminars, etc.	:	10 marks
<i>60% weightage for theory + 40% weightage for PBL/Practical</i>		

Any variation, other than the above distribution, requires the approval of the pertinent DUGC and Academic Council.

11.4 The letter grade awarded to a student in a 0-0-P (Practical) course, is based on an appropriate continuous evaluation scheme that the course instructor shall evolve, with the approval of the pertinent DUGC and the performance in SEE held on the specified period in a semester.

11.5 Evaluation Scheme (*Refer to Appendix-B for detailed evaluation guidelines*): The course Instructor shall announce in the class and/or display at the Notice board/faculty door/website the details of the Evaluation Scheme, including the distribution of the weightage for each of the components and method of conversion from the raw scores to the letter-grades within the first week of the semester in which the course is offered so that there are no ambiguities in communicating the same to

all the students concerned.

i) **Internship:** Mandatory Internship is in two parts. Internship-I (2 weeks) and Internship-II (8 weeks)

ii) **Internship-I**

- All the students admitted to the 1st semester of engineering programs shall have to undergo Internship-I of 02 weeks (or 80 to 90 hrs duration) during the first year. The internship shall include Inter / Intra Institutional activities. A viva – voce examination (Presentation followed by question-answer session) shall be conducted during the 2nd semester (for lateral entry students, during the 3rd semester) and the prescribed credit shall be included in the 4th-semester grade card.
- All the students admitted to the 3rd semester of Engineering programs (Lateral Entry Category) shall have to undergo a mandatory internship of 02 weeks (during the 3rd semester or the intervening period of the 3rd and 4th semesters). The internship shall include Inter/Intra Institutional activities.
- The internship shall be considered as a head of passing and shall be considered for the award of a degree. Those, who do not take up / complete the internship shall be declared to fail and shall have to complete it during subsequent University examinations after satisfying the internship requirements. (The faculty coordinator or mentor has to monitor the student's internship progress and interact to guide them for the successful completion of the internship).
- **Procedure for the Evaluation of Internship-I**
 - a) Students should submit the reports immediately on completion of the Internship to the respective mentors.
 - b) The Examination of the internship will be carried out by the mentor.
 - c) The Internship-I shall be slated for 100 marks CIE only and will not have SEE.
 - d) Internship-I marks are based on CIE marks (25 marks for the first presentation, 25 marks for the second presentation, and 50 marks for the report and final presentation).
 - e) A Viva-Voce examination is conducted during I/II/III Semesters (Presentation followed by question-answer session) and the prescribed credit shall be included in the IV semester grade card.

iii) **Internship-II**

- All the students admitted to engineering programs shall have to undergo Internship-II of 08 weeks during the second and third year of their Engineering studies.
- During the intervening period of the IV & V semesters and VI & VII semesters, students shall be ready for industrial experience. Therefore, they shall choose to undergo 8 weeks Internship involving Innovation / Entrepreneurship/ or short-term (about 2 weeks) societal-related activities and 6 weeks Industry Internship.

iv) **Project work evaluation:** The evaluation of CIE of the project work shall be based on the progress of the student in the work assigned by the project

supervisor, periodically evaluated by him/her together with a department committee constituted for this purpose. Seminar presentation, project report, and final oral examination conducted by the project evaluation committee at the department level shall form the SEE of the projectwork.

- v) In the case of other requirements, such as seminar, field work, or comprehensive viva voce, if any, the assessment shall be made as laid down by the DUGC/Academic Council.
- vi) There shall be no re-examination for any course in the credit system.

However, students

- who have abstained from attending CIE or SEE without valid reasons (“N” grade), or
- who have failed (F grade) to meet the minimum passing standards prescribed for CIE and/or SEE or
- who have been detained for shortage of attendance or who have withdrawn (W grade) who have dropped any course shall be required to re-register for such course(s) and go through CIE and SEE again and obtain a grade equal to or better than “P” Grade in each case.
- While such students should re-register for the same course(s) if core, they can re-register for the alternative course(s) from among the elective courses, as the case may be. The re-registration shall be possible when the particular course is offered again either in a main (Odd/Even) or summer semester.

11.6 Qualifying standards

Evaluation Method	Qualifying Standard
Sessional (CIE)	Score: $\geq 40\%$ (≥ 20 marks)
Terminal (SEE)	Score: $\geq 40\%$ (≥ 20 marks)
For securing a final Pass	Total 40 % of the Course maximum marks (100) i.e., the sum of the CIE and SEE marks prescribed for the Course is desired.

11.7 Grading System

The letter grade awarded to a student for his/her performance in a course is based on Absolute Grading.

- i) Absolute Grading – Letter Grade and its range

The grade point scale for absolute grading

Marks Range (%)	Grade Point	Letter Grade	Descriptor
90 & above	10	O	Outstanding
80-89	9	A+	Excellent
70-79	8	A	Very Good
60-69	7	B+	Good

CGPA	Classification
7.00-& above	First Class with Distinction
6.00-6.99	First Class

55-59	6	B	Above Average
50-54	5	C	Average
40-49	4	P	Pass
00-39	0	F	Fails
Absent	0	Ab	Absent

5.00-5.99	Second Class
CGPA < 5.00*	Academic Probation / Non-compliance

*If a student secures CGPA < 5.0 at any point time during his/her studies, he/she will be on Academic Probation/Noncompliance (refer to sections 14.2 and 17.3 for more details.)

- i) **Grade “N”:** A candidate having a shortage of attendance (<75%) in any course(s) or CIE marks less than 40% shall not be allowed to appear for SEE of such course(s). Such students will be awarded an ‘N’ grade in these courses with a grade point of 0.
- ii) The grade points are given above help in the evaluation of credit points earned by the student in a course as the credit points are equal to the number of credits assigned to the course multiplied by the grade points awarded to the student in that course. This shall be used in Arriving at the credit index of the student for that semester, as it is the total of all the credit points earned by the student for all the courses registered in that semester.

11.8 Earning of Credits

A student shall be considered to have completed a course successfully and earned the credits if he/she secures an acceptable letter grade in the range of O-P. The letter grade “F” in any course implies the failure of the student in that course and no credits earned.

- i) The Transitional Grades “I”, “W” and “X” would be awarded by the teachers in the following cases. These would be converted into one or the other of the letter grades (O-F) after the student completes the course requirements.
- ii) **Grade “I”:** To a student having attendance $\geq 85\%$ and CIE $\geq 70\%$, in a course, but remained absent from SEE for valid & convincing reasons acceptable to the College, like:
 - a) Illness or accident, which disabled him/her from attending SEE.
 - b) A calamity in the family at the time of SEE required the student to be away from the College.
 - c) However, the committee chaired by the Principal is authorized to relax the requirement of CIE $\geq 70\%$ if the student is hospitalized or advised long-term rest after discharge from the hospital by the Doctor.
 - d) Students who remain absent for Semester End Examinations due to valid reasons and those who are absent due to health reasons are required to submit the necessary documents along with their request to the Controller of Examinations to write Makeup Examinations within 2 working days of that examination for which he or she is absent, failing which they will not

be given permission.

- iii) **Grade “W”:** To a student having satisfactory attendance at classes, but withdrawing from that course before the prescribed date in a semester under Faculty Advice
- iv) **Grade “X”:** To a student having attendance $\geq 85\%$ and CIE $\geq 70\%$, in a course but SEE performance could result in an F grade in the course. **(No “F” grade will be awarded in this case, but the student’s performance record is maintained separately).**

11.9 Summer / Fast Track semester

- i) The students who have satisfied CIE and Attendance requirements for the course/s and obtained an F grade in SEE are permitted to appear directly in ensuing examination/s as backlog paper/s. The students need not re-register for such course/s in the summer / fast track semester. In case the student wishes to improve CIE/ he/she has to re-register for the summer / regular semester as and when offered next.
- ii) The student who obtains required attendance and CIE in the summer semester, but obtains an 'F' grade in SEE; is permitted to appear for SEE subsequently as backlog course/s. The student need not repeat the course for Attendance and CIE.
- iii) The course/s for which the student does not possess satisfactory attendance and CIE score shall be marked as ‘N’ on the Grade sheet. Such students are not permitted to SEE for the Courses marked as ‘N’ on the Grade sheet. The students have to re-register only for course/s marked as ‘N’ in the summer/ subsequent semester whenever that course is offered and obtain the required CIE and attendance. Subsequently, they are eligible to appear for SEE in such course/s.
- iv) Courses with Transitional Grades viz "W", "I", and "X" are also eligible to register in the summer semester in case they wish to improve their score in CIE.
- v) All courses may not be offered in the summer semester. It is the discretion of the University to offer the courses based on the availability of resources. The Institutes shall notify timetable for the summer semester well in advance.
- vi) Summer Semester is optional; it is for the student to make the best use of the opportunity.
- vii) A student is permitted to register for a maximum of 16 credits in the Summer / fast track semester.
- viii) A student has to choose those courses which are offered by the Institution in a given summer Semester.
- ix) In the summer semester, each course needs to be offered for the required number of lectures/ tutorial/ laboratory hours as prescribed in the syllabus.

11.10 Grade Card

Each student shall be issued a GradeCard at the end of each semester. This will have a list of all the courses registered by a student in the semester, together with their credits, the letter grades with grade points awarded. Only those courses registered for credit and having grade points shall be included in the computation of

the students' performance like SGPA and CGPA and the courses are taken for audit will not form part of this computation. The results of mandatory courses, which are of the non-credit type shall also be reflected in the Grade card as PP (for Passed) or NP (for not passed). **Each UG student shall have to obtain the grade PP in each mandatory course to qualify for the Degree awarded by the university.**

11.11 Re-evaluation and paper seeing.

Re-evaluation is permitted only for theory papers. The University, on receiving application within the stipulated time and remittance of a prescribed fee for re-evaluation, shall permit re-evaluation for the course/s applied. The marks obtained after re-evaluation shall be the final marks awarded.

11.12 The Make-Up Examination

The Make-Up Examination facility would be available to students who may have missed attending the SEE of one or more course(s) in a semester for valid reasons and given the "I" grade; Also, students having the "X" grade shall be eligible to take advantage of this facility. **The makeup examination would be held as per dates notified in the Academic Calendar during the summer semester.** However, it would be possible to hold a makeup examination at any other time in the semester with the permission of the Academic Council of the College. In all these cases, the standard of makeup examinations shall be the same as the regular SEE for the course(s).

- a) All the "I" and "X" grades awarded to the students would be converted to appropriate letter grades after the make-up examinations. Any outstanding "I" and "X" grades after the last scheduled make-up examinations shall be automatically converted to "F" grades.
- b) All the "W" grades awarded to the students would be eligible for conversion to the appropriate letter grades only after the concerned students re-register for these courses in a main/ Summer semester and fulfill the passing standards for their CIE and (CIE+SEE).

11.13 Rules for grace marks

Grace marks up to 1% of the maximum total marks of the courses for which he/she is eligible and have registered (non-credit courses excluded) in the examination or 10 marks whichever is less shall be awarded to the failed course(s), (with a restriction of a maximum of 5 marks per course) provided on the award of such grace marks the candidate passes in that course(s).

12. EVALUATION OF PERFORMANCE

The overall performance of a student will be indicated by two indices:

SGPA; which is the Semester Grade Point Average, and CGPA which is the Cumulative Grade Point Average.

SGPA for a semester is computed as follows.

$$SGPA = \frac{\sum[(Course\ Credits) \times (Grade\ Point)](\text{for all courses in that semester})}{\sum[Course\ Credits]}$$

CGPA is computed as follows:

$$CGPA = \frac{\sum[(Course\ Credits) \times (Grade\ Point)]}{\sum[Course\ Credits]}$$

(for all courses excluding those with F grades until that semester)

13. COMMUNICATION OF GRADES

The SGPA and CGPA respectively, facilitates the declaration of academic performance of a student at the end of a semester and the end of successive semesters. Both would be normally calculated to the second decimalposition.

14. REQUIREMENTS FOR VERTICAL PROGRESSION (PROMOTION / ELIGIBILITY TO HIGHER SEMESTERS)

14.1 All students are promoted to the next semester or year of their program, irrespective of their academic performance.

14.2 However, at any stage of his/her study, if a student reaches a CGPA below 5.00, the student will be on **Academic Probation** and is permitted to register for a maximum of 16 credits during odd semester of an academic year. However, the student has the choice to re-register for the courses/courses in which he/she has obtained an 'F' / 'N' grade.

14.3 A Student shall be declared fail if he/she

- (i) Has not satisfied the CIE requirements of anyCourse/s.
- (ii) Has not appeared for the SEE even after satisfying the attendance and CIE requirements.

14.4 Vertical Progression for regular students who have taken admission to the first year:

Normally a student is expected to complete a minimum of 85% of credits by the end of the 7th semester. However, **for submission of B.Tech. Major Project in 8th semester, the student should have completed at least 122 credits.**

14.5 Vertical Progression in case of Diploma students admitted to Second year (lateral entry):

- i) Lateral entry students should complete at least 85% of credits by the end of the 7th semester. However, **for submissionof B.Tech. Major Project in 8th semester, the student should have completed at least 88 credits.**
- ii) Diploma students should register for mandatory non-credit Mathematics Courses Bridge Courses (i) Calculus and Laplace Transforms and (ii) Probability and Differential Equations prescribed during III and IV semesters respectively. They shall attend these bridge course classes during the respective semesters to satisfy attendance and CIE requirements.
- iii) Completion of Mathematics Courses Bridge Courses (i) Calculus and Laplace Transforms and (ii) Probability and Differential Equations shall be mandatory for the award of the degree.

14.6 Termination from the program

A student shall be required to withdraw (discontinue) from the program and leave the college on the following grounds.

- i) Failure to secure a minimum CGPA of 5.0 at the end of the 8 years (6 years for Lateral entry students).
- ii) Failure to earn 164 credits (124 for lateral entry students) in 8 years (6 years for lateral entry students) of duration from the year of admission including the duration of temporary withdrawal (leave of absence).
- iii) Absence from classes for more than **six weeks at a time** in a semester without leave of absence being granted by competent authorities.
- iv) Failure to meet the standards of discipline as prescribed by the college from time to time.

15. AWARD OF CLASS

Sometimes, it would be necessary to provide equivalence of these averages, viz., SGPA and CGPA with the percentages and/or classes awarded as in the conventional system of declaring the results of university examinations. This can be done by prescribing certain specific thresholds in these averages for Distinction, First Class and Second Class. This can be seen in the following Table.

Percentage Equivalence of Grade Points (For a 10-Point Scale)

Grade Point	Percentage of Marks*	Class
≥ 7.00	$\geq 70\%$	First class with Distinction
≥ 6.00	$\geq 60\%$	First Class
$5.0 \geq \text{CGPA} < 6.00$	$50 \geq \text{Percentage} < 60\%$	Second Class

$$\text{Percentage} * = (\text{CGPA}) \times 10$$

16. APPEAL FOR REVIEW OF GRADES

- a) The entire process of evaluation shall be made transparent and the course instructor shall explain to a student why he/she gets whatever grade he/she is awarded, if and when required. A mechanism for the review of grades is incorporated into the evaluation system. However, before appealing for such review, a student shall first approach the concerned course Instructor and then the concerned DUGC, with the request to do the needful; and only in situations where satisfactory remedial measures have not been taken, the student may then appeal to the Department Academic Appeals Boards (DAAB) before the date specified in Academic Calendar, by paying the prescribed fees.
- b) The fee for such an appeal will be decided by the Senate from time to time. If the appeal is upheld by DAAB, then the fee amount will be refunded to the student.

17. AWARD OF DEGREE

17.1 B.Tech. Degree

- a) Students shall be declared to have completed the Program of B.Tech. degree and is eligible for the award of degree provided the students have undergone the stipulated Course work of all the semesters under the Scheme of Teaching and Examinations and have earned the prescribed number of credits (164 credits for regular students registered for 4-year degree programs & 124 for lateral entry students).

- b) For the award of a degree, a CGPA ≥ 5.00 at the end of the Program shall be mandatory.
- c) Completion of Additional Mathematics I and II shall be mandatory for the award of degree to lateral entry diploma students.
- d) **Earning of Activity Points:**
 - i. Every student entering 4-year degree program should earn 100 activity points & every student entering 4-year degree program through Lateral Entry should earn 75 activity points as per the AICTE Activity Point Program for the award of an Engineering degree.
 - ii. The activities can be spread over the years (duration of the program) at any time during the semester weekends and holidays, as per the interest & convenience of the students from the year of entry to the program.
 - iii. The Activity Points earned shall be reflected on the student's eighth-semester Grade Card.
 - iv. Activity Points (non-credit) do not affect SGPA/CGPA and shall not be considered for vertical progression.
 - v. In case students fail to earn the prescribed activity Points before the commencement of 8th-semester examinations, the eighth-semester Grade Card shall be issued only after earning the required activity Points. Students shall be admitted for the award of a degree only after the release of the Eighth semester Grade Card.

17.2 Honours/ Minors Degree

17.2.1 B.Tech. (Honours) Degree

- i. Students must earn a minimum of 18 additional credits in his/her major program discipline entitles a student to get an 'Honours' credential.
- ii. Students have to pay additional fees for all the courses registered for 'Honours'.
- iii. Students with a minimum of 7.5 CGPA and no backlog at the end of the 4th semester will qualify for registering for courses under the 'Honours' credential.
- iv. Students shall register for 'Honours' courses from the 5th semester onwards.
- v. Students should register for additional courses and plan to take courses that are prescribed under that 'Honours' list as per 'pre-requisite' courses to earn the 'Honours' credential.
- vi. Students who wish to acquire an 'Honours' credential need to carry out 'Honours' course registration along with their regular semester course registration.
- vii. He/she accumulates credits by registering for the required courses, and if the requirements for 'Honours' are met within the prescribed minimum time limit of the program, the 'Honours' will be awarded along with the degree.
- viii. Also, the student should meet the following **requirements to become eligible for the 'Honours' award.**
 - Minimum CGPA of 7.5 in this major discipline at the end of the 8th semester.

- Minimum CGPA of 7.0 in the registered 'Honours' courses.
- ix. In case a student withdraws from the 'Honours' registration in the middle of the program, the 'Honours' courses completed will be converted to 'Audit' courses and indicated accordingly in subsequent Grade Sheets and Consolidated Grade Sheets.
- x. It must be noted that the 'Honours' award will be mentioned in the Degree Certificate as **"Bachelor of Technology in (specialization) with Honours"**.
- xi. This fact will also be reflected in the Consolidated Grade Sheet under a separate heading 'Honours' with similar details shown for other credited courses and the CGPA for 'Honours' will be indicated at the end of the list of courses under 'Honours'.
- xii. The grades obtained in the courses credited towards the 'Honours' award are not counted and shall not influence the GPA/ CGPA of the 'program' student has registered.

17.2.2 Minor Degree

- i. Students have to earn a min of 18 additional credits from the courses focused on discipline other than his/her major program discipline entitles a student to get a 'Minor' credential.
- ii. Students have to pay additional fees for all the courses registered for 'Minor'.
- iii. Students with a minimum of 5.0 CGPA and no backlog at the end of the 3rd semester will only qualify for registering for the course under the 'Minor' credential.
- iv. Students shall register for 'Minor' degree courses from the 4th semester onwards.
- v. All Departments will offer 'Minors' in their varied disciplines and will prescribe what set of courses and/or projects is necessary for earning a minor in that discipline.
- vi. Students should register for additional courses and plan to take courses that are prescribed under that 'Minors' list as per 'pre-requisite' courses to earn the 'Minor' credential.
- vii. If any of the courses listed under the 'minor' option is a course listed under his/her curriculum as PCC then the student cannot opt for that 'Minor', since all minor courses need to be earned as additional courses to his/her program curriculum and depts decision is final and binding.
- viii. Students who wish to acquire a 'Minor' can register for 'Minor' courses along with their regular semester course registration.
- ix. Also, the student should have a minimum **CGPA of 5.0 in the 'Minor' courses registered to become eligible for the Minor credential**. This fact will also be reflected in the Consolidated Grade Sheet under a separate heading 'Minor in (specialization)'.
- x. If the course requirements for a particular 'Minor' are met within the prescribed minimum time limit of the program, the minor will be awarded along with the degree, and it will be mentioned in the **Degree Certificate as "Bachelor of Technology in (Major discipline) with**

Minor in (specialization).”

- xi. In case a student withdraws from the ‘Minor’, the ‘Minor’ courses completed, will be converted to ‘Audit’ courses and indicated accordingly in subsequent Grade Sheets and Consolidated Grade Sheets.
- xii. The grades obtained in the courses credited towards the ‘Minor’ award are not counted and shall not influence the GPA/ CGPA of the program the student has registered for.

17.2.3 Additional norms for Honours/Minors

- i. Students shall register for additional courses to earn Honours/Minors in consultation with their Class Advisor from the list of courses suggested by the DUGC.
- ii. DUGC may recommend Massive Open Online Courses (MOOCs)/ SWAYAM/ NPTEL courses to students who wish to register for Honours/Minors after justifying and establishing the equivalence of the curriculum. The decision of DUGC should be communicated to the Dean of Academics and Controller of Examinations for seeking approval.
- iii. A maximum of 40% credits prescribed for Honors/Minors may be earned through MOOCs/ SWAYAM/ NPTEL
- iv. Students may choose to take up additional course work, from the MOOCs courses list suggested by various departments (which can be from SWAYAM/NPTEL) with proctored examinations as approved by the University and complete the same before the last working day of the VIII semester with a final score (online assignments: 25 % + Proctored examination: 75 %) leading to the following certificates: Completed the course (40-59)– ELITE (60 to 75 %) or ELITE + SILVER (76 to 89 %) or ELITE + GOLD (≥ 90 %)
- v. In case, in MOOCs (ex: Coursera), there is no proctored examination, the University will conduct a SEE as deemed to be fit for the award of Credits.
- vi. The Credit equivalence for online courses shall be as follows –
 - 4 weeks of online course duration – 1 credit (approx. 13-14 hours)
 - 8 weeks of online course duration – 2 credits (approx. 26-28 hours) and
 - 12 weeks of online course duration – 3credits (approx. 39-42 Hours)

17.3 Noncompliance**17.3.1 Noncompliance of CGPA ≥ 5.00 at the end of the Program**

- a) Students, who have completed all the courses of the Program but do not have a CGPA ≥ 5.00 at the end of the Program, shall not be eligible for the award of the degree.
- b) In the cases of 17.3 (1), a student shall be permitted to appear again for SEE in course/s (other than Internship, Technical seminar, Project (Mini and Major), and Laboratories) of any Semester/s without the rejection of CIE marks for any number of times, subject to the provision of a maximum duration of the Program to make up the CGPA equal to or greater than 5.00 for the award of the Degree.
- c) Students shall obtain written permission from the Controller of Examinations to reappear in SEE to make up the CGPA equal to or

greater than 5.00.

- d) In case, the students earn improved grade/s in all the reappeared course/s, the CGPA shall be calculated considering the improved grade/s. If it is ≥ 5.00 , the students shall become eligible for the award of the degree. If $CGPA < 5.00$, the students shall follow the procedure laid in 17.3.1 (b).
- e) In case, the students earn improved grade/s in some course/s and the same or lesser than the previously earned pass grade/s in the other reappeared course/s, the CGPA shall be calculated considering the improved grade/s and the pass grades earned before the reappearance. If it is ≥ 5.00 , the students shall become eligible for the award of the degree. If $CGPA < 5.00$, the students shall follow the procedure laid in 17.3.1 (b).
- f) In case, the students earn improved grade/s in some courses and fail in the other reappeared course/s, the CGPA shall be calculated by considering the improved grade/s and the previously earned pass grade/s of the reappeared course/s in which the students have failed. If it is ≥ 5.00 , the students shall become eligible for the award of the degree. If $CGPA < 5.00$, the students shall follow the procedure laid in 17.3.1 (b).
- g) In case, the students fail (i.e., earns an F grade) in all the reappeared course/s, pass grade/s of the course/s earned by the students before reappearance shall be retained. In such cases, the students shall follow the procedure laid in 17.3.1 (b).

17.3.2 Noncompliance with Project/ Mini project

The project/mini project shall be considered as a head of passing and shall be considered for the award of degree. Those, who do not take up/complete the mini-project shall be declared to fail in that course and shall have to complete the same during subsequent University examinations after satisfying the Mini-project requirements.

17.3.3 Noncompliance of Internship

All the students of B. Tech shall have to undergo mandatory Internship-I and Internship-II for a total of 10 weeks to earn a total of 10 credits in parts during the vacations at the end of the 1/2/3 academic year. The evaluation of Internship shall be during IV and VIII semesters. The internship shall be considered mandatory for the award of a degree. Those, who do not take up/complete the internship shall be declared to fail in that Course and shall have to complete the same during subsequent University examinations after satisfying the internship requirements.

The maximum duration for a student for complying with the Degree requirements is 16 – semesters from the date of first registration for his/ her first semester (8 years from the date of admission to the first year, (12 semesters / 6 years from the date of admission for lateral entry student)).

18. GRADUATION REQUIREMENTS AND CONVOCATION

- 18.1** A student shall be declared to be eligible for the award of the degree if he/she has.
- Fulfilled “Award of Degree” Requirements
 - No Dues to the College, Departments, Hostels, Library, Central Computer Centre and any other centers
 - No disciplinary action is pending against him/her.
- 18.2** The award of the degree must be recommended by the Governing council.
- 18.3 Convocation:** Degree will be awarded to the students who have graduated during the preceding academic year. Students are required to apply for the Convocation along with the prescribed fees, after having satisfactorily completed all the degree requirements (refer to “Award of Degree”) within the specified date to arrange for the award of the degree during convocation.

19. AWARD OF PRIZES, MEDALS, CLASS & RANKS

- 19.1** For the award of Prizes and Medals, the conditions stipulated by the Donor may be considered as per the statutes framed by the University for such awards. Sometimes, it would be necessary to provide equivalence of these averages, viz., SGPA and CGPA with the percentages and/or Class awarded as in the conventional system of declaring the results of University examinations. This can be done by prescribing certain specific thresholds in these averages for Distinction, First Class, and Second Class as described in Section 15.
- 19.2** An attempt means the appearance/registration of a candidate for an examination in one or more courses either in part or failing a particular examination.
- A candidate who fails/remains absent (after submitting exam application) in the main examination and passes one or more subjects/courses or all subjects/courses in the supplementary/Make-up examination such candidates shall be considered as taken more than an attempt.
- 19.3** Merit Certificates and University Medals/ will be awarded based on overall CGPA, governed by the specific selection criteria that may be formulated by the University for such Medals / Awards
- Only those candidates who have completed the Program and fulfilled all the requirements in the minimum number of years prescribed (i.e., 3 years for Diploma lateral entry students or 4 years for students who joined after the 12th standard) and who have passed each semester in the **first attempt** are eligible for the award of Merit Certificates and /or University Medals.
 - Candidates with W, N, I, X & F grades and who passes the courses in the subsequent/ supplementary/ make up examinations are not eligible for the award of Gold Medal or Merit Certificate.

20. CONDUCT AND DISCIPLINE

- 20.1** Students shall conduct themselves within and outside the premises of the College in a manner befitting the students of an Institution of National Importance.
- 20.2** **As per the order of the Honorable Supreme Court of India, ragging in any form is considered a criminal offense and is banned. Any form of ragging will be severely dealt with.**

20.3 The following acts of omission/ or commission shall constitute a gross violation of the Code of Conduct and are liable to invoke disciplinary measures:

- i. Ragging.
- ii. Lack of courtesy and decorum; indecent behavior anywhere within or outside the campus.
- iii. Willful damage or stealthy removal of any property/ belongings of the College/ Hostel or fellow students/ citizens.
- iv. Possession, consumption, or distribution of alcoholic drinks or any kind of hallucinogenic drugs.
- v. Mutilation or unauthorized possession of Library books.
- vi. Noisy and unseemly behavior, disturbing studies of fellow students.
- vii. Hacking in computer systems (such as entering into another Person's area without prior permission, manipulation and/or Damage of computer hardware and software, or any other Cybercrime, etc.).
- viii. Plagiarism of any nature.
- ix. Any other act of gross indiscipline as decided by the Senate from time to time.
- x. Use of Mobile in the college Academic area.
- xi. Smoking in College Campus and supari chewing.
- xii. Unauthorized fundraising and promoting sales.
- xiii. Commensurate with the gravity of the offense the punishment may be: reprimand, expulsion from the hostel, debarring from an examination, disallowing the use of certain facilities of the College, rustication for a specified period or even outright expulsion from the College, or even handing over the case to appropriate law enforcement authorities or the judiciary, as required by the circumstances.

20.4 For an offense committed in (i) a hostel (ii) a department or a classroom and (iii) elsewhere, the Chief Warden, the Head of the Department, and the Dean (Academics), respectively, shall have the authority to reprimand or impose fine.

20.5 All cases involving punishment other than reprimand shall be reported to the principal.

20.6 Cases of adoption of unfair means and/or any malpractice in an examination shall be reported to the Controller of Examinations for taking appropriate action.

20.7 Note: Students are required to be inside the examination hall 20 minutes before the commencement of the examination. This is applicable for all examinations (Semester end/ Supplementary/ makeup) henceforth. Students will not be allowed inside the examination hall after the commencement, under any circumstances.

21. APPENDIX - A

Definitions, terminology, and abbreviations

1. Nitte DU / University

- a. Refers to Nitte (Deemed to be University)

2. EC

- a. Refers to Executive Council of Nitte (Deemed to be University)

3. BoS

- a. Refers to the Board of Studies in Computer Science and Engineering

4. Institute/Institution

- a. Refers to NMAM Institute of Technology, Nitte

5. Program

- a. A range of learning experiences over a specified period, leading to the award of a degree/diploma/certificate. A program is completed when the courses that make up the program are completed, and other requirements as specified in the program regulations are met.

6. Course

- a. A unit of learning that typically lasts one semester, led by one or more teachers, for a fixed roster of students. Often referred to as a “subject”. A course has identified course outcomes, modules/units of study, specified teaching-learning methods, and assessment schemes. A course may be designed to include lectures, tutorials, practical, laboratory work fieldwork, project work, internship experiences, seminars, self-study components, online learning modules, etc. in any combination.

7. Semester

- a. An academic session, usually of 16 weeks duration, with a minimum of 90 working days during which coursework and assessments are to be completed. Typically, two semesters make up an academic year, with the first of these referred to as the Odd Semester and the second as the Even Semester.
- b. An additional short semester (usually 8 weeks) may be offered between an even semester and subsequent odd semester (in the interval between two academic years) and is termed a summer semester. The summer semester is offered to enable students to register for:
 - i. Fast-tracked courses required for clearing backlog courses.

- ii. Fast-tracked courses for earning additional credit / completing non-credit mandatory requirement.
- iii. Value added courses.
- iv. The courses offered in summer semesters are bound by the same regulations as that of regular semesters, except that they are run at an accelerated pace to provide the required contact hours and conduct assessments within the 8 weeks.

8. Credit

- a. A unit by which the course work is measured. It determines the number of hours of formal learning (contact hours) required per week. Credits are calculated based on the concept of “notional learning time”. Notional learning time is the number of hours that a learner is expected to spend, on average, to achieve the specified learning outcomes of the course. This may comprise a variable combination of scheduled learning activities, (lectures, seminars, labs, etc.) and self-directed learning time (reading required before classes, working on assignments, examination preparation, and completion of assessments).

9. Credit equivalence of notional learning time for different types of activities

- a. The credit values assigned to various teaching-learning activities are as follows:

Type of teaching-learning	Nature of activity	No. of contact hours per week equivalent to one credit	The total number of contact hours over a 16-week semester is equivalent to one credit
Lectures / Seminars / synchronous virtual classes / synchronous webinars	Scheduled instruction	1:1	16
Tutorials	Scheduled instruction	2:1	32
Supervised Demonstrations / Laboratory sessions / Studio / Workshops / Workplace simulation / Skill Practice Sessions	Scheduled instruction	2:1	32
Supervised Field visits/community visits/Internships	Scheduled instruction	3:1	48
Scheduled self-directed study (individual or group)	Scheduled instruction	2:1	32
Asynchronous E-Learning modules	Independent learning	2:1	32

(structured self-directed study			
Student Seminar	Independent /small group learning	2:1	32
Project work/dissertation	Independent /small group learning	3:1	48
Internship for credit	Industry placement/ Research Internship	3:1	48

10. Choice-based credit system (CBCS)

A program structure for higher education requires students to earn a minimum of credits by completing various types of courses, including electives, which facilitate a student to have some freedom in selecting his/her own choices, within as well as across disciplines.

11. Course Registration

Refers to formal registration of the Courses in the study every semester (Credits and Audit) by every student under the supervision of a faculty advisor. The institution will maintain records of the same and communicate them to the University.

12. Learning outcomes

- Program Outcomes (PO) - Statements defining the skills, knowledge, and attitude that graduates of a program will be able to demonstrate upon completing the program.
- Course Outcomes (CO) - Statements defining the skills, knowledge, and attitude that students will be able to demonstrate upon completing the course. COs are mapped to the POs such that attaining the course outcomes leads to the attainment of program outcomes.
- Attainment of POs-COs is mapped to the POs such that attaining the course outcomes leads to the attainment of program outcomes.

13. Evaluation

For all courses, the evaluation will be based on both formative assessment (Continuous Internal Evaluation, CIE) and summative assessment (Semester End Evaluation, SEE). Weightage for CIE and SEE will be 50% each.

13.1 Continuous Internal Evaluation (CIE)

Refers to the periodic and continuous *formative assessment* of students' performance during the semester by the teacher(s) of the course to provide timely feedback to students and for guiding "course corrections" by the teachers. The assessment methods may include tests, quizzes, assignments, project evaluations,

portfolio evaluations, seminar assessments, etc. CIE will have a weightage of 50% in the determination of the final grading of the course.

13.2 Semester End Evaluation (SEE)

Refers to a *summative assessment* that covers the entire course syllabus, conducted by the University, at the end of the semester. Appropriate assessment methods aligned with the learning domain and teaching-learning methods are to be used. CIE will have a weightage of 50% in the determination of the final grading of the course.

14. Grading

Course Grade refers to a qualitative measure of performance of a student in each course, based on the percentage of marks secured in Continuous Internal Evaluation (CIE) and Semester End Evaluation (SEE). A Letter grade is awarded for each course.

15. Semester Grade Point Average (SGPA)

Refers to the measure of a student's academic performance in a semester. It is calculated based on the credits and the grades obtained in the courses offered in the semester.

16. Cumulative Grade Point Average (CGPA)

Refers to the measure of the cumulative performance of a student in all the previous semesters and is computed from the 2nd semester onwards. It is calculated based on the credits and the grades obtained in all the courses taken.

17. Academic Bank of Credits (ABC)

The Academic Bank of Credits is a national-level facility for “credit transfer”. It is provided by the Ministry of Education, Govt. of India, to promote the flexibility of the curriculum framework and interdisciplinary/multidisciplinary academic mobility of students across the Higher Education Institutions in the country. The banking and redemption of credits through ABC will be governed by the University's guidelines.

22. APPENDIX-B

Evaluation Guidelines

CIE and SEE details for various types of courses

1. Theory: PCC/IPCC/PEC/OEC

1.1. Scheme of examinations: CIE+SEE =50+50=100 marks

1.2. Continuous internal evaluation (CIE):

1.2.1. CIE (PCC/PEC/OEC)

Type of Questions	Questions to be set (Can have sub-questions a and b)	Questions to Be answered	Marks per question	Total marks
Mid Sem Exam-1				
40% of the total syllabus (Unit-1) (15 Teaching hours)				
Descriptive Part-1	2	1	10	10
Descriptive Part-2	2	1	10	10
Mid Sem Exam-2				
40% of the total syllabus (Unit-2) (15 Teaching hours)				
Descriptive Part-1	2	1	10	10
Descriptive Part-1	2	1	10	10
TASKS				
TASK	The task comprises 5 class tests/quizzes/assignments conducted for each unit for a max mark of 10. All tests/quizzes/Assignments are compulsory			10
Maximum Marks				50

1.2.2 CIE (IPCC/PBL)

Type of Questions	Questions to be set (Can have sub-questions a and b)	Questions to be answered	Marks per question	Total marks
Mid Sem Exam-1				
40% of the total syllabus (Unit-1) (15 Teaching hours)				
Descriptive Part-1	2	1	10	10
Descriptive Part-2	2	1	10	10
Mid Sem Exam-2				
40% of the total syllabus (Unit-2) (15 Teaching hours)				
Descriptive Part-1	2	1	10	10
Descriptive Part-1	2	1	10	10
Task	The task comprises 5 class tests/quizzes/assignments conducted for each unit for a max mark of 10. All tests/quizzes/Assignments are compulsory.			10
Maximum Marks				50
60% weightage, converted to 30 marks				
Practical/Project Based Learning (PBL)				
Practical/PBL	Practical/PBL (comprises of implementation of theoretical concepts through projects/problem solving)			50
40% weightage, converted to 20 marks				
Maximum Marks [30 (Theory)+ 20 (Practical/PBL)]				50

1.2.3 Semester End Evaluation (SEE): 3 Hours Duration

Type of Questions	Module & Teaching hours	Questions to be set (Can have sub-questions a, b, and c)	Questions to be answered	Marks per question	Total marks
MCQ	Entire Syllabus	10 or 20	All Questions	2 or 1	20
Descriptive	• Unit-1 • 15 teaching hours	3	2	16	32
Descriptive	• Unit-2 • 15 teaching hours	3	2	16	32
Descriptive	• Unit-3 • 10 teaching hours	2	1	16	16
				Maximum Marks	100
SEE Marks with 50% Weightage					50

1.2.4 CIE & SEE for various types of courses

Sl. No.	Courses		Evaluation scheme			
			CIE (Minimum eligibility marks 40% of Max marks to appear for SEE)		CIE (Minimum eligibility marks 40% of Max marks to appear for SEE)	
			Max Marks	Min eligibility marks required	Max Marks	Minimum passing marks required
1	Integrated Professional Core Course(IPCC)	Theory	30	12	50	20
		Practical	20	08	---	---
		Total	50	20	50	20
2	PCC with PBL component	Theory	30	12	50	20
		PBL component	20	08	--	--
		Total	50	20	50	20
3	PCC/PEC/OEC		50	20	50	20
4	Laboratory		50	20	50	20
5	Drafting		50	20	50	20
6	MiniProject		100	40	---	---
7	Inter/IntraInstitutionalInternship (2 weeks)		100	40	---	---
8	Industrial/Govt./ NGO/MSME/ Rural Internship/ Innovation / Entrepreneurship (In single or two stretches =Total of 8 weeks)		100	40	100	40
9	Research Internship/ Advanced IndustryInternship/Project work		100	40	100	40
10	Seminar		100	40	---	---

All university examinations (SEE) shall be conducted for a maximum of 100 marks. For assigning the letter grade the university examination marks secured by a student, except in the case of serial no. 06, 07, and 10 shall be reduced to 50 marks and added to CIE marks. If the total marks result in a fraction during reduction, it shall be rounded off to the nearest higher value.

2 Laboratory/Practical Course

2.1 Split-up of Marks for evaluation of Practical for 50 CIE marks and 50 SEE marks.

2.2 Split-up of Marks for evaluation of Laboratory work:

2.2.1 Laboratory in-charge faculty will follow rubrics given in the Tables below for an evaluation of laboratory courses

2.2.2 In the case of Practical, the IA marks shall be based on laboratory observation, records, viva, and at least one practical test.

2.2.3 Continuous Evaluation in every lab session will be done using the format mentioned in the Table to evaluate PO9 (Individual and teamwork) and PO10 (Communication).

2.2.4 Rubrics used for continuous Evaluation of **laboratory courses involving experiments with hardware**

Lab conduction and Record			LabInternal Assessment		
Split-up: 60% (30 Marks) of Maximum CIE marks (50). Each experiment is to be evaluated for conduction with an observation book and record write-up (30 marks per experiment). The final marks for conduction and record are the average of all the specified experiments in the syllabus.			Split-up: 40% (20 Marks) of Maximum CIE marks (50). One test of 20 Marks In the test, conduction of the experiment and acceptable result with viva-voce will carry a weightage of 60% per experiment, with the rest 40% for procedural knowledge and regularity of the student.		
Rubrics per Experiment	Marks Distribution	Remarks	Rubrics	Marks distribution	Remarks
Circuit	02	Evaluation ofRecord write-up to includeweightage for submission on time, neatness,etc.	Write-up	04	
Design	02		Conduction	10	
Procedure	02				
Conduction	06				
Viva	06				
Record write-up	12				
Total Marks	30		Total Marks	20	

2.2.5 Split-up of Marks used for continuous Evaluation of laboratory involving experiments with software

Rubrics for Split up of Marks	Methodology / Process Steps per Experiment	Marks
#R1	Observation, Write up of Procedure / Algorithm/ Program execution, and Conduction of experiment	12
#R2	Viva – Voce	06
#R3	Record writing	12
	Total Marks for each experiment	30
#R4	Internal Test: Lab Internal Assessment	
	(i) Write-up of Procedure/Program/Algorithm	04
	(ii) Conduction/Execution	10
	(iii) Viva-Voce	06
	Total Marks	20

3. Internship and Evaluation

3.1 Introduction

The rise in global competition has prompted organizations to devise strategies to have a talented and innovative workforce to gain a competitive edge. Developing an internship policy is an impactful strategy for creating a future talent pool for the industry. The internship (a form of experiential learning) program not only helps fresh pass-outs in gaining professional know-how but also benefits corporate sectors. The internship also enhances the employability skills of the student passing out from Technical Institutions.

The following list provides a brief illustrative overview of the knowledge, skills, work habits, and character traits commonly associated with 21st-century skills and to be acquired by graduates:

- Critical thinking, problem solving, reasoning, analysis, interpretation, and synthesizing information.
- Scientific literacy and reasoning, the scientific method.
- Research skills and practices, interrogative questioning.
- Creativity, artistry, curiosity, imagination, innovation, and personal expression.
- Information and communication technology (ICT) literacy, media and internet literacy, data interpretation and analysis, and computer programming.
- Oral and written communication, public speaking and presenting, listening.
- Economic and financial literacy, entrepreneurial skills.
- Global awareness, multicultural literacy, humanitarianism.
- Environmental and conservation literacy, ecosystems understanding.
- Civic, ethical, and social-justice literacy.

- Leadership, teamwork, collaboration, cooperation, and facility in using virtual workspaces.
- Perseverance, self-direction, planning, self-discipline, adaptability, initiative.
- Health and wellness literacy, including nutrition, diet, exercise, and public health and safety.

The internship experience will augment the outcome-based learning process and inculcate various attributes mentioned above in a student in line with the graduate attributes defined by the NBA as well as NEP 2020

Following are the intended objectives of internship training.

- (i) Expose Technical students to the industrial environment, which cannot be simulated in the classroom, and hence create competent professionals in the industry.
- (ii) Provide possible opportunities to learn, understand and sharpen the real-time technical/managerial skills required at the job.
- (iii) Expose to the current technological developments relevant to the subject area of training.
- (iv) Use the experience gained from the industrial internship in discussions held in the classrooms.
- (v) Create conditions conducive to the quest for knowledge and its applicability on the job.
- (vi) Learn to apply technical knowledge in real industrial situations.
- (vii) Gain experience in writing reports on technical works/projects.
- (viii) Expose students to the engineer's responsibilities and ethics.
- (ix) Familiarize with various materials, processes, products, and their applications along with relevant aspects of quality control and safety measures.
- (x) Promote academic, career, and/or personal development.
- (xi) Expose the students to future employers.
- (xii) Make students available to the industry for employment.
- (xiii) Understand the psychology of the workers and their habits, attitudes, and approach to problem-solving.
- (xiv) Understand the social, economic, and administrative considerations that influence the working environment of industrial organizations.

3.2 Academic credit framework for the internship and project work undergone as part of the B.Tech. program.

- A minimum of 20 credits of Internship/ Entrepreneurial activities / Project work/ Seminar and Inter/ Intra Institutional Training may be counted towards B. Tech. degree program
- Here, 1 credit is equivalent to a minimum of 40-45 hours of work. Therefore, a full-time intern is expected to spend 40 - 45 hours per week on Internship, Training, Project work, Seminar activities, etc. This will result in about 800 to 900 hours of total internship and project duration for the B. Tech program.
- To derive the benefits of an internship, it is introduced in two/ three stages of the B.Tech. program.
- Internships may be full-time or part-time; they are full-time during the summer vacation

and part-time during the academic session. The curriculum is flexible to adjust internship duration. Therefore, opportunities must be provided for experiences that cannot be anticipated when planning the course.

- The departments have the flexibility to schedule internships, Project work, Seminars, etc. according to the availability of the opportunities. However, the suggested minimum requirement regarding Internship duration and credits are as given in Table -B1

Table-B1 Suggested Credit Framework for Internship and Project work

Sl. No.	Title	Schedule	Duration	Activities	Credits
1	Internship-I	Ongoing First-year academic session/ Summer vacation after 2nd Semester/ vacation during 3 rd semester (for lateral entry students	02 weeks	Inter/ Intra Institutional Activities (Evaluation in 4 th semester)	02
2	Internship-II	a) Summer vacation after 4th Semester	02-04 weeks	Industrial/Govt./ NGO/ MSME/ Rural Internship/ Innovation / Entrepreneurship/ social internship	---
		b) Summer vacation after 6th Semester	04-06 weeks	Industrial/Govt./ NGO/ MSME/ Rural Internship/ Innovation / Entrepreneurship	
		c) Total of a) and b) at the beginning of the 8th semester	08 weeks	Evaluation in 8 th Semester	08
3	Project work	6 th Semester	6 hours/week	Mini -Project	02
		8th Semester	16 weeks	Extended Industry Internship /Research Internship/ Project work	10
				Report preparation and writing	
				Seminar	01
Total Credits					23

Table-1 states that during the ongoing/ summer vacations after the 2nd Semester, students are required to be involved in Inter/ Intra Institutional Activities viz; Training with higher Institutions; Soft skill training organized by Training and Placement Cell of the respective institutions; contribution at incubation/ innovation /entrepreneurship cell of the institute; participation in conferences/ workshops/ competitions, etc.; Learning at Departmental Lab/ Tinkering Lab/ Institutional workshop; Working for consultancy/ research project within the institutes and Participation in all the activities of Institute's Innovation Council for eg: IPR workshop/Leadership Talks/ Idea/ Design/ Innovation/ Business Completion/ Technical Expos, etc.

During the summer vacation after the 4th/ 6th semester, students are ready for industrial experience. Therefore, they may choose to undergo Internship / Innovation / Entrepreneurship related activities. Students may choose either to work on innovation or entrepreneurial activities resulting in start-up or undergo internship with industry/NGO/ Government organizations/ Micro/ Small/ Medium enterprises to make themselves ready for the industry.

3.3 Internship Supervision

- i) The internship shall be carried out under the supervision of a faculty mentor. The faculty mentor/guide should,
- ii) Serve as a teacher, mentor, trainer, critic, leader, and boss.
- iii) Provide sufficient time to guide the interns. (Interns are students or a trainee who does a job to gain work experience)
- iv) Play a vital role, along with the Training and Placement Officer, in providing internship opportunities for the students.
- v) Exhibit qualities such as leadership, strong communication skills, and patience.
- vi) Provide a letter of recommendation in due consultation with students and the industrial organization (if possible) where the internship is intended to be carried out, endorsed by the authority (Principal/Institution Internship Coordinator).

3.3.1 Each faculty mentor shall supervise the students/Student batches allotted to them. Often, the supervision may be by an external expert. In such cases, the faculty mentor shall jointly guide the student/s without causing miscommunications/embarrassment to either side.

3.3.2 Depending on the activity taken up by the students, the internship shall be carried out individually or in batches having not more than three students.

3.3.3 Faculty Mentor, along with the external expert, shall scrupulously evaluate the work of an individual student or students of a batch and maintain the relevant documents.

3.3.4 For allotment of CIE marks, the institutions shall prepare the rubrics for each activity offered by the institution as given in Table - B2. The marks shall be allotted by the Internship committee designated by HOD in consultation with the mentors.

3.3.5 For all activities conducted by the institution, the attendance of the students shall be maintained by the faculty and maintained in their respective departments.

3.4 Internship-I (Activity based Internship)

While intra-activities are within the institution, inter-activities shall be between the concerned institution and neighboring institutions. Intra and Inter activities are the activities that are the impetus to learning techniques. It adds to the comprehensive growth of the mind and associated activities.

As the students are on the verge of learning technical aspects and have a limited period of internship, it is preferable to expose students to polygonal activities instead of one type of activity. Therefore, activities completed by the students shall not be one type of activity but can be few within the period of the internship. In this regard, Intra and Inter-Institutional activities shall be completed under the supervision of a faculty on a self-learning basis.

The faculty have to kindle the latent abilities of the students, encourage, guide, supervise and shape them to achieve the desired result. Therefore, a learning agenda in the form of specific learning objectives and outcomes shall be prepared before the start of the internship.

Whatever the activity/activities that are/are done under Intra and Inter-Institutional activities, should ignite the inquisitiveness to learn, enhance the knowledge, thinking ability and

imagination, planning, application of mind, execution ability, innovation attitude, listening and understanding, vocabulary, personal expression, public speaking, written communication, oral presentation of the subject matter, acquire leadership qualities and teamwork requirements, responsiveness, ethics, etc.

3.4.1 List of proposed activities

- a. Activities concerned with the works of Indian scholars like Charaka and Susruta, Aryabhata, Bhaskaracharya, Chanakya, Madhava, Patanjali, Panini, and Thiruvalluvar, among numerous others
- b. Activities such as training with higher Institutions or Soft skill training
- c. Contribution at incubation/ innovation /entrepreneurship cell of the institute.
- d. Learning at Departmental Lab/Tinkering Lab/ Institutional workshop.
- e. Working for consultancy/ research projects within the institute.
- f. Learning MS Word, Excel, Microsoft equations, MS drawing tools, MS Powerpoint, etc.
- g. Coding.
- h. Mini projects using commercially available assembled electronic products.
- i. Debates, quizzes, and group discussions: On technical topics already studied (both in Kannada and English).
- j. Essay competitions: Both in Kannada and English on technical topics already studied.
- k. Survey and study of published literature on the assigned topic: Technical paper survey, Preparation of synopsis. Exposure to technical paper publications.
- l. Photography.
- m. Short film production: Contemporary aspects, technical aspects, etc.
- n. Internship in Disaster Management.
- o. Solar energy connected activities that help the common man.
- p. Working with Smart City Administration.
- q. Hackathon (it is a design sprint-like event in which computer programs and others involved in software development, including graphic designers, interface designers, project managers, and others, often including domain experts collaborate intensively on software projects).
- r. Industrial Safety, Fire Safety, Electrical Safety, Chemical Process Safety, Food Safety, etc.
- s. Internship and project work in Indian Knowledge System related Areas/Topics.
- t. Industrial visits to Small Scale Industries/ Factories/ Cottage Industries/substation visits etc., and submission of the report.

3.5 Documents to be submitted by Students for Internship Evaluation

3.5.1 Student's Diary

The main purpose of writing a daily diary is to cultivate the habit of documenting and to encourage the students to search for details. It develops the student's thought process and reasoning abilities. The students shall record in the daily training diary the day-to-day account of the observations, impressions, information gathered, suggestions given, if any, and activities carried out. It should contain sketches and drawings related to the observations made by the students. The daily training diary should be signed after every day or at least twice a week by the faculty/ in charge of the section (external expert) where the student has been working.

Student's Diary should be submitted by the students along with attendance records. It shall be evaluated based on the following criteria:

- i) Regularity in the maintenance of the diary.
- ii) Adequacy and quality of information recorded.
- iii) Drawings, sketches, and data were recorded.
- iv) Thought processes and recording techniques were used.
- v) Organization of the information

3.5.2 Internship report

After completion of the Internship, the student shall prepare, with a daily diary as a reference, a comprehensive report in consultation with the mentor/s to indicate what he/she has observed and learned in the training period along with the internship outcomes. The training report should be signed by the mentor. The Internship report shall be evaluated based on the following criteria and/or other relevant criteria about the activity completed.

- i) Originality.
- ii) Adequacy and purposeful write-up.
- iii) Organization, format, drawings, sketches, style, language, etc.
- iv) Practical applications, relationships with basic theory, and concepts taught in the appropriate course.
- v) Variety and relevance of learning experience.

Procedure for the Evaluation of Internship-I

- a) Students should submit the reports immediately on completion of the Internship to the respective mentors
- b) The Examination of the internship will be carried out by the mentor
- c) The Internship-I shall be slated for 100 marks CIE only and will not have SEE.
- d) Internship-I marks are based on CIE marks (25 marks for the first presentation, 25 marks for the second presentation, and 50 marks for the report and final presentation).
- e) A Viva-Voce examination conducted during the I/II/III Semesters (Presentation followed by question-answer session) and the prescribed credit shall be included in the IV semester grade card.

3.5.3 Assessment Rubrics for evaluation of Internship-I (Intra and Inter-Institutional Activities)

Table – B2 Internship-I Assessment Rubrics Scheduled during the first year (Prescribed Period 02 weeks and Prescribed credits: 02)					
Sl. No.	Sub Activity Head	Performance/ Appraisal	Assessment Rubrics (Allotted marks decide the letter	Proposed Document as Evidence	Evaluated by
1	Inter/ Intra Institutional Workshop/ Training.	Excellent	80 to 100	(i) Student's Diary and (ii) Internship Report along	Institute
		Good	60 to 79		
		Satisfactory	40 to 59		
		Unsatisfactory and fail	< 39		

2	Working for consultancy/ Research project.	Excellent	80 to 100	with the certificate issued from the relevant authorized Authority	Faculty (mentor) together with External Expert, if any.
		Good	60 to 79		
		Satisfactory	40 to 59		
		Unsatisfactory and fail	< 39		
3	Festival (Technical / Business / Others) Events.	Excellent	80 to 100		
		Good	60 to 79		
		Satisfactory	40 to 59		
		Unsatisfactory and fail	< 39		
4	Contribution in Incubation/ Innovation/ Entrepreneurship Cell.	Excellent	80 to 100		
		Good	60 to 79		
		Satisfactory	40 to 59		
		Unsatisfactory and fail	< 39		
5	Learning at Departmental Lab/Tinkering Lab/Institutional workshop.	Excellent	80 to 100		
		Good	60 to 79		
		Satisfactory	40 to 59		
		Unsatisfactory and fail	< 39		
6	Other than the above five activities	Excellent	80 to 100		
		Good	60 to 79		
		Satisfactory	40 to 59		
		Unsatisfactory and fail	< 39		
Note: The total CIE marks shall be the sum of marks allotted to completed activities by the student.					

3.6 Internship-II: (Societal internship and Research/Industry Internship) (08 weeks)

[Scheduled during the intervening period of IV & V semester and VI & VII semester]

During the intervening period of the IV & V semesters and VI & VII semesters, students shall be ready for industrial experience. Therefore, they shall choose to undergo an Internship involving Innovation / Entrepreneurship/short-term (about 2 weeks) societal-related activities. Students may choose to work on innovation or entrepreneurial activities, or both resulting in start-up or undergo internship with industry/NGO/ Government organizations/ Micro/ Small/ Medium enterprises to make themselves ready for the industry.

3.6.1 Innovation

Innovation refers to a new or improved product or process or a combination thereof that differs marginally or significantly from the unit's previous product. An innovation

center is a place where students are encouraged to implement the innovative ideas formed through imagination, brainstorming sessions, design thinking, and associated activities to bring them to reality. It is a place, where creative minds are shaped.

3.6.2 Entrepreneurship

Entrepreneurship refers to setting up a new business or business and taking on financial risks in the hope of profit. It involves investment to undertake production along with arranging inputs like land, labour, material, and capital, introducing new techniques and products, identifying new sources for the enterprise, etc.

3.6.3 Incubation Center

An organized unit designed for innovation as well as to accelerate the growth and success of new entrepreneurial companies through mentorship and an array of business support resources and services that could include physical space, capital, coaching, common services, and networking connections.

3.6.4 Startup

An entity that develops a business model based on either product innovation or service innovation and makes it scalable, replicable, and self-reliant.

An entity shall be considered a Startup

- i) Up to ten years from the date of incorporation/ registration, if it is incorporated as a private limited company (as defined in the Companies Act, 2013) or registered as a partnership firm (registered under section 59 of the Partnership Act, 1932) or a limited liability partnership (under the Limited Liability Partnership Act, 2008) in India.
- ii) Turnover of the entity for any of the financial years since incorporation/ registration has not exceeded one hundred crore rupees.
- iii) The entity is working towards innovation, development, or improvement of products or processes, or services, or if it is a scalable business model with a high potential for employment generation or wealth creation.
- iv) Provided that an entity formed by splitting up or reconstruction of an existing business shall not be considered a Startup.

3.6.5 Societal (Social) related activities

Short-term internships (about 2 weeks) in villages, slums, or urban areas can be under social internship. The internship will be more fruitful if students work in teams. The teams can select one or more fields to do their best in the field of agriculture, watershed management, wastelands development, non-conventional energy, low-cost housing, sanitation, nutrition and personal hygiene, schemes for skill development, income generation, blood bank, government schemes such as

- i) (Swachh Bharat: Swachh Bharat Mission, Swachh Bharat Abhiyan, or Clean India Mission is a country-wide campaign to eliminate open defecation and improve solid waste management.
- ii) Accessible India: Accessible India Campaign or Sugamya Bharat Abhiyan is a program to serve the differently able community of the country.
- iii) Digital India: A campaign to ensure the Government's services are made available to citizens electronically by improved online infrastructure and by increasing Internet connectivity or making the country digitally empowered in the field of technology.
- iv) Beti Bachao and Beti Padhao: A campaign of the Government of India that aims to generate awareness and improve the efficiency of welfare services intended for

girls in India.

- v) Environment and Energy Conservation and Education, legal aid, consumer protection, and allied field including Indian Red Cross Society, National Cadet Corps, Bharat Scouts, and Guides.

Societal activities are one of the NBA graduate attributes that are part of PO6 and PO7, which are reproduced below.

- vi) PO-6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- vii) PO-7: Environment and Sustainability: Understand the impact of the professional engineering solution in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development. The long-term goal under Societal (social work) related activities, particularly in a rural area, results in a rural internship. In urban areas, the student may adopt slum/ economically weaker section areas for short duration social internship to uplift the living conditions.

Given the above, internship coordinators should encourage students to take up a societal internship as far as possible.

3.6.6 Places for Innovation/Entrepreneurial Activities

Students shall carry out Innovation or Entrepreneurial activities or both at the Incubation Center and Entrepreneurship Cell of the parent institution or elsewhere such as ATAL Incubation Centers [A flagship of Atal Innovation Mission (AIM), NITI Aayog for promoting the culture of innovation and entrepreneurship in India], institutes of national importance, public sector units, IT companies, government organizations, and non-governmental organizations, industries including MSME, etc.

- **Institutes should deter students to opt for internships at places established for commercial benefits.**

3.6.7 Industrial Internships

The gap between the theoretical knowledge obtained in the classrooms and the practical skills required in the actual workplace scenarios is fast growing. This has put forth varied challenges to graduating students when it comes to job placements. As institutes cannot have a relevant facility to expose students to a real-time industrial environment, an industrial internship is an appropriate solution.

The main objective of the industry internship is to ensure that the intern is exposed to a real job world environment and gains practical experience. Often, it may be a practical exposure to the theory that has been learned during the academic period. The industry internship helps students understand analytical concepts and tools, hone their skills in real-life situations, and build confidence in applying the skills learned.

3.6.7.1 Industry Internship Benefits

- i) Have ample opportunities to attend seminars, symposiums, workshops, etc. This in turn provides an opportunity to establish rapport with professionals and pioneers in their respective fields for further growth.
- ii) Have wide scope to publish paper/s in journals.
- iii) Good recommendation letter/s that increase the prospectus for further internships, higher studies, and placements.
- iv) Helps to acquire team spirit, motivated acts, techniques to resolve conflicts, etc.
- v) Helps to develop a lot of leadership skills.
- vi) Increases the prospect of placement in the same concern, provided the intern has

exhibited a clear understanding of basics and completed the internship.

vii) Fosters to substantiate the issues with facts and figures.

For AICTE Internship opportunities refer to <https://internship.aicte-india.org/>

3.6.8 Assessment Rubrics for Innovation / entrepreneurship/ Societal Internship Activities

Once the internship begins, the students are required to maintain a diary/journal and submit a report regularly to the guide. These reports should summarize the activities in which the student was involved during the previous week's period. At the end of the internship, each student is required to submit a hard copy of the consolidated diary/journal and report for evaluation. The report should indicate the learning and achievements of the internship.

3.7 Research Internships / Extended Industry Internships

Table – B3 Innovation/entrepreneurship/ Societal Internship Activities and Assessment Rubrics Scheduled during the intervening period of IV & V semester and VI & VII Sem (Prescribed Period 08 weeks: Credits 08)

Sub Activity Head	Performance/ Appraisal	Assessment Rubrics	Proposed Document as Evidence	Evaluated by
(1) Development of new product/ Business Plan/ registration of start-up/societal internship	Excellent	80 to 100	(i) Student's Diary and (ii) Internship Report or the activity report along with Certificate or Declaration from relevant Authorized Authority. Wherever only Certificate is issued, Assessment shall be at the institute as per (i) and (ii) to decide the letter grade.	(i)Institute Faculty (mentor) together with External Expert if any.
	Good	60 to 79		
	Satisfactory	40 to 59		
	Unsatisfactory and fail	< 39		
(2) Internship with Industry/ Govt. / NGO/ PSU/ Any Micro/ Small/Medium Enterprise.	Excellent	80 to 100	(i) Student's Diary and (ii) Internship Report or the activity report along with Certificate or Declaration from relevant Authorized Authority. Wherever only Certificate is issued, Assessment shall be at the institute as per (i) and (ii) to decide the letter grade.	(i)Institute Faculty (mentor) together with External Expert if any.
	Good	60 to 79		
	Satisfactory	40 to 59		
	Unsatisfactory and fail	< 39		

Note:

(i) The total CIE marks shall be the sum of marks allotted to successfully completed activities by the student.

3.7.1 Research Internship /Extended Industry Internship of sufficient duration encourages students early on in their careers. Its main goal is to allow improving their analytical and technical skills in an international environment. An internship can be in an industry or at an appropriate workplace.

3.7.2 Research internships and industrial internships have different purposes and come with a set of benefits. A prior experience in any field is always preferred over a fresh start. Therefore, one of them can be selected depending on the interest the students have. Internships pose unexpected challenges and make students think appropriately, tackle

difficulties with ease, and act in a scholarly way to get past the hurdles and practical constraints. An internship is always beneficial however good or bad it is.

- 3.7.3 Internships not only enhance one's learning but also identifies him/her as someone who commits to approaching a project and completing it with or without guidance. Internship learning is an impetus for professional development.
- 3.7.4 While a research internship is a stepping stone to higher studies, an industry internship is a pathway to a placement. Those who are self-motivated and interested in searching for new things that are original and unique can choose a research internship. Those who are interested in real industry- experience and aspire to get a job soon after graduation can choose an industry internship.
- 3.7.5 Research Internships (Also known as dissertation internships) are focused research projects that push students' intellectual abilities beyond those driven by the classroom. Often, a research internship typically helps solve problems that are usually part of major research projects. It involves a short theoretical or experimental research project supervised by a researcher.
- 3.7.6 The research internships, under the advice of a faculty supervisor, can be one's own selected project or a project on which a Researcher is researching, or a new project/real-world project offered by an organization. The research area may be about single or multidisciplinary fields such as science, technology, engineering, mathematics, management, and business studies. Research internships can be carried out either individually or in teams (not exceeding 3 or 4 students).
- 3.7.7 Research internship opportunities, before graduation, maybe in a laboratory of college, a research institute, or a company's R & D department. Apart from fixed working hours of the day of an organization, the researcher can devote sufficient time to other research-related activities for early and successful completion of the Research Internship.

3.7.8 Necessary Skills for Research Internship and Industrial Internship

For the internships to progress without hurdles and for successful completion, the Researchers should maintain a harmonious relationship with the guide/s, administrators, co-workers, and others, and strictly adhere to the rules and regulations of the workplace. The other skills required or acquirable during the Internship are,

1. Good Communication skills.
2. Attention to detail.
3. Planning and scheduling.
4. Documentation.
5. Critical thinking.
6. Data collection.
7. Data analysis.
8. Ability to maintain quality, safety, and/or infection control standards.
9. Appreciating and practicing ethical issues.

3.7.9 Responsibilities of an Intern

Interns,

1. If working with a researcher, shall assist the researcher in an ongoing research project or work collaboratively in designing a new project of mutual interest.
2. Shall engage in literature survey and get an insight of the research work at the initial

stages.

3. Shall compile data, sort, file, implement ideas with minimal guidance and assist write papers.
4. Shall become familiar with several tools [meters (Electrical and Electronics, mechanical, computer, etc.)] used in data collection, software, graphic software, Statistical Package for the Social Sciences (SPSS) software [IBM's statistical software platform], etc.
5. Shall attain skills with Microsoft Word Office, Excel, PowerPoint, Outlook, etc.
6. Shall give a mid-term oral presentation to a committee for review and feedback.
7. Shall attend discussions, meetings, symposiums, classroom lectures, etc., to learn new scientific techniques, design experiments, analyze results, and formulate different hypotheses.
8. Shall learn to write reports and be able to correspond independently.
9. Shall manage time effectively.
10. Shall keep a track of the progress of the project.
11. Shall develop integrative thinking.

3.7.10 Research internship Outcomes

1. Generating technical paper/s and publishing in refereed journal/s.
2. Possibility of acquiring intellectual ownership and patent.
3. Build a prototype for an idea on which the research was carried out.
4. File patent/s.
5. Add academic knowledge to the field.
6. Enhanced ability in arranging meetings, presentations, seminars, training, etc.
7. Improved conscientiousness and ethics.

3.7.11 Research internships Benefits

1. Are a great way to pursue an academic career in teaching and research, as a Research Scientist at a Research Organization, Company, Industry sector, etc.
2. Establish professional networks for a future career.
3. Pave the way to join a research team and work alongside leading experts in the field.
4. Introduced to new ideas through interaction with like-minded students and others.
5. Develop research skills and knowledge in a specific area of interest.
6. Provide opportunities for growth, achievement, and personal development.
Offer an opportunity to publish a research paper that will boost the resume while applying for Post Graduate Studies

4. Evaluation Procedure of UC3001-1 Research Internship /Extended Industry Project/Internship/Project work (16 weeks)

- 4.1** The students pursuing the course UC3001-1 shall submit the diary recordings of day-to-day activities to the concerned guide, reporting progress achieved in the course and seeking guidance to proceed with the internship. The interns should provide all the details to the guide so that he/she can discuss with the employer to make the internship successful.
- 4.2** The intern should constantly update the guide about the progress of the internship. The guide

should know the intern's internship tasks, duties, responsibilities, and potential projects. The evaluation of interns and their internship progress should be honest and constructive.

4.3 The hardcopy or softcopy of the diary maintained by the interns must be signed at regular intervals by the guide.

4.4 Regarding the intern's feedback, the guides should propose changes in internship activities so that they are helpful to the internship.

4.5 Illustrations, drawings, photos, forms, samples, classified materials, etc., are to be included in the report only after obtaining the consent of the concerned authorities and should indicate the source of all such material. The final report should also be submitted to the place where the internship was carried out. The report should avoid a tone that is predominantly cynical or unduly critical of the employer or of those with whom the student intern has worked. The content of the report must be based on interns' own work.

4.6 Continuous Internal Evaluation (CIE)

The guides should evaluate the interns using the following as well as any other appropriate methods;

- a) Punctuality of intern.
- b) Conduct and character.
- c) Tactfulness and politeness with colleagues and the public.
- d) Attitude regarding professionalism.
- e) Inquisitiveness and eagerness to learn.
- f) Research attitude.
- g) Problem-solving techniques.
- h) Innovation mindset.
- i) Time management and meeting deadlines.
- j) Receptiveness to feedback and critiques.
- k) Ability to work in a team as a member.
- l) Ability to work without supervision.
- m) Supervisory skills and leadership skills.
- n) Judgment and decision-making skills.
- o) Writing skills, oral communication skills, technical communication skills, computer skills, analysis skills, and business writing skills.
- p) Appropriateness of technical skills.
- q) Familiarization with writing technical papers, standards, codes, etc.
- r) Reading Behavioural attitude.
- s) Outcomes.
- t) Successes and failures experienced

4.7 Recommendation letter

The guide must state whether the intern,

- a) Exceeded the expectations of the internship.
- b) Met the expectations of the internship.
- c) Did not meet the expectations of the internship.
- d) Did work to a satisfactory level.
- e) Did an unsatisfactory internship.

In the end, the guide should issue a recommendation letter.

4.8 Assessment of CIE marks

- 4.8.1 **Single discipline:** The CIE marks shall be awarded by a committee consisting of the Head of the concerned Department and two faculty members of the Department, one of whom shall be the Guide. The CIE marks awarded for the internship, shall be based on the evaluation of the diary, presentation skill, and viva-voce in the ratio of 50:25:25.
- 4.8.2 **Interdisciplinary:** The CIE marks awarded for the internship, shall be group-wise at the institution level with the participation of all guides of the internship. Participation of external guide/s, if any, is desirable.
- 4.8.3 The CIE marks awarded for the internship, shall be based on the evaluation of the diary, presentation skill, and viva-voce in the ratio of 50:25:25.

4.9 Assessment of SEE marks

- 4.9.1 Single discipline: Contribution to the internship and the performance of each group member shall be assessed individually in the semester-end examination (SEE) conducted at the department. Marks shall be awarded based on the evaluation of the report, presentation skill, and viva-voce in the ratio of 50:25:25.
- 4.9.2 Interdisciplinary: Contribution to the internship and the performance of each group member shall be assessed individually in the semester-end examination (SEE) conducted separately at the departments to which the student/s belongs. Marks shall be awarded based on the evaluation of the report, presentation skill, and viva-voce in the ratio of 50:25:25.

4.10 Evaluation of research Internship/Extended Industry Internship/Project Work:

Split-up of marks for evaluation of Project work for 100 CIE marks and 100 SEE marks

Split up	Rubrics		Marks
Report (50 Marks)	Content Development	Abstract/ Synopsis Write-up	10
		Selection of Topic/ Relevance of the subject to the concerned discipline	05
		Problem Identification	05
		Objectives and Methodology	05
	Problem-Oriented Exposition	Literature Survey (Papers/Sites/Sources Surveyed)	10
		Documentation/ Systematic Approach	10
		Results (with inferences, Conclusions, etc.)	05
Project Presentation Skill (25 Marks)		Quality of preparation of presentation	05
		Communication Skills	05
		Technical knowledge and awareness	05
		Individual involvement	10

Viva- Voce (25 Marks)		The clarity in answering questions relating to fundamentals and concepts	10
		The clarity in answering the questions related to the project	05
		The understanding ability of the questions asked	05
		The confidence in answering the questions asked.	05
		Total Marks	100

B.Tech. Syllabus

Effective from
Academic Year
2024 – 2025

Curriculum for Acquiring Professional Skills (CAPS)
With Scheme of Teaching & Examination

Course Numbering Scheme							
Branch Code		Course Level	Course Code			Separator	Version
Letter	Letter	Number	Number	Number	Number	-	Number
Branch Code	CS is 2 Letter code for the Department of Computer Science and Engineering						
Course Level	Course Level is a 1-digit number that can have a value between 1-4 and indicates the prerequisite of a course. Level-1 courses are basic courses with no courses as prerequisites Level-2 course(s) have Level-1 course(s) as prerequisites Level-3 course(s) have Level-2 course(s) as prerequisites Level-4 course(s) have Level-3 course(s) as prerequisites						
Course Code	Course Code is a 3 Digit number that can have a value between 001-999 and indicates the number assigned to a course based on the following guidelines 001-199 is assigned to Professional Core Courses 001-099 for Integrated Professional Core Courses [4 Credit] 101-199 for Professional Core Theory Courses [3 Credit] 201-499 for Professional Elective Courses 201-299 Electives under Group I 301-399 Electives under Group II 401-499 for future use 501-550 for Open Elective Courses 551 – 599 for Vocational Education Courses 601-650 for Professional Core Lab Courses [1 Credit] 651-699 for Ability Enhancement Courses 701-799 for Courses offered to Honours Program						
Separator	“_” is used as a separator between the Course code and the version						
Version	Version is a 1-digit number that can have a value between 1-9 and indicates minor revisions of the same course.						

I SEMESTER (AI&DS, AI&ML, CC, CS, IS, RI)

SI No.	Course and Course code		Course Title	Teaching Department	Teaching hours/Week			SL #	Examination				Credits
					Theory Lecture	Tutorial	Practical/ Drawing		Duration in hours	CIE	SEE	Total Marks	
					L	T	P						
1	BSC	MA1002 – 1	Calculus and Differential Equations	MA	3	0	0	2	3	50	50	100	3
2	BSC	PH1004-1	Quantum Computing and Modern Physics	PH	3	0	2	2	3	50	50	100	4
3	ESC	CS1001-2	Problem Solving through Programming	CS	3	0	2	2	3	50	50	100	4
4	ESC	EC1001-1	Basic Electronics	EC	3	0	0	2	3	50	50	100	3
5	ETC	IS1101-1	Fundamentals of Cyber Security	CS	3	0	0	2	3	50	50	100	3
6	HSMC	HU1001-1	Technical English	HU	1	0	2	-	3	50	50	100	2
7	UCC	HU1002-1	Constitution of India	HU	1	0	0	-	0	50	0	50	1
8	BSC	MA1006 – 1	Mathematics with MATLAB	MAT	0	0	2	-	0	50	0	50	1
TOTAL					17	0	8	10	18	400	300	700	21
Additional Academic activities beyond class hours					-	-	-	5					
Learning hours per week					17	-	8	15					
Total notional learning hours (L+T+P+J+SL)			Per week		40								
			Per Semester		40x16=640								

CALCULUS AND DIFFERENTIAL EQUATIONS			
Course Code	MA1002-1	Course Type	BSC
Teaching Hours/Week (L: T:P)	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Mathematics			
Course Objectives:			
1.	This course will enable the students to master the basic tools of differential calculus, partial differentiation, vector differentiation, differential equations, multiple integrals and become skilled for solving problems in science and engineering.		
UNIT-I			
Differential Calculus			7 Hours
Polarcurves, angle between the radius vector and the tangent, angle of intersection of two curves, derivatives of arcs and radius of curvature-cartesian, parametricandpolarforms. Rolle'stheorem(withoutproof), mean value theorems and applications to simple problems, Taylor's theorem for functions of single variable.			
Partial Differentiation			8 Hours
Partial derivatives of simple functions, total differentiation -differentiation of composite and implicit functions. Jacobians. Taylor's theorem for functions of two variables, maxima and minima for functions of two variables, Lagrange's method of undetermined multipliers (with one subsidiary condition).			
UNIT-II			
Vector Differential Calculus			7 Hours
Vector algebra(review), scalar and vector valued functions, gradient, directional derivative and hessian of multi-variable function, Divergence, and curl of a vector valued function. Solenoidal and irrotational vectors.			
Ordinary and Partial Differential Equations			8 Hours
Ordinary differential equations(review), linear and nonlinear differential equations. Second and higher order linear differential equations with constant coefficients. Formation of partial differential equations by eliminating arbitrary constants and arbitrary functions. Classification of second order PDES. Solution of P.D.E by the method of separation of variables.			
UNIT-III			
Multiple Integrals			10 Hours
Double integrals and triple integrals, evaluation by change of order of integration, change of variables and applications to area and volume. Beta and Gamma functions and their properties.			
Course Outcomes: At the end of the course student will be able to			
1.	Apply the concept of radius of curvature and mean value theorems.		
2.	Learn the concept of partial differentiation of a function with two or more independent variables, apply them to solve engineering problems and examine the given function for its extrema.		
3.	Solve the vector functions and their derivatives for engineering applications.		
4.	Apply the concepts of ordinary and partial differential equations in engineering problems.		
5.	Apply the notion of multiple integrals to find areas and volumes.		
Course Outcomes Mapping with Program Outcomes & PSO			

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
MA1002-1.1	3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
MA1002-1.2	2	2	-	-	-	-	-	-	-	-	-	-	-	-	-
MA1002-1.3	3	1	-	-	-	-	-	-	-	-	-	-	-	-	-
MA1002-1.4	3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
MA1002-1.5	3	2	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

Textbooks:

1. Erwin Kreyszig, “Advanced Engineering Mathematics”, John Wiley and Sons, 10th Edition (Reprint), 2016.
2. B.S. Grewal, “Higher Engineering Mathematics”, Khanna Publications, 43rd Edition, 2015.
3. Murray R. Spiegel, Seymour Lipschutz, Dennis Spellman, “Vector Analysis”, Schuam’s outlines series, 2nd edition, 2009.

Reference Books:

1. G.B. Thomas and R.L.Finney, “Calculus and Analytic geometry”, Pearson, 2002.
2. T. Veerarajan, “Engineering Mathematics”, McGraw-Hill, New Delhi, 2008.
3. B. V. Ramana, “Higher Engineering Mathematics”, Tata Mc Graw –Hill, New Delhi,2010.
4. N.P. Bali and M.Goyal, “A text book of Engineering Mathematics”, Laxmi Publications, 2010.
5. W.E. Boyce and R.C. DiPrima, “Elementary Differential Equations and Boundary Value Problems”, Wiley India, 2009.
6. E.A. Coddington, “An Introduction to Ordinary Differential Equations”, Prentice Hall India, 1995.
7. G.F. Simmons and S.G. Krantz, “Differential Equations”, McGraw Hill, 2007.
8. Shanthi Narayan, “Differential Calculus, 6th edition, Shyam Lal Charitable Trust, Delhi.

E Books / Moocs/ NPTEL

1. <http://nptel.ac.in/courses/111106100/>
2. <https://nptel.ac.in/courses/122101003>
3. <http://nptel.ac.in/courses/111106100/>

QUANTUM COMPUTING AND MODERN PHYSICS			
Course Code:	PH1004 -1	Course Type:	IPCC
Teaching Hours/Week (L: T: P: S)	3:0:2:0	Credits:	04
Total Teaching Hours:	40+0+26	CIE + SEE Marks:	50+50
Teaching Department: Physics			
Course Objectives:			
1.	To study the principles of quantum mechanics and its application in quantum computing		
2.	To study the concepts of semiconductors and semiconductor devices		
3.	To study the properties of superconductors and their applications		
4.	To understand the principle, working and applications of lasers.		
5.	To understand the principle, working and applications of optical fibers.		
UNIT-I			
Quantum Computing			15 Hours
Introduction to Quantum Computing and Quantum Mechanics			
Moore’s law and beyond, Classical computing, Quantum Computing, differences between classical and Quantum computing, Origins of Quantum mechanics. Different formalisms in Quantum Mechanics: Schrodinger, Heisenberg's formalisms and Dirac's formalisms (Qualitative).			
Linear vector spaces			
Revision of matrix theory (matrix operations, Eigen matrix, transpose, conjugate, unitary, adjoint of a matrix). Linear vector spaces - Definition of a linear vector space, linear independence, basis and dimensions, orthogonality and orthonormality. Hilbert’s space: Definition of Hilbert space, matrices as members of Hilbert space. Dual Space, row and column matrices transpose and conjugate, adjoint of matrices, Hermitian matrix.			
Quantum mechanics in Dirac formalism and fundamentals of Quantum Mechanics			
Dirac notations: Bra and Ket notations for vectors. Dirac representation and matrix operations: Familiarization of matrices as members of Hilbert space in Dirac notation: Matrix representation of 0 and 1 States, Identity Operator I, Applying I to 0⟩ and 1⟩ states, Pauli Matrices and their operations on 0⟩ and 1⟩ states, Unitary matrix and its operations on 0⟩ and 1⟩ states.			
Fundamentals of Quantum Mechanics: Postulates of Quantum Mechanics (Representation of state, concept of operators, Eigenvalue problem: Eigenvalues and Eigenfunctions. normalization of eigenfunctions. Physical interpretation of Eigenvalues, Eigenfunctions and expansion coefficients. Schrodinger's equation in Dirac's notation, superposition, discussion of measurements). Entanglement and its importance in Quantum Computing (Qualitative)			
Principles of Quantum Information & Quantum Computing:			
Concept of qubit and its properties. Representation of qubit by Bloch sphere. Single and Two qubits. Extension to N qubits (Mention only)			
Quantum Gates: Concepts of gates, Single Qubit Gates: Quantum Not Gate, Pauli – X, Y and Z Gates, Hadamard Gate, Phase Gate (or S Gate), T Gate. Action of gates on Qubits, Numerical problems.			

UNIT-II	
Electrical Properties of Materials: Semiconductors	11 Hours
Semiconductors: Band structure - classification of solids. Semiconductors - intrinsic and extrinsic semiconductors, carrier generation. Direct and indirect band gap semiconductors. Fermi-Dirac Statistics, Fermi factor, Fermi energy level in intrinsic and extrinsic semiconductors and effect of temperature on Fermi level, intrinsic effect - maximum device temperature. Conductivity of intrinsic and extrinsic semiconductors - derivation. Effect of temperature on conductivity of intrinsic and extrinsic semiconductor. Hall effect - derivation of Hall coefficient, carrier concentration and mobility. Applications of Hall effect. Numerical examples. p-n junction: Junction formation, Unbiased and biased p-n junction, Devices: LED, Photodiode and solar cell.	
Electrical Properties of Materials: Superconductors	4 Hours
Introduction to superconductors, characteristic properties. Type-I and Type-II superconductors. BCS theory (qualitative). High temperature superconductors, Applications of superconductors. Numerical examples.	
UNIT-III	
Photonics: Lasers	05 Hours
Introduction to lasers, Characteristics of LASER, Interaction of radiation with matter, Einstein's coefficients, Requisites of a Laser System. Conditions for Laser action. Principle, Construction and Working of Nd:YAG laser and Semiconductor laser. Application of Lasers. Numerical problems.	
Photonics: Optical Fibers	05 Hours
Introduction to optical fibers, Principle of Optical Fibers (TIR), Propagation mechanism in optical fibers - Angle of Acceptance and Numerical Aperture (N.A.), Expression for NA, Fractional index change, Modes of propagation, Number of modes and V-Number, Types of Optical Fibers, Attenuation and Mention of Expression for Attenuation coefficient, Attenuation Spectrum of an Optical Fiber- Optical Windows. Discussion of Block Diagram of Point-to-Point Communication, Numerical problems.	
List of Experiments	
1.	Energy gap of a semiconductor by four-probe technique.
2.	Hall effect
3.	Photo-Diode characteristics
4.	Semiconductor laser - Determination of wavelength by diffraction.
5.	LED characteristics and determination of Planck's Constant using LEDs.
6.	Solar cell characteristics
7.	Photo electric effect – Determination of the work function of the material of the emitter of a photocell.
8.	Determination of acceptance angle and numerical aperture of the given Optical fiber.
9.	I-V characteristics of Zener diode
10.	Dielectric constant by charging and discharging of a capacitor.

Course Outcomes: At the end of the course student will be able to

1.	Describe the fundamental principles of the Quantum Mechanics and quantum computing
2.	Summarize the properties of semiconductors and the working principles of semiconductor devices.
3.	Summarize the essential properties of superconductors and its applications.
4.	Describe the principles of LASERS and their relevant applications.
5.	Describe the principles of Optical fibers and their relevant applications.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes															
PH1004-1.1	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
PH1004-1.2	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
PH1004-1.3	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
PH1004-1.4	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
PH1004-1.5	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Parag K Lala, "Quantum Computing – A Beginner's Introduction", Indian Edition, McGraw Hill, Reprint 2020.
2. B. G. Streetmann, "Solid State Electronic devices", 6th edition, Prentice Hall India Learning Private Limited.
3. A. Ghatak, "Optics", Tata McGraw Hill Pub., 5th Edition, 2012.

REFERENCE BOOKS:

1. Michael A. Nielsen & Isaac L. Chuang, "Quantum Computation and Quantum Information", Cambridge Universities Press, 2010 Edition.
2. Vishal Sahani, "Quantum Computing", McGraw Hill Education, 2007 Edition.
3. Maria Luisa Dalla Chiara, Roberto Giuntini, Roberto Leporini, Giuseppe Sergioli, "Quantum Computation and Logic: How Quantum Computers Have Inspired Logical Investigations", Trends in Logic, Volume 48, Springer.
4. Gupta and Kumar, "Solid State Physics", K. Nath & Co., Meerut.
5. A. J. Dekker, "Electrical Engineering Materials", Prentice Hall India Pub., New Delhi, Reprint 2011.
6. S. O. Pillai, "Solid State Physics", New Age International Private Limited, 8th Edition, 2018.
7. M. Ali. Omar, "Elements of Solid State Physics: Principles and Applications", Pearson Publishers.
8. Arthur Beiser, "Concepts of Modern Physics", Tata McGraw Hill Education Private Limited, Special Indian Edition, 2009.
9. Kenneth Krane, "Modern Physics", Wiley International, 3rd Edition, 2012.
10. Michael Tinkham, "Introduction to Superconductivity", II Edition, McGraw Hill, INC

E Books / MOOCs/ NPTEL/ Web links

1.	LASER: https://www.youtube.com/watch?v=WgzynezPiyc
2.	Superconductivity : https://www.youtube.com/watch?v=MT5XI5ppn48
3.	Optical Fiber : https://www.youtube.com/watch?v=N_kA8EpCUQo
4.	Quantum Mechanics : https://www.youtube.com/watch?v=p7bzE1E5PMY&t=136s
5.	Quantum Computing : https://www.youtube.com/watch?v=jHoEjvuPoB8
6.	Quantum Computing : https://www.youtube.com/watch?v=ZuvCUU2jD30
7.	Physics of Animation : https://www.youtube.com/watch?v=kj1kaA_8Fu4
8.	Statistical Physics Simulation : https://phet.colorado.edu/sims/html/plinko-probability/latest/plinkoprobability_en.html
9.	NPTEL Superconductivity: https://archive.nptel.ac.in/courses/115/103/115103108/
10.	NPTEL Quantum Computing : https://archive.nptel.ac.in/courses/115/101/115101092
11.	Virtual LAB : https://www.vlab.co.in/participating-institute-amrita-vishwa-vidyapeetham
12.	Virtual LAB : https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1
Activity Based Learning (Suggested Activities in Class)/ Practical Based learning	
1.	http://nptel.ac.in
2.	https://swayam.gov.in
3.	https://virtuallabs.merlot.org/vl_physics.html
4.	https://phet.colorado.edu
5.	https://www.mypphysicslab.com

PROBLEM SOLVING THROUGH PROGRAMMING				
Course Code:		CS1001-2	Course Type:	ESC
Teaching Hours/Week (L: T: P)		3:0:2	Credits:	04
Total Teaching Hours:		40+0+26	CIE + SEE Marks:	50+50
Teaching Department: Computer Science & Engineering				
Course Objectives:				
1	Make students learn the basics of C programming language including the basic data types, Operators and Evaluating expressions in C.			
2	Apply the concepts of decision making and looping in problem solving to demonstrate its usage using simple programs.			
3	Apply the concepts of Arrays, User-defined functions and code reusability in problem solving along with parameter passing and returning with the help of user defined functions.			
4	Demonstrate the usage of Strings and Structures			
5	Demonstrate the usage of Pointers, and File handling that are essential for understanding the concepts with simple examples.			
UNIT-I				
Introduction To Computer and C language:				10 Hours
Introduction to computers, input, and output devices, Algorithm, Flowchart, Designing efficient programs. Introduction to C, Structure of C program, Compilers, Compiling and executing C programs, variables, constants, Input/output statements in C.				
Operators:				
Operators in C, Type conversion and typecasting.				
Decision control and Looping statements:				
Introduction to decision control, Conditional branching statements, iterative statements, nested loops, break and continue statements, goto statement.				
UNIT-II				
Arrays:				10 Hours
Declaration of arrays, accessing the elements of an array, storing values in arrays, Operations on arrays, Passing arrays to functions, Two dimensional arrays, operations on two-dimensional arrays, two-dimensional arrays to functions, multidimensional arrays. Example Programs on arrays				
Functions:				
Introduction using functions, Function definition, function declaration, function call, return statement, passing parameters to functions, scope of variables, storage classes, recursive functions.				
Introduction to strings:				
Declaring and Initializing strings, String manipulation functions.				
UNIT-III				
				05 Hours
Structures: Defining Structure, Declaration of Structure Variable, Accessing Structure members, copying and comparing structure variable, operation on individual member,				

nesting of structures, Array of structures. Application of pointers and function on Structures.

Files: Basic file operations: Open, Close, Read, Write.

Introduction to pointers: declaration and initialization of pointers, accessing the address of the variable, accessing the variable through the pointer.

Suggested List of Experiments

PART A

1. Write a C program to find the roots of a quadratic equation $ax^2+bx+c=0$
2. Write a C program to find the sum of all the digits and occurrence of a digit in the number.
3. Write a C program to find the GCD and LCM of given two numbers using Euclid's method.
4. Write a C program to print the prime numbers in a given range.
5. Write a C program to find if a given string is a palindrome or not using string manipulation functions.
6. Write a C program to input N real numbers in 1-D array. Compute mean, variance and Standard Deviation.
[Mean= sum/N, Variance = $\sum (Xi-\text{mean})^2 / N$, STD Deviation= $\sqrt{\text{variance}}$.]
7. Write a C program to read N integers into an array A and find the sum of elements using pointers.
8. Write a C program to copy contents of one file to another file.

PART B

1. Write a C program to perform a binary search for a given key integer in a single dimensional array of numbers in ascending order and report success or failure in the form of a suitable message.
2. Write a C program to input N integer numbers into a single dimension array, sort them in to ascending order using selection sort technique, and then to print both the given array and the sorted array with suitable headings.
3. Write a C program to transpose a matrix of order M x N and find the trace of the resultant matrix.
4. Write a C program using functions to read two matrices A (M x N) and B (P x Q) and to compute the product of A and B if the matrices are compatible for multiplication.
5. Write a C program using functions readmat(), rowsum (), colsum (), totsum () and printmat() to read the values into a two dimensional array A, find the sum of all the elements of a row, sum of all the elements of a column, find the total sum of all the elements of the two dimensional array A and print the results.
6. Write a C program to perform a linear search for a given key integer in a single dimensional array of numbers and report success or failure in the form of a suitable message using functions.
7. Write a C program to enter the information like name, register number, marks in 6 subjects of N students into an array of structures, and find the average & display grade based on average for each student.

Average	Grade
80-100	Distinction

		60-79	First Class	
		40-59	Second Class	
		<40	Fail	

8. Write a C program, to implement a bubble sort technique using function to sort given N integers in ascending/ descending order as per user's preference.

Course Outcomes: At the end of the course student will be able to

- 1. Describe** the basics of computer system, basics of C and the process of problem-solving aspects using algorithmic solution for a given problem.
- 2. Apply** the knowledge to develop the C program using control statements such as branching and looping constructs for a given problem.
- 3. Apply** the knowledge of code re-usability, parameter passing and returning values to develop a maintainable C program using these concepts including arrays and functions.
- 4. Understand** the concepts of user defined and built-in functions using strings.
- 5. Identify and describe** the use of structures, pointers and file handling mechanisms in a C program.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→ ↓Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
													1	2	3
CS1001-2.1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CS1001-2.2	2	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CS1001-2.3	2	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CS1001-2.4	2	2	3	-	-	-	-	-	-	-	-	-	-	-	-
CS1001-2.5	2	3	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1. E. Balaguruswamy, "Programming in ANSI C", Tata McGraw Hill, 3rd Edition, 2004.
2. Jacqueline A. Jones & Keith Harrow, "C Programming with Problem Solving", Pearson,

REFERENCE BOOKS:

1. Kernighan & Ritchie, "The C Programming (ANSI C)", Prentice Hall; 2nd Edition, 1998.
2. Rajiv Khanna, "Computer Concepts and C Programming", New Age International Pvt Ltd Publishers, 1st Edition, 2006.
3. Yashwant Kanetkar, "Let Us C", 5th Edition, BPB Publications, New Delhi, 2004.

E Books / MOOCs/ NPTEL

1. <http://www.lysator.liu.se/c/bwk-tutor.html#introduction>
2. http://www.acm.uiuc.edu/webmonkeys/book/c_guide/
3. C programming Tutorial by Mark Burgers <http://markburgess.org/CTutorial/C-Tut-4.02.pdf>
4. <http://nptel.ac.in/courses/106105085/4>
5. <https://www.lynda.com/C-training-tutorials/1249-0.html>

BASIC ELECTRONICS			
Course Code:	EC1001-1	Course Type:	ESC
Teaching Hours/Week (L: T: P):	3:0:0	Credits:	03
Total Teaching Hours:	40+0+0	CIE + SEE Marks:	50+50
Teaching Department: Electronics & Communication Engineering			
Course Objectives:			
1.	To familiarize the student with Semiconductor devices like Diodes, Transistors and their applications		
2.	To analyze the working of simple electronic circuits involving Op-amps, 555 Timer and Linear Regulator ICs.		
3.	To understand the fundamentals of Modern communication system.		
4.	To introduce the fundamentals of Embedded Systems		
UNIT-I			
Diodes and their Applications			07 Hours
Semiconductor Diode, Diode Equivalent circuits, Half Wave Rectifier, Full wave Bridge Rectifier, capacitor filter circuit (only qualitative approach). Zener Diode and its use in Voltage Regulation			
Transistors and their Applications			09 Hours
Bipolar Junction Transistor: Construction and operation, Common Emitter Characteristics RC coupled amplifier (frequency response excluded), BJT as a switch.			
Field Effect Transistor: Construction, Drain and Transfer Characteristics of Enhancement mode MOSFET, CMOS Inverter.			
UNIT-II			
Op-Amp & Linear IC Applications			11 Hours
Introduction, Op-Amp Specifications, Differential & Common-Mode operation, Op-Amp applications: Inverting/Non-Inverting Amplifier, Summing, Integrator, Differentiator, Comparator. 555 Timer IC in Astable mode. 78XX series IC Voltage Regulators.			
Feedback and Oscillator Circuits			05 Hours
Feedback– Principle and advantages of negative feedback, Voltage series feedback amplifier. Concept of positive feedback, Op-Amp Oscillators – RC phase shift, Hartley and Colpitts's Oscillator			
UNIT-III			
Fundamentals of Communication and Embedded Systems			08 Hours
Modern communication system scheme (Block scheme), Information source, Input Transducers, Transmitter, Channels, Receivers, Noise, Fundamentals of Cellular communication.			
Embedded system definition, Elements of Embedded systems, Core of Embedded systems, Microprocessor v/s Microcontroller, Sensors and Actuators with examples.			
Course Outcomes: At the end of the course student will be able to			

1.	Explain the operation of Rectifiers; Design a rectifier circuit, given the specification for output Voltage, PIV, and ripple factor; Design a Zener voltage regulator for the given specification of output voltage and Power;
2.	Explain the construction and operation of Bipolar transistor in CE or CB Mode; Explain the use of BJT in Amplification as well as switching operations; Explain the construction and operation of JFET or MOSFET; Explain the operation of a CMOS Inverter;
3.	List the ideal and practical parameters for an Op-Amp; Define Op-amp Specifications; Explain the use of Op-Amp in Amplification, Summing, Integration, Differentiation and comparison; Design an Astable Multivibrator, using 555 Timer IC, for the given frequency and duty cycle;
4.	List the advantages and disadvantage of Negative Feedback; Explain the impact of negative feedback on Amplifier gain, Input and Output Impedance for a Series Voltage Negative feedback; Explain the operation of Op-Amp based RC Phase-shift, Hartley, and Colpitts Oscillator
5.	Explain the scheme of a Modern Communication System; List the differences between a general computing system and Embedded System; Describe the differences between Harvard and Von-Neuman, RISC and CISC system architectures

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
EC1001-1.1	3	-	-	-	-	-	-	-	-	-	-	-
EC1001-1.2	3	-	-	-	-	-	-	-	-	-	-	-
EC1001-1.3	3	-	-	-	-	-	-	-	-	-	-	-
EC1001-1.4	3	-	-	-	-	-	-	-	-	-	-	-
EC1001-1.5	3	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Robert L. Boylestad, Louis Nashelsky, "Electronic Devices and Circuit Theory", 11 th Edition, PHI, 2016
2.	Simon Haykin, "Introduction to Analog and Digital Communications", Wiley Publishers, 2 nd Edition, 2019
3.	Theodore Rappaport, "Wireless Communications: Principles and Practice", Pearson, 2 nd Edition, 2016
4.	Shibu K V, "Introduction to Embedded Systems", TATA Mc Graw Hill Edu., 2 nd Edition, 2016

E Books / MOOCs/ NPTEL

1.	https://nptel.ac.in/courses/117107095
2.	https://nptel.ac.in/courses/117103063
3.	https://www.coursera.org/learn/electronics?#syllabus
4.	https://www.coursera.org/learn/diode-pn-junction-metal-semiconductor-contact?specialization=semiconductor-devices#syllabus
5.	https://www.coursera.org/learn/transistor-field-effect-transistor-bipolar-junction-transistor?specialization=semiconductor-devices

FUNDAMENTALS OF CYBER SECURITY			
Course Code:	IS1101-1	Course Type:	ETC
Teaching Hours/Week (L: T: P: S):	3:0:0:0	Credits:	03
Total Teaching Hours:	40	CIE + SEE Marks:	50+50
Teaching Department: Information Science & Engineering			
Course Objectives:			
1.	Define the area of cybercrime and forensics.		
2.	Explain the motive and causes for cybercrime, detection, and handling.		
3.	Investigate Areas affected by cybercrime.		
4.	Illustrate tools used in cyber forensic		
UNIT-I			
Introduction to Cybercrime			15 Hours
Cybercrime - Definition and Origins of the Word, Cybercrime and Information Security, Who are Cybercriminals? Classifications of Cyber Crimes. [T1: 1.1-1.5]			
Cyber offenses: How Criminals Plan Them			
How Criminals Plan the Attacks, Social Engineering, Cyberstalking, Cybercafe and Cybercrimes, Botnets: The Fuel for Cybercrime. [T1: 2.1-2.6]			
Mobile and Wireless Devices			
Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit Card Frauds in Mobile and Wireless Computing Era, Security Challenges Posed by Mobile Devices, Attacks on Mobile/Cell Phones, Mobile Devices: Security Implications for organizations, Organizational Measures for Handling Mobile, Organizational Security Policies and Measures in Mobile Computing Era. [T1: 3.1-3.11]			
UNIT-II			
Tools and methods used in Cybercrime			14 Hours
Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking, Keyloggers and Spywares, Virus and Worms, Trojan-horses and Backdoors, Steganography, DoS and DDoS Attacks, SQL Injection, Buffer Overflow, Attacks on Wireless Networks. [T1: 4.1-4.12]			
Phishing and Identity Theft			
Introduction to Phishing, Identity Theft (ID Theft). [T1: 5.1-5.3]			
UNIT-III			
Understanding Computer Forensics			11 Hours
Introduction, Digital Forensics Science, The Need for Computer Forensics, Cyberforensics and Digital Evidence, Forensics Analysis of E-Mail, Digital Forensics Life Cycle, Chain of Custody Concept, Setting up a Computer Forensics Laboratory: Understanding the Requirements, Computer Forensics and Steganography, Challenges in Computer Forensics, Special Tools and Techniques, Forensics Auditing, Antiforensics. [T1: 7.1-7.19]			
Course Outcomes: At the end of the course student will be able to			
1.	Comprehend the Cybercrime and its origin		
2.	Analyse the cybercrimes in mobile and wireless devices		

3.	Apply tools and methods used in Cyber crimes
4.	Analyse Phishing and ID Theft
5.	Comprehend Digital Forensics

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
IS1101-1.1	2	-	-	-	-	1	-	3	-	-	-	-	-	-	-
IS1101-1.2	-	3	-	1	-	2	-	-	2	-	-	-	-	-	-
IS1101-1.3	-	3	2	-	-	-	-	-	-	-	-	-	-	-	-
IS1101-1.4	2	-	-	-	-	2	-	-	-	-	-	-	-	-	-
IS1101-1.5	-	-	-	-	-	-	-	3	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Sunit Belapure and Nina Godbole, "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", Wiley India Pvt Ltd, ISBN: 978-81-265-21791, Publish Date 2013.
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REFERENCE BOOKS:

1.	Thomas J. Mowbray, "Cybersecurity: Managing Systems, Conducting Testing, and Investigating Intrusions", John Wiley & Sons, Inc, ISBN: 978 -1-118 -84965 -1, 2014.
2.	James Graham, Ryan Olson, Rick Howard, "Cyber Security Essentials", CRC Press, 15-Dec 2010. Anti- Hacker Tool Kit (Indian Edition) by Mike Shema, Publication Mc Graw-Hill.
3.	Santosh B. J., K. V. S. S. S. Sairam, Shubham Kumar, Chandu Jagan Sekhar M., "Information and Cyber Security", Scientific International Publishing House, ISBN-978-93-5625-694-1.

TECHNICAL ENGLISH			
Course Code	HU1001-1	Course Type	HSMC
Teaching Hours/Week (L:T:P:S)	1:0:2:0	Credits	02
Total Teaching Hours	15+0+26	CIE + SEE Marks	50+50
Teaching Department: Humanities			
Course Objectives:			
1.	Identify the nuances of Phonetics, Intonation and enhance pronunciation skills		
2.	Understand Technical Communication along with the barriers and application of effective Interpersonal Communication Skills		
3.	Enhance basic English grammar and essentials of language skills		
4.	Improve sentence structure through the utilization of cohesive devices, thereby fostering the development of both spoken and writing skills		
UNIT – I			
Phonetics & Pronunciation			8 Hours
Introduction to Phonetics; Word Stress, Rhythm and Intonation; Weak Forms and Strong Forms, Role of IPA in past tense and plural forms of words, Awareness of Different Accents			
Communication Skills			8 Hours
Introduction to Communication, Greeting and Introducing, Making Requests, asking for and Giving Permission, Offering Help			
Understanding Telephone Communication, Handling Calls, asking for and Giving Information, Telephone Etiquette			
UNIT – II			
Language Skills			16 Hours
Basic English Grammar, Ability to identify, Analyze, Interpret and Describe the critical ideas, values, and themes through literary works			
UNIT – III			
Writing Skills			9 Hours
Paragraph writing, Refutations, Linkers, Types of Letters			
Course Outcomes: At the end of the course students will be able to			
1.	Identify the nuances of phonetics, intonation, and pronunciation to appreciate and incorporate Received Pronunciation		
2.	Interpret and assess nuances of oral communication skills and non-verbal communication for professional usage		
3.	Identify, interpret, and describe the critical ideas, values, and themes to appreciate literary pieces for its language and social interpretations		
4.	Implement English vocabulary in both personal and professional contexts, enhancing language proficiency while also honing effective writing skills adaptable to various		

forms of written communication.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes													1	2
HU1001-1.1	1	1	-	-	-	-	-	2	-	2	-	3	-	-
HU1001-1.2	2	-	-	-	-	2	-	-	-	3	-	3	-	-
HU1001-1.3	-	2	-	-	-	-	3	2	-	3	-	3	-	-
HU1001-1.4	-	2	-	-	-	2	-	-	2	2	-	2	-	-

1: Low 2: Medium 3: High

TEXTBOOKS

1.	Tidings, Victor, R. et al. "A Textbook of English Language & Communication Skills", 2022.
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REFERENCE BOOKS

1.	Jones, D, "English Pronunciation Dictionary", 2011.
2.	Woods, "A Remedial English Grammar for Foreign Students", 2016.
3.	Kumar, S., "Communication Skills". Oxford University Press, 2011.
4.	CIEFL, Hyderabad. Exercises in Spoken English Part I. Oxford University Press.
5.	CIEFL, Hyderabad. Exercises in Spoken English Part II. Oxford University Press.
6.	CIEFL, Hyderabad. Exercises in Spoken English Part III. Oxford University Press.
7.	Zinsser, W. "On Writing Well", 1976.
8.	Swan, "Practical English Usage". Oxford University Press, 2014.
9.	Lyons, L. H., "Study Writing". Cambridge University Press, 2006.
10.	Subhashini, Victor, R. et al. , "A Textbook of English Language & Communication Skills", 2022.

E Resources

1.	https://www.macmillandictionary.com/dictionary/british/
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CONSTITUTION OF INDIA																	
Course Code				HU1002-1			Course Type				UCC						
Teaching Hours/Week (L:T:P:S)				1:0:0:0			Credits				01						
Total Teaching Hours				15+0+0			CIE + SEE Marks				50+0						
Teaching Department: Humanities																	
Course Objectives:																	
1.	Understand their Obligations, Responsibilities, Privileges and Rights, Duties and the Role that they have to play in deciding the Administrative Machinery of the country.																
2.	Understand the nature and character of relations between union and state governments and sensitize the students about the statutory institutions and policies.																
UNIT - I																	
Evolution of the Indian Constitution													6 Hours				
1909 Act, 1919 Act, 1935 Government of India Act, Constituent Assembly: Composition and Functions, Basic structure of Indian Constitution, Salient Features of Indian Constitution, Fundamental Duties & Rights																	
UNIT - II																	
Structure of Government													6 Hours				
Union Government: Legislature; Executive-President, Prime Minister, Council of Ministers; Judiciary, Judicial Review																	
State Government: Executive: Governor, Chief Minister, Council of Ministers																	
Local Government: Panchayat Raj Institutions, Urban Governance																	
UNIT - III																	
Statutory Institutions													3 Hours				
Elections - Election Commission of India, National Human Rights Commission, National Commission for Women																	
Course Outcomes: At the end of the course student will be able to																	
1.	Analyze the legalities and related issues of drafting, adoption, and enforcement of the Indian Constitution as a fundamental law of the nation and the provisions and privileges of Indian Citizenship																
2.	Understand and judiciously use the fundamental rights, fundamental duties and privileges envisaged in the constitution propagating social harmony and equality and respecting the rights and liberties of other people.																
Course Outcomes Mapping with Program Outcomes & PSO																	
Program Outcomes→				1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes																	1 2

HU1002-1.1	-	-	-	-	-	-	-	3	-	-	1	1	-	-
HU1002-1.2	-	-	-	-	-	-	-	2	-	-	1	1	-	-
1: Low 2: Medium 3: High														
REFERENCE BOOKS														
1.	Basu, D.D., "Introduction to the Constitution of India", Twentieth Edition, Reprint 2011. LexisNexis Butterworths Wadhwa, Nagpur, Haryana, India, 2011.													
2.	Pylee, M. V. "Introduction to Constitution of India", Fourth Revised Edition. Vikas Publishing House Pvt. Ltd., New Delhi, 2005.													
3.	Sharma, B. K. "Introduction to Constitution of India", Second Edition. Prentice Hall of India Pvt. Ltd., New Delhi, 2004.													
4.	Venkatesh, B. R. & Merunandan, K. B., "An Introduction to Constitution of India and Professional Ethics", Second Edition. Merugu Publications, Bangalore, 2007.													
E Resources														
1.	NPTEL course: "Constitution of India". http://nptel.ac.in/courses/109104032/													
2.	"Constitution of India" eBook. https://pothi.com/pothi/book/ebook-ministry-law-and-justice-constitution-india .													
3.	iasplanner.blogspot.com/2010/11/free-ebook-download-constitution-of.html													
4.	www.iasabhiyan.com													
5.	Samvidhaan, Documentary by Prasaar Bharathi													

Security

MATHEMATICS WITH MATLAB																				
Course Code:				MA1006-1				Course Type				BSC								
Teaching Hours/Week (L: T: P: S)				0:0:2:0				Credits				01								
Total Teaching Hours				0+0+26+0				CIE Marks				50								
Teaching Department: Mathematics																				
Course Objectives:																				
1.		Understand the use of the basic operators, and some built-in functions of MATLAB.																		
2.		Create and work with arrays																		
3.		Create and display simple plots																		
4.		Solve by Symbolic and Numerical computation techniques																		
List of Experiments																				
1		Introduction to MATLAB: Basic Operators: Arithmetic, Logical and Relational Operators. Elementary math functions such as algebraic, trigonometric, logarithmic, exponential functions, Conditions and Loops.																		
2		Symbolic Computation, plotting curves, surfaces and vector fields.																		
3		Differentiation of composite and implicit functions.																		
4		Taylor's/ Maclaurin's series expansion of a function of a single variable.																		
5		Computation of partial derivatives and Jacobians																		
6		Evaluation of double/triple integrals with constant/variable limits.																		
7		Computation of angle between (a) radius vector and tangent ; (b) two curves																		
8		Computation of radius of curvature																		
9		Computation and visualization of (a) gradient of a scalar function ; (b) divergence and curl of a vector function																		
10		Solution (with solution curve) of first-order ordinary differential equation																		
11		Solution (with solution curve) of second and higher-order linear differential equation with constant coefficients																		
12		Computation of (a) eigenvalues and eigenvectors of a square matrix; (b) largest eigenvalue and the corresponding eigenvector of a square matrix; (c) rank of a square matrix																		
13		Solution of a system of linear equations by Gauss elimination method																		
14		Solution of a system of linear equations by Gauss-Seidel method																		
Course Outcomes: At the end of the course student will be able to																				
1.		Write and compile simple MATLAB codes. Implement basic operators and conditions and loops effectively.																		
2.		Write MATLAB programs gradually for the theoretically studied mathematical concept. Develop illustrations to demonstrate mathematical issues and their answers.																		
Course Outcomes Mapping with Program Outcomes & PSO																				
		Program Outcomes→			1	2	3	4	5	6	7	8	9	10	11	12	PSO↓			
		↓ Course Outcomes																1	2	3
		MA1006-1-1.1			3	2		-	-	-	-	-	-	-	-	-	-	-	-	-

MA1006-1-1.2															3	2	-	-	-	-	-	-	-	-	-	-	-	Security
1: Low 2: Medium 3: High																												
REFERENCE BOOKS:																												
1.	Rudra Pratap, “MATLAB”, OXFORD University Press, 2010.																											
2.	Dorothy C. Attaway Ph.D., A practical introduction to prog. And problem-solving, 5 th edition.																											
E-Resources																												
1.	https://www.mathworks.com › matlab › matlab_prog																											
2.	https://www.coursera.org/specializations/mathematics-engineers																											
3.	https://www.coursera.org/specializations/matlab-programming-engineers-scientists																											
4.	https://www.coursera.org/learn/matlab																											

II SEMESTER (AI&DS, AI&ML, CC, CS, IS, RI)

Course and	Course Title	Te ac	Teaching hours/Week		Examination
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Course code				Theory Lecture	Tutorial	Practical/ Drawing	SL #	Duration in hours	CIE	SEE	Total Marks	
				L	T	P						
BSC	MA1007 – 1	Discrete Mathematics and Transform Techniques	MA	4	0	0	3	3	50	50	100	
BSC	CY1003-1	Materials Chemistry for Computer Systems	CY	3	0	2	2	3	50	50	100	
ESC	EC1002-2	Applied Digital Logic Design	EC	2	0	2	1	3	50	50	100	
PLC	CS1005-2	Introduction to Python Programming	CS	2	0	2	1	3	50	50	100	
ESC	EE1001-2	Basic Electrical Engineering	EE	2	0	2	1	3	50	50	100	
AEC	BT1651-1	Biology for Engineers	BT	1	0	0	-	1	50	50	100	
ESC	ME1004-1	Engineering Visualization	ME	0	0	2	-	0	50	0	50	
MNC	CV1005-1	Environmental Studies and Sustainability	CV	1	0	0	-	0	50	0	50	
TOTAL				15	0	10	8	16	400	300	700	
Additional Academic activities beyond class hours				-	-	-	3					
Learning hours per week				15	-	10	11					
Total notional learning hours (L+T+P+J+SL)		Per week		36								
		Per Semester		36x16=576								

+Mandatory Internship-I*								
1.	INT	UC1001-1	Internship – I	Mandatory Intra Institutional Internship of duration (80 - 90 Hours) to be completed during I & II Semesters. *The grades will be included in the IV semester grade card (Refer 11.5.2 for details)	100	--	100	2

DISCRETE MATHEMATICS AND TRANSFORM TECHNIQUES			
Course Code:		MA1007 - 1	Course Type: BSC
Teaching Hours/Week (L: T: P):		4:0:0	Credits: 04
Total Teaching Hours:		50+0+0	CIE + SEE Marks: 50+50
Teaching Department: Mathematics			
Course Objectives:			
1.	This course will enable the students to master the basic tools of set theory and relations, propositional and predicative logics, numerical methods, Fourier series and transforms and become skilled for solving problems in science and engineering.		
UNIT-I			
Logics			8 Hours
Propositional logic, logical operations, Rules of inference, Predicates calculus. Methods of Proof: Direct, Indirect and Proof by Contradiction and Contrapositive. Proofs by Mathematical Induction (both weak and strong inductions).			
UNIT-II			
Set Theory And Graph Theory			11 Hours
Relations- Relations and Digraphs, Properties of Relations, Equivalence Relations, Transitive Closure and Warshall's Algorithm. Functions- permutations functions, functions for computer science. Graphs: Basic terminologies, simple graphs, complete graphs, bipartite graphs. Adjacency matrices, incidence matrices and graph isomorphism. Connectivity- vertex and edge connectivity. Euler and Hamiltonian graphs and their applications. Planar graphs, graph coloring and their applications.			
UNIT-III			
Numerical Methods			11 Hours
Roots of algebraic and transcendental equations- Newton Raphson method, Regula Falsi method. Numerical solution of ordinary differential equations- Taylor's series method, Modified Euler's method and Runge –Kutta method of fourth order. Numerical solution of partial differential equations-classification of partial differential equations, examples, solution of Laplace and Poisson equations by standard five point formulae, solution of heat and wave equations.			
UNIT-IV			
Fourier Series and Fourier Transform			10 Hours
Periodic functions, Euler's formulae, Fourier series of odd and even functions, functions with arbitrary period, half range series. Fourier transform, inverse Fourier transform, convolution theorem, Fourier sine and cosine transforms. Discrete Fourier transform(DFT) and Fast Fourier transform (FFT)- applications.			
UNIT-V			
Z-Transforms			10 Hours
Z-transforms of standard functions, Bilateral Z- Transform. ROC, linearity, Time shift, Convolution, Scaling & Differentiation in Z-Domain, Time reversal property, Initial and Final Value Theorems. Inverse Z-transform: Partial Fraction Method, Power series/ division method, Contour integral Method. Unilateral Z-Transform: Properties, Solution of difference equations.			
Course Outcomes: At the end of the course student will be able to			
1.	Establish by deduction the validity of an argument using inference rules.		

2.	Represent a relation in terms of matrix and digraph, apply permutation functions for encoding and decoding simple text messages. Identify suitable data structure for representing a graph, apply the concept of connectivity in real life problems.
3.	Apply numerical methods to find solutions of algebraic equations and ordinary differential equations and partial differential equations.
4.	Apply the analytical technique to express periodic function as a Fourier sine and cosine series and apply the concepts of Fourier- transforms to solve engineering problems.
5.	Apply the concepts of Z- transforms to solve engineering problems.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes													1	2
MA1007 - 1.1	3	2	-	-	-	-	-	-	-	-	-	-	-	-
MA1007 - 1.2	3	2	-	-	-	-	-	-	-	-	-	-	-	-
MA1007 - 1.3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
MA1007 - 1.4	2	2	-	-	-	-	-	-	-	-	-	-	-	-
MA1007 - 1.5	3	2	-	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Kenneth H. Rosen, "Discrete Mathematics and its applications", Tata McGraw Hill, V Edition, 2003.
2.	B.S. Grewal, J. S. Grewal, "Numerical Methods in Engineering and Science", Khanna Publishers, 6 th edition, 2002.
3.	Martin Vetterli, Jelena Kovacevic and Vivek Goyal, "Foundations of Signal Processing", Cambridge University Press, 2014.

REFERENCE BOOKS:

1.	Erwin Kreyszig, "Advanced Engineering Mathematics", John Wiley and Sons, 10 th Edition (Reprint), 2016.
2.	Bernard Kolman, Robert C. Busby, Sharon Ross, "Discrete Mathematical Structures" III edition, PHI 2001.
3.	Ralph P. Grimaldi, "Discrete and Combinatorial Mathematics", Pearson Education, Asia, IV Edition-2002.
4.	J. P. Tremblay, R. Manohar, "Discrete Mathematical Structures with applications to computer Science", Tata McGraw Hill-1987.
5.	S. S. Sastry, "Introductory methods of Numerical Analysis", Prentice Hall, 2nd edn.1990.
6.	M. K. Jain, S.R.K. Iyengar and R.K. Jain "Numerical methods for Scientific and Engineering computations", Wiley Eastern, edn.1985.

E Books / MOOCs/ NPTEL

1.	http://www.nptelvideos.in/2012/11/discrete-mathematical-structures.html
2.	http://cglab.ca/~discmath/notes.html
3.	http://ocw.mit.edu/courses/mathematics/ (online course material)

MATERIALS CHEMISTRY FOR COMPUTER SYSTEMS

Course Code:	CY1003-1	Course Type:	BSC
Teaching Hours/Week (L: T:P: S):	3:0:2:0	Credits:	04
Total Teaching Hours:	40+0+26	CIE + SEE Marks:	50+50

Teaching Department: Chemistry

Course Objectives:

1	To understand the basic components of Electrochemical cells and relate their principles to battery technology.
2	To Identify the different types of polymers and polymer composites, their synthetic routes and applications, and Analytical techniques.
3	To understand the basic components of sensors, PCB, and synthesis and properties of nanomaterials.
4	To Identify the synthetic approaches undertaken for designing nanomaterials, and materials used in display devices.
5	To understand the prime problems associated with e-waste and subsequent remedial measures undertaken.

UNIT-I

Electrode & Energy Systems

8 Hours

Electrode System: Introduction to galvanic cell. Reference electrode - Introduction, calomel electrode – construction, working and applications. Concentration cell –Definition, construction, working, and numerical problems. Ion selective electrode–definition, construction, and advantages of glass electrode, determination of pH using glass electrode.

Energy Systems: Introduction to batteries, construction, working, and applications of Lithium-ion batteries. Fuel cells, Construction, working and applications of methanol-oxygen fuel cell.

Polymers & Analytical Techniques

07 Hours

Polymers: Introduction, Molecular weight –Number average, weight average, and numerical problems. Preparation, properties, and commercial applications of carbon fibre. Conducting polymers– synthesis and conducting mechanism of polyacetylene and commercial applications.

Analytical Techniques: Principle and instrumentation of Conductometry; its application in the estimation of weak acid and strong acid. Principle and instrumentation of Potentiometry; its application in the estimation of iron.

UNIT-II

Sensors, PCB and Nanomaterials

08 Hours

Sensors: Introduction, working, principle, and applications of Optical sensors and electrochemical sensors. Sensors for the measurement of dissolved oxygen (DO). Electrochemical gas sensors for SO_x and NO_x.

Printed Circuit Boards: Electroless plating – Introduction, Electroless plating of copper in the manufacture of double-sided PCB and its applications.

Nanomaterials: Introduction, classification, preparation of nanomaterials by sol-gel and co-precipitation method with example. Properties, synthesis, and applications of carbon nanotubes.

Memory Devices and Display Systems		07 Hours
Memory Devices: Introduction, Basic concepts of electronic memory. Classification of electronic memory devices, types of organic memory devices (organic molecules, polymeric materials, organic-inorganic hybrid materials).		
Display Systems: Photoactive and electroactive materials, Nanomaterials, and organic materials used in optoelectronic devices. Liquid crystals (LCs) - Introduction, classification of LCs, and types of mesophases. Applications in Liquid Crystal Displays (LCDs)-Electro-optic effect. Light emitting electrochemical cells.		
UNIT-III		
E-Waste Management & Green Fuels		10 Hours
E-waste Management: Introduction, sources of e-waste, Composition, Characteristics, and Need of e-waste management. Toxic materials used in manufacturing electronic and electrical products; health hazards due to exposure to e-waste. Recycling and Recovery: Different approaches of recycling (separation, thermal treatments, hydrometallurgical extraction, pyro metallurgical methods, direct recycling). Extraction of gold from E-waste. Role of stakeholders in environmental management of e-waste (producers, consumers, recyclers, and statutory bodies).		
Green Fuels: Introduction, construction and working of solar photovoltaic cell, advantages, and disadvantages. Generation of energy (green hydrogen) by electrolysis of water by PEM method.		
List of Experiments		
1.	Determination of strength of an acid in Pb-acid battery (Demonstration).	
2.	Determination of Total Hardness of a water sample using disodium salt of EDTA.	
3.	Estimation of iron in TMT bar by diphenylamine/external indicator method.	
4.	Synthesis of polyurethane (Demonstration).	
5.	Conductometric estimation of a strong acid with standard NaOH solution.	
6.	Potentiometric estimation of FAS using standard K ₂ Cr ₂ O ₇ solution.	
7.	Determination of pKa of vinegar using pH sensor (Glass electrode).	
8.	Determination of the viscosity coefficient of a given liquid using Ostwald's viscometer.	
9.	Estimation of Copper present in electroplating effluent by optical sensor (colorimetry).	
10.	Colorimetric determination of iron.	
11.	Conductometric estimation of a weak acid using standard NaOH solution.	
12.	Estimation of Sodium present in soil/effluent sample using a flame photometer.	
13.	Synthesis of biodiesel (Demonstration).	
14.	Synthesis of Iron-oxide Nanoparticles (Demonstration).	
Course Outcomes: At the end of the course student will be able to		
1	Understand the basic components of Electrochemical cells and relate their principles on sensors and Analytical techniques.	
2	Identify the different types of polymers and polymer composites, their synthetic routes and applications and Analytical techniques.	
3	Understand the basic components of Electrochemical cells and relate their principles to sensors.	
4	Identify the synthetic approaches undertaken for designing nanomaterials, materials used in display devices.	
5	Understand the prime problems associated with e-waste and subsequent remedial measures undertaken.	

Course Outcomes Mapping with Program Outcomes & PSO

	Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
	↓ Course Outcomes															
	CY1003-1.1	3	1	-	-	-	-	-	-	-	-	-	-	-	-	-
	CY1003-1.2	3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
	CY1003-1.3	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
	CY1003-1.4	2	1	-	-	-	-	-	-	-	-	-	-	-	-	-
	CY1003-1.5	2	1	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1	P. C. Jain & Monica Jain, "Engineering Chemistry", Dhanpat Rai Publications, New Delhi, 2015.
2	R. V. Gadag and Nityananda Shetty, "A Text Book of Engineering Chemistry", 2 nd Edition, I. K. International Publishing house, 2016.
3	S. S. Dara & S. S. Umare, "A Textbook of Engineering Chemistry", 12 th Edition, S. Chand & Company Ltd., 2011.

REFERENCE BOOKS:

1	Baskar, "Wiley Engineering Chemistry", 2 nd Edition, Wiley India Pvt. Ltd, New Delhi, 2013.
2	Satya Prakash & Manisha Agrawal, "Engineering Chemistry", Khanna Book Publishing, Delhi.
3	Bahl & Tuli, "Essentials of Physical Chemistry", S. Chand Publishing.
4	Sunita Rattan, "Applied Chemistry", Kataria.
5	D. Groul Krishana, "Engineering Chemistry – I", Vikas Publishing.
6	F. W. Billmeyer, "Text Book of Polymer Science", John Wiley & Sons, 4 th Edition, 1999.
7	G. A. Ozin & A. C. Arsenault, "Nanotechnology A Chemical Approach to Nanomaterials", RSC Publishing, 2005.
8	Kirby W. Beard, "Linden's Handbook of Batteries", Fifth Edition, Mc GrawHill, 2019.
9	Takatoshi Tsujimura, "OLED Display Fundamentals and Applications", Wiley-Blackwell, 2012.
10	MaxLu, Francois Beguin, Elzbieta Frackowiak, "Super capacitors: Materials, Systems, and Applications", Wiley-VCH;1st edition, 2013.
11	H. Panda, "Handbook on Electroplating with Manufacture of Electro-chemicals", ASIAPACIFIC BUSINESS PRESS Inc., 2017.
12	Sudharani, "Laboratory manual in Engineering Chemistry", Dhanpat Rai Publishing Company, New Delhi.
13	"Expanding the Vision of Sensor Materials", National Research Council 1995, Washington, DC: The National Academies Press. doi:10.17226/4782.
14	Mahesh B and Roopa Shree B, "Engineering Chemistry", Sunstar Publisher, Bengaluru, ISBN978-93-85155-70-3, 2022
15	F. H. Froes, et al., "High Performance Metallic Materials for Cost Sensitive Applications", John Wiley & Sons, 2010.
16	K. R. Mahadik and L. Satyanarayana, "Instrumental Methods of Analysis", Nirali Prakashan, 2020.
17	Douglas A. Skoog, F. James Holler, Stanley R. Crouch, "Principles of Instrumental Analysis", Seventh Edition, Cengage Learning, 2020.
18	V R Gowariker, N V Viswanathan, Jayadev, Sreedhar, "Polymer Science", NewageInt. Publishers, 4 th Edition, 2021.

19	Hari Singh, "Nanostructure materials and nanotechnology", Nalwa, Academic press, 1 st Edition, 2002.
20	O. G. Palanna, "Engineering Chemistry", Tata McGraw Hill Education Pvt. Ltd. New Delhi, Fourth Reprint, 2015.
E-Books / MOOCs/ NPTEL	
1	http://libgen.rs/
2	https://nptel.ac.in/downloads/122101001/
3	https://nptel.ac.in/courses/104/103/104103019/
4	https://ndl.iitkgp.ac.in/
5	https://www.youtube.com/watch?v=faESCxAWR9k

APPLIED DIGITAL LOGIC DESIGN			
Course Code:		EC1002-2	Course Type: ESC
Teaching Hours/Week (L: T: P)		2:0:2	Credits: 03
Total Teaching Hours:		25+0+26	CIE + SEE Marks: 50+50
Teaching Department: Electronics & Communication Engineering			
Course Objectives:			
1.	To understand the basics of Number Systems, Logic Gates and Boolean Functions.		
2.	To understand simplification of the Boolean Equations using Boolean Algebra, Karnaugh Maps and QM method.		
3.	To design combinational Logic Circuits like Adders/Subtractors, Binary Comparators, Decoders, Encoders, and Multiplexers.		
4.	To understand the operation of Flip-Flops, Master-Slave Flip-Flops and Conversion of Flip Flops.		
5.	To design Shift Registers and Counters.		
UNIT-I			
Fundamentals of Digital Design			10 Hours
Difference between Analog and Digital Signals, Number Systems: Decimal, Binary, Octal and Hexadecimal. Binary Addition and Subtraction, Digital Logic Gates, Boolean Algebra, Boolean Functions: Canonical Forms, Completely and Incompletely Specified Functions, Simplification of Boolean Functions using Boolean Algebra, Karnaugh Map and Quine-McCluskey Method, Realization of Boolean functions using Basic Gates and Universal Gates.			
UNIT-II			
Combinational Logic and Sequential Logic Circuits			10 Hours
Introduction to Combinational Logic Circuits, Half/Full Adders/Subtractors, Parallel Adders/Subtractors, Binary Comparators, Decoders, Encoders, Multiplexers. Basic Bistable Element, SR Flip-Flop, D Flip Flop, JK Flip Flop, T Flip Flop, Master Slave JK Flip Flop, Characteristic Equations, Conversion of Flip Flops.			
UNIT-III			
Applications of Flip Flops			05 Hours
Design of Shift Register using D- flip flop, Design of Counters: Asynchronous counters using T-flip flop, Synchronous Counters using D-flip flop and T Flip Flop.			
Suggested List of Experiments			
1.	Introduction to Digital Circuit Simulation Software.		
2.	Introduction to Basic gates, Universal gates.		
3.	Realization of Logic Circuits using Universal gates.		
4.	Realization of Combinational Logic Circuits.		
5.	Realization of Sequential Logic Circuits.		
Course Outcomes: At the end of the course student will be able to			
1.	Compare Analog & Digital Signals; Convert the number from one numbering system to another; Analyze Boolean functions.		
2.	Simplify the logic expressions using Boolean Algebra or K-Map or QM Method; Realize the logic expressions using Basic/Universal Gates.		
3.	Analyze and Design different Combinational Logic Circuits such as Adders, Subtractors, Binary		

	Comparators, Decoders, Encoders and Multiplexers.
4.	Describe the operation of Flip Flops, Master-Slave Flip Flops and Conversion of Flip Flops.
5.	Make use of Flip Flops to design Shift Registers and Synchronous/Asynchronous Counters.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
EC1002-1.1	3	-	-	-	-	-	-	-	-	-	-	-
EC1002-1.2	3	1	1	-	3	-	-	-	3	1	-	-
EC1002-1.3	3	2	1	-	3	-	-	-	3	1	-	-
EC1002-1.4	3	-	-	-	3	-	-	-	3	1	-	-
EC1002-1.5	3	1	1	-	3	-	-	-	3	1	-	-

1: Low 2: Medium 3: High
TEXTBOOKS:

1. Morris Mano, "Digital Design", Prentice Hall of India, 3rd Edition.
2. Donald D. Givone, "Digital Principles and Design", McGraw Hill, 2002.

REFERENCE BOOKS:

1. John M Yarbrough, "Digital Logic Applications and Design", Thomson Learning, 2001.
2. D. P. Kothari and J. S Dhillon, "Digital Circuits and Design", Pearson, 2016.
3. Charles H Roth, "Fundamentals of Logic Design", Cengage Learning.

E Books / MOOCs/ NPTEL

1. <https://nptel.ac.in/courses/117106086>

INTRODUCTION TO PYTHON PROGRAMMING				
Course Code:		CS1005-2	Course Type:	ESC
Teaching Hours/Week (L: T: P):		2:0:2	Credits:	03
Total Teaching Hours:		25+0+26	CIE + SEE Marks:	50+50
Teaching Department: Computer Science & Engineering				
Course Objectives:				
1.	Construct python programs using data types and looping.			
2.	Make use of python operators for manipulating lists, dictionaries and files.			
3.	Design function-based Python programs.			
4.	Design list, tuple related programs in Python.			
5.	Write string handling programs in python.			
UNIT-I				
Introduction to Python and Function:				10 Hours
Python Concepts: Introduction to Python, Variables, Keywords, Identifiers, Literals, Operators, Comments, Control statements: if, if else, else if, nested if, for loop, while loop, do while, break, continue, pass.				
Design with functions: Functions Overview, arguments and return values; formal vs actual arguments, named arguments, Recursive functions, Lambda functions, Modules				
UNIT-II				
Data structure and Files:				10 Hours
Data Structures: Strings, Lists, Tuples, Sets and Dictionaries (basic operations)				
Introduction to Object oriented concepts: Class, object.				
Files: Create a file, read and write operations with text and CSV files				
UNIT-III				
				05 Hours
Exception handling: Introduction to Exceptions and Errors, Handling exceptions using try-except-else-finally.				
Introduction to Pandas				
Data Visualization using Matplotlib				
Suggested List of Experiments				
1.	Experiments related to basic operation, data types and variables.			
2.	Experiments related to operations of Lists, tuples and dictionaries.			

3.	Experiments on writing functions and parameter passing.
4.	Experiments related to working with strings.
5.	Experiments related to file handling.

Course Outcomes: At the end of the course student will be able to

1.	Understand and experiment with the basics of python programming like data types and looping.
2.	Describe the concept of functions in python for solving real-world problems.
3.	Apply the Python operations for manipulating strings, lists, tuples and dictionaries.
4.	Demonstrate the proficiency in handling File Systems.
5.	Understand the basic knowledge on Exception handling and plotting graphs.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→ ↓Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
													1	2	3
CS1005-2.1	1	2	1	2	-	-	2	-	-	-	-	2	-	1	1
CS1005-2.2	-	2	-	-	-	1	-	-	-	-	-	1	-	2	-
CS1005-2.3	1	-	-	-	-	-	-	-	-	-	-	1	-	-	1
CS1005-2.4	-	1	-	2	-	1	-	-	-	-	-	1	-	-	-
CS1005-2.5	-	-	1	3	-	-	2	-	-	-	1	2	-	2	-

1: Low 2: Medium 3: High

Textbooks:

1	Kenneth A. Lambert, "The Fundamentals of Python: First Programs", Cengage Learning, 2nd Edition, 2019.
2	Yashwanth Kanetkar, Adithya Kanetkar, "Let us Python", 3rd Edition.
3	Mark Summerfield, "Programming in Python 3 - A Complete Introduction to the Python Language", Second Edition, Addison-Wesley, 2009.
4	Y. Daniel Liang, "Introduction to Programming Using Python", Pearson, 2013.

Online Resources:

1	https://www.w3schools.com/
2	https://www.programiz.com/python-programming/guide

BASIC ELECTRICAL ENGINEERING			
Course Code:	EE1001-2	Course Type:	BSC
Teaching Hours/Week (L: T: P)	2:0:2	Credits:	03
Total Teaching Hours:	25+0+26	CIE + SEE Marks:	50+50
Teaching Department: Electrical & Electronics Engineering			
Course Objectives:			
1.	To familiarize the student with the DC circuit analyses.		
2.	To analyze single and three-phase AC circuits.		
3.	To understand the working principle of electrical machines.		
4.	To introduce fundamental concepts in EV, basic converters and special motors, electrical wiring protective devices and safety measures		
UNIT-I			
Circuit Fundamentals			02 Hours
Introduction to DC circuits, Basic nodal and mesh analysis excited by independent DC voltage sources, Power and Energy.			
			08 Hours
AC Fundamentals: Generation of sinusoidal voltage, frequency of generated voltage, definition and numerical values of average value, root mean square value, form factor and peak factor of sinusoidally varying voltage and current, phasor representation of alternating quantities.			
A.C. Circuits: Analysis of R, L, C, R-L, R-C and R-L-C series. Phasor Diagrams. Real power, reactive power, apparent power and power factor. Three-phase balanced circuits, voltage and current relations in star and delta connections. Measurement of three phase power using two wattmeter.			
UNIT-II			
DC Machines			03 Hours
Faradays Laws, self and mutually induced emfs. Constructional details, Principle of operation of generator and motor, Expression for back emf, Types of dc motors, Characteristic of dc motors (shunt and series motors only) and Applications.			
Single-Phase Transformers			03 Hours
Necessity of transformer, Principle of operation. Types of Transformers, Emf equation, losses, efficiency, problems on emf equation and efficiency, Autotransformer, Applications.			
Induction Motors			03 Hours
Concept of rotating magnetic field, Construction and working of a three-phase Induction Motor, Slip and its significance, Torque slip characteristics (qualitative). Necessity of a starter, Principle of operation Single Phase Induction Motor. Applications			
UNIT-III			
Electric Vehicles			04 Hours
Fundamentals, Block diagram of EV and its components. Motors used in EV – BLDC, Permanent Magnet Synchronous Machine (PMSM) -Working principle			
SMPS: Concept of step up and step-down converter (Basic equation and Block diagram representation), Applications. Block diagram of UPS and applications.			
Domestic Wiring			02 Hours
Types of wiring. Two-way and Three-way control of lamp. Elementary discussion on Circuit protective devices: Fuse and Miniature Circuit Breaker (MCB's).			

Personal safety measures: Electric Shock and Precautions against shock. Potential between neutral and ground. Necessity of Earthing, Earthing types- Pipe and Plate earthing.

*One additional tutorial class will be allotted every week

Suggested List of Experiments

1.	Verification of KVL and KCL for DC circuits.
2.	Measurement of current, power and power factor of incandescent lamp, fluorescent lamp, CFL and LED lamp.
3.	Sinusoidal steady state response of R-L, and R-C circuits- impedance calculation and verification
4.	Voltage and Current relationships of three phase star/delta circuits.
5.	Measurement of three-phase power using two wattmeter method
6.	Load test on a single-phase Transformer.
7.	Speed load characteristic of a 3-phase Induction Motor.
8.	Time characteristic of fuse

Demonstration Experiments

1.	Demonstration of fuse, MCB by creating a fault.
2.	Two-way and Three-way Control of lamp and formation of truth table.
3.	Demonstration of cut out sections of electrical machines (DC machines, Induction machines and Synchronous machines).
4.	Demonstration of EV and its Components.

Course Outcomes: At the end of the course student will be able to

1.	Analyze the DC Circuits using mesh & node methods to compute power and energy.
2.	Analyze voltage & current phasor relationships in single phase & three phase AC circuits to compute circuit parameters.
3.	Describe the fundamentals of electromagnetism, construction, operating principle of DC & Induction motor to study performance characteristics.
4.	Apply principle of single-phase transformer to compute transformer efficiency.
5.	Describe fundamental concepts in EV, converters, domestic wiring, protection and safety schemes

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes													1	2
EE1001-2.1	2	3	-	-	-	-	-	-	-	-	-	-	-	-
EE1001-2.2	2	3	-	-	-	-	-	-	-	-	-	-	-	-
EE1001-2.3	2	3	-	-	-	-	-	-	-	-	-	-	-	-
EE1001-2.4	2	3	-	-	-	-	-	-	-	-	-	-	-	-
EE1001-2.5	2	3	-	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	D. P. Kothari and I. J. Nagrath, "Basic Electrical Engineering", Tata McGraw Hill, 2010.
2.	S. K. Sahdev, "Basic Electrical Engineering (with Lab Manual)", January 2022
3.	Lecture Notes on Basic Electrical Engineering, Department of E&E, NMAMIT, Nitte. (New version)
4.	Hughes, Edward, "Electrical Technology", Pearson Education Publications, 10 th Edition, 2010.
5.	A. Chakrabarti, M. L. Soni and P. V. Gupta, U. S. Bhatnagar, "Power system engineering", Gagan Kanur, Dhanapat Rai and Co Pvt. Ltd, 2013.

REFERENCE BOOKS:

1.	Vincent Del Toro, "Electrical Engineering Fundamentals", 2 nd Edition, Pearson, 2015.
2.	H. Cotton, "Electrical Technology", CBS, 7 th Edition, 2005.
3.	A. Mittle and V. N. Mittle, "Basic Electrical Engineering", Tata McGraw Hill, 2005.
4.	Debashisha Jena, "Basic Electrical Engineering", Wiley India Private Limited, 2012.
5.	M.V. Deshpande, "Elements of Power Station Design", 1 st edition, PHI learning, 2009.

E Books / MOOCs/ NPTEL

1.	http://nptel.ac.in/downloads/108105053/
2.	http://www.textbooksonline.tn.nic.in/books/11/stdxi-voc-ema-em-1.pdf
3.	Basic Electrical Technology Lectures by Dr. L Umanand Department of Power Electronics Group, CEDT IISC Bangalore available at http://www.nptelvideos.in/2012/11/basic-elerical-technology.html

BIOLOGY FOR ENGINEERS													
Course Code:				BT1651-1			Course Type			AEC			
Teaching Hours/Week (L: T: P: S):				1:0:0:0			Credits			01			
Total Teaching Hours				15			CIE + SEE Marks			50+50			
Teaching Department: Biotechnology													
Course Objectives:													
1.	To learn the types of cells, biomolecules and life processes												
2.	To know the applications inspired by nature in various streams												
3.	To be updated application of biology in real life scenarios.												
UNIT-I													
Introduction for Biology for Engineers										05 Hours			
Why Biology for Engineers? Cell Types & Properties: Prokaryotes - Bacteria, Viruses and Fungi, Eukaryotes - Plant and Animal Cells, Biomolecules, Life Processes at Cellular Level.													
UNIT-II													
Applications Inspired by Nature										05 Hours			
Composites in Construction, Termite Mound architecture, Counter current heat exchangers, Design of aeroplane, helicopter and submarine, Information Theory and Biology, SONAR, Medical Devices.													
UNIT-III													
Real Life Scenarios										05 Hours			
Recent scenarios in Environment, Agriculture and Medical Technology.													
Course Outcomes: At the end of the course student will be able to													
1.	Ascertain the importance of biology and interpret the basics of cell and life processes.												
2.	Summarize the applications inspired by nature and understand the recent advances in application of biology.												
Course Outcomes Mapping with Program Outcomes & PSO													
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes													
BT1651-1.1		3	-	-	-	-	-	-	-	1	-	-	1
BT1651-1.2		3	3	-	-	-	-	2	-	1	-	-	1
1: Low 2: Medium 3: High													
TEXTBOOKS:													
1.	Suraishkumar, G. K. "Biology for Engineers", Oxford University Press India, 2019.												
2.	Chakraborty, T, Akthar N, "Biology for Engineers", PHI learning Print Book ISBN: 9789391818142 eBook ISBN: 9789391818197												
REFERENCE BOOKS:													

1.	Rao C. V., "Biology for Engineers", 2021.
2.	Raven, P. H. and Johnson G. B. "Biology", 4 th Ed. WCB publishers, 2010.
3.	Ethier, R. S. and Simmons C. A. "Introductory biomechanics- From cells to organisms", Cambridge University Press, 2012.

ENGINEERING VISUALIZATION																
Course Code:				ME1004-1		Course Type				PCC						
Teaching Hours/Week (L: T: P: S)				0:0:3:0		Credits				1						
Total Teaching Hours				26		CIE + SEE Marks				50+00						
Teaching Department: Mechanical Engineering																
Course Objectives:																
1.	To impart and inculcate understanding of the concept of orthographic projection and projection of plane surfaces and solids in different position in first angle projection system.															
2.	To develop the lateral surfaces of solid objects and to draw the isometric projection of simple solids.															
UNIT-I																
Orthographic Projection										02 Hours						
Introduction to orthographic projection, Quadrants, principal planes, principal views, Difference between First angle and third angle projection, Dimensioning, Conventions employed for drawing.																
Projection of plane surface										06 Hours						
Triangle, Square, Rectangle, Pentagon, Hexagon and Circle in simple position (Resting on HP with inclination to HP and VP, true length with true inclination only)																
UNIT-II																
Projection of Solids										06 Hours						
Prisms, Pyramids, Cones and Cylinders in simple position (Resting on HP with inclination to HP and VP, true length with true inclination only)																
Orthographic projection of simple components using their isometric projection.																
UNIT-III																
Development of Lateral surfaces of solids										06 Hours						
Right regular Prisms, Pyramids, Cylinders and cones (with single section plane)																
Isometric projection										06 Hours						
Isometric scale, Isometric dimensions, to draw Isometric views of simple solids and machine components using their orthographic projections.																
Course Outcomes: At the end of the course student will be able to																
1.	Draw the orthographic projections of a plane and solids for a given position using Solid Edge software.															
2.	Draw the development of lateral surfaces of standard solid objects using Solid Edge software. Draw isometric projection of solid objects individually or in combination using Solid Edge software.															
Course Outcomes Mapping with Program Outcomes & PSO																
	Program Outcomes→			1	2	3	4	5	6	7	8	9	10	11	12	PSO↓

↓ Course Outcomes														1	2												
ME1004-1.1														3	1							1	1		2	2	1
ME1004-1.2														3	1							1	1		2	2	1
1: Low 2: Medium 3: High																											
TEXTBOOKS:																											
1.		N. D. Bhat & V. M. Panchal, Pramod R. Ingle, "Engineering Drawing", 53 rd Edition, Charotar Publishing House, Gujarat, 2014.																									
2.		K. R. Gopalakrishna, "Engineering Drawing", Subhas publishers, Bangalore , 32 nd Edition, 2012.																									
REFERENCE BOOKS:																											
1.		"A Primer on computer aided Engineering Drawing", VTU, Belgaum, 8th edition, 2011.																									
2.		Shah, "Engineering Drawing and Computer Graphics", Pearson, 2010.																									
3.		Agarwal & Agarwal, "Engineering Graphics", TMH, Second edition, 2013.																									
4.		P. S. Gill, "A Text book of Engineering Graphics and Drafting", 11 th Edition, S. K. Kataria & sons, New Delhi, 2009.																									

ENVIRONMENTAL STUDIES AND SUSTAINABILITY			
Course Code:	CV1005-1	Course Type	MNC
Teaching Hours/Week (L: T: P: S)	1:0:0:0	Credits	00
Total Teaching Hours	15	CIE + SEE Marks	50+50
Teaching Department: Civil Engineering			
Course Objectives:			
1.	To study about environment and ecosystems		
2.	To study about environment and ecosystems and to study about different types of natural resource.		
3.	Basic knowledge and concept of causes, effect and control of different type of environmental pollution.		
4.	Students will be able to define sustainability and identify major sustainability challenges.		
UNIT-I			
The Environment and Ecosystem			03 Hours
Environment and Environmental studies: Definition, concept, components and importance. Global concept of biosphere: Subdivisions of the biosphere – lithosphere, Atmosphere, Hydrosphere. Ecosystem and Ecology: Structure and Function of ecosystem, Brief concept of Autecology and Synecology. Food chain, food web and ecological pyramids. Biogeochemical cycles in an ecosystem: (Carbon, Nitrogen and Phosphorous cycle)			
Natural Resources and their Conservation			03 Hours
Forest Resources: Uses and overexploitation of forests and consequences of deforestation. Water Resources: Use and consequences of over-utilization, concept of rain water harvesting and watershed management, water conflicts. Food Resources: Sources of food, food problems- Indian scenario, Impacts of modern agriculture on environment (Fertilizer - pesticide problem, water logging and salinity), Organic farming. Energy Resources: Renewable and Non-Renewable energy sources, growing energy needs and alternate energy sources. Land Resources: Global land use patterns, Soil erosion, Desertification, Wasteland Reclamation.			
UNIT-II			
Environmental Pollution			03 Hours
Definition, causes, effects and control measures of: a. Air pollution b. Water pollution (thermal and marine pollution) c. Land pollution d. Radiation pollution and Nuclear hazard.			

e. Noise pollution

Solid waste management: Causes, effects and control measures.

Global warming and climate change Ozone depletion;

Acid rain: Causes, effects and control measures

Sustainable Development in theory and practice

03 Hours

Concept of Sustainability, Sustainable Development and its different constituents.

Global Responses to Sustainable Development

- Sustainable Development Goals (vs Millennium Development Goals)
- Triple Bottomline of Sustainability: Food, Water, Energy nexus
- Potential and Barriers to Sustainable Business
- Sustainable rural and urban livelihoods

UNIT-III

Sustainable Practices

03 Hours

Zero waste and R concept - concept of 5R (Refuse, Reduce, Reuse, Repurpose, Recycle), Circular economy, ISO 14000 Series, Material Life cycle assessment, Environmental Impact Assessment. Sustainable habitat: Green buildings and rating system in India, Green materials, Energy efficiency, Sustainable transports. Sustainable energy: Non-conventional Sources, Energy Cycles carbon cycle, emission and sequestration, Green Engineering: Sustainable urbanization- Socioeconomical and technological change.

Course Outcomes: At the end of the course students will

1.	Understand environment and ecosystems, types of natural resources and concept of causes, effect and control of different type of environmental pollution.
2.	Be able to define sustainability and identify major sustainability challenges and Practices.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes															
CV1005-1.1	2	2	3	-	-	2	2	1	-	2	1	1	2	3	-
CV1005-1.2	1	2	1	-	-	2	3	2	-	2	1	1	1	3	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Benny Joseph, "Environmental Studies", Tata McGraw Hill Publ. Co., New Delhi, 2005.
2.	Rajagopalan, R., "Environmental Studies: From Crisis to Cure", Oxford University Press, London, 2005.
3.	Allen, D. T. and Shonnard, D. R., "Sustainability Engineering: Concepts, Design and Case Studies", Prentice Hall. 2015.
4.	R. L. Rag and Lekshmi Dinachandran Remesh, "Introduction to Sustainable Engineering", 2 nd Edition, PHI Learning Pvt. Ltd., 2016.

REFERENCE BOOKS:

1.	Balasubramanya, N and Chatwal, Gurdeep R., "Environmental Studies", Himalaya Publishing House, Mumbai, 2007.
2.	Barucha, E., "Environmental Studies", University Grants Commission, New Delhi, 2004.
3.	A.S. Bradley, A. O. Adebayo, P. Maria, "Engineering applications in sustainable design and development", 1 st Edition, Cengage learning, 2016.

4.	Bhatia, S. C., "Environmental Chemistry", CBS Publishers, New Delhi, 2005.
5.	De, A.K. and De, A. K., "Environmental Studies", 2006.
6.	Keller, Edward A., "Environmental Geology", CBS Publishers and Distributors, Delhi, 1985.

INTERNSHIP-I															
Course Code	UC1001-1					CIE Marks					100				
Teaching Hours/Week (L: T: P)	-					SEE Marks					-				
Total Hours of Pedagogy	80-90 Hours (During I/II semesters)					Total Marks					100 (Evaluation in I/II/III Semester and grades earned shall be included in IV Semester grade card)				
Credits	2					Exam Hours					--				
Course objective															
1. This course is meant to provide students an opportunity to be involved in Inter/ Intra Institutional Activities viz; Training with higher Institutions; Soft skill training organized by Training and Placement Cell of the institution; contribution at incubation/ innovation /entrepreneurship cell of the institution; participation in conferences/ workshops/ competitions etc.; Learning at Departmental Lab/ Tinkering Lab/ Institutional workshop; Working for consultancy/ research projects within the institution and Participation in all the activities of Institute’s Innovation Council.															
Activities: Refer Appendix B - 3.4 for details															
Course outcomes															
1. Experience the working in Inter / Institutional activities															
2. Work in teams and communicate efficiently both written and oral.															
3. Develop the ability to do work in different activities, which will provide the necessary understanding and contribute to the same and provide a foundation to undergo higher level training in subsequent internships.															
Course Outcomes Mapping with Program Outcomes & PSO															
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
UC2001-1.1	3	1	-	-	1	-	-	-	2	3	1	-	-	-	-
UC2001-1.2	3	1	-	-	1	-	-	-	2	3	1	-	-	-	-
UC2001-1.3	3	1	-	-	1	-	-	-	2	3	1	-	-	-	-
1: Low 2: Medium 3: High															

HOLISTIC COMPONENTS

HUMANITIES

Holistic education is not only about teaching the basic subjects, but it is more about redefining the way a student should be taught. The purpose of holistic language teaching is the development of the learners' ability to handle both their language oral skills as well as maximizing their life skills. The department contributes to educational life and work spaces that are creative and meaningful. Multidisciplinary and holistic learning is an ancient method used in



Indian education system as well as the other parts of the world. This is the reason that such type of education system was advocated by scholars like Kautilya, Banabhatta, Plato, and Aristotle among many others. Holistic approach is essentially a student centered strategy rather than a teacher centered one.

Holistic education through courses allied to Humanities is created within the inclusive connections of social and human experience. A curriculum built around such stages is considered holistic if they involve the practices that integrates language acquisition and fills multiple cognitive demands in interlocking activities that spiral learning. Through the applied learning style of a person--mind, body and spirit students will learn more effectively the nuances of language, responsibilities towards social fabrics and ethics.

The approach strives to make a learner construct his own understanding of the text he/she interacts with and converses with others according his understanding. Intensive experiential and group sessions, a co-created learning ambience and hands-on engagement through real-life cases, field trips and internships to make learning exciting, rigorous and transformative. As a part of the holistic approach and its philosophy, a student is educated beyond core academics providing him/her virtuous and holistic education. This helps the students to discover their individuality and comprehend the significance of life purposefully, creatively, and morally in a complex world. Krishnamurti writes If the unity of life and the oneness of its purpose could be clearly taught to the young, how much brighter would be our hopes for the future! (Krishnamurti, J. 1974).

MATHEMATICS

INDIAN MATHEMATICIANS

It is essential to know about the ancient, medieval and modern time Indian mathematicians and their contribution to Science and Mathematics. Ancient Indian mathematicians have contributed immensely to the field of mathematics. The invention of zero is attributed to Indians and this contribution outweighs all other made by any other nation since it is the basis of the decimal number system, without which no advancement in mathematics would have been possible. The number system used today was invented by Indians and it is still called Indo-Arabic numerals because Indians invented them and the Arab merchants took them to the western world.

Here we are introducing some of the important Indian mathematicians from ancient times.

Aryabhata: (500 A. D.) - Studied at the University of Nalanda, which was considered as a great centre of learning. Aryabhata was a great Indian mathematician. He gave the value of " π " as 3.1416, claiming for the 1st time, that it was approximation. Aryabhata also dealt with other aspects of mathematics and Astronomical calculations, namely

Geometry, Mensuration, Squareroot, Cuberoor, Progression and Celestialsphere. He presented a method to solve an intermediate equation of certain type that are important in astronomy and computer science.

Bhaskara : (1100 A. D.) - was a great Mathematician and Astrologer. He was the first Mathematician to declare confidently that any term divided by ZERO is infinity and the sum of any term and infinity is infinity. His concept of “Tatkalikagati”, which means instantaneousmotion, used by astronomers to determine the motion of the planet accurately brought credit to him. He explained the solutions of quadratic and cubic equations. He stated the Rolle’s theorems in analysis, the mean value theorem.

Srinivas Ramanujan was an Indian Mathematician who made significant contributions to mathematical analysis, Number theory and continued fractions. He made many important contributions in the field of mathematics with his wonderful and unique knowledge. That’s why his birthday is celebrated as Mathematics Day.

PHYSICS

The ancient world had considered Physical Sciences, Chemical Sciences, Earth Sciences, Biological Sciences, Mathematical Sciences etc. as study of nature, which were all studied under the banner of Philosophy. Even today,



the philosophers are studying Metaphysics which connects physical attributes to mind. Physics is a branch of science which deals with the study of matter and energy. The Physical Science was a matter of interest for all the civilizations including Vedic era of India dating back to over 3000 years. The physical science in ancient India was majorly restricted to Astronomy and Astrology. It was **Kanada**(600 B.C.) who presented holistic approach of physics, by blending science, philosophy and religion through ‘Vaisesika Sutra’. Their essence is the atomic theory of matter. He gave the name ‘Paramanu’ (Atom), to be the indivisible entity of matter. The idea of chemical change was also put forward by Kanada.



Bharadwaja is credited with teaching missile technology. Aryabhata(500 A.D.) was a great astronomer. He was the first to state that the earth is round and it rotates on its own axis, creating day and night. He declared that the moon is dark and shines only because of sunlight. Aryabhatta contributed greatly to the field of science particularly astronomy.

Varaha mihira (500 A.D.) studied astrology and astronomy and declared that the earth was spherical. He also proposed that the moon and planets are lustrous not because of their own light but due to sunlight. Bhaskra (1100 A. D.) was a great scientist his concept of “Tatkalikagati”, which means instantaneous motion, used by astronomers to determine the motion of the planet accurately brought credit to him. Brahmagupta(598 A.D.) calculated the instantaneous motion of a planet, gave correct equations for parallax, and some information related to the computation of eclipses and is widely regarded as one of the most accomplished of the ancient Indian astronomers.

“If you wish to make an apple pie from scratch, you must first invent the universe.” So said astronomer Carl Sagan in an episode of his landmark television series, Cosmos. Embedded in Sagan’s memorable quip is a certain holistic understanding of the universe — a notion that the existence of any one thing is intimately tied to the existence of everything else. There are no apple pies without apples; there are no apples without the proper climate for growing apple trees; there is no proper climate for growing apple trees without a planet on which the apple trees can grow — and so on, all the way back to the Big Bang. Pythagoras and his followers held mathematics in an almost holy regard, and they saw numbers as a basic form of matter. According to their view, all things had numbers, and the objects of the universe — including human societies — were arranged in harmonious mathematical relationships with one another.

All sciences were originated from philosophy. Physics was called natural philosophy until the 19th century, but once it was proven to be correct it was no longer philosophy and became a science. Physics is the science of the natural

world, more specifically dealing with the matter, energy, space-time, and fundamental forces that govern the physical world. In physics we study a wide range of physical phenomena from subatomic particles to large galaxies of the material universe, and use empirical data and mathematics to find results and conclusions. Physics is also deeply concerned with arriving at knowledge about the ultimate nature of reality. Since we cannot know whether we have discovered everything which would affect our theories of the universe, all such theories are perpetually subject to modification or change. Mathematics is a language and a tool that we use in physics to explain the universe. Quantum physics is a mathematical description that rules the tiny world of atoms and subatomic particles in our universe. Without quantum physics, much of the information technology that we rely on, from microcircuits to lasers, would not exist. Today many scientists argue that metaphysics plays an important role in quantum mechanics at a deeper level; the nature of reality is all mathematical. This could be an example of how metaphysical assumptions can get in the way of our understanding the paradoxical nature of quantum mechanics. But even when quantum mechanics appears a mystical science of metaphysics, it is not metaphysics but productive science.

Thus, the Physics though has many branches and uses many other branches of science and philosophy, in the past and the present, its aim is to understand the whole universe which is nothing but matter and energy which is seen or unseen.

CHEMISTRY

ANCIENT SEERS OF INDIA – CHEMISTRY

In ancient India, chemistry was called **Rasayan Shastra**, **Rasa-Vidya**, **Rasatantra** and **Rasakriya** all of which roughly mean '**Science of Liquids**'. There also existed chemical laboratories which were called **Rasakriya-nagaram/Rasakriya-shala**, which literally mean '**School where liquids are activated**'. Rigveda (earlier than 1500 BCE) mentions many fermented drinks and methods of fermentation, apart from various metals. Soma juice from the stems of the soma plant was considered a divine drink. The Vedic Indians were acquainted with the art of dyeing with certain natural vegetable colouring matters. A type of pottery, now known as 'Painted Grey Ware', is also associated with the Vedic period.

Ancient chemistry in India grew out of the early efforts to develop an elixir; to turn base metals into gold and on metallurgy. Chemical techniques in India can be traced back all the way to the Indus valley or Harappan civilisation (3rd millennium BCE). Pre-Harappan Indians were acquainted with the art of making baked or burnt clay pottery as well as painting the same with two or more colours (by addition of iron oxide, manganese oxide, etc.). Kautilya's Arthashastra (3rd or 4th century BCE) has a lot of information on prevailing chemical practices. Apart from mines and minerals, it discusses the details of precious stones (pearl, ruby, beryl, etc.); preparation of fermented juices (sugarcane, jaggery, honey, jambu, jackfruit, mango, etc.) and oil extraction.



It is said that **Maharshi Kanada** was the first to propound that the *Parmanu* (atom) was an indestructible particle of matter and that Universe is made up of *Kana*. When matter is divided and subdivided, we reach a stage beyond which no division is possible, the undivisible element of matter is *Parmanu*. Kanada explained that this indivisible, indestructible *y* cannot be sensed through any human organ.



Nagarjuna (931 A.D.) from Somnath in Gujarat was a chemist/alchemist, who concentrated his efforts in transforming the base metals into gold. His reputation was such that people believed Nagarjuna to be in communion with gods and goddesses who had blessed him with the power of changing base metals into gold and extracting the 'elixir of life'.



Prafulla Chandra Ray (1861-1944), an Indian chemist, is often referred to as the Father of Chemistry in India. He received his BS in 1882 and his PhD in 1887 from University of Edinburgh. In 1896, he announced a major discovery of a new compound, mercurous nitrite.

Today's Science and Technology has been greatly inspired by the contributions of these wise seers. Indians have continued to show their global impact in the Field of Science.



In the 21st century, biochemist **Har Gobind Khorana** won the Nobel Prize (1968) for demonstrating how the nucleotides in nucleic acids control the synthesis of proteins. Thus, the seers of ancient India have contributed significantly in the development of Modern Chemistry.

BIOTECHNOLOGY

Biology for Engineers

Science deals with matter. It is based on starting from scratch with what a human can observe, test, and rationalize. Ancient sages have worked hard to be seen as the only reliable providers of knowledge to the world. In 1875, the VymaanikaShaastra, a 4th Century BC text written by Sage Bharadwaj was discovered in a temple in India. It contains 3000 shlokas in 8 chapters which was physically delivered by the ancient Hindu Sage Bharadwaj. The book greatly deals with the operation of ancient vimanas and included information on steering, precautions for long flights, protection of the airships from storms and lightning and how to switch the drive of solar energy or some other form of energy. One of the chapter will reveal the secrets of constructing aeroplanes that cannot be broken or cut, that is indestructible, that is fire resistant. It also deals with the secret of making planes motionless and invisible. It also describes how to defeat the enemy planes etc. as per the Sage Bharadwaj the vimanas were classified as per the Yugas. During the period of Krita Yuga, Dharma was established firmly. The pushpak Vimana which was used by Ravan was an Aerial vehicle. He used this vehicle to kidnap Sita from jungle and took him to his Kingdom Srilanka. Ramayana was during the Treta Yug in which the Vimanas were highly discovered. During this period "Laghima" gave them the power to lighten their vehicle so they can travel freely in the air.

COMPUTER, INFORMATION SCIENCE & ENGINEERING

The Indians (**Aryabhata**, 476 BC - 550 BC) contributed **Zero (0)** to the number system. So that numeric system and computing world found an ease in solving numerical problems using computer programs.

Acharya **Pingala** was an ancient Indian mathematician who lived around 300 BCE. He wrote the *Chandaḥśāstra*, where he analysed **Sanskrit poetry mathematically**. It also contained the first known explanations of **digit zero, binary numbers, Fibonacci numbers and Pascal's triangle**.

Baudhayana (8th century BCE) composed the *Baudhayana Sulba Sutra*, which contains examples of Pythagorean triples, such as: (3,4,5), (5,12,13), (8,15,17), (7,24,25) and (12,35,37) as well as a statement of the Pythagorean theorem for the sides of a square: "The rope which is stretched across the diagonal of a square produces an area double the size of the original square."

In Indian astronomy, the study of **trigonometric functions** flourished in the Gupta period, especially due to **Aryabhata (sixth century CE)**, who discovered the **sine function**.

Quadratic equation of the form $ax^2 + bx + c = 0$, $a \neq 0$ and is given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. was discovered by **Sridharacharya** in the 11th century.

The largest numbers the Greeks and Romans used were 106. In 5000 BC **Indians used numbers as big as 10^{53}** (10 to the power 53) with specific names. The largest used number today is **Tera 10^{12}** .

Katāpayadi numerical notation is an ancient Indian system to depict letters to numbers for easy remembrance of numbers as **words or verses**. For example: क(Ka)=1 ख(Ka)=2 ग(Ga)=3 घ(Ga)=4 ङ(Gnya)=5 च(Cha)=6 छ(Cha)=7 ज(Ja)=8 झ(Ja)=9 ञ(Nya)=0. The modern **Hasing technique in computing system** which is resembling was then being used in the **Indian Katāpayadi system**. For example, the hashing number based on Katāpayadi system would be as follows for '**Gurudev**' Gu=Ga(is the consonant)=3, Ru=Ra(is the consonant)=2, De=Da(is the consonant)=8 Va=Va(is the consonant)=4, So Gurudeva = 4823.

In the recent decades, following are the few of the major contributors to the computing world:

1. In 1996 the USB port invented by the **Ajay Bhatt**, an Indian at Intel Oregon which involved low level **programs delt with embedded C Language** to perform flexible IO transfer and opened up an area to use plug-and-play devices efficiently.
2. The Pentium chip invented by **Vinod Dham**, that **made C compiler to speed up the program execution** and do well with **GUI applications (both System and User Level) that are wiritten in C language**.
3. **Amit Singhal** is an Indian who rewrote (search engine in 2001) the **google algorithm** (C language coding embedded with Assembly Language service routins in Windows and Unix/Linux). Then on the Google processes over 40,000 search queries every second on average which translates to over **3.5 billion searches per day** and **1.2 trillion searches per year** worldwide.

Few of the contribution as Author of CP and Educators of C language:

1. **Yashavant Kanetkar** is an Indian computer science author, known for his varieties of C Programming books.
2. **E. Balagurusamy : An Computer scientist** known for **Programming in ANSI C**.

ELECTRONICS AND COMMUNICATION ENGINEERING

The idea of a holistic approach to engineering design and education has been envisioned to meet the perceived and emerging needs for innovation in the 21st century. Many engineering educators, practicing engineers and engineering students have already recognized the gaps and areas of potential improvements in the knowledge acquisition process implemented in current engineering degree programs when compared to current societal and technological issues and developments.

Society and humanity have progressed drastically over the past few generations. Engineers as a network of professional problem solvers have been heavily involved in these global communities and the engineering profession is evolving from one that focuses on targeted, isolated issues, to one that embraces challenges that incorporate physical, economic, environmental, and humanitarian aspects.

Currently, engineering students are required to take classes on ethics, liberal studies and technology and society courses, however engineering students are not prefaced with the importance of rounding out their education with these topics, and while social issues are discussed, they are not related to engineering specifically. That being said, explicitly linking the technical aspects of engineering to society is paramount in training effective problem solvers for the 21st century. With some exposure to multi-disciplinary, inter-disciplinary and trans-disciplinary approaches to engineering and design, students will be better prepared for their future careers in industry or research fields.

The functional requirements for the perceived solution were determined by the expected outcomes and what students should take away after experiencing the new educational product. Some of them are:

- students will be inspired and driven to seek opportunities in engineering for environmental, social, medical, and human development/poverty issues.
- students will be able to identify the issues that are emerging from new technology, how to mitigate the negative aspects and reduce the amount of impact, while leveraging the positive outcomes.
- students will have respect and knowledge of the importance of ethics and policy matters in the field of engineering and be able to determine between unethical and an ethical situation in a proactive manner.

The courses should overcome the challenges of the current engineering educational system. Approaching the degree from a holistic perspective. The integrated system that fosters collaboration among faculty and students. A new organizational and pedagogical model, which emphasizes knowledge integration and interweaves thematic content threads throughout the curriculum should be proposed.

- Foundations thread (math and science) Key mathematical concepts lay the foundation for understanding the anchoring concepts in courses throughout the ECE curriculum. The foundations thread unpacks mathematics and physics concepts to help students learn fundamentals in ECE topics like circuits, signals and systems, and electromagnetics. The foundations thread champion spearheads the collaboration between the math and ECE departments to introduce and promote the value and utility of mathematics in ECE courses, as well as the importance of mathematical thinking.
- Creativity thread (research, design, and optimization tools) The creativity thread is intended to integrate research and design throughout the undergraduate experience. By showing the impact of research, students will see the practical applications and potential breakthroughs of fundamental ECE concepts. Likewise, exposing students to design at every level of the undergraduate experience allows them to experience the excitement of engineering by applying their foundational knowledge to a tangible product.

- Professional formation thread (communications, cultural adaptability, ethics, leadership, and teamwork) Partnering with faculty and industry leaders to ensure students develop professional skills meaningfully and effectively to enhance student-industry interactions.

ELECTRICAL AND ELECTRONICS ENGINEERING

Agastya Samshita available at Prince's Library of Ujjain in India, dates back to the first millennium BC, contains a detailed description construction of an electric battery/cell along with way to utilize the battery to 'split' water into its constituent gasses. The method of generating electricity using modern battery cell resembles Agastya's method. The materials used by Sage Agastya for generating electricity were an earthen pot, copper plate, copper sulphate, wet saw dust, zinc amalgam. As quoted in *Agastya Samhita* the open circuit voltage and short circuit current of the prepared cell are 1.138 volts and 23 mA respectively. He articulates 100 earthen pots on water, has the power to change the form of water to oxygen and hydrogen. If hydrogen is contained in an air tight cloth, it can be used in aerodynamics, i.e. it will fly in air. In an iron vessel and in a strong acidic medium, gold or silver nitrate covers copper with a layer of gold or silver. The copper that is covered by gold is called *Shatakumbha* or artificial gold.

Rao Saheb Krishnaji Vajhe, an engineer from Pune while reading books related to science found the pages of *Agastya Samhita* with Damodar Tryambak Joshi of Ujjain. Dr. M. C. Sahastrabuddhe, the Head of the Sanskrit Department in Nagpur, when reading *Agastya Samhita* found the similarity of it with of Daniel Cell. He requested P.P. Hole, the Professor of Engineering at Nagpur to investigate on the same.

On the basis of the descriptions in *Agastya Samhita* Mr. Hole and his friend started preparing the apparatus for the experiment. While preparing the set up they could not understand the meaning of *shikhigreeva* and while checking the Sanskrit dictionary, they understood that it meant the neck of a peacock. They went to Maharaja Park and asked the chief when a peacock would die. The chief was very angry and asked them to give in an application. After few days during a conversation with an Ayurveda expert he confirmed that *shikhigreeva* is copper sulphate, which solved their problem. Thus, a cell was formed and it had an open circuit voltage of 1.38 volts and short circuit current of 23 milli amperes. The results of the experimentation were communicated to Dr. M.C. Sahastryabuddhe. It was exhibited fourth general meeting at the Swadeshi Vigyan Sanshodhan Sanstha, Nagpur on August 7, 1990 to the scholars. It was concluded that the description was of an electric cell

On the basis on *Agastya Samhita* and other scriptures, Rao Saheb Vajhe, who spent his life in rummaging the Indian scientific scriptures, gave different names to electricity. The six ancient terminologies for electricity are:

- Tadit—produced by friction from leather or silk,
- Saudamini—produced by friction from gems or glass,
- Vidyut— from clouds or steam,
- Shatakoti alias *Shatakumbhi*—produced from a battery of hundreds of cells,
- Hradini—obtained from storage cells,
- Ashani—the one emanating from a magnetic rod.

MECHANICAL ENGINEERING

Mechanical engineering is one of the oldest disciplines of engineering, which requires the knowledge of mathematics, materials, physics and other engineering technologies. It is concerned with materials, processes and machines and requires the concepts of forces, moments, energy, entropy, work etc. The developments that are visible in all spheres

of life have connection to mechanical engineering. Engineering has made a significant contribution in the development of civilizations and contribution of mechanical engineering in areas like construction of large scale structures including for irrigation, architecture, military etc. is significant. Difficult problems of the society have been solved using simple concepts of mechanical engineering, say for eg. use of lever principle to move heavy objects. In fact, mechanical engineering made a significant contribution to the first cycle of industrial revolution, i.e., industrial revolution 1.0 during the 18th century. James Watt is often called the 'Father of Mechanical Engineering', as his invention of steam engine led to significant developments during the industrial revolution and beyond. The earliest computers were mechanical devices with electronics.

Significant contributions have been made during the Vedic ages and the first ever mechanical device that was invented was wheel and potter. SurmyamSuiramiva identified metals like Fe, Cu, Ag, Au etc., during the Vedic times. People knew about materials and material processing during those times and identified terminologies for the same in Sanskrit and produced gold and silver coins.

Seers like Tritala, Jalayan, Karaa, Vayurathaa and Vidyutrathaa discovered about aerodynamics during Rig Veda period, much before Wright Brothers discovered about aero planes. Computational Fluid Dynamics (CFD) analysis, which we are talking about today for different analysis, was there in the Vimana Shastra slokas.

Mechanical and manufacturing technology of ancient India ensured processing of natural products and their transformation into goods of trade, commerce and export.

Many scientists have made significant contributions to this domain. Leonardo da Vinci (16th century) studied and designed many mechanical systems that were related to transportation and warfare. In 17th century, Isaac Newton contributed the Laws of Motion used in several applications. Rudolf Diesel (18th century) was a German inventor, who created the first successful diesel engine and today diesel engines play a very important role in the transport and power sector in the world. Carl Frederick Benz (18th century) was a German automotive engineer, who developed the first practical automobile.

Mechanical engineering has evolved over the years and today the advent of computer and IT tools has facilitated better mechanical engineering in terms of design, analysis, and manufacturing. A mechanical engineer needs to work in multiple domains and needs to possess multiple skills like design, redesign, analyze, test, manufacture etc. It has been one of the founding disciplines of engineering and has contributed and will keep contributing to the growth and developments in this physical world.

CIVIL ENGINEERING

Indian civilization was the oldest civilization in the world and has a strong tradition of science and technology. It was the land of sages, seers, scholars, and scientists. Hinduism is a knowledge-based civilization, the Vedic texts should not be ignored dismissed as mythologies or as the work of imagination or just containing some moral stories. The Veda means knowledge and they contain relevant knowledge otherwise these texts would not have survived the millennia years of the historic storm. Let us know some of the great work done in ancient times.

Ancient India not only practised scientific methods of design and construction but also documented them for future generations. Here are some tips given by ancient sages on selection of site and construction

(1) Vishwakarma Vastu Shastra- Vishwakarma explains the first point of construction in the ancient book VastuShastra – 'पूर्वभूमिपरिक्षयेतपश्चात्वास्तुप्रकल्पयेत्', This means that before construction one should test the land. Vishwakarma further says that construction should not be done on the land which is very mountainous or on land with large cracks. Vastu shastra literally "science of architecture" are texts on the traditional Indian system of architecture. These texts describe principles of design, layout, measurements, ground preparation, space arrangement, and spatial geometry. The designs aim to integrate architecture with nature, the relative functions of various parts of the structure, and ancient beliefs utilising geometric patterns (yantra), symmetry, and directional alignments.

(2) Kashyap Shilpa (Craft) – In this ancient book, Kashyap Rishi has said that the foundation should be dug until water is seen because this way you would ensure that you have reached the rock level and the foundation would be strong.

(3) Bhrigu Samhita – In this scripture saint Bhrigu says that before buying land, one should test it for form, colour, juice, smell and touch. Rishi Bhrigu also explains its methods in his book.

Ancient cities of India found on the basis of archaeological discoveries:

- Rama was the world's first king to build a bridge across the sea. But he did not do it on his own. He sought the help of a great engineer called Nala according to Valmiki Ramayana. Any wise man will seek local knowledge when he ventures into new places. Nala knew the shallow areas across the sea in and around Tamilnadu. American space agency NASA also confirmed that there was a bridge through the satellite pictures. Any wise engineer will use such naturally elevated areas instead of deep waters to build a bridge.
- Bageeratha changed the course of the mighty river Ganges. The vast forest areas of modern Bihar, Uttar Pradesh, and West Bengal were made into fertile lands by his marvelous engineering feat. In those days very few people lived in those jungles. Puranas say that Bageeratha did penance for several thousand years to do this that too 'standing in one foot'. This is a phrase Indians use very often. Even the great Tamil poet Tiruvalluvar uses the simile of Stork that stands in one foot to catch a fish. This is the hidden language to say that he tried for a very long time with focused attention.
- Vedic Saint Agasthya discovered the land route to South India via Vindhya. The Puranas say that he "subdued the arrogance of the hills", this is hidden language. Till Agastya's this great discovery kings and travellers used only sea routes. Since they knew the secret of monsoon winds they can travel to West Bengal or Maharashtra from Sri Lanka in a few months' time.
- Uparichara Vasu, an ancient king made mountain passes for the benefit of land travellers. He was a Vasu king ruling over the Chedi kingdom. Mahabharata says that he kicked the Kolahal Mountain which was blocking the flow of the Shaktimati River. This is a hidden language to say that he diverted the river for irrigation by cutting the hills.
- In short Bageerathan, Agastya, and Uparichara Vasu are the earliest engineers who built dams across the rivers. But unlike modern engineers, they did not use cement or mortar but they used the hills themselves. To avoid the force they made checks and balances. They use a hidden language saying that Shiva bore the force when Ganga came down from heaven.
- Parasuraman retrieved a lot of lands and gave it to Indians. A Pandya king called Nilam Tharu VilNediyon built sea walls to prevent the sea from invading the land.
- Balraman always travelled with an axe to clear the forests and make them cultivable. He was a great agriculturist. When Krishna spent most of his time in politics, his brother Balarama did constructive work.
- The Mohanjodaro, created 3000 years ago, is considered as a wonderful piece of civil engineering. Found in archaeological excavations even the ruins prove that this town was well settled and its buildings and roads – all were made using symmetry and geometrical measurements. The roads found in this city were straight and were made from east to west and north to south and surprisingly they were at an angle of 90 degrees from each other. Buildings were also constructed in proportion. The intersection of the corners, the heights of the walls was equal. The city had public buildings, gardens, a restaurant, a large public bath as well as residential buildings. There was a provision for bathroom, living room etc in the residential buildings. The public buildings were 11.82m long, 7.01m wide and 2.44m high, and there were two streams of water. The building material and bricks of the walls were coated with a substance on which there was no effect of water. Archaeological research shows that people living here were well-versed in the construction techniques.
- Indus Valley Cities such as Harappa, Mohanjodaro, Lothal, Dholavira, Kalibangan need no new interpretations. The well-laid cities with uniform brick structures, Great Bath, most hygienic drainage systems, grain storage barns, and wells are all already well known to the world.
- Dwarka, also known as Lord Krishna's city, also narrates a similar story. Dr S R Rao discovered Dwarka in the archaeological excavation and found that the ancient city (Dwarka Nagar) was well built and settled. There was a wall around the city. The stones used for the construction of buildings did not erode despite the

fact that the city was very close to the sea. Two-storey buildings, roads and water system are also found in the city. Copper, bronze and some alloys with zinc mixed up to 34 percent have also been found during the excavation. The size of columns, windows, etc reveals that they were designed with a complete mathematical precision.

- South Indian Tamil saint Appar always travelled with a pickaxe to clear the bushes from the temple towers. He simply followed Balarama. Great Chola king Karikalan built a dam across river Cauvery in Kal Anai. The Grand Anicut was an engineering wonder of ancient Tamils. It was built around the 1st century AD. Big temples of India, the number of which runs into thousands, stand as monumental proof for the engineering skills of Indians. Mamallapuram and other Pallavacave temples are well-known milestones in Indian architecture.
- The Group of Monuments at Hampi are also recognized as a UNESCO World Heritage Site. The Vittala temple—the stone chariot – is the most iconic symbol of Hampi. The Virupaksha Temple at Hampi was built in the seventh century by the Chalukya rulers.



Virupaksha and Vithala Temple in Hampi

**Scheme & Syllabus for
B. Tech. (Computer Science and Engineering (Cyber
Security))**

HIGHER SEMESTER COURSES

DEPARTMENT OF CYBER SECURITY 2024-25

B. Tech. in Computer Science & Engineering (Cyber Security)

VISION:

To be a premier Department of Computer Science and Engineering in the field of Cyber Security by providing high quality Education, Research and Employability to enable graduates to be ethical, competent professionals and become entrepreneurs and partnership for community development.

MISSION:

- To develop technically competent security professionals with strong foundations, capable of adapting with the changing technologies for developing world-class software.
- To inculcate values of professional ethics, leadership qualities and lifelong learning.
- To prepare professionals for employment in industry, research, higher education, and entrepreneurship to benefit the society.

Program Educational Objectives (PEOs):

PEO 1: Graduates shall have the ability to apply knowledge across the disciplines Computer Science and Engineering in emerging areas of Cyber Security for higher studies, research, employability, product development and handle the realistic problems.

PEO 2: Graduates shall have good communication skills, possess ethical values, sense of responsibility to serve the society, and protect the environment.

PEO 3: Graduates shall possess academic excellence with innovative insight, managerial skills, leadership qualities, knowledge of contemporary issues and understand the need for lifelong learning for a successful professional career.

Program Outcomes (POs):

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to

provide valid conclusions.

5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs):

PSO 1: Ability to apply software technical skills to design, develop and debug optimized solutions for all real-world problems.

PSO 2: Ascertain knowledge in diverse areas of Cyber Security and experience an environment conducive in novel skills for successful career, entrepreneurship and higher studies.

B.Tech. (CSE (CS)): Scheme of Teaching and Examinations 2023-27
Outcome Based Education (OBE) and Choice Based Credit System (CBCS)
(Effective from the academic year 2024-25)
2nd Year Scheme

III SEMESTER								
Sl. No.	Course and Course code		Course Title	Teaching Dept.	Contact Hours/week			
					L	T	P	
1.	BSC	MA2001-1	Statistics and Probability Theory	MA	3	0	0	
2.	IPCC	CS2001-1	Data Structures	CB	3	0	2	
3.	IPCC	CS2002-1	Object Oriented Programming	CB	3	0	2	
4.	PCC	CS2101-1	Computer Organization and Architecture	CB	3	0	0	
5.	PCC	CB1102-1	Fundamentals of Data Communications	CB	3	0	0	
6.	PCC	CB1602-1	UNIX and Shell Programming	CB	0	0	2	
7.	HSMC	HU2002-1	Developing Professional Skills	HU	1	0	0	
8.	HEC	HU1005-1	Essence of Indian Culture	HU	1	0	0	
9.	AEC	ME1654-1	Innovation and Design Thinking	Any Dept.	1	0	0	
10.	MNC	HU1003-1	Kannada (Balake / Samskrithika)	Any Dept.	1	0	0	
TOTAL					19	0	6	
Additional Academic activities beyond class hours					-	-	-	
Learning hours per week					19	0	6	
Total notional learning hours (L+T+P+J+SL)				Per week	38			
				Per Semester	38 X 16 = 608			

*SL: Supervised Learning

Course prescribed to lateral entry Diploma holders admitted to III semester of Engineering programs													
11	MNC	MA1012 -1	Bridge course - Calculus & Differential Equations	MA	3	0	0	0	3	100	0	100	0

IV SEMESTER											
Sl. No.	Course and Course code		Course Title	Teaching Dept.	Contact Hours/week						
					L	T	P				
1.	BSC	MA2005-1	Linear Algebra and its Applications	MA	3	0	0				
2.	IPCC	CS3004-1	Design and Analysis of Algorithms	CB	3	0	2				
3.	IPCC	CS3005-1	Microprocessor and Embedded Systems	CB	3	0	2				

4.	PCC	CS2103-1	Software Engineering and Project Management	CB	3	0	0
5.	PCC	CB2102-1	Fundamentals of Computer Networks	CB	3	0	0
6.	PCC	CB2601-1	UNIX System Programming and Networking Lab	CB	0	0	2
7.	HSMC	HU1004-1	Universal Human Values	Any Dept.	1	0	0
8.	HSMC	HU2001-1	Enhancing Self Competence	HU	2	0	0
9.	VEC	CB1551-1	Department specific Vocational Education Course - NPTEL course 4 weeks	-	-	-	-
10.	UCC	UC1001-1	Internship – I (Activity based Internship)	-	-	-	-
TOTAL					18	0	6
Additional Academic activities beyond class hours					-	-	-
Learning hours per week					18	0	6
Total notional learning hours (L+T+P+J+SL)				Per week	45		
				Per Semester	45 X 16 = 720		

*SL: Supervised Learning

B.Tech. (CS): Scheme of Teaching and Examinations 2023-27
Outcome Based Education (OBE) and Choice Based Credit System (CBCS)
(Effective from the academic year 2023-24)
3rd Year Scheme

V SEMESTER							
Sl. No.	Course and Course code		Course Title	Teaching Dept.	Contact Hours/week		
					L	T	P
1.	IPCC	CB3001-1	Information and Network Security	CB	3	0	2
2.	IPCC	CB2004-1	Principles of Database Management Systems	CB	3	0	2
3.	PCC	CB3003-1	Essentials of Operating Systems	CB	3	0	0
4.	PCC	CB3603-1	Secure Full Stack Web Development	CB	0	0	2
5.	PEC	CSxxxx-x	Professional Elective -1 [Group 1]	CB	3	0	0
6.	HSMC	HU1006-1	Introduction to IPR	Any Dept.	1	0	0
7.	AEC	CS3601-1	Program Specific Ability Enhancement Course Back End Development	CB	1	0	2
		HU1010-1	Research Methodology	Any Dept.	2	0	0
8.	AEC	HU1007-1	Social Connect & Responsibility	Any Dept.	1	0	0
9.	AEC	UM1003-1	Employability Skill Development	Any Dept.	1	0	0
TOTAL					16	0	8
Additional Academic activities beyond class hours					-	-	-

Learning hours per week		16	0	8
Total notional learning hours (L+T+P+J+SL)	Per week	38		
	Per Semester	38 X 16 = 608		

*SL: Supervised Learning

VI SEMESTER							
Sl. No.	Course and Course code		Course Title	Teaching Dept.	Contact Hours/week		
					L	T	P
1.	IPCC	CB2003-1	Machine Learning for Cyber Security	CB	3	0	2
2.	PCC	CB3101-1	Firewall and UTM Architecture	CB	3	0	0
3.	PCC	CB3602-1	Ethical Hacking and Network Defense	CB	0	0	2
4.	PEC	CBxxxx-1	Professional Elective -II [Group 1]	CB	3	0	0
5.	PEC	CBxxxx-1	Professional Elective -III [Group 2]	CB	3	0	0
6.	OEC	XXX5XX-1	Open Elective –I	Any Dept.	3	0	0
7.	HSMC	MG1003-1	Management & Entrepreneurship	Any Dept.	3	0	0
8.	AEC	HU1008-1	Life Skills for Engineers	Any Dept.	1	0	0
TOTAL					19	0	4
Additional Academic activities beyond class hours					-	-	-
Learning hours per week					19	0	4
Total notional learning hours (L+T+P+J+SL)					Per week	39	
					Per Semester	39 X 16 = 624	

Value added Course												
Value added Course		Introduction to DevOps		0	0	2	0	3	-	0	-	PP/ NP

B.Tech. (CS): Scheme of Teaching and Examinations 2023-27
Outcome Based Education (OBE) and Choice Based Credit System (CBCS)
(Effective from the academic year 2023-24)
4th Year Scheme

VII SEMESTER							
Sl. No.	Course and Course code		Course Title	Teaching Dept.	Contact Hours/week		
					L	T	P
1.	IPCC	CB3002-1	Cyber Forensics	CB	3	0	2
2.	PCC	CB3603-1	Secured Mobile Application Development	CB	0	0	2
3.	PEC	CBxxxx-x	Professional Elective -IV [Group 1]	CB	3	0	0
4.	PEC	CBxxxx-x	Professional Elective -V [Group 2]	CB	3	0	0
5.	OEC	XXX5XX-1	Open Elective –II	Any Dept.	3	0	0
6.	HSMC	MG1002-1	Financial Management	Any Dept.	3	0	0
7.	HEC	HU1009-1	Indian Knowledge Systems	Any Dept.	1	0	0
8.	UCC	UC3001-1	Major Project Phase I	CB	-	-	4
TOTAL					16	0	8
Additional Academic activities beyond class hours					-	-	-
Learning hours per week					16	0	8
Total notional learning hours (L+T+P+J+SL)				Per week	38		
				Per Semester	38 X 16 = 608		

VIII SEMESTER							
Sl. No.	Course and Course code		Course Title	Teaching Dept.	Contact Hours/week		
					L	T	P
1.	UCC	UC2001-1	Internship- II (Societal internship and Research/Industry Internship)		Mandatory Societal internship (80 – 90 h) and Internship / Industry Internship (240 – 270 h) or Internship / Industry internship (240 – 270 h) for a total of 8 weeks (320 – 360 h) completed in one/two streams during the vacation periods between the VII semesters		
2.	UCC	UC3002-1	Major Project Phase II	CB	Students should carry out research at an institute/industry or research institute/industry Center of Excellence. The contact hours /week for the project should be between the project guidelines for the students.		
TOTAL					-	-	-
Additional Academic activities beyond class hours					-	-	-
Learning hours per week					-	-	-
Total notional learning hours (L+T+P+J+SL)				Per week	-		
				Per Semester	608		

B.Tech. (CS): Scheme of Teaching and Examinations 2023-27
Outcome Based Education (OBE) and Choice Based Credit System (CBCS)
(Effective from the academic year 2023-24)

Program Specific Ability Enhancement Courses	
Course Code	Course Title
CB3601-1	Secured Web Application Development

B.Tech. (CS): Scheme of Teaching and Examinations 2023-27

Outcome Based Education (OBE) and Choice Based Credit System (CBCS)

(Effective from the academic year 2023-24)

List of Professional Elective Courses (PEC)			
Group – 1		Group – 2	
Architecture, Embedded Systems & General			
Course Code	Course Name	Course Code	Course Name
CS2201-1	Cloud Computing	CS3301-1	Business Intelligence and its Applications
CS3201-1	Advanced Computer Architecture	CS3302-1	Internet of Things
CS3202-1	Fundamentals of Image Processing	CS3303-1	Multicore Architecture and Programming
CS3203-1	Operations Research	CS3304-1	Speech Processing
CS3204-1	Robotic Process Automation Design & Development		
Software Engineering & Development			
Course Code	Course Name	Course Code	Course Name
CS2211-1	Internet & Web Programming	CS2311-1	Angular and ReactJS
CS3211-1	Advanced Java Programming	CS2312-1	User Interface Design
CS3212-1	Agile Technology	CS3311-1	Full Stack Development
CS3213-1	Game theory and applications	CS3312-1	Object Oriented Modelling and Design
CS3214-1	NoSQL database	CS3313-1	Software Architecture and Design Patterns
CS3215-1	Competitive Programming	CS3314-1	Software Testing
Systems, Networks & Security			
Course Code	Course Name	Course Code	Course Name
CS2221-1	System Modelling and Simulations	CS3321-1	Blockchain Technology
CS3221-1	Adhoc Wireless Networks	CB3322-1	Security and Resilience
CB3222-1	Malware Analysis and Detection	CS3323-1	Software Design
Intelligent Systems & Analytics			
Course Code	Course Name	Course Code	Course Name
CS1231-1	Introduction to Data Science	CS1331-1	Bio Informatics
CS2231-1	Artificial Intelligence	CS2331-1	Big Data Analytics
CS3231-1	Computer Vision	CS2332-1	Soft Computing
CS3232-1	Data Mining	CS3331-1	Human Computer Interaction
CS3233-1	Deep Learning	CS3332-1	Natural Language Processing
CS3234-1	Social and Web Analytics	CS3333-1	Virtual Reality

Courses from Basic Science

STATISTICS & PROBABILITY THEORY

Course Code:	MA2001-1	Course Type:	BSC
Teaching Hours/Week (L: T: P):	3:0:0	Credits:	03
Total Teaching Hours:	40+0+0	CIE + SEE Marks:	50+50
Prerequisite	MA1002-1		

Teaching Department: Mathematics

Course Objectives:

1. Understand the basic principles of probability, Bayes theorem, understand the definitions of discrete, continuous, and joint random variables, compute the mean, variance and covariance of random variables.
2. Define the binomial, uniform, Poisson, exponential and normal random variables use these principles in problem solving situations.
3. Understand the concepts of statistical population and sample, variables and attributes. Learn about moments and their use in studying various characteristics of data and various distributions.

UNIT-I

PROBABILITY THEORY 16 Hours

Finite sample space, probability, conditional probability and independence, Bayes' theorem. One dimensional random variable: discrete and continuous random variable, probability functions, cumulative distribution function, expectation and variance.

Two-dimensional random variable: joint pdf, marginal pdf's, covariance (CO1)

Distributions: Binomial, Poisson, Uniform, Normal and exponential distributions.

Moment generating function- properties and simple problems.(CO2)

UNIT-II

SAMPLING DISTRIBUTION AND ESTIMATION 14 Hours

Random Sample, Sample mean, sample variance, sampling distribution of mean, Central limit theorem, sampling distributions of proportions and sums. Student's t-distribution, Chi-square distribution. Sample distribution of variance.

Estimation: Point estimation, interval estimation, confidence intervals for means and variance. (CO3)

CURVE FITTING AND REGRESSION

Least square principle, fitting of straight lines, polynomials and exponential curves.

Correlation, Rank correlation, Coefficient of correlation, Linear regression. (CO4)

UNIT-III

STOCHASTIC PROCESS 10 Hours

Stochastic processes, stochastic matrices, fixed points, regular stochastic matrices, Markov chains, transition probabilities, Birth-death process, Queuing theory – M/M/1 Model, simple problems.(CO5)

Course Outcomes: At the end of the course student will be able to

1. Apply the concepts of probability, including discrete and continuous random variables, probability distributions, conditioning, independence, expectations, and variances.
2. Define and explain the different statistical distributions (e.g., Normal, Binomial, Poisson) and the areas of their application.
3. Explain the concept of correlation and the difference between positive and negative correlation. Compute the correlation coefficient, r , Explain and apply the least square errors method numerically and algebraically to find the curve of best fit.
4. Able to apply the central limit theorem to sampling distribution. Translate real-world problems into probability models.
5. Identify and apply the most appropriate stochastic process technique for a given applied problem. Calculate probabilities of absorption and expected hitting times for discrete time Markov chains with absorbing states.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes													1	2
MA2001-1.1	3	2	-	-	-	-	-	-	-	-	-	-	-	-
MA2001-1.2	2	2	-	-	-	-	-	-	-	-	-	-	-	-
MA2001-1.3	3	1	-	-	-	-	-	-	-	-	-	-	-	-
MA2001-1.4	3	2	-	-	-	-	-	-	-	-	-	-	-	-
MA2001-1.5	3	2	-	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

- Paul L Meyer, "Introductory Probability and Statistical Applications", Addison-Wesley Publishing Company, 2nd Edition (Reprint), 1970.
- Hogg and Craig, "Introduction to mathematical Statistics", Pearson Education, New Delhi, 6th Edition.

REFERENCE BOOKS:

- Schaum Outlines, "Probability and Statistics", Mc Graw Hill, 3rd edition, 2010.
- T. Veerarajan, "Engineering Mathematics", McGraw-Hill, New Delhi, 2008.
- B. V. Ramana, "Higher Engineering Mathematics", Tata Mc Graw –Hill, New Delhi, 2010.

E Books / MOOCs/ NPTEL

- <https://nptel.ac.in/courses/110107114>
- <https://nptel.ac.in/courses/111105090>
- <https://nptel.ac.in/courses/111102098>

LINEAR ALGEBRA AND ITS APPLICATIONS

Course Code:	MA2005-1	Course Type:	BSC
Teaching Hours/Week (L: T: P):	3:0:0	Credits:	03
Total Teaching Hours:	40+0+0	CIE + SEE Marks:	50+50
Prerequisites	MA1002-1, MA1007-1		

Teaching Department: Mathematics

Course Objectives:

- Learn to apply elementary row operations to solve linear systems of equations and find the eigenvalues and eigenvectors of a matrix.
- Find the eigenvalues and eigenvectors of a square matrix using the characteristic polynomial and will know how to diagonalize a matrix, when this is possible
- Understand real vector spaces and subspaces, linear independence and dependence, and find basis and dimension of a vector space, row space, column space and null space of a matrix.
- Define a linear transformation and find the matrix associated with it; determine the kernel and range of a transformation; find inner product of vectors, orthogonal and an orthonormal basis.
- Learn basic concepts of real quadratic forms, decomposition of matrices and solve problems on the same.

UNIT-I

Matrices **15 Hours**

Elementary transformation of a matrix, Echelon form and rank of a matrix. Consistency and solution of system of linear equations - Gauss elimination method, LU Decomposition method and approximate solution by Gauss Seidel method.

Trace, relation between trace and Eigen values of a matrix, Eigen values and Eigen vectors of symmetric matrices, Rayleigh's power method to find the largest eigen value and eigen vector of square matrices. Diagonalization.

UNIT-II

Vector Space **08 Hours**

Vector spaces, subspaces, linearly dependent and independent vectors, basis and dimension, coordinates, row space, column space and null space.

Security

Linear Transformations	07 Hours																																																																																																									
Linear transformations, algebra of linear transformations, representation of transformations by matrices, isomorphism, Range and Null space of a linear transformation. Rank – nullity theorem. Inner products, orthogonal sets of projections, Gram-Schmidt’s orthogonalization process.																																																																																																										
UNIT-III																																																																																																										
Matrix Decompositions	10 Hours																																																																																																									
Quadratic forms, QR-factorization, least-squares problems, singular value decomposition and principal component analysis.																																																																																																										
Course Outcomes: At the end of the course student will be able to																																																																																																										
<table><tr><td>1.</td><td>Solve the system of linear equations for exact or approximate solutions.</td></tr><tr><td>2.</td><td>Compute and use eigenvectors and eigenvalues.</td></tr><tr><td>3.</td><td>Analyze finite dimensional vector spaces and subspaces over a field and their properties, including the basis structure of vector spaces.</td></tr><tr><td>4.</td><td>Relate matrices and linear transformations, apply the properties of inner product and determine orthogonality on vector spaces and orthogonal bases.</td></tr><tr><td>5.</td><td>Derive and utilize Quadratic forms, SVD and QR factorization of the matrix for efficiently solving problems in practice.</td></tr></table>		1.	Solve the system of linear equations for exact or approximate solutions.	2.	Compute and use eigenvectors and eigenvalues.	3.	Analyze finite dimensional vector spaces and subspaces over a field and their properties, including the basis structure of vector spaces.	4.	Relate matrices and linear transformations, apply the properties of inner product and determine orthogonality on vector spaces and orthogonal bases.	5.	Derive and utilize Quadratic forms, SVD and QR factorization of the matrix for efficiently solving problems in practice.																																																																																															
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Course Outcomes Mapping with Program Outcomes & PSO																																																																																																										
<table><tr><th>Program Outcomes→</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th><th>10</th><th>11</th><th>12</th><th colspan="2">PSO↓</th></tr><tr><th>↓ Course Outcomes</th><th></th><th></th><th></th><th></th><th></th><th></th><th></th><th></th><th></th><th></th><th></th><th></th><th>1</th><th>2</th></tr><tr><td>MA2005-1.1</td><td>3</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>MA2005-1.2</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>MA2005-1.3</td><td>2</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>MA2005-1.4</td><td>2</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>MA2005-1.5</td><td>3</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr></table>		Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		↓ Course Outcomes													1	2	MA2005-1.1	3	1	-	-	-	-	-	-	-	-	-	-	-	-	MA2005-1.2	2	3	-	-	-	-	-	-	-	-	-	-	-	-	MA2005-1.3	2	1	-	-	-	-	-	-	-	-	-	-	-	-	MA2005-1.4	2	2	-	-	-	-	-	-	-	-	-	-	-	-	MA2005-1.5	3	2	-	-	-	-	-	-	-	-	-	-	-	-
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1: Low 2: Medium 3: High																																																																																																										
TEXTBOOKS:																																																																																																										
1.	Kenneth Hoffman And Ray Kunze, “Linear Algebra”, Prentice-Hall, 2 nd edition, 1971																																																																																																									
2.	David C. Lay, “Linear Algebra and Its Applications”, Pearson Education, Inc., 5 th edition, 2016.																																																																																																									
REFERENCE BOOKS:																																																																																																										
1.	Seymour Lipschutz And Marc Lars Lipson, “Schaum’s outlines - Linear Algebra”, McGraw-Hill, 4 th Edition, 2002.																																																																																																									
2.	Gilbert Strang, “Introduction to Linear Algebra”, Wellesley-Cambridge Press, 5 th Edition, 2016.																																																																																																									
3.	Gerald Farin, Dianne Hansford, “Practical Linear Algebra, A Geometry Toolbox”, Chapman and Hall, 4 th edition, 2021.																																																																																																									
4.	Sheldon Axler, “Linear Algebra Done Right”, Springer Nature, 3 rd Edition, 2015.																																																																																																									
E Books / MOOCs/ NPTEL																																																																																																										
1.	https://nptel.ac.in/courses/111101115																																																																																																									
2.	https://archive.nptel.ac.in/courses/111/106/111106135/																																																																																																									
3.	https://nptel.ac.in/courses/110104024																																																																																																									

Bridge Courses for Lateral Entry Students

Bridge course - Calculus & Differential Equations (common to AM\CC\CS\IS\DS\RI\CY)			
Course Code	MA1012-1	Course Type	MNC
Teaching Hours/Week (L:T:P: S)	3:0:0	Credits	00
Total Teaching Hours	40+0+0	CIE (NO SEE) Marks	100
Teaching Department: Mathematics			
Mandatory Non – credit course (MNC): This course is prescribed to the lateral entry Diploma holders admitted to III semester of BE/B.Tech., programs, they shall attend the classes during the respective semesters to complete all the formalities of the course and appear for the Continuous Internal Evaluation (CIE). In case, any student fails to register for the said course/fails to secure the minimum 40 % of the prescribed CIE marks, he/she shall be deemed to have secured an F grade. In such a case, the student has to fulfill the requirements during subsequent semester/s to appear for CIE. MNC Courses shall not be considered for vertical progression as well as for the calculation of SGPA and CGPA, but completion of the courses shall be mandatory for the award of degree Course Objectives: This course will enable the students to master the basic tools of differential calculus, partial differentiation, Vector differentiation and Integration and become skilled for solving problems in science and engineering.			
UNIT-I			
DIFFERENTIAL CALCULUS Polar curves, angle between the radius vector and the tangent, angle of intersection of two curves, derivatives of arcs and radius of curvature -cartesian, parametric and polar forms (No Derivation). Taylor's theorem for functions of single variable. Mean value theorems. 7 Hours			
PARTIAL DIFFERENTIATION Partial derivatives of simple functions, Total differentiation - differentiation of composite and implicit functions. Taylor's theorem for functions of two variables, maxima and minima for functions of two variables.			8 Hours
UNIT-II			
VECTOR DIFFERENTIAL CALCULUS Vector algebra(review), scalar and vector valued functions, gradient, directional derivative and hessian of multivariable function, Divergence and curl of a vector valued function.			7 Hours

Security

ORDINARY AND PARTIAL DIFFERENTIAL EQUATIONS Ordinary differential equations(review), linear and nonlinear differential equations. Second and higher order linear differential equations with constant coefficients. Formation of partial differential equations by eliminating arbitrary constants and arbitrary functions. Solution of P.D.E by the method of separation of variables.													8 Hours	
UNIT-III														
MULTIPLE INTEGRALS Double integrals and triple integrals, Evaluation by change of order of integration, change of variables and applications to area and volume.													10 Hours	
Course Outcomes: At the end of the course student will be able to														
1.	Apply the concept of radius of curvature, mean value theorems, partial differentiation of a function to solve engineering problems													
2.	Solve the vector functions, ordinary and partial differential equations and multiple integrals													
Course Outcomes Mapping with Program Outcomes														
	Program Outcomes →	1	2	3	4	5	6	7	8	9	10	11	12	
	↓ Course Outcomes													
	MA1012-1.1	3	2											
	MA1012-1.2	2	2											
1: Low 2: Medium 3: High														
Assessment Details (CIE) The weightage of Continuous Internal Evaluation (CIE) is 100% (No Semester End Exam (SEE)) . The student must obtain minimum of 40% in CIE to pass.														
Continuous Internal Evaluation:														
1.	Methods recommended: Two Tests (80%), Written Quiz (10%) and module assignments (10%).													
2.	The class teacher must decide the topics for closed book test and Written Quiz. The methods of CIE for the subject must be announced at the beginning of the course.													
TEXTBOOKS:														
1.	B.S. Grewal, "Higher Engineering Mathematics", Khanna Publications, 43 rd Edition, 2015.													
2.	Erwin Kreyszig, "Advanced Engineering Mathematics", John Wiley and Sons, 10 th Edition (Reprint), 2016.													
3.	Murray R. Spiegel: "Vector Analysis", Schuam publishing Co.													
REFERENCE BOOKS:														

1.	G.B. Thomas and R.L.Finney, "Calculus and Analytic geometry", Pearson, 2002.
2.	T. Veerarajan, "Engineering Mathematics", McGraw-Hill, New Delhi, 2008.
3.	B. V. Ramana, "Higher Engineering Mathematics", Tata Mc Graw –Hill, New Delhi, 2010

Bridge course - Discrete Mathematics & Numerical Methods (common to AM\CC\CS\IS\DS\RI\CY)			
Course Code	MA1014-1	Course Type	MNC
Teaching Hours/Week (L:T:P: S)	3:0:0	Credits	00
Total Teaching Hours	40+0+0	CIE (NO SEE) Marks	100
Teaching Department: Mathematics			
Mandatory Non – credit course (MNC): This course is prescribed to the lateral entry Diploma holders admitted to III semester of BE/B.Tech., programs, they shall attend the classes during the respective semesters to complete all the formalities of the course and appear for the Continuous Internal Evaluation (CIE). In case, any student fails to register for the said course/fails to secure the minimum 40 % of the prescribed CIE marks, he/she shall be deemed to have secured an F grade. In such a case, the student has to fulfill the requirements during subsequent semester/s to appear for CIE. MNC Courses shall not be considered for vertical progression as well as for the calculation of SGPA and CGPA, but completion of the courses shall be mandatory for the award of degree			
Course Objectives: This course will enable the students to master the basic tools of set theory and relations, propositional and predicative logics, numerical methods, Fourier series and transforms and become skilled for solving problems in science and engineering.			
UNIT-I			
Set Theory and Logic Sets- operations on sets, product sets and partitions (review) Relations- representation of relations as matrices and digraphs, equivalence relations. Functions- permutations functions, functions for computer science. Fundamentals of logic- Propositional logic, logical operations(review), rules of inference Predicates calculus. 7 Hours			
Graph Theory Graphs: Basic terminologies, some special simple graphs, bipartite graphs, adjacency matrices, incidence matrices, graph isomorphism, connectivity- vertex and edge connectivity, Euler and Hamiltonian graphs and their applications, planar graphs, graph coloring and their applications.			

													8 Hours	
UNIT-II														
Numerical Methods														
Roots of algebraic and transcendental equations- Newton Raphson method, Regula Falsi method. Numerical solution of ordinary differential equations- Taylor's series method, Modified Euler's method and Runge –Kutta method of fourth order.														
Numerical solution of partial differential equations- Classification of partial differential equations, examples, solution of Laplace and Poisson equations by standard five-point formulae, solution of heat and wave equations by explicit method.														
													15 Hours	
UNIT-III														
Fourier series and Transforms														
Periodic functions, Euler's formulae, Fourier series of odd and even functions, functions with arbitrary period, half range series. Fourier transform, inverse Fourier transform, Convolution theorem, Fourier sine and cosine transforms.														
													10 Hours	
Course Outcomes: At the end of the course student will be able to														
1	Represent a relation in terms of matrix and digraph, apply permutation functions for encoding and decoding simple text messages and establish by deduction the validity of an argument using inference rules. Identify suitable data structure for representing a graph, apply the concept of connectivity in real life problems													
2	Apply numerical methods to find solutions of algebraic equations, ordinary differential equations, partial differential equations and the analytical technique to express periodic function as a Fourier sine and cosine series and apply the concepts of Fourier- transforms to solve engineering problems.													
Course Outcomes Mapping with Program Outcomes														
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	
↓ Course Outcomes														
MA1014-1.1		3	2											
MA1014-1.2		3	2											
1: Low 2: Medium 3: High														
Assessment Details (CIE)														
The weightage of Continuous Internal Evaluation (CIE) is 100% (No Semester End Exam (SEE)) . The student must obtain minimum of 40% in CIE to pass.														
Continuous Internal Evaluation:														
1.	Methods recommended: Two Tests (80%), Written Quiz (10%) and module assignments (10%).													
2.	The class teacher must decide the topics for closed book test and Written Quiz. The methods of CIE for the subject must be announced at the beginning of the course.													

TEXTBOOKS:

- | | |
|----|--|
| 1. | B.S. Grewal, "Higher Engineering Mathematics", Khanna Publications, 43 rd Edition, 2015. |
| 2. | Erwin Kreyszig, "Advanced Engineering Mathematics", John Wiley and Sons, 10 th Edition (Reprint), 2016. |

REFERENCE BOOKS:

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|----|---|
| 1. | T. Veerarajan, "Engineering Mathematics", McGraw-Hill, New Delhi, 2008. |
| 2. | B. V. Ramana, "Higher Engineering Mathematics", Tata Mc Graw –Hill, New Delhi, 2010 |

Integrated Core Courses Professional

DATA STRUCTURES			
Course Code:	CS2001-1	Course Type:	IPCC
Teaching Hours/Week (L: T: P):	3:0:2	Credits:	04
Total Teaching Hours:	40+26	CIE + SEE Marks:	50+50
Prerequisite	CS1001-1/CS1004-1		
Teaching Department: Computer Science & Engineering			
Course Objectives:			
1.	Outline the concepts of data structure, it's operations, Memory allocation functions and design the programs using arrays and structures, pointers, pointer to structure.		
2.	Implement linear data structure stack and usage of stacks in various applications.		
3.	Implement linear data structure Ordinary Queue, Circular Queue and priority queues		
4.	Implement the operations of singly linked lists and circular linked lists, doubly linked list and circular doubly lists.		
5.	Identify and differentiate different types of binary trees and binary search trees data structures and also implement them and illustrate threaded binary trees, expression trees, graph representation and techniques of hashing.		
UNIT-I			
			15 Hours
Introduction: Data Structure Definition, Classification (Primitive and non-primitive), data structure operations, Pointers and Dynamic Memory Allocation functions with programming examples Arrays and Structures: Arrays in C, Dynamically allocated arrays, Structures and Union, Array of Structures and Pointer to Structure, Programming Example. Linear Data structures-Stack: Introduction and Definition, Representation of stack: Array and structure representation of stacks, Primitive operations on stacks Applications of Stack: Conversion of Expressions Algorithms and C programs with tracing Examples: For evaluating postfix expression, infix to postfix conversion. Recursion: Definition, Implementation, Examples on Recursion with tracing: Factorial function, Fibonacci sequence and Tower of Hanoi.			
UNIT-II			
			15 Hours
Linear Data structures-Queue: Introduction and Definition, Representation of Queue: Array and Structure representation of Queue, Other queue structures: circular queue, priority queue. Linear Data structures-Linked List: Singly Linked List and chains, Representing chain in C using dynamic variables ,Inserting and deleting nodes, Other list Operations on singly Linked List, Linked Stack and Queues, Header Nodes, Representation of Linked list using arrays. Circular Linked List, Doubly Linked List and Circular doubly Link list : Representation and Operations			
UNIT-III			
			10 Hours
Non linear Data structures – Tree data structures: Introduction: Tree definition, Terminology, Binary Trees: Definition, Types, Properties, Representation			

of Binary Tree: Array representation, Linked representation, Binary Tree traversals- Preorder, Inorder and postorder. **Threaded binary Trees:** Definition, types, Data structure and memory representation of threaded tree, **Binary Search Tree:** Definition, Construction- Searching, Insertion operations, deletion process, Traversal examples.

Expression Tree: Constructing expression tree for a given expression, traversals, Evaluation of expression, programming examples.

Non linear Data structures – Graphs:

Representation of graphs: Definition, types and terminology, Matrix representation, Adjacency list chain and sequential representation.

Hashing: Hash Table organizations, Hashing Functions, Over flow handling.

Suggested List of Experiments

1.	Programs on arrays and structures using Pointers
2	Stack and Ordinary Queue implementation using array and structure.
3	Application of stack data structure- (i) Evaluation of post fix and Tower of Hanoi problem using recursion. (ii) Conversion infix to postfix
4	Circular and priority queues.
5	Operation on Singly Linked list implementation using dynamic variables.
6	Dynamic implementation of stack and Queue data structure.
7	Circular linked list
8	Doubly linked list implementation.
9	Binary Search Tree Construction and Tree traversal operations.
10	Hashing- Searching and collision handling

Course Outcomes: At the end of the course student will be able to

1.	Acquire the fundamental knowledge of various types of data structures, dynamic memory allocation and design the programs using arrays, structures and pointers
2.	Apply the fundamental knowledge of data structures to design stack and use them for solving problems.
3.	Apply the fundamental knowledge of data structures to design queues and use them for solving problem
4.	Design and develop singly linked lists, circular linked lists and doubly linked list.
5.	Acquire the knowledge of trees and employ binary trees and binary search tree data structure, advanced tree representation of graphs and hashing techniques.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS2001-1.1	3	1	3	-	-	-	-	-	-	-	-	1	3	-	-
CS2001-1.2	3	1	3	-	-	-	-	-	-	-	-	1	3	-	-
CS2001-1.3	3	1	3	-	-	-	-	-	-	-	-	1	3	-	-
CS2001-1.4	3	1	3	-	-	-	-	-	-	-	-	1	3	-	-
CS2001-1.5	3	1	3	-	-	-	-	-	-	-	-	1	3	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Aaron M. Tenenbaum, YedidyahLangsam& Moshe J. Augenstein, "Data Structures using C", Pearson Education/PHI, 2009.
2.	Ellis Horowitz and Sartaj Sahni, "Fundamentals of Data Structures in C", 2nd edition, Universities Press, 2014.

REFERENCE BOOKS:

1. Seymour Lipschutz, "Data Structures, Schaum's Outlines", Revised 1st edition, McGraw Hill, 2014.

E Books / MOOCs/ NPTEL

1. Data Structures Using C, ISRD Group, Tata McGraw Hill, 2006.
2. Data Structures Using C, Reema Thareja, 2nd edition, Oxford University Press, 2014
3. Introduction to Data Structures by edx , URL: <https://www.edx.org/course/>
4. Data structures by Berkley, URL: <https://people.eecs.berkeley>
5. Advance Data Structures by MIT OCW , URL: <https://www.mooclab.club/>
6. Data Structure by Harvard Extension School, URL: <http://www.extension.harvard>

OBJECT ORIENTED PROGRAMMING

Course Code:	CS2002-1	Course Type:	IPCC
Teaching Hours/Week (L: T: P):	3:0:2	Credits:	04
Total Teaching Hours:	40+0+26	CIE + SEE Marks:	50+50
Prerequisite	CS1001-1/ CS1004-1		

Teaching Department: Computer Science & Engineering

Course Learning Objectives:

1. Learn fundamental features of object-oriented language and JAVA programming constructs.
2. Develop and run simple Java programs using OOPS concepts of java.
3. Create multi-threaded programs and event driven Graphical User Interface (GUI) programming using swing package.

UNIT-I

INTRODUCTION:	15 Hours
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Introducing Classes—Class Fundamentals, Declaring Objects, Assigning Object Reference Variables, Introducing Methods, Constructors, this keyword, Method overloading, using objects as parameters, Argument passing, returning objects, Access control, static, final

Inheritance – Inheritance Basics, super keyword, create a Multilevel Hierarchy, when constructors are called? Method Overriding, using abstract classes, Using final with Inheritance.

Packages and Interfaces – Packages, Access protection, Importing Packages, Interfaces.
 (Textbook 1, Chapter – 6-9)

UNIT-II

Exception Handling and Multithread Programming:	15 Hours
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Exception Handling – Exception-Handling Fundamentals, Exception Types, Uncaught Exceptions, Using try and catch, multiple catch Clauses, Nested try statements, throw, and throws, finally.

Multithreaded Programming – The Java Thread Model, The Main Thread, creating a Thread, Creating Multiple Threads, Thread synchronization, using isAlive() and join(), Thread Priorities.

File Handling – Reading and writing Serial Access Files, basic File Methods.

Event Handling - Two Event Handling Mechanisms, Delegation Event Model, Event Classes, Sources of Events, Event Listener Interfaces.

(Textbook 1, Chapter – 10-11, 19, 22)

UNIT-III

FRAMEWORKS:	10 Hours
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Generics-What are Generics? A Simple Generics Example, the general form of a generic class, Creating a Generic method.

Collections framework- Collection Interfaces – Stack and Queue, Collection classes – ArrayList, LinkedList, Stack and Queue.

(Textbook 1, Chapter – 14, 17)

Swings – The origins of Swing, Two key Swing features, The Swing Packages: A simple Swing Application; Event Handling, JLabel, JTextField; The Swing Button: JList, JRadioButton, JCheckBox, JComboBox, JTable.

(Textbook 1, Chapter - 29, 30)

Suggested List of Experiments

1. Use java program to demonstrate the OOP concepts.
2. Implement the java programs that uses the concepts of exception handling
3. Develop multi-threaded java programs
4. Demonstrate the file handling using Java
5. Develop user interfaces using Java Swing

Course Outcomes: At the end of the course student will be able to

1.	Demonstrate a solid understanding of object-oriented programming, including creating classes, declaring objects, implementing methods and constructors, and using access control and keywords effectively.
2.	Understand and implement inheritance concepts, including creating hierarchies and using relevant keywords, as well as managing packages and implementing interfaces.
3.	Develop programs to effectively manage exceptions in Java and master event handling mechanisms, including the delegation event model and event listener interfaces.
4.	Demonstrate proficiency in multithreading, including thread management and synchronization, and in file handling and basic file operations in Java.
5.	Effectively apply generics, utilize the collections framework, and create GUI applications using Swing in Java.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS2002-1.1	3	1	3	-	1	-	-	-	-	-	-	2	2	3	-
CS2002-1.2	3	1	3	-	2	-	-	-	-	-	-	2	-	3	-
CS2002-1.3	3	1	3	-	3	-	-	-	-	-	-	2	-	3	-
CS2002-1.4	3	1	3	-	3	-	-	-	-	-	-	2	2	3	-
CS2002-1.5	3	1	3	-	3	-	-	-	-	-	-	2	-	3	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Herbert Scheldt, Java the Complete Reference, 7th Edition, Tata McGraw Hill, 2007.
2. Jan Graba, An Introduction to Network Programming with Java, 2007, Springer Publications.

REFERENCE BOOKS:

1. Mahesh Bhav and Sunil Patekar, "Programming with Java", First Edition, Pearson Education, 2008, ISBN:9788131720806
2. Rajkumar Buyya, SThamarasiselvi, xingchen chu, Object oriented Programming with Java, Tata McGraw Hill education private limited.
3. Richard A Johnson, Introduction to Java Programming and OOAD, CENGAGE Learning.
4. E Balagurusamy, Programming with Java A primer, Tata McGraw Hill companies.

E Books / MOOCs/ NPTEL	
1.	Online course material by Oracle : http://docs.oracle.com/javase/tutorial/index.html
2.	https://www.udemy.com/courses/search/?q=java&price=pricefree&view=grid
3.	Oracle: www.oracle.com/events/global/en/java.../java-a-beginners-guide-1720064.pdf
4.	NPTEL: www.nptelvideos.com/java/java_video_lectures_tutorials.php

DESIGN AND ANALYSIS OF ALGORITHMS

Course Code:	CS3004-1	Course Type:	IPCC
Teaching Hours/Week (L: T: P):	3:0:2	Credits:	04
Total Teaching Hours:	40+0+26	CIE + SEE Marks:	50+50
Prerequisite	CS2001-1, CS1001-1/CS1004-1		

Teaching Department: Computer Science & Engineering

Course Objectives:

1.	Understand the notion of algorithms, Algorithm design and analysis process, asymptotic notations and Analyze the non-recursive and recursive algorithms and to represent efficiency of these algorithms in terms of the standard asymptotic notations.
2.	Devise the Brute Force and Divide and Conquer techniques to design the algorithms and apply these methods in designing algorithms to solve a given problem.
3.	Apply the Decrease and Conquer, Transform and Conquer algorithm design techniques to solve a given problem.
4.	Get idea of Time versus Space Trade-offs and Apply and Analyse dynamic programming methods in designing algorithms to solve a given problem.
5.	Describe and illustrate the idea of Greedy method, Backtracking and Branch and Bound algorithm design techniques to solve a given problem and to describe P, NP and NP Complete problems.

UNIT-I

15 Hours

INTRODUCTION: What is an Algorithm? Fundamentals of Algorithmic Problem Solving

(Text Book-1: Chapter 1: 1.1 to 1.2)

FUNDAMENTALS OF THE ALGORITHMS EFFICIENCY: Analysis Framework, Asymptotic Notations and Basic efficiency classes, Mathematical Analysis of Non-recursive and Recursive Algorithms,

(Text Book-1: Chapter 2: 2.1 to 2.4)

BRUTE FORCE: Background, Selection Sort and Bubble sort, Sequential search and Brute-Force String Matching algorithms with complexity analysis, Exhaustive search

(Text Book-1: Chapter 3: 3.1, 3.2, 3.4)

DIVIDE AND CONQUER: General Method, Merge sort, Quick sort, Binary Search algorithms with Complexity analysis

(Text Book-1: Chapter 4: 4.1 to 4.3)

UNIT-II

15 Hours

DECREASE & CONQUER: General method, Insertion Sort algorithm, **Graph algorithms:** Depth First Search, Breadth First Search, Topological Sorting with complexity analysis
TRANSFORM AND CONQUER: General method, **Balanced Search Trees:** AVL trees, 2-3 trees, Heaps and Heap sort algorithms with complexity analysis
TIME AND SPACE TRADEOFFS: Sorting by counting, Input Enhancement in String Matching: Horspool's algorithm and analysis
(Text Book-1: Chapter 5: 5.1 to 5.3, Chapter 6: 6.3 to 6.4, Chapter 7: 7.1, 7.2)
DYNAMIC PROGRAMMING: General method, The Floyd-Warshall Algorithm, The Knapsack problem and memory function with complexity study
(Text Book-1: Chapter 8: 8.2 and 8.4).
 The Bellman-Ford algorithm, **(Text Book-2: Chapter 24: 24.1).**

UNIT-III

10 Hours

GREEDY TECHNIQUE:
 General method of Greedy technique, **Minimum Spanning Trees:** Prim's Algorithm, Kruskal's Algorithm, Single-Source Shortest Paths using Dijkstra's Algorithm, Huffman Trees
(Text Book-1: Chapter 9: 9.1 to 9.4)
BACKTRACKING: General method, State space trees and algorithms for N-Queens problem, Subset-sum problem
(Text Book-1: Chapter 12: 12.1 selected topics)
BRANCH AND BOUND: General method, Solving job Assignment Problem, Travelling Salesman problem, Knapsack Problem using branch and bound method
(Text Book-1: Chapter 12: 12.2)
 P, NP and NP Complete Problems **(Text Book-1: Chapter 11: 11.3)**

Suggested List of Experiments

- | | |
|---|--|
| 1 | Various Sorting/Searching algorithms |
| 2 | Graph traversals –DFS and BFS, Connectivity and Reachability of graphs |
| 3 | Topological Sorting |
| 4 | Descending Priority Queue using Heap |
| 5 | Horspool string matching algorithm |
| 6 | Binomial coefficient, Warshall's algorithm, Floyd's algorithm, Knapsack problem using Dynamic Programming and by using memory functions. |
| 7 | Prim's, Kruskal's, Dijkstra's algorithms |
| 8 | N-Queens problem. |

Course Outcomes: At the end of the course student will be able to

1.	Analyze the performance of the algorithms, state the efficiency using asymptotic notations and analyze algorithms mathematically for the complexity of the algorithm.
2.	Apply Brute force method, divide and conquer approaches in solving the problems and analyze the same.
3.	Apply the appropriate algorithmic design technique like decrease and conquer approaches, transform and conquer approaches and compare the efficiency of algorithms to solve the given problem.
4.	Apply and analyze dynamic programming approaches to solve some problems. And improve an algorithm time efficiency by sacrificing space.
5.	Apply and analyze greedy method, backtracking, branch and bound methods to solve problems and to describe P, NP and NP Complete problems.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→ ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CS3004-1.1	2	3	2	2	-	-	-	-	-	-	-	1	-	2	-
CS3004-1.2	2	2	2	2	-	-	-	-	-	-	-	1	-	3	-
CS3004-1.3	2	3	2	2	-	-	-	-	-	-	-	1	-	3	-
CS3004-1.4	2	2	2	2	-	-	-	-	-	-	-	1	-	3	-
CS3004-1.5	2	3	2	2	-	-	-	-	-	-	-	1	-	2	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Anany Levitin, "Introduction to the Design & Analysis of Algorithms", 2 nd Edition, Pearson Education, 2011.
2.	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, "Introduction to Algorithms", 3 rd Edition, PHI, 2014.

REFERENCE BOOKS:

1.	Horowitz E., Sahni S., Rajasekaran S, "Computer Algorithms", Galgotia Publications, 2001.
2.	R.C.T. Lee, S.S. Tseng, R.C. Chang & Y.T. Tsai, "Introduction to the Design and Analysis of Algorithms A Strategic Approach", Tata McGraw Hill, 2005.

E Books / MOOCs/ NPTEL

1.	http://www.facweb.iitkgp.ernet.in/~sourav/daa.html
2.	http://nptel.ac.in/courses/106101060/https://www.coursera.org/specializations/algorithms
3.	http://nptel.ac.in/courses/106101060/https://www.coursera.org/specializations/algorithms
4.	http://elearning.vtu.ac.in/econtent/courses/video/CSE/06CS43.html
5.	http://elearning.vtu.ac.in/econtent/courses/video/FEP/ADA.html
6.	http://openclassroom.stanford.edu/MainFolder/CoursePage.php?course=IntroToAlgorithms

MICROPROCESSOR AND EMBEDDED SYSTEMS

Course Code:	CS3005-1	Course Type:	IPCC
Teaching Hours/Week (L: T: P):	3:0:2	Credits:	04
Total Teaching Hours:	40+0+26	CIE + SEE Marks:	50+50

Prerequisite		CS2101-1
Teaching Department: Computer Science & Engineering		
Course Objectives:		
1.	Outline the internal architecture of 8086 microprocessor, concept of addressing modes, instruction set and develop and execute basic programs.	
2.	Develop and execute assembly level language program for 8086.	
3.	Build and examine the 8086 modular programming, Understand the fundamentals of ARM embedded systems and analyze communication device components.	
4.	Apply embedded C programming for micro controller operations and to interface with peripherals.	
5.	Describe the hardware components of 8086 microprocessor, Construct machine code for 8086 instructions and Outline the concepts of interrupts.	
UNIT-I		
		15 Hours
8086 INTERNAL ARCHITECTURES: The Programming Model, Multipurpose registers, Special purpose Registers, Segment registers		
8086 ADDRESSING MODES: Register addressing, Immediate addressing, Direct addressing, Register Indirect addressing, Based addressing with displacement, Indexed addressing with displacement, Based Indexed addressing, Based Indexed addressing with displacement.		
8086 ASSEMBLER DIRECTIVES: ASSUME, DB, DW, DD, END, PROC, ENDP, SEGMENT, ENDS, EQU, INCLUDE, OFFSET, MACRO, ENDM.		
8086 INSTRUCTIONS SET: Data transfer instructions (including I/O transfers), Binary arithmetic instructions, Decimal arithmetic instructions (DAA and AAM only), Logical instructions, Shift and rotate instructions, Control transfer instructions.		
8086 PROGRAMMING BASED ON INSTRUCTION SET: Programs based on data transfer instructions, binary arithmetic instructions, logical instructions, shift and rotate instructions, control transfer instructions.		
UNIT-II		
		15 Hours
MODULAR PROGRAMMING: Using procedures, Using macros, Comparison between procedure and macro.		
DATA CONVERSIONS: ASCII to Hexadecimal, Hexadecimal to ASCII.		
USING THE KEY BOARD AND VIDEO DISPLAY: DOS and BIOS interrupts, Example Programs.		
ARM Embedded Systems: RISC design philosophy, ARM design philosophy, Embedded system hardware, Embedded system software.		
ARM processor fundamentals: Registers, Program Status Register, Pipeline.		
Introduction to Embedded C Programming for Micro Controller operations: Arduino Due Micro controller board, Pinout Diagram, Features of Arduino Due board, Arduino IDE, Program Structure. Built in functions: pinMode(), Digital and Analog I/O operations, Delay, Serial Communication. Interfacing peripherals with micro controller: LED, Switches, Buzzer, Potentiometer, 16x2 LCD Display, Seven Segment Display, Stepper Motor, 4x4 Keypad.		
UNIT-III		

10 Hours
8086 HARDWARE SPECIFICATIONS:

8086 memory addressing, 8086 pin functions (minimum mode, maximum mode and both minimum and maximum mode).

8086 INSTRUCTION FORMAT (MOV INSTRUCTION ONLY):

Generating machine code for register-to-register data transfer, memory/register to register/memory data transfer.

8086 INTERRUPTS: 8086 Interrupts and Interrupt responses, 8086 Interrupt types.

Suggested List of Experiments
Part-A

1.	Searching
2.	String manipulation
3.	usage of Macros and subroutines
4.	DOS interrupt usage
5.	BIOS interrupt usage

Part-B

7.	Interfacing LED/Buzzer on / off using Potentiometer and switches
8.	Reading and displaying over serial monitor
9.	Interfacing Stepper motor -Clockwise, anticlockwise, delay
10.	Interfacing 16x2 LCD – scrolling and flickering
11.	Interfacing Seven segment display
12.	Interfacing 4x4 keypad
13.	Interfacing basic sensors.

Course Outcomes: At the end of the course student will be able to

1.	Illustrate the internal architecture of 8086 microprocessor, concept of addressing modes and assembler directives.
2.	Understand the 8086 instructions set and Develop and execute assembly level language program using 8086 instructions set.
3.	Build and examine the 8086 Modular programming. Describe the internal architecture of ARM processor and analyze schematics of communication devices.
4.	Understand ARM Processor fundamentals. Interface microcontroller to external I/O devices namely logic controller, stepper motor, seven segment display, DAC, keypad, and basic sensors.
5.	Describe the hardware components of 8086 and Construct machine code for 8086 and understand interrupts.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3005-1.1	3	-	-	-	-	-	-	-	-	-	-	2	3	-	-
CS3005-1.2	3	2	-	-	-	-	-	-	-	-	-	1	3	-	-
CS3005-1.3	3	3	-	-	-	-	-	-	-	-	-	2	3	-	-
CS3005-1.4	3	-	-	-	-	-	-	-	-	-	-	2	3	-	-
CS3005-1.5	-	2	3	-	-	-	-	-	-	-	-	3	-	3	-

1: Low 2: Medium 3: High
TEXTBOOKS:

1.	Barry B Brey, "The Intel Microprocessors", 8 th Edition, Pearson Education, 2009.
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2.	Douglas V Hall, "Microprocessors and Interfacing", 2 nd Edition, Tata McGraw-Hill Publication, 2006.
3.	Andrew N. Sloss, Dominic Symes, Chris Wright, "ARM System Developer's Guide Designing and Optimizing System Software", Morgan Kaufmann Publisher, 2004
4.	Julien Bayle, "C Programming for Arduino", Packt Publishing, 2013
REFERENCE BOOKS:	
1.	Liu & Gibson, "Microcomputer Systems: The 8086/8088 Family Architecture, Programming and Design", PHI, 2006.
2.	Muhammad Ali Mazidi, Rolin McKinlay, Janice Gillispie Mazidi, Microcontroller And Embedded Systems Using Assembly And C, Pearson, 2007.
3.	Muhammad Ali Mazidi, Rolin McKinlay, Danny Causey, PIC Microcontroller and Embedded Systems: Using assembly and C, Pearson, 2008.
E Books / MOOCs/ NPTEL	
1.	https://www.intel.com/content/www/us/en/products/processors/atom.html
2.	https://en.wikipedia.org/wiki/Embedded_system
3.	http://nptel.ac.in/courses/108107029/39

INFORMATION AND NETWORK SECURITY				
Course Code:		CB2102-1	Course Type	IPCC
Teaching Hours/Week (L: T: P):		3:0:0	Credits	04
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Prerequisite		CS1002-1		
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives:				
1.	Comprehend the cryptographytechniquesdevelopment strategy.			
2.	Apply the Knowledge of number theory in Public Key Crypto Systems.			
3.	Identify the Key management issues and resolveit.			
4.	Analyze the security issues in the network and solution forit.			
5.	Apply security mechanisms using rigorousapproaches			
UNIT-I				
Classical Encryption Techniques:				15 Hours
Symmetric Cipher Model, Cryptography, Cryptanalysis and Brute-Force Attack, Substitution Techniques, Caesar Cipher, Monoalphabetic Cipher, Playfair Cipher, Hill Cipher, Polyalphabetic Cipher, One Time Pad. Block Ciphers And The Data Encryption Standard: Traditional block Cipher structure, stream Ciphers and block Ciphers, DES, Block cipher design principles, number of rounds, design of function F, key schedule algorithm, AES algorithm introduction. Public-Key Cryptography and RSA: Principles of Public-key cryptosystems. Public-key cryptosystems. The RSA algorithm, description of the algorithm, computational aspects, the security of RSA.				
UNIT-II				
Other Public-Key Cryptosystems:				15 Hours

Diffie-Hellman key exchange, The algorithm, key exchange protocols, man in the middle attack, Elgamal Cryptographic systems, Elliptic curve arithmetic, Elliptic curve cryptography, Analog of Diffie-Hellman key exchange, Elliptic curve encryption/decryption.

Key Management And Distribution: Symmetric key distribution using Symmetric encryption, Hierarchical key control, Decentralized key control, Symmetric key distribution, public key authority, public keys certificates, X-509 certificates.

User Authentication: Remote user Authentication principles, Kerberos, Remote user Authentication using Asymmetric encryption, identity management.

UNIT-III

Web Security Considerations:

10 Hours

Web Security Threats, Web Traffic Security Approaches. Secure Sockets Layer: SSL. Cipher Suites, Secure Shell (SSH). IP Security: IP Security overview, IPSec, Security associations, IP traffic processing, Encapsulating Security payload, encryption and authentication algorithms, Padding, Anti replay service, transport and tunnel modes.

Suggested List of Experiments

1. To implement Caesar/Cipher Encryption/Decryption
2. To implement monoalphabetic Encryption/Decryption
3. To implement polyalphabetic Encryption/Decryption
4. To implement Playfair Cipher algorithm
5. To implement Hill Cipher algorithm
6. To implement Diffie Hellman key exchange algorithm
7. To implement RSA Encryption/Decryption algorithm
8. To implement Triple-DES Encryption/Decryption algorithm

Course Outcomes: At the end of the course student will be able to:

- | | |
|----|--|
| 1. | Comprehend the cryptography techniques development strategy. |
| 2. | Apply the Knowledge of number theory in Public Key Crypto Systems. |
| 3. | Identify the Key management issues and resolve it. |
| 4. | Analyze the security issues in the network and solution for it. |
| 5. | Apply security mechanisms using rigorous approaches. |

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→ ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CB2102-1.1	3	3											2	3	
CB2102-1.2		2	2										2	2	
CB2102-1.3	2	3											3	3	
CB2102-1.4	2	2	2										2	3	
CB2102-1.5	2	3											2	3	

1: Low 2: Medium 3: High

TEXTBOOKS:

1. William Stallings: Cryptography and Network Security, Pearson 6th Edition, 2013.
2. Cryptography & Network Security by Behrouz A. Forouzan, McGraw-Hill Education, 2007.

REFERENCE BOOKS:

1. V K Pachghare: Cryptography and Information Security, PHE, 2013.

E Books / MOOCs/ NPTEL

1. <https://www.mooc-list.com/tags/cryptography>
2. <https://www.udemy.com/topic/cryptography/>
3. https://onlinecourses.nptel.ac.in/noc22_cs90/

PRINCIPLES OF DATABASE MANAGEMENT SYSTEMS

Course Code:	CB2102-1	Course Type	IPCC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	CS1002-1		

Teaching Department: Computer Science & Engineering (Cyber Security)

Course Objectives:

1.	Provide a strong foundation in database concepts, design and application.
2.	Understand the concepts of relational model and relational algebra in database design.
3.	Learn structured query language (SQL) to an intermediate/advanced level and evaluate the result set.
4.	Understand the use of normalization techniques for building effective database design. Learn the concepts of NOSQL Systems to manage bigdata
5.	Demonstrate the use of File organization and Indexing, Concurrency Control and transactions in databases.

UNIT-I

Databases and Database users, Database System Concepts	15 Hours
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Databases and Database Users: Introduction, An Example, Characteristics of the database approach.

Database System Concepts and Architecture: Three-Schema Architecture and data Independence, Database languages and interfaces.

Data Modeling Using the Entity–Relationship (ER) Model: Entity Types, Entity Sets, Attributes and Keys, Relationship types, Relationship Sets, Roles and Structural Constraints, Weak Entity Types, Refining the ER Design for the COMPANY Database, ER Diagrams, Naming Conventions, and Design Issues. **The Relational Data Model and Relational Database Constraints:** Relational Model Concepts, Relational Model Constraints and Relational Database Schemas, Update Operations, transactions, and dealing with constraint violations.

The Relational Algebra and Relational Calculus: Unary Relational Operations: SELECT and PROJECT, Relational Algebra Operations from Set Theory. Binary Relational Operations: JOIN and DIVISION, Additional Relational Operations; Examples of Queries in Relational Algebra.

Relational Database Design by ER-to-Relational Mapping: Relational Database Design Using ER- to-Relational Mapping.

(T1: 1.1, 1.2, 1.3, 2.2, 2.3, 3.3-3.7, 5.1-5.3, 8.1-8.5 ,9.1)

UNIT-II

Basic SQL:	15 Hours
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Basic SQL: SQL Data Definition and Data Types, Specifying Constraints in SQL, Basic retrieval queries in SQL, Insert, Delete and Update Statements in SQL,

More SQL: Complex Queries, Views, and Schema Modification: More complex SQL retrieval queries, Specifying constraints as assertions and Actions as Triggers, Views in SQL, Schema Change Statements in SQL.

Basics of Functional Dependencies and Normalization for Relational Databases: Informal design guidelines for relation schemas, Functional Dependencies, Normal Forms Based on Primary Keys, general definitions of Second and Third Normal Forms, Boyce-Codd Normal Form.

Relational Database Design Algorithms and Further Dependencies: Inference Rules, Equivalence, and Minimal cover, Properties of Relational Decompositions, Algorithms for Relational Database Schema.

(T1: 6.1-6.4 ,7.1-7.4, 14.1-14.5, 15.1, 15.2,15.3)

UNIT-III

Storage and Indexing, Query Evaluation, Transaction Management	10 Hours
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Storage and Indexing: File Organizations and Indexing, Index Datastructures, Comparison of File Organizations.

Tree Structured Indexing: B+ Tree: A Dynamic Index Structure.

Overview of Query Evaluation: Introduction to Query Optimization, What a Typical Optimizer Does.

Overview of Transaction Management: The ACID Properties, Transactions and Schedules. Concurrent Execution of Transactions, Lock-Based Concurrency Control.

Concurrency Control: 2PL, Serializability and Recoverability.

(T2: 8.2, 8.3, 8.4, 10.2, 12.4, 12.6, 14.4, 16.1-16.4, 17.1)

Suggested List of Experiments

1. Execute various SQL queries in a DBMS. Use front end tool to connect to the backend database system.
2. Create Insurance database. Create related tables and execute queries on this Insurance database.
3. Create Order Processing database. Create related tables and execute queries on this Order Processing database.

4. Create Student Enrollment in Courses & Books adopted for each Course database. Create related tables and execute queries on this database.
5. Create Book Dealer database. Create related tables and execute queries on this database.
6. Create Banking Enterprise database. Create related tables and execute queries on this database.

Course Outcomes: At the end of the course student will be able to

1.	Illustrate the concepts of database objects for the given problem.
2.	Identify and enforce integrity constraints on a database using RDBMS.
3.	Apply structured query language for (SQL) for database manipulation.
4.	Model normalized database structures by creating simple database systems. Understand the concepts of NOSQL Systems to manage bigdata
5.	Illustrate the concepts of File organization and Indexing, Concurrency Control and transactions in databases.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS2102-1.1	2									1		1			
CS2102-1.2	2	2								1		1			
CS2102-1.3	2	3								1		1		3	
CS2102-1.4	2	2	3							1		1		2	
CS2102-1.5	2									1		1		2	

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Database Systems Models, Languages, Design and Application Programming, RamezElmasri and Shamkant B. Navathe, 7th Edition, 2017, Pearson.
2. Database Management Systems, Raghu Ramakrishnan, JohannesGehrke, Indian Edition, Mc Graw Hill Education.

REFERENCE BOOKS:

1. Database Systems: Models, Languages, Design and Application Programming, RamezElmasri and Shamkant B. Navathe, 6th Edition, 2017, Pearson.
2. Database System Concepts, SilberschatzKorth and Sudharshan, 6th Edition, McGraw Hill, 2013.

E Books / MOOCs/ NPTEL

1. <https://www.udemy.com/course/introduction-to-basic-database-concepts/>, Introduction to Basic Database Concepts (Udemy).
2. <https://www.udemy.com/course/database-management-systems-mysql/>, Database Management Systems – MySQL (Udemy).
3. https://swayam.gov.in/nd1_noc19_cs46/preview, Database Management System (Swayam).

MACHINE LEARNING FOR CYBER SECURITY

Course Code:	CB2003-1	Course Type:	IPCC
Teaching Hours/Week (L: T: P):	3:0:2	Credits:	04
Total Teaching Hours:	40+0+26	CIE + SEE Marks:	50+50

Teaching Department: Computer Science & Engineering (Cyber Security)

Course Objectives:

1.	Understand the need and basics of machine learning and learn the Decision Tree model.
2.	Learn ANN and Genetic Algorithm along with its applications.
3.	Explore the various learning algorithms using Supervised Learning.
4.	Understand the important aspects of Analytical Learning and difference between Analytical and Inductive Learning Algorithms.
5.	Explore the difference between Analytical and Inductive Learning Algorithms and analyse the

	techniques related to reinforcement learning.	Security
UNIT-I		
Introduction: Machine learning: what and why?- Types of machine learning, Supervised learning and Unsupervised learning.		
Decision tree: Representation, Appropriate problems for decision tree learning and basic decision tree learning algorithm.		
Artificial Neural Networks: Introduction, Neural Network Representations, Appropriate problems for neural network learning, Perceptrons and basics of back propagation algorithm.		
15 Hours		
UNIT-II		
Bayesian Learning: Bayes theorem, Bayes theorem and concept Learning, Minimum Description Length, Bayes Optimal Classifier, Naive Bayes Classifier.		
Instance Based Learning: k-nearest neighbour learning and Locally Weighted Regression.		
Analytical Learning: Inductive and Analytical learning problems, PROLOG-EBG.		
15 Hours		
UNIT-III		
Combining Inductive and Analytical Learning: Motivation, Inductive –Analytical Approaches to Learning and Using Prior Knowledge to Initialize the Hypothesis.		
Reinforcement Learning:		
Introduction, Learning Task, Q Learning.		
10 Hours		
Suggested List of Experiments		
PART-A		
1.	Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.	
2.	For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples.	
3.	Develop a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.	
4.	Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.	
5.	Develop a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.	
6.	Apply EM algorithm to cluster a set of data stored in a .CSV file. Use the same data set for clustering using k-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. You can add Java/Python ML library classes/API in the program.	
7.	Implement and demonstrate the working of k-Nearest Neighbour algorithm and apply it to classify the iris data set.	
8.	Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs.	
9.	Build a model to classify email as spam or ham. First, download examples of spam and ham from Apache SpamAssassin’s public datasets and then train a model to classify email.	
PART-B		
Mini Project on Machine Learning:		
The main goal is to prepare students to apply machine learning algorithms to real-world tasks, or to leave students well-qualified to start machine learning or AI research. The mini project is intended to start in these directions.		
Students shall carry out either one of the following three kinds of projects:		
1. Application project. Pick an application that is of interest and explore how best to apply learning algorithms to solve it.		
2. Algorithmic project. Pick a problem or family of problems, and develop a new learning algorithm, or a novel variant of an existing algorithm, to solve it.		
3. Theoretical project. Prove some interesting/non-trivial properties of a new or an existing learning algorithm. (This is		

often quite difficult, and so very few, if any, projects will be purely theoretical.)

Course Outcomes: At the end of the course student will be able to

1.	Explain the fundamental concept and importance of machine learning, identify, analyze and categorize applications using decision tree algorithm.
2.	Demonstrate the application of ANN and Genetic algorithm for real world problems.
3.	Design and implement algorithms for supervised learning system.
4.	Design and implement algorithms for Analytical and Inductive Learning.
5.	Develop machine learning algorithm and reinforcement techniques for real world problems.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS2003-1.1	2	2	3										3	3	
CS2003-1.2	2	2	3										3	3	
CS2003-1.3	2	2	3										3	3	
CS2003-1.4	2	3											1	3	
CS2003-1.5	2	2	3										2	3	

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Sumeet Dua, Xian Du, Data Mining and Machine Learning in Cybersecurity, CRC Press, 2011.
2. T. M. Mitchell, "Machine Learning", McGraw Hill, 2017.

REFERENCE BOOKS:

1. EthemAlpaydin, "Introduction to Machine Learning", Second Edition, The MIT Press, 2004.
2. C. M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
3. R. O. Duda, P. E. Hart and D. G. Stork Pattern Classification, Wiley Publications, 2001
4. T. Hastie, R. Tibshirani, J. Friedman. The Elements of Statistical Learning, 2e, 2008.
5. P. Flach, "Machine Learning: The art and science of algorithms that make sense of data", Cambridge University Press, 2012.
6. K. P. Murphy, "Machine Learning: A probabilistic perspective", MIT Press, 2012.
7. M. Mohri, A. Rostamizadeh, and A. Talwalkar, "Foundations of Machine Learning", MIT Press, 2012.
8. S. Russel and P. Norvig, "Artificial Intelligence: A Modern Approach", Third Edition, Prentice Hall, 2009.

E Books / MOOCs/ NPTEL

1. <https://in.mathworks.com/>
2. <https://www.kdnuggets.com/>
3. <https://blog.cambridgespark.com/>
4. <https://www.udemy.com/topic/>
5. <https://www.mooc-list.com/>
6. <https://peltarion.com>

CYBER FORENSICS

Course Code:	CB3002-1	Course Type:	IPCC
Teaching Hours/Week (L: T: P):	3:0:2	Credits:	04
Total Teaching Hours:	40+0+26	CIE + SEE Marks:	50+50

Teaching Department: Computer Science & Engineering (Cyber Security)

Course Objectives:

1. To understand the basics of Cyber Forensics.

2.	To understand and analyze forensic data.
3.	To analyze network logs.
4.	To study cyber laws.
5.	To apply cyber forensic skills to find out malicious users.

UNIT-I

INTRODUCTION: 15 Hours

Digital forensic evidence collection and processing Framework, Fundamentals of end point forensics for Microsoft windows - Kernel and device driver architecture, Registry, Auditing and security architecture. File system handling - Reconstruction of files and directory structures on the FAT and NTFS.

UNIT-II

DIGITAL FORENSICS: 15 Hours

Fundamentals of host forensics for Unix derivatives - Linux operating system, Kernel and device driver architecture, Security and audit mechanisms, File system and pseudo file systems, Reconstruction of file and directory structures using UFS and EXT2/3/4 file systems as exemplars.

Forensic analysis of database systems, Database tampering, Forensic analysis of database components, Table storage, Transaction logs, indexes, Forensic recovery for table storage.

UNIT-III

NETWORK FORENSICS AND CYBER LAWS: 10 Hours

Network device forensics, investigating logs, Network traffic and web attacks, Mobile device, Social media and wireless forensics, Steganography and image file forensics, Email investigation.

Cyber laws in India, Case studies and tools.

Suggested List of Experiments

1. Collect Forensic data from Windows machine and create report.
2. Cyber forensic on Linux device.
3. Perform forensics on MySQL database system.
4. Capture network data and perform network forensics using Wireshark Tool.

Course Outcomes: At the end of the course student will be able to

1. Understand how to perform cyber forensic on Windows devices.
2. Understand how to perform cyber forensic on Linux devices.
3. Demonstrate how to capture and analyze network traffic.
4. Demonstrate how to create a forensic report.
5. Understand Cyber Laws and latest Cyber Forensic use cases.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3002-1.1	2	3							1			1	2	3	
CS3002-1.2	2	3							1			1	2	3	
CS3002-1.3	2	3							1			1	2	3	
CS3002-1.4	2	3							1			1	2	3	
CS3002-1.5	2	3			3				1			1	2	3	

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Digital Forensics and Incident Response: Incident response techniques and procedures to respond to modern cyber threats, 2nd Edition by Gerard Johansen
2. Brian Carrier, File System Forensic Analysis, Pearson, 2006.
3. E. Casey, Handbook of Digital Forensics and Investigation, Academic Press, 2010.

REFERENCE BOOKS:

1. Practical Cyber Forensics: An Incident-Based Approach to Forensic Investigations by Niranjan Reddy.

E Books / MOOCs/ NPTEL

1. <https://www.coursera.org/learn/digital-forensics-concepts>

2.	https://www.udemy.com/topic/digital-forensics/
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Professional Core Courses (Theory)

COMPUTER ORGANIZATION & ARCHITECTURE				
Course Code:		CS2101-1	Course Type	PCC
Teaching Hours/Week (L: T: P):		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Prerequisite		CS1001-1/ CS1004-1		
Teaching Department: Computer Science& Engineering				
Course Objectives:				
1.	Learn the basic structure and operation of a digital computer			
2.	Learn arithmetic unit and perform fixed point and floating-point addition, subtraction, multiplication and division in binary 2's complement number system			
3.	Understand the basic processing unit in terms of control unit, execution of instructions, write control sequences for instructions. Learn the instruction and thread level parallelism.			
4.	Explore the design of hierarchical memory system including cache memories and virtual memory. Compare the performance.			
5.	Discuss serial and parallel communication with I/O devices and standard I/O interfaces available.			
UNIT-I				
BASIC COMPUTER ORGANIZATION:				15 Hours
Functional units, Basic Operational Concepts, Performance, Instructions execution and straightline sequencing, Branching, condition codes, Addressing Modes				
ARITHMETIC OPERATIONS:				
Multiplication of Positive Numbers, Signed Operand Multiplication, Integer Division, IEEE standard for Floating-point Numbers.				
UNIT-II				
BASIC PROCESSING UNIT:				15 Hours
Some Fundamental Concepts, Execution of a Complete Instruction, Multiple Bus Organization, Hard-wired Control, Micro programmed Control.				

MEMORY SYSTEMS:																
Memory System: Basic Concepts, Semiconductor RAM Memories, Read Only Memories, Speed, Size, and Cost, Cache Memories –Mapping Functions, FIFO and LRU replacement policies, Performance Considerations, Virtual Memories.																
UNIT-III																
INPUT/OUTPUT ORGANIZATION														10 Hours		
Accessing I/O Devices, Interrupts –Interrupt Hardware, Enabling and Disabling Interrupts, Exceptions, Handling Multiple Devices, Controlling Device Requests, Buses, Direct Memory Access, PCI Bus and USB (Basics only).																
Course Outcomes: At the end of the course student will be able to																
1.	Outline the basic structure and operation of a digital computer.															
2.	Learn the functioning of the arithmetic unit, perform multiplication and division of signed and unsigned numbers.															
3.	Understand the fine grain details of basic processing unit in terms of control unit, execution of instructions and learn the scope for instruction and thread level parallel-ism. Implementation of instructions for single and multiple bus configuration.															
4.	Demonstrate the computer architecture concepts in the design of hierarchical memory system including cache memories and virtual memory.															
5.	Explain different ways of communication with I/O devices and standard I/O interfaces.															
Course Outcomes Mapping with Program Outcomes & PSO																
	Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
	↓ Course Outcomes													1	2	3
	CS2101-1.1	3	-	-	-	-	1	-	-	1	1	-	1	3	-	-
	CS2101-1.2	3	2	-	-	-	1	-	-	1	1	-	1	3	-	-
	CS2101-1.3	3	2	1	-	-	1	-	-	1	1	-	1	3	-	-
	CS2101-1.4	3	2	2	-	-	1	-	-	1	1	-	1	3	-	-
	CS2101-1.5	3	1	-	-	-	1	-	-	1	1	-	1	3	-	-
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.	Carl Hamacher, Zvonko Vranesic, Safwat Zaky, “Computer Organization”, 5 th Edition, TMH, 2011.															
REFERENCE BOOKS:																
1.	William Stallings, “Computer Organization & Architecture”, 7 th Edition, PHI, 2006.															
2.	Vincent P. Heuring& Harry F. Jordan, “Computer Systems Design and Architecture”, 2nd Edition, Pearson Education, 2004.															
3.	David A. Patterson, John L. Hennessy, “Computer Organization and Design”, 4th Edition Elsevier, 2012.															
4.	John P.Hayes, “Computer Architecture”, 2nd edition, McGraw Hill, 1988.															
5.	John L. Hennessey and David A. Patterson, “Computer Architecture, A Quantitative Approach”, 6th Edition, Elsevier, 2017															
6.	Shameem Akhter and Jason Roberts, “Multicore programming- Increasing performance through software multithreading”, Intel press, 2006.															
E Books / MOOCs/ NPTEL																

1.	https://dcs.abu.edu.ng/staff/sani-ahmad-hassan/course materials/COSC303 LEC.pdf
2.	http://www.cse.iitm.ac.in/~vplab/courses/comp_org/
3.	http://www.ddegjust.ac.in/studymaterial/msc-cs/ms-07.pdf
4.	http://nsec.sjtu.edu.cn/data/MK.Computer.Organization.and.Design.4th.Edition.Oct.2011.pdf
5.	http://nptel.ac.in/courses/106103068/
6.	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-823-computer-system-architecture-fall-2005/syllabus/

FUNDAMENTALS OF DATA COMMUNICATIONS			
Course Code:	CB1102-1	Course Type	PCC
Teaching Hours/Week (L: T: P)	1:1:3	Credits	03
Total Teaching Hours	15+39	CIE + SEE Marks	50+50
Prerequisite	NIL		
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
1.	To Focus on information sharing and networks.		
2.	To Introduce flow of data, categories of network, different topologies.		
3.	To Focus on different coding schemes		
4.	To understand the protocols and standards.		
5.	To give clear idea of signals, transmission media, errors in data communications and their correction, networks classes and devices.		
UNIT-I			
Introduction to data communication and networking:			15 Hours
Why study data communication? Data Communication, Networks, Protocols and Standards, Standards Organizations. Line Configuration, Topology, Transmission Modes, Categories of NetworksInternet works. Study of OSI and TCP/IP protocol suit: The Model, Functions of the layers, TCP/IP Protocol Suites. Study of Signals: Analog and Digital, Periodic and Aperiodic Signals, Analog Signals, Time and Frequency Domains, Composite Signals, Digital Signals.			
UNIT-II			
Digital and Analog Transmission:			15 Hours
Study of Digital transmission: Digital to Digital Conversion, Analog to Digital Conversion. Study of Analog transmission: Digital to Analog Conversion, Analog to Analog Conversion. Study ofMultiplexing: Many to one/one to Many, Frequency division Multiplexing, Wage division Multiplexing, Time division Multiplexing, Multiplexing applications.			
Types of transmission media: Guided Media, Unguided Media, Transmission Impairments, Performance Wavelength, Shannon Capacity, Media Comparison, PSTN, Switching.			

Security

UNIT-III																
Error Detection and Correction and Introduction to networks and devices:														10 Hours		
Error Detection and Correction: Types of Errors, Detection, Parity Check, Vertical Redundancy Check Longitudinal Redundancy Check, Cyclic Redundancy Check, Checksum, Error Correction. Introduction to networks and devices: Network classes, Repeaters, Hub, Bridges, Switches, Routers, Gateways Routers Routing Algorithms, Distance Vector Routing, Link State Routing.																
Course Outcomes: At the end of the course student will be able to																
1.	Understand the information sharing and networks.															
2.	Analyze the flow of data, categories of network, different topologies.															
3.	Understand different coding schemes.															
4.	Understand the protocols and standards.															
5.	Get clear idea of signals, transmission media, errors in data communications and their correction, networks classes and devices.															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CB1102-1.1		3	2										2	3		
CB1102-1.2		3	2	1									2	3		
CB1102-1.3		3	2	1									3	3		
CB1102-1.4		3	2	1									1	3		
CB1102-1.5		3	2	2									2	3		
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.	Behrouz A. Forouzan, "Data Communications and Networking", 5 th edition, Tata McGraw Hill.															
REFERENCE BOOKS:																
1.	Alberto Leon-Garcia & Indra Widjaja, "Communication Networks - Fundamental Concepts and Key Architectures", 2nd edition, Tata McGraw Hill.															
2.	William Stallings, Data and Computer Communication, 5th Edition, Prentice Hall India.															
E Books / MOOCs/ NPTEL																
1.	http://eng.uok.ac.ir/abdollahpouri/Network/A.Leon-Garcia_Communication_Networks.pdf															
2.	https://vtucsenotes.wordpress.com/2014/02/23/computer-networks/															
3.	http://highered.mheducation.com/sites/0073376221/student_view0/index.html															
4.	http://nptel.ac.in/downloads/106105080/															
5.	http://www.nptelvideos.in/2012/11/data-communication.html															
6.	http://nptel.ac.in/courses/106105082/															
7.	https://www.youtube.com/playlist?list=PL374944B232C0B48E															
8.	http://freevideolectures.com/Course/2278/Data-Communication															

UNIX and Shell Programming

Course Code	CB1602-1	Course Type	PCC
Teaching Hours/Week (L: T: P)	0:0:2	Credits	01

Total Teaching Hours	0+0+26	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
1.	Recognize, understand and make use of various UNIX commands.		
2.	Gain hands on experience of UNIX commands and shell programs.		
3.	Develop a simple application based on the knowledge gained in Shell scripting.		
			26 Hours
Use of basic Unix Shell Commands: ls, mkdir, rmdir, cd, cat, banner, touch, file, wc, sort, cut, grep, dd, dfspace, du, ulimit. Commands related to inode, I/O redirection, piping, process control commands, mails.			
Shell Programming: shell script exercise based on: Interactive shell script, Positional parameters, Arithmetic, If-then-fi, if-then-else-fi, nested if-else, Logical operators, Else + if equals elif, case structure, While, for loop, Meta characters. Applications of these concepts to develop shell scripts to solve various problems.			
List of Experiments			
PART-A (Shell Scripts)			
1. Write a shell program to find and display largest and smallest of three numbers. 2. Write a shell program to check the number n is divisible by m or not. Where m and n are supplied as command line argument or read from key board interactively. 3. Write a shell program to check the year is the leap year or not. Display appropriate message. 4. Write a shell program to search a pattern in a file that will take both pattern and file name from the command line arguments. Write a second shell program to read the pattern and the file name from the key board interactively if the pattern and file names are not passed in command line arguments. If they are passed in command line argument, then the variables pattern and file name are set to these values. Finally invoke the first shell file to search a pattern in the file if the file exists. 5. Write a shell program that takes two file names, checks the permissions for these files are identical and if they are identical, output the common permissions; otherwise output each file name followed by its permissions. 6. Write a shell program to display the length of the name and also display first three characters and last three characters in the name in two different lines if the name contains at least 6 characters. 7. Write a shell program to implement simple calculator operations. 8. Write a menu driven shell script for Copy a file, Remove a file, Move a file. 9. Write a Shell Program that takes the any number of arguments and print them in same order and in reverse order with suitable messages. (For example, if the script m named rargs, then executing rargs A B C should produce A B C and C B A on the standard output). 10. Shell script that accepts path names and creates all the components in that path names as directories. For ex, if the script name is mpe, then the command mpe a/b/c/d should create directories a, a/b, a/b/c, and a/b/c/d. 11. Write a shell script that accepts filename as arguments. For every filename, it should first check whether it exists in the current directory and then convert its name to uppercase, but only if a file with new name doesnt exist. 12. Write shell script that performs following string handling operations i) Calculate the length of the string ii) locate a position of a character in a string iii) extract last three characters from string			
PART-B (C + AWK + PERL)			
1. C program to do the following: Using fork() create a child process. The child process prints its own process-id and id of its parent and then exits. The parent process waits for its child to finish (by executing the wait()) and prints its own process-id and the id of its child process and then exits.			

2. C program to prompt the user for the name of an environment variable and print its value if it is defined and a suitable message otherwise; and to repeat the process if user wants it.
3. C program that creates a child process to read commands from the standard input and execute them (a minimal implementation of a shell - like program). You can assume that no arguments will be passed to the commands to be executed.
4. C program that accepts a valid directory names as a command line argument and lists all the files in the given directory as well as all the subsequent subdirectories (The solution can be recursive or non-recursive).
5. Write an Awk script to delete duplicated lines from a text file. The order of the original must remain unchanged.
6. Write an Awk script that folds long line into 40 columns. Thus, any line that exceeds 40 characters must be broken after 40* and is to be continued with the residue. The inputs to be supplied through a text file created by the user.
7. Write an awk program to provide extra spaces at the end of the line (if required) so that the line length is maintained as 127.
8. Write an AWK program to find the transpose of a given matrix.
9. Write a PERL program to convert a decimal number (supplied as argument) to an unsigned binary number (for example if the argument is 22, then the output should be 10110). If an argument is present, it can be a valid decimal number and if no argument is present, the program should display an error message.
10. PERL script that echoes its command line arguments, one per line after translating all lower-case letters to upper case.
11. PERL program to find the sum of digits of an unsigned number passed through argument.
12. Write a PERL program that prompts user to input the string and a number, and prints the string that many times, with each string on separate line.

Course Outcomes: At the end of the course student will be able to

1.	Understand the basic concepts of UNIX Architecture and basic Commands.
2.	Understand different types of Files, File system and basic file system commands.
3.	Understand the commands related to Shell basics, vi editor and regular expression commands.
4.	Understand the concepts of advance file concepts, commands related to Shell script and filter commands.
5.	Understand the concepts of process and commands related to Perl script.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS1602-1.1	3	2											1	1	2
CS1602-1.2	3	2	2										1	1	2
CS1602-1.3	3	2	2										1	2	3
CS1602-1.4	3	2	3										1	2	3
CS1602-1.5	3	2	3										1	2	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1. The Unix programming Environment by Brain W. Kernighan & Rob Pike, Pearson.
2. Introduction to Unix Shell Programming by M.G. Venkateshmurthy, Pearson.

REFERENCE Books:	
1.	Unix and shell programming by B.M. Harwani, OXFORD university press.
2.	UNIX: Concepts and Applications by Sumitabha Das, 4th Edition, 2017, McGraw Hill Education
E Resources	
1.	https://www.geeksforgeeks.org/introduction-linux-shell-shell-scripting/
2.	https://www.tutorialspoint.com/unix/shell_scripting.htm
3.	https://www.shellscript.sh/first.html

SOFTWARE ENGINEERING AND PROJECT MANAGEMENT				
Course Code:		CS2103-1	Course Type	PCC
Teaching Hours/Week (L: T: P)		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Prerequisite		CS1002-1		
Teaching Department: Computer Science & Engineering				
Course Objectives:				
1.	Outline software engineering principles and activities involved in building large software programs.			
2.	Explain the importance of software process models in designing the software.			
3.	Describe the process of Agile project development.			
4.	Recognize the importance of software testing and describe the intricacies involved in software evolution.			
5.	Identify several project planning and estimation techniques and explain the importance of software quality.			
UNIT-I				
Title				15 Hours
Introduction: Need for Software Engineering, Professional Software Development, Software Engineering Ethics				
Software Processes: Models: Waterfall Model, Incremental Model and Spiral Model; Process activities.				
Requirements Engineering: Functional and non-functional requirements, Software requirements document, Requirements specification, Requirements engineering processes, Requirements Elicitation and Analysis, Requirements validation				
UNIT-II				
System Models:				15 Hours

System Models:

Context models, Interaction models, Structural models, Behavioral models.

Design and implementation:

Object oriented Design using UML.

Agile Software Development:

Agile methods, Plan-driven and agile development, Extreme Programming, Agile project management.

Software Testing: Development Testing, Test-driven development, Release testing, User testing

UNIT-III
Project Management:
10 Hours
Project Management:

Risk management, Teamwork

Project Planning:

Software pricing, Plan-driven development, Project Scheduling

Quality Management: Software quality, Reviews and inspections, Software measurement and metrics, Software standards.

Course Outcomes: At the end of the course student will be able to

1.	Recognize the basics of software system, component, process and Software Requirement Specification to meet desired needs within realistic constraints and outline the professional and ethical responsibility.
2.	Describe the waterfall, incremental and iterative models in implementing the software.
3.	Make use of software testing techniques, modern engineering design tools and agile methods necessary for engineering practice.
4.	Describe the methods for maintaining software system.
5.	Discuss project planning and management and illustrate the quality of software products.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes →	1	2	3	4	5	6	7	8	9	10	11	12	PSO ↓		
↓ Course Outcomes													1	2	3
CS2103-1.1	-	3	1	-	-	-	-	2	-	-	-	-	1	2	-
CS2103-1.2	1	3	1	-	-	-	-	-	-	-	-	-	1	2	-
CS2103-1.3	1	1	3	-	-	-	-	-	-	-	-	-	2	3	-
CS2103-1.4	1	3	2	-	-	-	-	-	-	-	-	-	1	2	-
CS2103-1.5	1	2	2	-	-	-	-	-	-	-	-	-	1	2	-

1: Low 2: Medium 3: High
TEXTBOOKS:

1. Ian Sommerville, "Software Engineering", 9th Edition, Pearson Education, 2012.

REFERENCE BOOKS:

1. Roger S. Pressman: "Software Engineering-A Practitioners approach", 7th Edition, Tata McGraw Hill, 2017.
2. Pankaj Jalote: "An Integrated Approach to Software Engineering", Wiley, India, 2010.

E Books / MOOCs/ NPTEL

1. <http://agilemanifesto.org/>
2. <http://www.jamesshore.com/Agile-Book/>
3. <https://www.mooc-list.com/course/uml-class-diagrams-software-engineering-edx>
4. <https://www.mooc-list.com/course/enterprise-software-lifecycle-management-edx>

FUNDAMENTALS OF COMPUTER NETWORKS

Course Code:	CB2102-1	Course Type:	PCC
Teaching Hours/Week (L: T: P):	3:0:2	Credits:	03
Total Teaching Hours:	40+0+26	CIE + SEE Marks:	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
1.	Outline the principles of computer networks and its application		
2.	Illustrate the concept of types of network		
3.	Identify the issues in network layer and solution for it		
4.	Analyze the process of congestion control algorithms.		
5.	Illustrate IP Packets and fragmentation process.		
UNIT-I			
Network Layer (Part-I):			15 Hours
Network layer design issues: Store and Forward packet Switching, Services Provided to the Transport Layer, Implementation of Connectionless Service, Implementation of Connection-Oriented Service, Comparison of Virtual Circuit and Datagram Subnets; Routing algorithms: The Optimality Principal, Shortest Path Routing, Flooding. Distance Vector Routing, Link state Routing, Hierarchical Routing, Broadcast Routing, Multicast Routing, Anycast Routing, Routing for Mobile Hosts, Routing in Ad hoc Networks. Congestion Control Algorithms: Approaches to Congestion Control, Traffic-AwareRouting, Admission Control, Traffic Throttling, LoadShedding; Quality of Service: Application Requirements, Traffic Shaping, Packet Scheduling, Admission Control, Integrated Services, Differentiated Services			
UNIT-II			
Network layer (Part – II):			15 Hours
Internetworking: How networks differ, How Networks Can Be Connected, Tunneling, Internetwork Routing, Fragmentation; The Network Layer in the Internet: The IP Version 4 Protocol, IP Addresses, IP Version 6, Internet Control Protocols, OSPF, BGP, Internet Multicasting, Mobile IP. The Transport Layer: The Transport Service: Services Provided to the Upper Layers, Transport Service Primitives, Elements of Transport Protocols: Addressing, Connection Establishment, Connection Release, Error Control and Flow Control, Multiplexing, Crash Recovery; The Internet Transport Protocols (UDP): Introduction to UDP			
UNIT-III			
The Internet Transport Protocols (TCP)			10 Hours
The Internet Transport Protocols (TCP): Introduction to TCP, The TCP Service Model, The TCP Protocol, The TCP Segment Header, TCP Connection Establishment, TCP Connection Release, TCP Connection Management Modeling, TCP Sliding Window, TCP Timer Management, TCP Congestion Control.			

The Application Layer:
Principles of Network Applications:

Network Application Architectures, Processes Communicating, Transport Services Available to Applications, Transport Services Provided by the Internet;

The Web and HTTP: Overview of HTTP, Non-Persistent and Persistent Connections, HTTP Message Format;

DNS: The Internet's Directory Service, Services Provided by DNS, Overview of How DNS Works, DNS Records and Messages;

Electronic Mail in the Internet: SMTP, Comparison with HTTP, Mail Message Formats, Mail Access Protocols.

Course Outcomes: At the end of the course student will be able to

1.	Express the basic concept of computer network.
2.	Design the network layer and the related issues.
3.	Explain the congestion control, and prevention methods.
4.	Explain different type of networks and protocols.
5.	Explain network applications and describe application layer protocols.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓			
↓ Course Outcomes													1	2	3	
CB3001-1.1	3	2			1					1			2			
CB3001-1.2	3	2	1							2		1	2			
CB3001-1.3	3	1			1		1			1		1	3			
CB3001-1.4	3	2	1		1		1			2		1	3	1		
CB3001-1.5	3	1			1	1				1		2	2	1		

1: Low 2: Medium 3: High

TEXTBOOKS:

- 1 Computer Networks, Andrew S. Tanenbaum David J. Wetherall, 5th Edition, Pearson, ISBN 10: 1292024224, 2014
- 2 Computer Networking. A Top-down Approach, James F. Kurose, Keith W. Ross, Pearson, ISBN: 1292153598, 2017

REFERENCE BOOKS:

- 1 Computer Networking. A Top-Down Approach, Kurose & Ross, 5th Edition, Mc Graw Hill, ISBN: 9780073376226, 2013
- 2 Data and Computer Communication, 8th Edition, William Stallings, Prentice Hall, 0132433109, 2007
- 3 An Introduction to Computer Networks, Peter L Dordal, OpenBook, <http://intronetworks.cs.luc.edu/>, 2020

E Books / MOOCs/ NPTEL

- 1 Computer Networks web course by Prof. Ajit Pal, IIT Khargpur, <https://nptel.ac.in/courses/106/105/106105080/>
- 2 Computer Networks, Fall 2019 by Peter Dordal, Loyola University Chicago Dept of Computer Science, <http://pld.cs.luc.edu/courses/443/now/>
- 3 Computer Networks Lecture Notes, MIT OpenCourseWare, Massachusetts Institute of Technology, <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-829-computer-networks-fall-2002/lecture-notes/>

UNIX System Programming and Networking Lab

Course Code	CB2601-1	Course Type	PCC
Teaching Hours/Week (L: T: P)	0:0:2	Credits	01
Total Teaching Hours	0+0+26	CIE + SEE Marks	50+50

Teaching Department: Computer Science & Engineering (Cyber Security)

Course Objectives:

1.	To be able to understand fundamentals of Unix operating system concepts.
2.	To be able to implement various system calls in Unix.
3.	To develop programs based on Unix /POSIX-compliant system call APIs.

List of Experiments

26 Hours

File management system calls, Directory management system calls, Parent process – Child process Relationship, Implementing IPC using pipes, Signal functions, unreliable signals, interrupted system calls, kill and raise, System call basic, opening and closing a file, Using I/O Calls, Umask, Handling Errors.

File sharing, Directory Navigation, modifying entries in Directory, Reading the Inode, Checking the file access right, modifying file attribute, Socket Programming.

List of Experiments

Part A

1. Design a program in which sender module should count the no of bytes in the frame and receive module should display each frame received.
2. Design a program to implement bit stuffing, encoding and decoding concept in data link layer.
3. Design and implement CRC error detection method used in data link layer..
4. Design a program to implement client server model (TCP) using socket programming.
5. Design a program to implement client server model (UDP) using socket programming.
6. Design and implement a program to route the packet in a network using distance vector algorithm.
7. Design a program for congestion control using leaky bucket algorithm.

Part B

1. Simulate peer-to-peer communication between a client and a server using Point-to-Point protocol. Apply NetAnim software to demonstrate the scenario graphically. Analyze packet parameters by creating trace file using Ascii trace metrics.
2. Simulate to implement a bus topology using Point-to-Point protocol between a client and a LAN with 4 nodes. The LAN use CSMA during packet transmission. Apply NetAnim software to demonstrate the scenario graphically. Analyze packet parameters by creating trace file using Ascii trace metrics.
3. Simulate peer-to-peer communication between a client and a server using CSMA protocol. Apply NetAnim software to demonstrate the scenario graphically. Analyze packet parameters by creating trace file using Ascii trace metrics.
4. Simulate the transmission of ping messages over a network topology consisting of 3 nodes n0, n1 and n2, where node n0 and n1 are the pingers.
5. Simulate to implement the star topology using Point-to-Point protocol. Apply NetAnim software to demonstrate the scenario graphically. Analyze packet parameters by creating trace file using Ascii trace

metrics.

Course Outcomes: At the end of the course student will be able to

1.	Understand fundamentals of Unix operating system concepts
2.	Implement various system calls in Unix.
3.	Develop programs based on Unix /POSIX-compliant system call APIs.
4.	Implement various signal calls.
5.	Implement Socket Programming.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS1602-1.1	3	2											1	1	2
CS1602-1.2	3	2	2										1	1	2
CS1602-1.3	3	2	2										1	2	3
CS1602-1.4	3	2	3										1	2	3
CS1602-1.5	3	2	3										1	2	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	W. Richard Stevens, Advaced Programming in the Unix Environment, Addison-Wesley, 1992.
2.	Sumitabha Das "Unix concepts and Applications", Tata McGraw Hill, Second Edition,2001.

REFERENCE Books:

1.	Y. Kanetkar "Unix shell programming", BPB Pub.
2.	B.W. Kernighan & R. Pike, "The UNIX Programming Environment", PHI Learning

E Resources

1.	https://alandix.com/academic/tutorials/courses/unixprog.html
2.	https://www.tutorialspoint.com/unix_sockets/index.htm

ESSENTIALS OF OPERATING SYSTEMS				
Course Code:		CB3003-1	Course Type	PCC
Teaching Hours/Week (L: T: P)		3:0:0	Credits	03
Total Teaching Hours		40+0+20	CIE + SEE Marks	50+50
Teaching Department: Cyber Security				
Course Objectives:				
1.	Explain the concepts, principles and services of operating system.			
2.	Identify fundamental operating system concepts such as processes, inter-process communication, threads, CPU scheduling and demonstrate them.			
3.	Assess the need of concurrency and synchronization and apply them to write concurrent programs and analyze the cause for the occurrence of deadlocks and determine solutions to overcome the deadlocks.			
4.	Study the concepts of main memory and virtual memory allocation methods and demonstrate them.			
5.	Analyze the need for file system concepts, directory implementation and disk scheduling algorithms and demonstrate them.			
UNIT-I				
Operating System structure:				15 Hours
Operating System structure: Operating System Services, User and Operating System interface, System calls,				

System Services, Linkers and Loaders, **Process Management:** Process concept; Process scheduling; Operations on processes; Inter-process communication. **Threads:** Multithreading Models. **CPU Scheduling:** Basic concepts, Scheduling criteria, Scheduling algorithms, Thread scheduling

UNIT-II

Process Synchronization:

15 Hours

Process Synchronization: The Critical section problem, Peterson's solution, Synchronization hardware, Semaphores, Monitors, Classical problems of synchronization. **Deadlocks:** System model; Deadlock characterization, Methods for handling deadlocks, Deadlock prevention, avoidance, detection and recovery from deadlock. **Main Memory:** Paging, Structure of page table, Swapping.

UNIT-III

Virtual Memory:

10 Hours

Virtual Memory: Demand paging, Copy-on-write, Page replacement, Allocation of frames. Implementing File System: File system Concepts, File System Structure, Operations and implementation, Directory implementation, Allocation methods, Free space management. disk scheduling algorithms

Course Outcomes: At the end of the course student will be able to

1.	Recognize the structural components of operating system	
2.	Demonstrate the creation and termination of the processes, threads and CPU scheduling algorithms.	
3.	Illustrate critical section problem and demonstrate the Peterson's solution. Investigate the Deadlock condition and determine the solution to avoid.	
4.	Summarize Main memory and Virtual Memory allocation methods and prepare a page replacement schedule to the given set of page requirement request.	
5.	Classify file systems based on operations and implementations and illustrate the disk scheduling algorithms.	

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS2004-1.1	3	2													2
CS2004-1.2	3	2											1	2	3
CS2004-1.3	3	2											1	2	3
CS2004-1.4	3	2											1	2	3
CS2004-1.5	3	2											1	2	3

1: Low 2: Medium 3: High

TEXTBOOKS:

- Operating System Concepts, Abraham Silberschatz, Greg Gagne, Peter B. Galvin, 10th Edition, John Wiley & Sons, 2018, ISBN: 9781119320913.

REFERENCE BOOKS:

- D.M Dhamdhare: Operating systems - A concept based Approach, 2nd Edition, Tata McGraw- Hill, 2002.
- P.C.P. Bhatt: Operating Systems, 2nd Edition, PHI, 2006.
- Harvey M Deital: Operating systems, 3rd Edition, Addison Wesley, 1990.

E Books / MOOCs/ NPTEL

- http://www.uobabylon.edu.iq/download/M.S%2020132014/Operating_System_Concepts,_8th_Editio%5BA4%5D.pdf
- <http://iips.icci.edu.iq/images/exam/Abraham-Silberschatz-Operating-System-Concepts--9th2012.12.pdf>
- http://www.uobabylon.edu.iq/download/M.S%2020132014/Operating_System_Concepts,_8th_Edition%5BA4D.pdf
- <https://freevideolectures.com/university/iit-bombay/>
- <https://www.cse.iitb.ac.in/~mythili/os/>

SECURE FULL STACK WEB DEVELOPMENT

Course Code:	CB3603-1	Course Type:	PCC
Teaching Hours/Week (L: T: P):	0:0:2	Credits:	01
Total Teaching Hours:	0+0+30	CIE + SEE Marks:	50+50
Prerequisite	CS2001-1,CS1001-1		

Teaching Department: Cyber Security

Course Objectives:

1.	Demonstrate proficiency in creating responsive and semantic web pages using HTML5, CSS3, and JavaScript.
2.	Develop dynamic and interactive web applications using ReactJS and state management techniques.
3.	Design and implement secure, scalable, and RESTful APIs using NodeJS and ExpressJS.
4.	Integrate frontend and backend systems to build full stack web applications with seamless data exchange
5.	Deploy full stack applications on cloud platforms and understand basic concepts of version control and CI/CD pipelines

16 Hours

Introduction to Web Development:

- Overview of full stack development
- Web architecture: Client-server model
- Tools and environment setup: VS Code, Node.js, npm, browser developer tools

HTML5 Essentials:

- HTML5 structure and semantic elements
- Forms and input validation
- Multimedia elements: <audio>, <video>, <canvas>
- Accessibility and ARIA roles

CSS3 and Responsive Design:

- CSS selectors, properties, and combinators
- Box model, flexbox, and grid layout
- Media queries for responsive design
- Animations and transitions in CSS
- CSS frameworks overview: Bootstrap and Materialize

JavaScript Fundamentals:

- ES6+ features: let, const, arrow functions, template literals, destructuring
- DOM manipulation and event handling
- JavaScript promises and async/await
- Introduction to Fetch API for HTTP requests

ReactJS Basics:

- Introduction to ReactJS and its ecosystem
- Setting up a React project with Create React App
- JSX syntax and components (functional and class components)
- Props and state management

Advanced React Concepts:

- Lifecycle methods and React Hooks (useState, useEffect)
- Handling forms and controlled components
- Routing with React Router
- Conditional rendering and list rendering

State Management with Context API :

- Introduction to Context API for global state management
- Creating and consuming context
- Prop drilling vs Context API

Building a React Application:

- Project: Creating a simple React app with features like navigation, forms, and data rendering from APIs
- Styling React components using CSS modules

Introduction to NodeJS:

- NodeJS runtime and architecture
- Setting up a NodeJS project with npm
- Working with built-in modules (e.g., fs, path, http)
- Event-driven programming and asynchronous operations

ExpressJS Basics:

- Setting up an Express server
- Middleware concepts: Built-in and third-party middleware
- Handling routes and query parameters
- Building a RESTful API with Express

Connecting Frontend and Backend:

- Fetching data from backend APIs in React
- Passing data between client and server using JSON
- Handling CORS in Express

Database Integration:

- Overview of databases (MongoDB or other)
- Connecting NodeJS to a database using mongoose or MySQL
- Performing CRUD operations

Deployment and Final Project:

- Deploying the full stack application on platforms like Heroku or Vercel
- Final project:
 - Full stack application combining ReactJS frontend and NodeJS/Express backend
 - Features: Authentication, data display, and interaction with backend APIs

Secure Coding Practices (Frontend & Backend):

- Input validation and sanitization
- Prevention of Cross-Site Scripting (XSS)
- Secure session management and authentication
- Securing database queries (avoiding SQL/NoSQL injection attacks)
- Implementing secure cookies and CSRF tokens
- Content Security Policy (CSP) and secure headers
- Preventing Clickjacking

Course Outcomes: At the end of the course student will be able to

1.	Demonstrate proficiency in creating responsive and semantic web pages using HTML5, CSS3, and JavaScript.
2.	Develop dynamic and interactive web applications using ReactJS and state management techniques.
3.	Design and implement secure, scalable, and RESTful APIs using NodeJS and ExpressJS.
4.	Integrate frontend and backend systems to build full stack web applications with seamless data exchange
5.	Deploy full stack applications on cloud platforms and understand basic concepts of version control and CI/CD pipelines.

Security

2.	Understand the difference between stateful and stateless firewalls.
3.	Deploy basic firewall along with IPS and URL Filtering feature.
4.	Demonstrate the capabilities of UTM over Basic Firewall.
5.	Able to perform basic Firewall troubleshooting.

Course Outcomes Mapping with Program Outcomes & PSO																
↓ Course Outcomes	Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
														1	2	3
CS2102-1.1		3	3											2	3	
CS2102-1.2			2	2										2	2	
CS2102-1.3		2	3											3	3	
CS2102-1.4		2	2	2										2	3	
CS2102-1.5		2	3											2	3	

1: Low 2: Medium 3: High

TEXTBOOKS:	
1.	Official Check Point Administration book (CCSA) - by Check Point Software Technologies.
2.	Learn Wireshark: Confidently navigate the Wireshark interface and solve real-world networking problems - by Lisa Bock.

REFERENCE BOOKS:	
1.	Fortigate Firewall Security Pocket Guide (Fortigate Pocket Guide Book 1) – by Ofer shmueli
2.	Fortigate Firewall Security Pocket Guide (Fortigate Pocket Guide Book 2) - by Ofer shmueli

E Books / MOOCs/ NPTEL	
1.	https://community.sophos.com/utm-firewall/f/recommended-reads/115519/sophos-utm-architecture
2.	https://www.networkworld.com/article/2288390/a-closer-look-at-utm-hardware-architecture.html

ETHICAL HACKING AND NETWORK DEFENSE

Course Code	CB1602-1	Course Type	PCC
Teaching Hours/Week (L: T: P)	1:0:2	Credits	02
Total Teaching Hours	15+0+26	CIE + SEE Marks	50+50

Teaching Department: Computer Science & Engineering (Cyber Security)

Course Objectives:

1. To understand the core concepts of Ethical Hacking.
2. To understand how security vulnerabilities are exploited.
3. To analyze the impact of security vulnerabilities in systems.
4. To understand popular Network Defense solutions deployed at large organizations.
5. To configure basic firewall and IDS solution.

List of Experiments

15 Hours

Fundamentals of Ethical hacking, how organizations gain from Ethical Hacking, Typical Life Cycle of Ethical Hacking, Types of Ethical Hacking- Red, Blue and Purple Teaming, Fundamentals of Vulnerability Analysis and Penetration Testing, Threat Modeling and Attack Surface Identification, Life Cycle of Penetration Testing, Using Kali Linux, and other tools for a penetration testing Assignment. Lab – Perform Vulnerability and Penetration testing on given Vulnerable system and generate report.

Networking Primer-understanding Security aspect of OSI Model, Active and passive Network Attacks, Network Layer and Cryptography, Single Sign On (SSO), Email encryption: PGP, STARTTLS; IPSec, SSL3.0, TLS 1.2, Attacks on SSL/TLS: Drown attack, Poodle attack, and Secure HTTP, DNSSEC. ARP Cache poisoning, MAC flooding, Port Stealing, DHCP attacks, DNS based attacks, VLAN hopping, Man in the middle attacks.

Web Application Security: Security threats, XSS, CSRF, SQL Injection attacks, RFI, DoS/DDoS.

Techniques for Network Intrusion Detection System: Snort, Signature-based and Anomaly-based detection; Firewalls: packet filters and stateful firewalls, application-aware firewalls, Proxies, NAT, VPN, Honeypots and Honeynets.

Lab – Deploy Snort IDS and create custom signatures to capture malicious traffic.

Course Outcomes: At the end of the course student will be able to

1.	To understand how to find security vulnerabilities in given system.
2.	To suggest the remediation steps for identified security bugs.
3.	To perform VAPT task on given system and submit professional report.
4.	To deploy IDS system.
5.	To develop custom IDS signatures.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS1602-1.1	3	2											1	1	2
CS1602-1.2	3	2	2										1	1	2
CS1602-1.3	3	2	2										1	2	3
CS1602-1.4	3	2	3										1	2	3
CS1602-1.5	3	2	3										1	2	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	The Web Application Hacker's Handbook: Discovering and Exploiting Security Flaws Book by Dafydd Stuttard and Marcus Pinto.
2.	Hacking: The Art of Exploitation Book by Jon Erickson.
3.	Hacking Exposed 7: Network Security Secrets and Solutions by Stuart McClure, Joel Scambray, George Kurtz.
4.	Snort Intrusion Detection and Prevention Toolkit by Brian Caswell, Jay Beale, Andrew Baker.

REFERENCE Books:

1.	Advanced Penetration Testing: Hacking the World's Most Secure Networks by Wil Allsopp
2.	Snort 2.1 Intrusion Detection, Second Edition by Brian Caswell, Jay Beale (2004), Publisher(s): Syngress

E Resources

1.	https://www.coursera.org/learn/ethical-hacking-essentials-ehe
2.	https://www.udemy.com/topic/ethical-hacking/
3.	https://onlinecourses.nptel.ac.in/noc22_cs13/preview

SECURED MOBILE APPLICATION DEVELOPMENT

Course Code:	CB3604-1	Course Type:	PCC
Teaching Hours/Week (L: T: P):	0:0:2	Credits:	01
Total Teaching Hours:	0+0+26	CIE + SEE Marks:	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
1.	Describe the architecture and overview of Android.		
2.	Develop a mobile application on android platform using UI components and Android Components.		
3.	Demonstrate data handling in Android and Develop a mobile application on Android platform using SQLite		
4.	Build an Android web service.		
5.	Develop application to demonstrate google map and navigation.		

List of Experiments									
PART-A									
1.	Design four checkboxes namely any four food items and one button. Find total amount of food items selected in Toast message after clicking the button.								
2.	Design simple calculator application that performs basic arithmetic operations. Use ADD, SUB, MUL, DIV buttons to perform operations, CLEAR button to reset the fields, and edit text widgets for reading operands, displaying result value.								
3.	Create an application which generates a random color on each click.								
4.	Implement the options menu concept in the application to choose between two activities (Give appropriate titles to activities).								
5.	Implement context menu concept in application to change the background color.								
6.	Design an application to send SMS using Intent class.								
7.	Design a phone call application that takes a phone number from the user.								
8.	Write an application to toast your joining date and course selected for engineering using a Date picker and List view/Spinner.								
9.	Write an application to make a dialogue box to confirm the change of background color or image.								
10.	Design an application that captures the image using a camera and set the captured image as the background for your application.								
PART-B									
1.	Implement a service concept to play the music in the background for long duration and perform a foreground job.								
2.	Implement an AsyncTask to count from 1 to 1000 in the background and the display the progress using progress bar on the screen.								
3.	Implement broadcast receiver to carry out the of following: Read battery charge of your mobile, display it using progress bar and change the background color as given in table. <table border="1" data-bbox="587 1064 1082 1272"> <thead> <tr> <th><u>Color</u></th><th><u>Battery Charge</u></th></tr> </thead> <tbody> <tr> <td>Red</td><td>0% To 20%</td></tr> <tr> <td>Blue</td><td>21% To 60%</td></tr> <tr> <td>Green</td><td>61% To 100%</td></tr> </tbody> </table>	<u>Color</u>	<u>Battery Charge</u>	Red	0% To 20%	Blue	21% To 60%	Green	61% To 100%
<u>Color</u>	<u>Battery Charge</u>								
Red	0% To 20%								
Blue	21% To 60%								
Green	61% To 100%								
4.	Write an application to insert the data entered by a user into a database and display all the values in database.								
5.	Write an application to search for a given USN from a student database and call to that student.								
6.	Design a simple resume builder application using two activities that take basic information namely Name, Email Id, Mobile No, Gender, Qualification, and Profile Picture from the user. Use appropriate text widgets for (Name, Email Id, Mobile No, Qualification), radio buttons for gender, Imageview for the profile picture.								
7.	Write an application that creates a notification message that will launch another activity after clicking on it.								
8.	Implement web view concept in application that contains two activities and opens default web page/user entered web page.								
9.	Implement an application to store and retrieve data by using shared preference. (Include save, delete and retrieve operations)								
10.	Implement the following animation concept i. Blink ii. Move the image object iii. Rotate. iv. Zoom In and Out								
Course Outcomes: At the end of the course student will be able to									
1.	Understand the tool like Android Platform and Android Studio Environment to familiarize with android development environment.								

2.	Design the user interface using the Android UI Components and Android Application Components.
3.	Apply the concepts such as SQLite, shared preference, files, broadcast, notifications, and other APIs for developing the android applications.
4.	Develop Application using Sensor telephony APIs.
5.	Understand the tool like Android Platform and Android Studio Environment to familiarize with android development environment.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3603-1.1	3	3	3		1	1		2	2	1	2	1	2	3	
CS3603-1.2	3	3	3		1	1		2	2	1	2	1	2	3	
CS3603-1.3	3	3	3		1	1		2	2	1	2	1	2	3	
CS3603-1.4	3	3	3		1	1		2	2	1	2	1	2	3	
CS3603-1.5	3	3	3		1	1		2	2	1	2	1	2	3	

1: Low 2: Medium 3: High

REFERENCE BOOKS:

1.	Anubhav Paradhan, Anil V Deshpande, "Mobile apps Development", First Edition, Wiley, 2014.
2.	Barry Burd, "Android Application Development All in one for Dummies", Second Edition Wiley, 2015.
3.	SAMS, "Teach Yourself Android Application Development in 24 Hours", First Edition, Sams Publishing, 2010.
4.	Wei-Meng Lee, "Beginning Android Application Development", Wrox Publication, 2011.
5.	Reto Meier, "Professional Android 4 Application Development", Wrox Publication, 2012..

E Resources

1.	https://www.tutorialspoint.com/android/index.htm
2.	https://www.javatpoint.com/android-tutorial
3.	https://developer.android.com/guide/
4.	http://nptel.ac.in/courses/106106156/
5.	https://www.youtube.com/watch?v=SYoNOvdZ3M&list=PLonJJ3BVjZW6CtAMbJz1XD8ELUs1KXaTD&index=19

Professional Elective Courses (Architecture, Embedded Systems & General Stream)

CLOUD COMPUTING				
Course Code:		CS2201-1	Course Type	PEC
Teaching Hours/Week (L: T: P) :		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives:				
1.	Explain the technology and principals involved in building a cloud environment.			
2.	Understand concepts of virtualization and cloud architecture			
3.	Choose appropriate cloud model for a given application			

4.	Understand advanced topics in cloud
5.	Use simulators and tools to understand working of cloud environment

UNIT-I

Introduction	15 Hours
Cloud computing at a glance Historical development (T1: 1.1,1.2) Principles of Parallel and Distributed Computing Eras of computing Parallel vs distributed computing Elements of parallel computing Elements of distributed computing Technologies for distributed computing (T1: 2.1, 2.2, 2.3, 2.4, 2.5) Virtualization Introduction, Characteristics of virtualized environments, Taxonomy of virtualization techniques, Virtualization and cloud computing, Pros and cons of virtualization. (T1: 3.1, 3.2, 3.3, 3.4, 3.5)	

UNIT-II

Cloud Computing Architecture	15 Hours
Introduction, The cloud reference model, Types of clouds, Economics of the cloud, Open challenges (T1: 4.1, 4.2, 4.3, 4.4, 4.5) Cloud Platforms in Industry Amazon web services, Google AppEngine, Microsoft Azure (T1: 9.1, 9.2, 9.3) Cloud Applications Scientific applications, Business and consumer applications (T1: 10.1, 10.2)	

UNIT-III

Advanced Topics in Cloud Computing:	10 Hours
Energy efficiency in clouds, Market-based management of clouds, Federated clouds/Inter Cloud, Third-party cloud services, (T1: 11.1, 11.2, 11.3, 11.4) Introduction to Software Defined Networks (R1), Network Function Virtualization. (R2) Introduction to Microservices in cloud (R3), Fog Computing (R4, R5).	

Course Outcomes: At the end of the course student will be able to

1.	Explain the concepts and terminologies of cloud computing
2.	Illustrate virtualization, cloud frameworks and technologies
3.	Interpret appropriate cloud model for a given application
4.	Explain energy efficiency and basics of software defined networks in cloud
5.	Explain micro services and fog computing

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS2201-1.1	1	2											1	2	
CS2201-1.2	1	3											1	2	
CS2201-1.3	1	3											1	2	
CS2201-1.4	1	3											1	2	
CS2201-1.5	1	3											1	2	

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Buyya, R., Vecchiola, C. and Selvi, S.T., 2013. "Mastering cloud computing: foundations and applications programming", Newnes.

REFERENCE BOOKS:

1. <https://www.opennetworking.org/images/stories/downloads/sdn-resources/technical-reports/SDN-architecture-overview-1.0.pdf>
2. https://web.archive.org/web/20161128071647/https://portal.etsi.org/NFV/NFV_White_Paper.pdf
3. <https://tsh.io/state-of-microservices-2020-by-tsh.pdf>
4. https://www.cisco.com/c/dam/en_us/solutions/trends/iot/docs/computing-overview.pdf

5.	Mahmud, Redowan, RamamohanaraoKotagiri, and Rajkumar Buyya. "Fog computing: A taxonomy, survey and future directions." In <i>Internet of everything</i> , pp. 103-130. Springer, Singapore, 2018.
E Books / MOOCs/ NPTEL	
1.	https://nptel.ac.in/courses/106105167/
2.	https://www.coursera.org/learn/cloud-computing-basics

ADVANCED COMPUTER ARCHITECTURE				
Course Code:		CS3201-1	Course Type	PEC
Teaching Hours/Week (L: T: P):		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives:				
1.	Outline the preamble of quantitative principles of computer architecture, various parallel computer models and fundamentals of parallel processing.			
2.	Make use of the concept of pipelining and apply in Linear and Non Linear pipelining processors			
3.	Learn how to enhance a MIPS processor’s ability by understanding challenges like hazards and techniques like static and dynamic scheduling.			
4.	Get an idea of Synchronization mechanism in Multiprocessors and Optimizations in Cache and memory.			
5.	Summarize the fundamental aspects of Instruction Level Pipelining and utilize in case studies of Itanium and Intel IA-64 Architecture along with the Hardware and Software.			
UNIT-I				
FUNDAMENTALS OF COMPUTER DESIGN				15 Hours
Introduction,Classes of Computers, measuring, reporting and summarizing performance, quantitative principles of computer design (Text 1, chapter: 1).				
PARALLEL COMPUTER MODELS:				
Shared memory multiprocessors, Distributed-Memory multi computers (Text 3: chap1.2). Introduction to Parallel processing: Concepts of concurrent and parallel execution, types and levels of parallelism. (Text 2: chapter 3)				
PIPELINING:				
Introduction, the major hurdle of pipelining- pipeline hazards, How is pipelining implemented. (Text 1, Appendix A). Linear pipeline processors and Non-linear pipeline processors (Text 3, chapter 6).				
UNIT-II				
EXPLOITING INSTRUCTION LEVEL PARALLELISM				15 Hours

Concepts and Challenges, Basic compiler techniques for exposing ILP, Reducing branch cost with prediction, overcoming data hazards with dynamic scheduling, hardware based speculation, exploiting ILP using multiple issues and static scheduling, exploiting ILP using Dynamic scheduling, multiple issue and speculation, advanced techniques for instruction delivery and speculation. (Text 1, chapter 2)

MEMORY HIERARCHY DESIGN:

Introduction; review of concepts. Basic six cache optimization. Eleven Advanced optimizations of Cache performance (self-study); Memory technology and optimizations. (Text 1, chapter 5: 5.1,5.2,5.3)

UNIT-III

HARDWARE AND SOFTWARE FOR VLIW AND EPIC

10 Hours

Introduction: Exploiting Instruction-Level Parallelism Statically; Detecting and Enhancing Loop- Level Parallelism; Scheduling and Structuring Code for Parallelism; Hardware Support for Exposing Parallelism: Predicated Instructions; Hardware Support for Compiler Speculation; The Intel IA-64 Architecture and Itanium Processor; Conclusions. (Text 1, Appendix G)

Course Outcomes: At the end of the course student will be able to

1.	Describe the principles of computer design using Amdahl's law, principle of locality and parallelism.
2.	Demonstrate instruction level parallelism in MIPS processor using instruction pipelining.
3.	Elaborate how processor performance is enhanced using software and hardware techniques.
4.	Compare cache optimization techniques and choose the suitable one to improve processor performance.
5.	Illustrate the hardware and software support for VLIW and EPIC with the case study of Intel IA-64 architecture.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→ ↓ Course Outcomes													PSO↓		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CS3201-1.1	2	3							1	1		1	3		
CS3201-1.2	2	3							1	1		1	3		
CS3201-1.3	2	3							1	1		1	3		
CS3201-1.4	2	3							1	1		1	3		
CS3201-1.5	2	3							1	1		1	3		

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	John L. Hennessey and David A. Patterson, "Computer Architecture, A Quantitative Approach", 4th Edition, Elsevier, 2007
2.	DezsoSima, Terence Fountain, Peter Kacsuk, "Advanced Computer Architectures- A Design space approach", Pearson Education 1997.
3.	Kai Hwang, "Advanced Computer Architecture Parallelism, Scalability", Tata Mc Grawhill, 2003

REFERENCE BOOKS:

1.	Computer Architecture and Organization, John P. Hayes, Volume 2, McGraw-Hill, 2012
2.	Computer Organization and Architecture: Designing for Performance, William Stallings, PHI 9 edition, 2012

E Books / MOOCs/ NPTEL

1.	Computer Architecture and Organization - Design Principles and Applications, B. Govindarajalu,
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	Mcgrawhill HED, Edition: 2nd Edition, 2010
2.	Fundamentals of computer organization and architecture, M Abd-El-Barr and Hesham El- Rewini, Wiley Interscience, 2005
3.	NPTEL course on Computer Architecture, by Prof. Madhu Mutyam, PACE Laboratory, Department of computer Science and Engineering, Indian Institute of Technology Madras. Online: https://www.youtube.com/watch?v=Tz7kMR-MAuk
4.	NPTEL course on Advanced Computer Architecture, by Dr. John Jose, Department of computer Science and Engineering, Indian Institute of Technology Guwahati. Online: https://www.youtube.com/watch?v=6oiKaIH7BKU
5.	NPTEL course on Parallel computer Architecture, by Dr. Mainak Chaudhuri, Department of Computer Science and Engineering, Indian Institute of Technology Kanpur. Online: https://nptel.ac.in/courses/106/104/106104024/

FUNDAMENTALS OF IMAGE PROCESSING				
Course Code:		CS3202-1	Course Type	PEC
Teaching Hours/Week (L: T: P):		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives:				
1.	Outline the theory behind the basics of digital image processing, the relation between the components of image processing system. Make use of Electromagnetic Spectrum, find the equivalence between pixels.			
2.	Make use of spatial and frequency domain, smoothing and sharpening filters.			
3.	Make use of Homomorphic Filtering and how to simplify Detection of Discontinuities.			
4	Get the idea of Models Elements of Information, find the equivalence between Dilation and Erosion, Opening and Closing, and identify the Hit-or-Miss Transformation. Understand different compression model.			
5	Tell how Components of an Image Processing System works, their design, and get the feeling of Histogram Processing.			
UNIT-I				
Introduction				15 Hours
What is Digital Image Processing? Examples of Fields that Use Digital Image Processing, Fundamental Steps in Digital Image Processing, Components of an Image ProcessingSystem. Digital Image Fundamentals - Elements of Visual Perception, Light and the Electromagnetic Spectrum,Image Sensing and Acquisition, Image Sampling and Quantization, Some Basic Relationships between Pixels. IntensityTransformationsandSpatialFiltering-Background, Some Basic Intensity Transformation Functions, Histogram Processing-Histogram Equalization, Histogram Matching. Local Histogram Processing, Fundamentals of Spatial Filtering, Smoothing Spatial Filters, Sharpening SpatialFilters.[Textbookchapters1,2,3]				
UNIT-II				
Filtering in Frequency Domain				15 Hours
Background, Sampling and the Fourier Transform of Sampled Functions, The Discrete Fourier Transform (DFT)of Functions of one continuous variable, Image smoothing using Frequency-Domain Filters – Ideal Lowpass Filters, Butterworth Lowpass Filters, Gaussian Lowpass Filters, Image Sharpening using Frequency Domain Filters -Ideal Highpass Filters, Butterworth Highpass Filters, Gaussian Highpass Filters, Homomorphic Filtering.				

[Text book chapter 4.1,4.3,4.4,4.8,4.9] Image Compression – Fundamentals- Coding Redundancy, Spatial and Temporal Redundancy, Image Compression Model. Some Basic Compression Model – Huffman Coding, Arithmetic Coding, LZW coding, Bit-Plane Coding, Run -Length Coding. [Text book chapter 8.1,8.2] Morphological Image Processing – Preliminaries, Dilation and Erosion, Opening and Closing, the Hit-or-Miss Transformation, Some Basic Morphological Algorithms -Boundary Extraction, Thinning, Thickening [Text book chapter 9.1,9.2,9.3,9.4,9.5]																
UNIT-III																
Image Segmentation														10 Hours		
Point, Line and Edge Detection – Background, Detection of Isolated Points, Line Detection, Edge Model, Basic Edge Detection, Edge Linking and Boundary Detection, Thresholding- Foundation, Basic Global Thresholding, Region Based Segmentation Region growing, splitting and merging. [Text book chapter 10.1,10.2,10.3,10.4]																
Course Outcomes: At the end of the course student will be able to																
1.	Apply the concept of Digital Image Processing andSteps in Digital Image Processing, Able to apply the Knowledge of Image Sampling and Quantization and Illustrate Some Basic Relationships between Pixels using Knowledge of 4-8 and Madjacency.															
2.	Design and Formulate Histogram processing. Analyze Smoothing Spatial Filters, Sharpening Spatial Filters by applying mathematical knowledge.															
3.	Explain Frequency domain and illustrate Smoothing Frequency-Domain Filters. Analyze Sharpening frequency-Domain Filters. Apply and Design Image Compression Standards andmodels.															
4.	Analyze the concept of Morphological ImageProcessing by applying mathematical knowledge.															
5.	Design and Formulate Image segmentation techniques and prove the properties Region-Based Segmentation.															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS3202-1.1		1	2	3										3	2	
CS3202-1.2			3											3		
CS3202-1.3		2	3											3		
CS3202-1.4		2		3										3	2	
CS3202-1.5			2	3										3	2	
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.	Rafael C. Gonzalez and Richard E. Woods, “Digital Image Processing”, Pearson Education, Third Edition.															
REFERENCE BOOKS:																
1.	DigitalImageProcessingWithMatlab&LabviewVipulaSinghPublishedbyReedElsevierIndiaPvt.Ltd Language -English Binding-Paper Back.															
2.	AnilKJain, “FundamentalsofDigitalImageProcessing”,Prentice-HallofIndiaPvt.Ltd.,1997.															
3.	MilanSonka,VaclavHlavacandRogerBoyle,“ImageProcessing,AnalysisandMachineVision”,Thomason Learning,Brooks/Cole,SecondEdition. 2001.															
4.	B.Chanda,DDuttaMajumder,“DigitalImageProcessingandAnalysis”,Prentice-Hall,India,2002.															
5.	StevenW.Smith,“TheScientistandEngineersGuidetoDigitalSignalProcessing“,CaliforniaTechnical															

	Publishing, Second Edition, 1999.
E Books / MOOCs/ NPTEL	
1.	iitlab.bit.edu.cn/HandbookofImageandVideoProcessing.pdf
2.	http://www.cs.ukzn.ac.za/~sviriri/Books/Image-Processing/book4.pdf
3.	https://nptel.ac.in/courses/117105079/
4.	https://swayam.gov.in/nd1_noc19_ee55/preview
5.	https://www.coursera.org/learn/image-processing
6.	https://www.coursera.org/learn/image-processing

OPERATIONS RESEARCH			
Course Code:	CS3203-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
This Course will enable students to:			
1.	Describe the scope and limitations of OR methods and outline the role of OR techniques in supporting the decisions.		
2.	Explain the concept of Linear Programming Model (LPM) and formulate Linear Programming problems.		
3.	Describe the various methods like Simplex Method, revised simplex Method, Big M Method, Two Phase Method, Dual Simplex M ethod and duality theory and use it on Linear Programming Problems.		
4.	Describe the formulation of Transportation problems, different methods in Transportation problems like North West Corner Rule, Row minima method, Column minima method, Matrix minima method, Vogel’s approximation method, U-V method and use those methods on the respective real-world problems.		
5.	Describe the formulation of Assignment problems, use Hungarian method in Assignment problems, CPM and PERT (project management techniques) and use it on the respective real-world problems.		
UNIT-I			
INTRODUCTION			15 Hours
Introduction to OR, nature and meaning, applications, modeling in OR, phases of OR study			
LINEARPROGRAMMING			
Introduction to Linear Programming through an example, graphical method, formulation of LP model from practical problems, assumptions and properties of linear programming, simplex method, Big M method, 2 phase method, Revised simplex method, Duality theory, Primal and dual relationship. (Text Book-1: Chapter 2,3,5,6,7,8)			
UNIT-II			
TRANSPORATIONPROBLEMS			15 Hours
Transportation problems, methods to find initial feasible solution and modification to obtain optimal solution (Degeneracy in transportation problems, unbalanced transportation problems			
ASSIGNMENTPROBLEM			
Mathematical formulation of an assignment problem, unbalanced assignment problem, Travelling Salesman Problem (TSP), Hungarian method.(Text Book-1: Chapter 15,16)			
UNIT-III			
CPM, PERT			10 Hours
Representation of a project by a network, activities and events, starting times, finishing times, floats, slacks, CPM, Idea of crashing probabilistic times and PERT analysis (Text Book-1: Chapter 31)			

Course Outcomes: At the end of the course student will be able to

1.	Describe the basics of OR, modelling and applications of OR and the linear programming model.
2.	Construct linear programming problem and apply methods like Simplex method, revised simplex method, Big M method, 2 phase method and Dual simplex method to solve the different use cases of linear programming problem.
3.	Apply the North West Corner Rule, Row minima method, Column minima method, Matrix minima method, Vogel's approximation method and U-V method to solve the Transportation Problems.
4.	Apply the Hungarian method to solve the Assignment Problems and Travelling Salesman Problems.
5.	Apply the CPM and PERT project management techniques on the respective use cases to solve the problems related to the use cases.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3203-1.1	3												3		
CS3203-1.2	2	3											3		
CS3203-1.3	2	3											3		
CS3203-1.4	2	3											3		
CS3203-1.5	2	3											3		

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Operations Research, S. D. Sharma, 17th Revised edition, 2014.

REFERENCE BOOKS:

1. Operations Research, Er. Premkumar Gupta, Dr. D.S. Hira, 4th edition, 2015.
2. Introduction to Operations Research - A Computer Oriented Algorithmic Approach, Gillelt B G, McGraw Hill, 2008.
3. Operations Research – An introduction, Hamdy A Taha, PHI, 8th edition, 2007.

E Books / MOOCs/ NPTEL

1. https://www.tutorialspoint.com/linear_programming/index.asp / Fundamentals of Operations Research IIT Madras Course, Prof. G. Srinivasan: <https://swayam.gov.in/>
2. <https://www.cs.toronto.edu/~stacho/public/IEOR4004-notes1.pdf>
3. <http://people.brunel.ac.uk/~mastjjb/jeb/or/contents.html>

ROBOTIC PROCESS AUTOMATION DESIGN & DEVELOPMENT

Course Code:	CS3204-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50

Teaching Department: Computer Science & Engineering (Cyber Security)

Course Objectives:

1.	Learn the basic concepts of Robotic Process Automation
2.	Understand processes which can be automated, associated business documentation basics, RPA journey of an organization
3.	Develop familiarity and deep understanding of UiPath tools and prepare students to be Junior RPA Developer

UNIT-I

FIRST WEEK – RPA Basics

History of Automation, Story of Work, Introduction to RPA, RPA vs. Automation, RPA and AI, RPA and emerging ecosystem, Industries best suited for RPA, Process that can be Automated

SECOND WEEK – Introduction to UiPath

UiPath and its products, Robots and their types, Studio Overview, Orchestrator, UiPath Studio Installation, The User Interface, Features of Studio – Managing Activities Packages, Managing Extensions, Reusing Automation Library, Version Control, Introduction to Automation Debugging, Activities Guide

Lab Experiments: Install UiPath Studio Extension in Browsers – Chrome Browser, Firefox Browser; Install Activity Packages in UiPath Studio – Excel Activity Package Installation, Email Activity Package Installation, PDF Activity Package Installation; Version Control using TFS; Build a workflow that prints “Hello World” in a message box.

THIRD WEEK – Variables and Arguments

Variables and its types, Variables Panel, Scope of Variable, Arguments, Arguments Panel, Argument Directions, Argument vs Variable

Lab Experiment: Build a workflow that swaps the values of two variables using a third variable

FOURTH WEEK – UI Automation & Selectors

UI interactions, Input actions and Input Methods – Input actions: Click, Type Into, Send HotKey; Input Methods – Default, SendWindowMessages, Simulate Type/Click, Containers, Recording & its types, Selectors, Types of Selectors – Full and Partial, Containers and Partial Selectors, Dynamic Selectors, Wildcards in Selectors, UI Explorer, Anchors, Debugging selectors

Lab Experiments: Build a workflow that uses different input methods to input data in a Notepad; Build a workflow that opens a browser and then opens UiPath’s website; Build a workflow using Web Recorder in UiPath Studio to Sign in to UiPath’s website; Build a workflow that fills the form on RPACHallenge.com website with organized data from an excel file; Build a workflow that replaces double spaces with single spaces from a text stored in multiple Notepad files with different names.

FIFTH WEEK – Control Flow

Sequences, Control Flow and its types, Decision control – IF, Switch, IF vs Switch, Loops – Do While, While, For Each, Other control flow activities – Delay, Break, Assign, Continue, Parallel, Flowcharts – Introduction, Decisions in flowcharts, Loops in flowcharts, nesting flowcharts and sequences, Sequences vs. Flowcharts, Error handling – Errors, Exceptions, Error handling approach, Try Catch, Retry Scope, Global Exception Handler, Continue On Error, Best Practice for Error Handling

Lab Experiments: Build a workflow using if statement which tells a user whether he will get the second Marshmallow or not; Build a workflow using Switch activity that asks users’ their eye colour and display their personality in a message box; Build a workflow for a ‘Guessing Game’; Build a workflow using While loop that tells the user if the input is a prime number or not; Build a workflow to display file names from a folder in the Output panel and also store names in an MS Word file; Build a workflow using Parallel activity; Build a workflow that asks user for his name and two-digit lottery number and displays if he is a winner; Build a workflow using Try Catch activity.

SIXTH WEEK – Data Manipulation

Data Manipulation and its importance – Introduction & operations, Data conversion, String Manipulations – Introduction & methods, RegEx, DataTable Manipulations, Collection, its types and manipulations – Lists, Dictionaries

Lab Experiments: Build a workflow using ToString method that converts an integer to string; Build a workflow using Format, Join, IndexOf, Split, and Substring methods that extracts key information from a text and prints in a different format; Build a workflow using Split and Contains methods that extract sentences containing “RPA” from a UiPath webpage

SEVENTH WEEK

Lab Experiments -Build a workflow using data table activities to join two library databases using

matching student ID and display output in a message box; Build a workflow using Concat and Join method that merges two lists containing the UK and Spain city names, sorts it, capitalizes the first letter of each item, and displays it in a message box.

EIGHT WEEK – Automation Concepts and Techniques

Extraction and its techniques – Screen scraping, Data scraping, PDF Extraction, Automation Techniques – Workbook and Excel automation (read/write), Email automation

Lab Experiments: Build a workflow using Screen Scrapper Wizard that scrapes text using Tesseract OCR scraping method from an image and stores in a Notepad

NINTH WEEK

Lab Experiments: Build a workflow using Screen Scrapper Wizard that scrapes text using Full-Text scraping method and stores in a Notepad file; Build a workflow using Data Scraping Wizard that scrapes blog post titles from UiPath Blog from multiple pages;

TENTH WEEK

Lab Experiments: Build a workflow using Read PDF Text activity and extract only Email IDs and Phone Numbers from a PDF file and store in an MS Word file; Build a workflow using Read Range and Append Range activity to read data from a workbook and append data to another workbook;

ELEVENTH WEEK

Lab Experiments: Build a workflow that calculates total monthly deposit of a bank from an Excel file and store output in a new sheet; Build a workflow that extracts attachments from the emails containing the word “Resume” in its subject.

TWELETH WEEK – Orchestrator

Orchestrator overview, Publishing a Robot to Orchestrator, Orchestrator Functionalities, Orchestrator User Interface – Categories of functionalities – Automations – Processes, Triggers, Queues, Transactions, Assets, Management – Folders, Users, Roles, Robots, Environments, Machines, Packages, Libraries, Monitoring – Robots, Jobs, Queues, Logs

Lab Experiments: Create an Asset of Credential type in Orchestrator and display the credential asset username in Studio;

THIRTEENTH WEEK

Lab Experiments - Create a Queue in Orchestrator and add excel data values in the queue.

Course Outcomes: At the end of the course student will be able to

1.	To prepare students to be Junior RPA Developers
2.	Learn the basic concepts of Robotic Process Automation
3.	Develop familiarity and deep understanding of UiPath tools
4.	Develop the ability to independently design and create robots for business processes
5.	Develop skills required to pass UiPath RPA Associate v1.0 Exam

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3204-1.1	1	2													
CS3204-1.2	1	2													
CS3204-1.3	1	2												2	
CS3204-1.4	1	2											2		
CS3204-1.5	1	2												2	

1: Low 2: Medium 3: High

TEXTBOOKS:

- Alok Mani Tripathi: Learning Robotic Process Automation, 1st Edition, Packt Publishing Ltd., 2018.

REFERENCE BOOKS:

- Power Point Presentations from UiPath.

2.	Lab Guide from UiPath
E Books / MOOCs/ NPTEL	
1.	https://academy.uipath.com/
2.	https://www.uipath.com/rpa/academic-alliance

BUSINESS INTELLIGENCE AND ITS APPLICATIONS				
Course Code:		CS3301-1	Course Type	PEC
Teaching Hours/Week (L: T: P):		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives: This Course will enable students to:				
1.	Comprehend the need of BI for a business enterprise			
2.	Summarize the types of Digital data & its operation			
3.	To outline the Need & Significance of data warehouse in BI applications			
4.	Identify the types and step involved in ETL process.			
5.	To understand the measurement concept to evaluate business performance and build enterprise reports			
UNIT-I				
BUSINESS VIEW OF INFORMATION TECHNOLOGY APPLICATIONS				15 Hours
Business Enterprise Organization, Its Functions, and Core Business Processes; Baldrige Business Excellence Framework; Key Purpose of Using IT in Business; The Connected World: Characteristics of Internet-ready IT Applications; Enterprise Applications (ERP/CRM, etc.) and Bespoke IT Applications; Information Users and Their Requirements; Case Studies (T1: Ch-1.1 to 1.6)				
TYPES OF DIGITAL DATA				
Introduction; Getting into “GoodLife” Database; Getting to Know Structured Data; Getting to Know Unstructured Data; Getting to Know Semi-Structured Data; Difference Between Semi-Structured and Structured Data. (T1: Ch-2.1 to 2.6)				
INTRODUCTION TO OLTP AND OLAP				
OLTP (On-Line Transaction Processing); OLAP (On-Line Analytical Processing); Different OLAP Architectures; OLTP and OLAP; Data Models for OLTP and OLAP; Role of OLAP Tools in the BI Architecture; Should OLAP be Performed Directly on Operational Databases? A Peek into the OLAP Operations on Multidimensional Data; Leveraging ERP Data Using Analytics (T1: Ch-3.1 to 3.9)				
UNIT-II				
GETTING STARTED WITH BUSINESS INTELLIGENCE				15 Hours
Using Analytical Information for Decision Support; Information Sources Before Dawn of BI? Business Intelligence (BI) Defined; Evolution of BI and Role of DSS, EIS, MIS, and Digital Dashboards; Need for BI at Virtually all Levels; BI for Past, Present, and Future; The BI Value Chain; Introduction to Business Analytics (T1: Ch-4.1 to 4.8)				
BI DEFINITIONS AND CONCEPTS				
BI Component Framework; Who is BI for? BI Users; Business Intelligence Applications; BI Roles and Responsibilities; Best Practices in BI/DW; The Complete BI Professional; Popular BI Tools (T1: Ch-5.1 to 5.8)				
BASICS OF DATA INTEGRATION				

Need for Data Warehouse; Definition of Data Warehouse; What is a Data Mart? What is Then an ODS? Ralph Kimball's Approach vs. W.H. Inmon's Approach; Goals of a Data Warehouse; What Constitutes a Data Warehouse? Extract, Transform, Load; What is Data Integration? Data Integration Technologies; Data Quality; Data Profiling

(T1: Ch-6.1 to 6.12)

MULTIDIMENSIONAL DATA MODELING

Introduction; Data Modeling Basics; Types of Data Model; Data Modeling Techniques; Fact Table; Dimension Table; Typical Dimensional Models; Dimensional Modeling Life Cycle.

(T1: Ch-7.1 to 7.8)

UNIT-III

MEASURES, METRICS, KPIS, AND PERFORMANCE MANAGEMENT

10 Hours

Understanding Measures and Performance; Measurement System Terminology; Navigating a Business Enterprise, Role of Metrics, and Metrics Supply Chain; "Fact-based Decision Making" and KPIs; KPI Usage in Companies; Where Do Business Metrics and KPIs Come From? Connecting the Dots: Measures to Business Decisions and Beyond

(T1: Ch-8.1 to 8.7)

BASICS OF ENTERPRISE REPORTING

Reporting Perspectives Common to All Levels of Enterprise; Report Standardization and Presentation Practices; Enterprise Reporting Characteristics in OLAP World; Balanced Scorecard; Dashboards; How Do You Create Dashboards? Scorecards vs. Dashboards; The Buzz Behind Analysis. **(T1: Ch-9.1 to 9.8)**

BI ROAD AHEAD

Understanding BI and Mobility; BI and Cloud Computing; Business Intelligence for ERP Systems; Social CRM and BI.

(T1: Ch-10.1 to 10.4)

Course Outcomes: At the end of the course student will be able to

1.	Interpret the business view of information technology applications
2.	Summarize the types of Digital data & its operation.
3.	Outline the Need & Significance of data warehouse in BI applications
4.	Explain the basics of data integration including data quality and data profiling and implement various data integration approaches
5.	Identify Key Performance Indicators, Business Metrics, Future of BI, creation of Enterprise Reports.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3301-1.1		2							1	1		1		1	
CS3301-1.2	2	2			2				1	1		1		3	
CS3301-1.3	2	2				2			1	1		1		3	
CS3301-1.4	2	2				2			1	1		1		3	
CS3301-1.5	2	2			2				1	1		1		2	

(L/1 = Low 30%-49%, M/2 = Medium 50%-69%, H/3 = High >70%)

TEXTBOOKS:

1. Prasad RN, Seema Acharya: Fundamentals of Business Analytics, First Edition, Wiley India Pvt. Ltd.
2. William H. Inmon: Building the Data Warehouse, 4th Edition, Wiley India Ed.
3. Infosys Reference Book on Business Intelligence

REFERENCE BOOKS:

1. David Loshin: Business Intelligence, First Edition, Elsevier Science, 2003.

2.	Mike Biere: Business Intelligence for the Enterprise, First Edition, IBM Press, 2003
3.	Larissa T. Moss and Shaku Atre: Business Intelligence Roadmap, Addison-Wesley Professional, 2003.

SEE Question Paper Pattern:

There will be **8** questions of **20** marks each in the question paper divided into **3 Units** as per the syllabi & contact hours and the student will have to answer **5** full questions, selecting **2** full questions from **Unit - I & Unit – II** and **1** full question from **Unit – II**.

INTERNET OF THINGS			
Course Code:	CS3302-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
This Course will enable students to:			
1.	Learn the IoT Definitions, Design aspects.		
2.	Identify the IoT hardware and software requirements.		
3.	Describe IoT logical and physical design concepts.		
4.	Implement Arduino based IoT Projects.		
5.	Implement Raspberry Pibased IoT Projects.		
UNIT-I			
Introduction			15 Hours
Introduction to IoT: Definition and characteristics, Physical design, Logical design, Enabling technologies, Levels and deployment templates, Examples: Domain specific IoTs			
IoT Design and System Engineering			
Discuss IoT Requirements, Hardware & Software; Study of IoT sensors, Tagging and Tracking, Embedded Products; IoT Design, (U) SIM Card Technology, IoT Connectivity and Management, IoT Security & IoT Communication.			
Python Programming			
Data types, Data structures, Control flow, Functions, Modules, Packages, File Handling, Date and time operation, Classes, Python packages of IoT			
UNIT-II			
IoT Logical Design			15 Hours
IoT Physical Design, Basic building blocks, Raspberry Pi, Linux on Raspberry Pi, Interfaces, Programming on Raspberry Pi with Python			
Arduino Based IoT Projects Development			
Arduino for Project development using components such as LED/Buzzer, Push button/Digital sensor (IR/LDR), Interface motor using relay, Sensing Temperature and Humidity smart phone using Bluetooth.			
Raspberry Pi			
Raspberry Pi for Project Development: Raspberry Pi platform, GPIO, Establishment and setting, of Raspberry Pi software			
UNIT-III			
Raspberry Pi based IoT Project Implementation			10 Hours
Developing projects using components such as LED/Buzzer, Push button/Digital sensor (IR/LDR), Interface motor using relay, Sensing Temperature and Humidity smart phone using Bluetooth.			
Course Outcomes: At the end of the course student will be able to			

1.	Explain IoT Definitions, Requirements, Systems Design, Sensors, Tags, security communications and apply IoT knowledge in understanding IoT systems and applications.
2.	Describe Python basics, Control structures, Functions, Modules, Packages, File Handling, Date and time operation, Classes, Python packages of IoTAnalyze and Develop Simple programs using Python.
3.	Outline IoT systems Logical and Physical Design Aspects, Develop Arduino simple programmes for LED, Buzzer, Push button, Digital sensors.
4.	Develop and Implement the simple IoT projects usingArduino boards.
5.	Develop and Implement the simple IoT projects usingRaspberry Pi boards.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3302-1.1	3	1						1	1			1		3	
CS3302-1.2	2	3						1	1			1		3	
CS3302-1.3	3	1						1	1			1		3	
CS3302-1.4	3	2			3			1	1			1	1	3	
CS3302-1.5	3	2			3			1	1			1	1	3	

1: Low 2: Medium 3: High

TEXTBOOKS:

1. ArshdeepBahga, Vijay Madiseti, "Internet of Things: A Hands-On Approach,VijayMadiseti", 2014.
2. Donald Norris, "The Internet of Things: Do-It-Yourself at Home Projects for Arduino,Raspberry Pi and BeagleBone Black", 1st Edition, McGraw Hill, 2015.

REFERENCE BOOKS:

1. Dr. SRN Reddy, RachitThukral and Manasi Mishra," Introduction to Internet of Things: A practical Approach", ETI Labs
2. Pethuru Raj and Anupama C. Raman,"The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press
3. Jeeva Jose,"Internet of Things", Khanna Publishing House,Delhi
4. Adrian McEwen,"Designing the Internet of Things", Wiley
5. Raj Kamal, "Internet of Things: Architecture and Design", McGraw Hill
6. Cuno Pfister, "Getting Started with the Internet of Things", O Reilly Media

E Books / MOOCs/ NPTEL

1. Object-Oriented Analysis and Design with Applications, Grady Booch, Robert A. Maksimchuk, Michael W. Engel,Bobbi J. Young, Jim Conallen,Kelli A. Houston, Third Edition The / <https://www.coursera.org/specializations/internet-of-things>
2. Addison-Wesley Object Technology Series, 2007/ <https://www.udemy.com/course/iot-internet-of-things-automation-using-raspberry-pi/>
3. Object-Oriented Modeling and Design with UML, James R Rumbaugh, Michael R. Blaha Pearson Education, 21-Nov-2011/ <https://www.udemy.com/course/arduino-iot-cloud/>

MULTICORE ARCHITECTURE AND PROGRAMMING

Course Code:	CS3303-1	Course Type	PEC
Teaching Hours/Week (L: T: P) :	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50

Teaching Department: Computer Science & Engineering (Cyber Security)

Course Objectives:
This Course will enable students to:

1.	Outline the principles of multi-core design and performance measurement
2.	Illustrate the concept of parallelization and develop parallel programs

3.	Identify the hurdles of parallelization and determine ways to handle these issues
4.	Analyze the process of code optimization
5.	Recognize the need and usage of multi-threading tools

UNIT-I

INTRODUCTION TO MULTI-CORE ARCHITECTURE

15 Hours

Introduction, Moore's law, Amdahl's law, Gustafson's law, Motivation for Multi-core processors, Types and levels of parallelism, Flynn's classification of multi-processors, Introduction to parallelization and vectorization: Data dependencies, SIMD technology, Hardware Multithreading vs. Software multi-threading, Hyper threading, SMT, Case Study of multi-core processors: Intel, AMD, IBM/Sony.

CONCEPTS AND DESIGN OF PARALLEL AND THREAD PROGRAMMING

Definition of thread and process, Parallel programming models, Parallel Programming constructs: Synchronization, Deadlock, Critical sections.

THREAD PROGRAMMING

Parallel programming using POSIX APIs, OpenMP- Directives, clauses, and environment variables. Introduction to intel TBB, Thread- Safeness, Cache related issues

UNIT-II

PARALLEL PROGRAMMING WITH DISTRIBUTED MEMORY PARALLEL COMPUTERS

15 Hours

MPI Model: Collective communication, Data decomposition, Communicators and topologies, point-to-point communication, MPI Library, Programs using MPI.

MULTITHREADED PROGRAM DEBUGGING

Benchmarks and other performance analysis tools, VTune Performance Analyzer, Thread Checker. Thread Profiler, hotspots, performance issues in algorithms, branch misprediction, cache organization, cache loads, efficiency, hardware and software prefetch.

UNIT-III

COMPILER OPTIMIZATIONS AND PARALLEL ALGORITHMS

10 Hours

Compilers for High performance Computing, compiler optimization, code and loop optimization, scalar and vector processing, temporal and spatial locality-matrix multiplication example. OS support to multi-core architectures. Parallel algorithms study and analysis - The Sieve of Eratosthenes, Floyd's algorithm, Matrix-Vector multiplication, Monte Carlo methods, Matrix Multiplication, Parallel Quicksort Algorithm.

Course Outcomes: At the end of the course student will be able to

1.	Describe the multi-core architecture and motivation behind it.
2.	Design and develop parallel program on shared memory parallel computers using Pthreads API and OpenMP.
3.	Design and develop parallel program on distributed memory parallel computers using MPI.
4.	Analyse the performance of multithreaded programs using performance analysis tools.
5.	Apply compiler optimization techniques for parallel programs.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3303-1.1	2	3							1	1		1	3		
CS3303-1.2			3						1	1		1		3	
CS3303-1.3			3						1	1		1		3	
CS3303-1.4				3					1	1		1	3		
CS3303-1.5				3					1	1		1	3		

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Shameem Akhter and Jason Roberts," Multicore programming- Increasing performance through
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	software multithreading”, Intel press, 2013
2.	Richard Gerber, AartJ.C.Bik, Kevin B.Smith, Xinmin Tian,”The software optimization cookbook”,High performance Recipes for IA-32 Platforms,Intel press, 2005.
REFERENCE BOOKS:	
1.	Steven S.Muchnick, Morgan Kaufman,“Advanced Compiler Design Implementation”,Publishing 2000.
E Books / MOOCs/ NPTEL	
1.	www. tutorials on introduction to parallel computing /http://nptel.ac.in/courses/106104025/2
2.	http://www.cs.cmu.edu/afs/cs/user/fp/www/courses/15213-s07/lectures/27-multicore.pdf / https://www.mooc-list.com/tags/parallel-programming
3.	www.openmp.org for OpenMP

SPEECH PROCESSING																
Course Code:				CS3304-1				Course Type				PEC				
Teaching Hours/Week (L: T: P):				3:0:0				Credits				03				
Total Teaching Hours				40+0+0				CIE + SEE Marks				50+50				
Teaching Department: Computer Science & Engineering (Cyber Security)																
Course Objectives:																
This Course will enable students to:																
1.		Understand the fundamentals of speech processing.														
2.		Study the models of speech processing.														
3.		Explain the linear predictive coding.														
4.		Illustrate the application of speech processing.														
UNIT-I																
INTRODUCTION														15 Hours		
Introduction, Fundamentals of Digital Speech Processing, Digital models for the speech signals, Time domain models for speech processing, Digital representation of the speech waveform, short term Fourier analysis.																
UNIT-II																
HOMOMORPHIC SPEECH PROCESSING														15 Hours		
Homomorphic speech processing, Linear predictive coding of speech: Introduction, Basic principles of LP analyse, Computation of gain for the model, solution of LPC equation, Comparison between the methods of solution of the LPC analysis equation, the prediction error signal.																
UNIT-III																
LINEAR PREDICTIVE CODING OF SPEECH														10 Hours		
Linear predictive coding of speech: Frequency domain interpretation of LP analysis, Relation of LP analysis, Relations between various speech parameters, applications Digital speech for man machine communication by voice.																
Course Outcomes: At the end of the course student will be able to																
1.		Explain the fundamentals of speech processing.														
2.		Summarize the models of speech processing.														
3.		Infer the linear predictive coding.														
4.		Illustrate the application of speech processing.														
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS3304-1.1		3							1	2					2	

CS3304-1.2	2							1	2					3	
CS3304-1.3		2						1	2				3		
CS3304-1.4	2								2					3	
1: Low 2: Medium 3: High															
TEXTBOOKS:															
1. Digital Processing of Speech Signals, Lawrence R. Rabiner , Ronald W. Schafer, Pearson															
REFERENCE BOOKS:															
1.Speech and Audio Signal Processing, A.R. JAYAN, PHI															
2.Speech and Audio Processing, Apte Shaila D, Wiley India Pvt. Ltd															

Professional Elective Courses (Software Engineering & Development Stream)

INTERNET & WEB PROGRAMMING

Course Code:	CS2211-1	Course Type:	PEC	
Teaching Hours/Week (L: T: P):	3:0:0	Credits:	03	
Total Teaching Hours:	40+0+0	CIE + SEE Marks:	50+50	
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives:				
1.	Gain exposure to a basic website repository, including its directory structure and HTML document contents.			
2.	Configure in-line, internal, and external CSS stylesheets.			

3.	Configure your website to track pageviews and user events.															
UNIT-I																
Basics of HTML: Formatting text by using tags, Creating hyperlinks and anchors. Basics of CSS: CSS formatting text using style sheets, Creating tables, formatting tables, incorporate sound and video: audio and video in HTML5, HTML multimedia basics, embedding video clips, incorporating audio on web page.		13 Hours														
UNIT-II																
What is JavaScript, Syntax, usage of super global variables, PHP syntax and variables, comments passing information with PHP, Integrating web forms and databases, displaying queries in table. Functional Components in React JS, what is JSX? , Inline Styling with JSX in React JS, JavaScript Variable in JSX, Handling Forms		13 Hours														
UNIT-III																
What is Node JS?, Advantages of Node JS, Node JS HTTP Module, File System Module, Node js Events: The EventEmitter Object, Upload Files, Send an Email, Nosql database: Introduction to MongoDB, Creating a Database, Insert, Find, Sort, Delete, Drop Collection		14 Hours														
Suggested List of Experiments																
1.	Implement a Web site for Information Technology department Using a) Frameset b) Tables c) Internal Linking d) Headers e) List Items f) Hyperlink for mailing															
2.	Develop a JavaScript program to get Register Number as Input and print the Student's total mark and grades.															
3.	Write a HTML code that takes name and email as input and display the entered details using \$_GET array/ \$_POST array															
4.	Write Java script code that will change the colour of the background to red when the user brought the focus to the object and change the color to green when user has left the focus on the object.															
5.	Write a PHP code to Connect to Database, Create a table Department (Dname, Dno, Number_Of_faculty)															
6.	Write a PHP program to connect to database. Create a table named “Student” with following fields (sno, sname, percentage). Insert 3 records of your choice. Display the names of the students whose percentage is between 35 to 75 in a tabular format															
7.	Use the createServer() method to create an HTTP server															
8.	Use File system Module to perform following operations on file Read files - Create files - Update files - Delete files - Rename files															
9.	Example programs on Inline Styling with JSX in React JS, JavaScript Variable in JSX, Handling Forms															
10	Mongo DB database creation with queries															
Course Outcomes: At the end of the course student will be able to																
1.	Adapt HTML and CSS syntax and semantics to build web pages. Construct and visually format tables and forms using HTML and CSS															
2.	To learn Client-Side programs using JavaScript and Server-Side programs using PHP. Understand the principles of object oriented development using PHP, Inline styling with JSX															
3.	To learn creating methods on HTTP server and file operations using Node.JS MongoDB database creation and queries															
Course Outcomes Mapping with Program Outcomes & PSO																
	Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
	↓ Course Outcomes													1	2	3
	CS2211-1.1	1	2	1	2			2					2		1	
	CS2211-1.2	1	2				1						1			
	CS2211-1.3			1	3			1				1	2		2	
1: Low 2: Medium 3: High																

TEXTBOOKS:	
1.	Randy Connolly, Ricardo Hoar, "Fundamentals of Web Development", 1st Edition, Pearson Education India. (ISBN:978-9332575271)
REFERENCE BOOKS:	
1.	Robin Nixon, "Learning PHP, MySQL & JavaScript with jQuery, CSS and HTML5", 4th Edition, O'Reilly Publications, 2015. (ISBN:978-9352130153)
2.	Luke Welling, Laura Thomson, "PHP and MySQL Web Development", 5th Edition, Pearson Education, 2016. (ISBN:978-9332582736)
3.	Nicholas C Zakas, "Professional JavaScript for Web Developers", 3rd Edition, Wrox/Wiley India, 2012. (ISBN:978-8126535088)
E Books / MOOCs/ NPTEL	
1.	https://www.coursera.org/learn/front-end-react#syllabus https://www.udemy.com/course/react-the-complete-2021-guide-with-nodejs-and-mongo-db/

ADVANCED JAVA PROGRAMMING				
Course Code:	CS3211-1	Course Type	PEC	
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03	
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50	

Teaching Department: Computer Science & Engineering (Cyber Security)

Course Objectives:

1.	Develop networking Java applications.
2.	Apply the database concept for a Java database applications.
3.	Design server side web applications using Java Servlets.
4.	Develop server side web applications using Java Server Pages.

UNIT-I

REVISIT TO OOP CONCEPTS

15 Hours

Class, Object and Inheritance in Java. String buffer and string builders, Java beans, Introspection, Bean APIs, EJB concepts, Collection interfaces and Collection classes.

FILE HANDLING:

Serial Access Files, File Methods, Redirection, Command Line Parameters, Random Access Files.

UNIT-II

JAVA DATABASE CONNECTIVITY(JDBC):

15 Hours

The Vendor Variation Problem, SQL and Versions of JDBC, Creating an ODBC DataSource, Simple Database Access, Modifying the Database Contents, Transactions, MetaData, Scrollable ResultSets in JDBC 2.0, Modifying Databases via Java Methods.

NETWORK PROGRAMMING WITH JAVA:

Basic Concepts, Protocols and Terminology, Clients, Servers and Peers, Ports and Sockets, The Internet and IP Addresses, Internet Services, URLs and DNS, TCP, UDP. The InetAddress Class, Using Sockets (TCP and UDP).

UNIT-III

JAVA SERVLETS

10 Hours

Benefits, A simple Java Servlet, Anatomy of a Java Servlet, Reading data from a client, Reading HTTP Request Headers, Sending data to a client, working with Cookies, Tracking Sessions. JAVA SERVER PAGES (JSP): JSP Tags, Form handling in JSP, User Sessions, Cookies, Session objects.

Course Outcomes: At the end of the course student will be able to	
1.	Apply the knowledge of Java Programming to demonstrate the OOP Concepts.
2.	Demonstrate the file handling using JAVA.
3.	Develop Java Program to store and retrieve data from the database.
4.	Apply Java Programming to establish Network Connectivity also Demonstrate TCP and UDP sockets.
5.	Communicate and function effectively with Modern Engineering and IT.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes → ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12	PSO ↓		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CS3211-1.1	3	2											2	1	
CS3211-1.2	2	3											2	2	
CS3211-1.3	2	3											2	2	
CS3211-1.4	3	3											2	2	
CS3211-1.5	3	3											2	1	

1: Low 2: Medium 3: High

TEXTBOOKS:

- Herbert Schildt, "The Complete Reference Java by Seventh Edition", Tata McGraw-Hill, 2007
- Jan Graba, "An Introduction to Network Programming with Java", Springer Publications, 2007
- Jim Keogh, "The Complete Reference J2EE", Tata McGraw-Hill, 2002.

REFERENCE BOOKS:

- H.M. Deitel, "Java – How to Program?", Prentice Hall, 2004.

E Books / MOOCs / NPTEL

- <http://www.mindview.net/Books/TIJ>
- <http://docs.oracle.com/javase/specs/jls/se8/html/index.html>

AGILE TECHNOLOGY

Course Code:	CS3212-1	Course Type:	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits:	03
Total Teaching Hours:	40+0+0	CIE + SEE Marks:	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
1.	Explain the principles involved in agile technologies		
2.	Explain XP lifecycle		
3.	Understand concepts of XP practices, collaborating in agile methods		
4.	Understand concepts of releasing, planning and developing in agile methods		
5.	Explain the process of mastering agility.		

Security

UNIT-I																
Why Agile?														15 Hours		
Understanding Success, Beyond Deadlines, The Importance of Organizational Success, Enter Agility, How to Be Agile?: Agile Methods, Don't Make Your Own Method, The Road to Mastery, Find a Mentor Understanding XP: The XP Lifecycle, The XP Team, XP Concepts, Adopting XP: Is XP Right for Us?, Go!, Assess Your Agility(T1:1,2,3,4)																
UNIT-II																
Practicing XP														15 Hours		
Thinking: Pair Programming, Energized Work, Informative Workspace, RootCause Analysis, Retrospectives, Collaborating: Trust, Sit Together, Real Customer Involvement, Ubiquitous Language, Stand-Up Meetings, Coding Standards, Iteration Demo, Reporting, Releasing: "Done Done", No Bugs, Version Control, Ten-Minute Build, Continuous Integration, Collective Code Ownership, Documentation. Planning: Vision, Release Planning, The Planning Game, Risk Management, Iteration Planning, Slack, Stories, Estimating. Developing: Incremental requirements, Customer Tests, Test-Driven Development, Refactoring, Simple Design, Incremental Design and Architecture, Spike Solutions, Performance Optimization, Exploratory Testing(T1:5,6,7,8,9)																
UNIT-III																
Mastering Agility: Values and Principles														10 Hours		
Commonalities, About Values, Principles, and Practices, Further Reading, Improve the Process: Understand Your Project, Tune and Adapt, Break the Rules, Rely on People: Build Effective Relationships, Let the Right People Do the Right Things, Build the Process for the People, Eliminate Waste :Work in Small, Reversible Steps, Fail Fast, Maximize Work Not Done, Pursue Throughput(T1:10,11,12,13)																
Course Outcomes: At the end of the course student will be able to																
1.	Explain XP Lifecycle															
2.	Illustrate the process of adopting XP															
3.	Interpret XP practices and explain collaborating in agile technologies.															
4.	Demonstrate the process of releasing, planning and developing in agile technologies.															
5.	Explain the concept of improving the process, eliminate waste.															
Course Outcomes Mapping with Program Outcomes & PSO																
	Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS3212-1.1		1	2											1	2	
CS3212-1.2		1	3											1	2	
CS3212-1.3		1	3											1	2	
CS3212-1.4		1	3											1	2	
CS3212-1.5		1	3											1	2	
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.	The Art of Agile Development James shore, Chromatic, O'Reilly 2007															
REFERENCE BOOKS:																
1.	Agile Software Development, Principles, Patterns, and Practices Robert C. Martin Prentice Hall 1st edition, 2002															
2.	Agile and Iterative Development A Manger's Guide Craig Larman Pearson Education First Edition, India, 2004															
3.	John R. Levine, Tony Mason, Doug Brown, "LEX and YACC", O'Reilly Publication,1999.															
E Books / MOOCs/ NPTEL																
1.	https://www.udemy.com/course/agile-fundamentals-scrum-kanban-scrumban/															

GAME THEORY AND APPLICATIONS

Course Code:	CS3213-1	Course Type	PEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50

Teaching Department: Computer Science & Engineering (Cyber Security)
UNIT-I
Introduction; Strategic Games
15 Hours

What is game theory? Four elements, Classification of games, The theory of rational choice; Interacting decision makers, Strategic games; Example: The prisoner's dilemma; Nash equilibrium; Examples of Nash equilibrium; Best- response functions; Dominated actions; Equilibrium in a single population: symmetric games and symmetric equilibria, Interpretation of Nash Equilibrium

Mixed Strategy Equilibrium

Introduction; Strategic games in which players may randomize; Mixed strategy Nash equilibrium; Finding mixed strategy by graphical method; Finding mixed strategy by analyzing subset of all actions; Dominated actions; Pure equilibria when randomization is allowed, examples; The formation of players beliefs; Eliminating dominated actions, Median Voter theorem.

UNIT-II
Extensive Games
15 Hours

Extensive games with perfect information; Strategies and outcomes; Nash equilibrium; Subgame perfect equilibrium; Finding subgame perfect equilibria of finite horizon games.

Extensions

Allowing for simultaneous moves, examples, Discussion: subgame perfect equilibrium and backward induction.

Strictly Competitive Games & Maximization

Maximization and Nash equilibrium; Strictly Competitive Games; Maximization and Nash equilibrium in strictly competitive games

Rationalizability

Iterated elimination of strictly dominated actions; Iterated elimination of weakly dominated actions; Dominance solvability.

UNIT-III
Applications Of Game Theory
10 Hours

Assumptions and issues in Game theory, Mechanism design problem and examples ,game theory and cryptography ,game theory and wireless Adhocnetworks ,game theory and network security, Pareto optimal, Selfish routing, Correlated equilibrium

Course Outcomes: At the end of the course student will be able to

1.	Apply the concept of strategic games on the use cases.
2.	Analyze the mixed strategy equilibrium and Apply in the Games.
3.	Apply the concepts of extensive games and extensions in solving the problems.
4.	Apply the concepts of strictly competitive games, maximization and rationalizability on the use cases.
5.	Describe the applications of game theory in different domains.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3213-1.1	3	3											3	2	
CS3213-1.2	3	3											3	2	
CS3213-1.3	3	3											3	2	
CS3213-1.4	3	3											3	2	
CS3213-1.5	3	1											3	2	

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Martin Osborne: An introduction to game theory, Oxford University Press, Indian Edition, 2004.
2.	An Introduction to Game Theory: Strategy, Joel Watson, W W Norton and Company. Algorithmic Game Theory, Noam Nisan, Tim Roughgarden, Eva Tardos, Vijay V Vazirani, Cambridge University Press.
REFERENCE BOOKS:	
1.	Roger B Myerson: Game theory: Analysis of Conflict, Harvard University Press, 1997.

NoSQL DATABASE																
Course Code:					CS3214-1			Course Type					PEC			
Teaching Hours/Week (L: T: P):					3:0:0			Credits					03			
Total Teaching Hours					40+0+0			CIE + SEE Marks					50+50			
Teaching Department: Computer Science & Engineering (Cyber Security)																
Course Objectives:																
The primary Course Learning Objective is to Prepare the background in abstraction, notation and critical thinking of mathematics related to computer science. The course will enable students to:																
1.	Understand the importance of NoSQL data management and compare with traditional relational database management system.															
2.	Understand the CAP theorem and compare with ACID properties of traditional relational database management system.															
3.	Understand the basics of MongoDB and Cassandra NoSQL database management systems.															
4.	Develop queries to store and retrieve the data using MongoDB NoSQL database management systems.															
5.	Develop queries to store and retrieve the data using Cassandra NoSQL database management systems.															
UNIT-I																
Introduction to NoSQL															15 Hours	
Why NoSQL? Types of NoSQL databases, Distribution models: single server, sharding, Master-Slave Replication, Peer-to-Peer Replication, Combining Sharding and Replication, Consistency: Update Consistency, Read Consistency, The CAP Theorem, MapReduce: Partitioning and Combining, Composing MapReduce calculations.																
UNIT-II																
Introduction to MongoDB															15 Hours	
What is MongoDB? Why MongoDB?, JSON, Creating Unique Key, Storing Binary data, Terms used in RDMS and MongoDB, Data types in MongoDB, MongoDB Query Language: Insert method, Save method, Update method, Remove method, Find method, Dealing with Null values, Count, Limit, Sort, Skip, Arrays, Aggregate Functions.																
UNIT-III																
Introduction to Cassandra															10 Hours	
Features of Cassandra, CQL data types, CRUD (Create, Update, Read and Delete) operations, Collections: Set collection, List collection, Map collection, Set and List, Map, Alter commands, Import and Export.																
Course Outcomes: At the end of the course student will be able to																
1.	Understand the importance of NoSQL data management and compare with traditional relational database management system.															
2.	Understand the CAP theorem and compare with ACID properties of traditional relational database management system.															
3.	Understand the basics of MongoDB and Cassandra NoSQL database management systems.															
4.	Develop queries to store and retrieve the data using MongoDB NoSQL database management systems.															
5.	Develop queries to store and retrieve the data using Cassandra NoSQL database management systems.															
Course Outcomes Mapping with Program Outcomes & PSO																
	Program Outcomes→			1	2	3	4	5	6	7	8	9	10	11	12	PSO↓

													Security		
↓ Course Outcomes													1	2	3
CS3214-1.1	3									1		1	3		
CS3214-1.2	3									1		1	3		
CS3214-1.3	3									1		1	3		
CS3214-1.4	3									1		1	3		
CS3214-1.5	3	2								1		1	3		
1: Low 2: Medium 3: High															
TEXTBOOKS:															
1.	Pramod J. Sadalage, Martin Fowler. “NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence”. 1st Edition, Addison-Wesley, 2012.														
2.	Seema Acharya, Subhashini Chellappan, “Big Data Analytics”, 1st Edition, Wiley, 2015.														
REFERENCE BOOKS:															
1.	Deepak Vohra, Nosql Web Development with Apache Cassandra, Cengage Learning, Inc; New edition, 2015.														
2.	Doug Bierer, MongoDB 4 Quick Start Guide: Learn the skills you need to work with the world's most popular NoSQL database, Packt, 2018.														
E Books / MOOCs/ NPTEL															
1.	https://www.mongodb.com/nosql-explained														
2.	https://cassandra.apache.org/_/index.html														

COMPETITIVE PROGRAMMING			
Course Code:	CS3215-1	Course Type	PEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
• Introduce the competitive programming. • Cover in details the algorithms. • Cover in details the data structure. • Make students aware of approaches applied at the world competitions.			
UNIT-I			
Recursion, Arrays and Arraylists, Linked Lists			15 Hours
Last non-zero digit of the factorial, Pascal Triangle, Two people meet each other, Next largest number, Count number of increasing subsequences. Minimum Swaps to get the sorted Array, Printing smallest k elements in the same order, Find duplicate and missing elements in O(n), Print the matrix in the spiral form, Implement own ArrayList using arrays, Minimum Value of "Max + Min" in a Subarray. Last nth Node of the Linked List, Remove a Loop in a Singly Linked List, Find the middle element of a linked list, Subtract two numbers, Flatten the Given Linked List, Find the Intersection Point.			
UNIT-II			
Strings, Stacks and Queues, Sets and Hashtables			15 Hours
Longest substring, Reverse the individual words of the string, cUSTOM cASE of the given String, Decimal Number to Roman Numeral, Longest Palindromic Substring, Reverse the string. Implement a stack using one queue, Find kth largest element, Implement queue using stack, Detect duplicate parenthesis, Reverse a stack, Identify a palindromic string, The annual weather report, Maximum of subarrays of size k, Optimize stack operations. Four Elements Such That a+b = c+d, Find Symmetric Pairs, Check Array of Contiguous Integers			

Section 1

UNIT-III																
Trees, Priority Queues and Heaps,Graphs and Graph Algorithms														10 Hours		
Left view of a Binary Tree, Flatten a Binary tree, The sum from root to leaf, Lowest common ancestor in a BST, Spiral order traversal, The sum from root to leaf.																
Merge K sorted linked list, Maximum Sum Pairs, Median of a stream of integers.																
Check Whether the Graph Is Bipartite, Shortest Distance Between Every Pair, Detect Loop in a Directed Graph																
Course Outcomes: At the end of the course student will be able to																
	1. Understand and use most common algorithms for competitive programming.															
	2. Analyze data structures for competitive upsolving.															
	3. Solve programming contest tasks related to strings.															
	4. Understand and analyse the algorithms of basic data structures															
	5. Understand and solve most common algorithms of tree and graph data structures															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS3215-1.1		2	2										2		2	
CS3215-1.2		2	2		2				2				2		3	
CS3215-1.3		2	2						2				2		2	
CS3215-1.4		2	2		2				2				2		2	
CS3215-1.5		2	2										2		2	
1: Low 2: Medium 3: High																
TEXTBOOKS:																
	1. Aaron M. Tenenbaum, YedidyahLangsam& Moshe J. Augenstein, “Data Structures using C”, Pearson Education/PHI, 2009.															
	2. Anany Levitin, “Introduction to the Design & Analysis of Algorithms”, 2 nd Edition, Pearson Education, 2011.															
REFERENCE BOOKS:																
	1. Horowitz E., Sahni S., Rajasekaran S, “Computer Algorithms”, Galgotia Publications, 2001.															
	2. R.C.T. Lee, S.S. Tseng, R.C. Chang &Y.T.Tsai, “Introduction to the Design and Analysis of Algorithms A Strategic Approach”, Tata McGraw Hill, 2005.															

ANGULAR AND REACT JS			
Course Code:	CS2311-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
UNIT-I			
INTRODUCTION TO ANGULARJS			15 Hours
AngularJS Overview, AngularJS MVC Architecture, AngularJS Expressions, Numbers, Strings, Objects, Arrays, AngularJS Modules, AngularJS Directives, AngularJS Model, Data Binding, AngularJS Controllers, Repeating HTML Elements, AngularJS Scope, AngularJS Filters, AngularJS Services.			

Security

UNIT-II																
INTRODUCTION TO REACTJS													15Hours			
Introduction to ReactJS, ReactJS vs AngularJS, Pros and Cons of ReactJS, React Components, React Class, React State, React Props, React Constructor, React Forms, React Events, React List, React Keys, React Fragments.																
UNIT-III																
INTRODUCTION TO REACT ES6, REACT JSX AND REACTDOM													10 Hours			
ES6 Variables and Arrays, ES6 Classes, ES6 Modules, React Elements as JSX, JSX Attributes, JSX Styling, Expression in JSX, Introduction to ReactDOM, Introduction to Babel.																
Course Outcomes																
1.	Explain the fundamentals of AngularJS and its Model-View-Controller (MVC) architecture, along with the concepts of data binding, expressions, modules, and directives															
2.	Compare and contrast ReactJS with AngularJS, analyze the advantages and disadvantages of using ReactJS, and gain proficiency in creating React components, handling state,															
3.	Demonstrate advanced features of ReactJS using ES6 syntax, including variables, arrays, classes															
4.	Perform hands-on experiments with AngularJS controllers, filters, services,															
5.	Develop practical skills in utilizing ReactDOM to render React elements into the DOM															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS2311-1.1		3	1	3		1							2	2	3	
CS2311-1.2		3	1	3		2							2		3	
CS2311-1.3		2	1	3		3							3		3	
CS2311-1.4		3	1	3		2							2	2	3	
CS2311-1.5		3	2	3		3							2		3	
1: Low 2: Medium 3: High																
TEXTBOOKS																
1. Deshmukh & Kothawade, Angular JS, 2 nd Edition, Nirali Prakashan.																
2. Vipul A M, Prathamesh Sonpatki, React JS by Example-Building Modern web Applications with React.																

USER INTERFACE DESIGN				
Course Code:	CS2312-1	Course Type	PEC	
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03	
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50	
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives:				
1.	To study the concept of menus, windows, interfaces			
2.	To study about user interface design, pitfalls, commandments of design and usability aspects.			
3.	To study system menus and windows and layouts for effective communication.			
4.	To study the psychology of users and their action			

Security

5.	To study about constraints, discoverability of features, feedback system and detecting and reporting error.
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UNIT-I	
The User Interface	15 Hours
Introduction, Overview, The importance of user interface Defining the user interface, The importance of Good design, benefits of good design, Introduction to graphical user interface, A brief history of screen design. The User Interface Design process- Obstacles, Usability, Important human characteristics in design,Human considerations in Design, Business definition and requirement analysis.	

UNIT-II	
System menus	15 Hours
Structures of menus, Functions of menus, Contents of menus, Formatting of menus, Phrasing the menu, Kinds of graphical menus. Windows - Characteristics, Components of window, Window presentation styles, Types of window, Window management, Window operations, Characteristics of device based controls. Screen based controls- Operable control, Text control, Selection control Provide effective feedback and Guidance assistance, Organize and Layout Windows and pages	

UNIT-III	
The psychology of everyday actions	10 Hours
How people do things: The gulfs of execution and evaluation, The seven stages of action,The seven stages of action and The three levels of processing , Seven fundamental design principles. Knowing what to do: constraints, discoverability, and feedback: four kinds of constraints: physical, cultural, semantic, and logical, constraints that force the desired behaviour, conventions the faucet: a case history of design, using sound as signifiers. Human error? no, bad design: understanding why there is error, two types of errors: slips and mistakes, social and institutional pressures, reporting error, detecting error, designing for error, design principles for dealing with error.	

Course Outcomes: At the end of the course student will be able to	
1.	Explain various Graphical User Interface design aspects & measures to improve them.
2.	Summarise the basic principles of user interface and design
3.	Outline the differences between usability and user experience and the need for human factors in design.
4.	Infer user-interface design process and introducing common design scenarios.
5.	To understand the user psychology and enunciate the design principles from psychological aspects.

Course Outcomes Mapping with Program Outcomes & PSO																
↓ Course Outcomes	Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
														1	2	3
CS2312-1.1		3	3							1	1		1	1		
CS2312-1.2		2	3							1	1		1	1		
CS2312-1.3		1	3	2						1	1		1	3	2	
CS2312-1.4		1	3	2						1	1		1	3	2	
CS2312-1.5			2	2						1	1		1	1	2	
1: Low 2: Medium 3: High																

TEXTBOOKS:	
1.	The Essential Guide to User Interface Design, Wilbert O. Galitz, 3rd Edition, 2007, John Wiley & Sons, Inc., ISBN: 0470146222.
2.	The design of Everyday Things, Don Norman, 2013, Basic Books Publication, ISBN: 978-0-465-00394-5.

E Books / MOOCs/ NPTEL	
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- | | |
|----|---|
| 1. | Coursera course: User Interface Design Specialization by Loren Terveen(16 weeks) |
|----|---|

FULL STACK DEVELOPMENT

Course Code:	CS3311-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50

Teaching Department: Computer Science & Engineering (Cyber Security)

Course Objectives:

1.	Design static and modern web pages using HTML5, Cascading Style Sheets (CSS) and Bootstrap.
2.	Develop client-side validations using JavaScript.
3.	Develop the server-side script using PHP.
4.	Design the server-side database using MySQL
5.	Develop the interactive web application using NodeJS framework and MongoDB.

UNIT-I

BASICS OF HTML5, CSS AND JAVASCRIPT **15 Hours**

Overview of HTML5, HTML5 elements, Introduction to CSS, Levels of style sheets, The Box Model, The basics of Javascript, General syntactic characteristics, Event Handling.

Title: BOOTSTRAP:

What is Bootstrap? Why use Bootstrap? Where to get Bootstrap? Bootstrap CDN, First Web Page with Bootstrap, Bootstrap Grid system, Contextual Colors and Backgrounds, Bootstrap Tables, Bootstrap Images, Bootstrap Jumbotron and Page Header, Bootstrap Wells, Bootstrap Alerts, Bootstrap Buttons, Bootstrap Badges and Labels, Bootstrap Progress Bars, Bootstrap List Groups, List Group With Badges, Tabs, Tabs With Dropdown Menu, Pills, Bootstrap Navigation Bar, Bootstrap Forms, Bootstrap Form Inputs, Bootstrap Media Objects, Bootstrap Carousel Plugin.

UNIT-II

INTRODUCTION TO PHP **15 Hours**

Origins and uses of PHP, Overview of PHP, General syntactic characteristics, Primitives, Operations and Expressions, Output, Control statements, Arrays, Functions, Pattern Matching, Form handling, Files, Cookies, Session tracking, Database access with PHP and MySQL.

UNIT-III

NodeJS: **10 Hours**

Introduction to Node.js- Installing Node.js - Node.js Modules, Node.js File System, Node.js URL Module, Node.js NPM, Node.js Events, Node.js Upload Files
Node.js Email.

NodeJS MySQL -

Create Database, Create Table, Insert into, select from, Where, Order by, Delete, Drop Table, Update, Limit, Join.

Introduction to Mongo DB- Node.js MongoDB, Create Database, Create Collection, Insert, Find, Query, Sort, Delete, Drop Collection, Update, Limit, Join.

Course Outcomes: At the end of the course student will be able to

1.	Describe the fundamental features of HTML5, CSS and Bootstrap and Design static web pages.
2.	Design and Implement the client-side validations using JavaScript.
3.	Illustrate the concept of PHP and Develop the server-side script using PHP.
4.	Design the server-side database using MySQL
5.	Develop the interactive web application using NodeJS framework and MongoDB.

Security

Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓			
↓ Course Outcomes													1	2	3	
CS3311-1.1	1	2	3											3		
CS3311-1.2	2	3												3		
CS3311-1.3	2	3												3		
CS3311-1.4	1	2	3											3		
CS3311-1.5	1	2												3		
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.	Robert W. Sebesta, “Programming the World Wide Web”, Fourth Edition, Pearson, 2014.															
	Jake Spurlock, “Bootstrap-Responsive Web Development”, O’Reilly publications, 2013.															
2.	Ari Lerner, Ng-book, “The complete book on Angular JS”, 2013.															
3.	Chris Northwood, The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer Paperback – Import, 20 November 2018.															
4.	David Herron, Node.js Web Development: Server-side web development made easy with Node 14 using practical examples, 5th Edition Edition, 2020															
REFERENCE BOOKS:																
1.	M. Deitel, P.J. Deitel, A. B. Goldberg,” Internet & World Wide Web: How to Program, 4e Paperback – 1 January 2009.															
2.	Chris Bates,”Web Programming Building Internet Applications”, Third Edition, Wiley India, 2006															
E Books / MOOCs/ NPTEL																
1.	https://www.cs.uct.ac.za/mit_notes/web_programming.html															
2.	http://www.multitech.ac.ug/uploads/IntroductiontoWebProgramming.pdf															
3.	https://www.w3schools.com/php/															
4.	https://www.w3schools.com/bootstrap/															
5.	https://www.w3schools.com/nodejs/															

OBJECT ORIENTED MODELLING AND DESIGN			
Course Code:	CS3312-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
1.	Recall the object-oriented concepts, three pillars of object-orientation and their benefits.		
2.	Illustrate the various models that can be used to demonstrate the objectoriented		

	design of any real world software systems.
3.	Make use of use-cases for interpreting the requirements and develop class diagrams that model both the domain state model and design model of a software system.
4.	Examine the dynamic aspects of a software system, model the interaction diagrams to justify those aspects.
5.	Relate how the UML constructs are used to represent various models.
UNIT-I	
Introduction:	15 Hours
What is Object Orientation? What is OO development? OO themes; Evidence for usefulness of OO development; OO modeling history.	
Modeling as Design Technique:	
Modeling; abstraction; The three models. Class Modeling: Object and class concepts; Link and associations concepts; Generalization and inheritance; A sample class model; Navigation of class models; Practical tips.	
Advanced Class Modeling:	
Advanced object and class concepts; Association ends; N-ary associations; (Text Book-1: Chapter 1 to 4.3)	
Advanced Class Modeling:	
Aggregation; Abstract classes; Multiple inheritance; Metadata; Reification; Constraints; Derived data; Packages; Practical tips.	
State Modeling:	
Events, States, Transitions and Conditions; State diagrams; State diagram behavior	
UNIT-II	
Advanced State Modeling:	15 Hours
Advanced State Modeling: Nested state, diagrams; Nested states; Signal generalization; Concurrency; A sample state model; Relation of class and state models	
Interaction Modeling:	
Use case models; Sequence models; Activity models; Use case relationships; Procedural sequence models; Special constructs for activity models (Text Book-1: Chapter 4.4,5,6 ,7)	
Process Overview, System Conception:	
Development stages; Development life cycle, Devising a system concept; Elaborating a concept; Preparing a problem statement.	
Domain Analysis:	
Overview of analysis; Domain class model; Domain state model; Domain interaction model; Iterating the analysis.	
Application Analysis:	
Application Analysis: Application interaction model; Application class model; Application state model; Adding operations. (Text Book-1: Chapter 10,11,12, 13)	
UNIT-III	
System Design:	10 Hours
Overview; Estimating performance; Making a reuse plan; Breaking a system in to sub- systems; Identifying concurrency; Allocation of sub-systems; Management of data storage; Handling global resources; Choosing a software control strategy; Handling boundary conditions; Setting the trade-off	

priorities; Common architectural styles; Architecture of the ATM system as the example.

Class Design:

Overview of class design; Bridging the gap; Realizing use cases; Designing algorithms; Re-cursing downwards, Re-factoring; Design optimization; Reification of behavior; Adjustment of inheritance; Organizing a class design; ATM example
 (Text Book-1: Chapter 14 and 15)

Course Outcomes: At the end of the course student will be able to

1.	Acquire Knowledge about different software systems modelling techniques, class design and associations by making use of concept diagrams.
2.	Illustrate Advanced Class, State and Interaction models of software systems utilizing class, state and interaction diagrams.
3.	Outline the system concepts, Development Life Cycle, Analyse and Define Problem Statement, Analyse the system domain, application, class, state and interaction models.
4.	Overview of system design, estimate performance, divide it into subsystems, managing resources, selecting appropriate architectural styles.
5.	Describe class design, Implementation modelling, Legacy systems and Reverse engineering concepts, realizing use cases, associations, Fine Tuning Classes, Constructing Interaction and State models.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes →	1	2	3	4	5	6	7	8	9	10	11	12	PSO ↓		
↓ Course Outcomes													1	2	3
CS3312-1.1	2	3			1							1	2		
CS3312-1.2	2	2	3		1							1	2		
CS3312-1.3	2	3			1							1	2		
CS3312-1.4	2	2	3		1							1	1		
CS3312-1.5	1	3			1							1	1		

1: Low 2: Medium 3: High

TEXTBOOKS:

- Object-Oriented Modeling and Design with UML, Michael Blaha, James Rumbaugh, 2nd Edition, Pearson Education, 2005

REFERENCE BOOKS:

- Object-Oriented Analysis and Design with Applications, Grady Booch et al, 3rd Edition, Pearson Education, 2007.
- Practical Object-Oriented Design with UML, Mark Priestley, 2nd Edition, Tata McGraw-Hill, 2003.
- Object-Oriented Design with UML and JAVA, K. Barclay, J. Savage, Elsevier, 2008.
- The Unified Modeling Language User Guide, Booch, G., Rumbaugh, J., and Jacobson I, 2nd Edition, Pearson, 2005.
- Object-Oriented Systems Analysis and Design Using UML, Simon Bennett, Steve McRobb and Ray Farmer, 2nd Edition, Tata McGraw-Hill, 2002.

E Books / MOOCs/ NPTEL

- Object-Oriented Analysis and Design with Applications, Grady Booch, Robert A. Maksimchuk, Michael W. Engel, Bobbi J. Young, Jim Conallen, Kelli A. Houston, Third Edition The Addison-Wesley Object Technology Series, 2007.
- Object-Oriented Modeling and Design with UML, James R Rumbaugh, Michael R. Blaha Pearson Education, 21-Nov-2011
- Object-Oriented Analysis and Design, Ramnath, Sarnath, Dathan, Brahma, ISBN 978-1-84996-522-4, Springer Publications, 2011.
- Object-Oriented Design, <https://www.coursera.org/learn/object-oriented-design> Object-Oriented Analysis and Design, <https://nptel.ac.in/courses/106/105/106105153/#>

SOFTWARE ARCHITECTURE AND DESIGN PATTERNS				
Course Code:		CS3313-1	Course Type	PEC
Teaching Hours/Week (L: T: P):		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives:				
This Course will enable students to:				
1.	Outline the various architectural influences and its qualities on the organizational requirements.			
2.	Make use of different case studies to critically evaluate the suitability of a software architecture.			
3.	Develop the architecture using different architecture styles.			
4.	Choose the different architectural pattern and design patterns to design the architecture that enhances the architectural capabilities.			
5.	Document the software architecture to communicate the system evolution strategy to the stakeholder.			
UNIT-I				

INTRODUCTION:													15 Hours			
The Architecture Business Cycle: Where do architectures come from? Software processes and the architecture business cycle; What makes a “good” architecture? What software architecture is and what it is not; Other points of view; Architectural patterns, reference models and reference architectures; Importance of software architecture; Architectural structures and views (<i>Text Book-1: Chapter 1: 1.1,1.2,1.3, Chapter 2: 2.1,2.2,2.3,2.4,2.5</i>)																
QUALITY: Functionality and architecture; Architecture and quality attributes; System quality attributes; Quality attribute scenarios in practice; Other system quality attributes; Business qualities; Architecture qualities. Achieving Quality: Introducing tactics; Availability tactics; Modifiability tactics; Performance tactics; Security tactics; Testability tactics; Usability tactics. (<i>Text Book-1: Chapter 4: 4.1, 4.2,4.3,4.4,4.5,4.6,4.7, Chapter 5:5.1,5.2,5.3,5.4,5.5,5.6, 5.7</i>).																
UNIT – II																
ARCHITECTURAL STYLES AND CASE STUDIES:													15 Hours			
Architectural styles; Pipes and filters; Data abstraction and object-oriented organization; Event-based, implicit invocation; Layered systems; Repositories; Interpreters; Process control; Other familiar architectures; Heterogeneous architectures. Case Study: Mobile robotics. (<i>Text Book-2: Chapter 2: 2.1, 2.2,2.3,2.4,2.5,2.6,2.7,2.8,2.9,2.10, Chapter 3:3.3</i>)																
ARCHITECTURAL PATTERNS: Introduction , Distributed Systems: Broker; Interactive Systems: MVC, Presentation-Abstraction-Control. Adaptable Systems: Microkernel. (<i>Text Book-2: Chapter 2: 2.1, 2.3,2.4,2.5</i>)																
UNIT-III																
DESIGNING AND DOCUMENTING SOFTWARE ARCHITECTURE:													10 Hours			
Architecture in the life cycle; designing the architecture; Forming the team structure; Creating a skeletal system. Uses of architectural documentation; Views; choosing the relevant views; Documenting a view; Documentation across views. (Text Book-1: Chapter 7: 7.1, 7.2,7.3,7.4, Chapter 9: 9.1,9.2,9.3,9.4,9.5)																
Course Outcomes:																
Upon Completion of this course students will be able to:																
1.	Identify the requirements which influence the architecture and development strategy.															
2.	Analyze the architecture using different case studies and quality attributes.															
3.	Recognize architecture styles to design the architecture.															
4.	Apply different architecture patterns and design patterns to develop architecture that yields the system that has new organizational capabilities and requirements.															
5.	Describe the different views to document the architecture.															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS3313-1.1		2	2											2	3	
CS3313-1.2		2	2	3										2	3	
CS3313-1.3		2	2												3	
CS3313-1.4		1	2	3											3	
CS3313-1.5		2	3												3	
3: Substantial (High)		2: Moderate (Medium)						1: Poor (Low)								
TEXTBOOKS:																
1.	Len Bass, Paul Clements, Rick Kazman,“Software Architecture in Practice”,SecondEdition, Pearson Education, 2003.															

2.	Frank Buschmann, Regine Meunier, Hans Rohnert, Peter Sommerlad, Michael Stal, "Pattern-Oriented Software Architecture", A System of Patterns -Volume 1 , John Wiley and Sons, 2006.
3.	Mary Shaw and David Garlan, "Software Architecture-Perspectives on an Emerging Discipline", Prentice-Hall of India, 2007
REFERENCE BOOKS:	
1.	E. Gamma, R. Helm, R. Johnson, J. Vlissides, "Design Patterns- Elements of Reusable Object-Oriented Software ", Addison- Wesley, 1995
E Books / MOOCs/ NPTEL	
1.	http://www.hillside.net/patterns/
2.	https://www.cs.cmu.edu/afs/cs/project/vit/ftp/pdf/intro_softarch.pdf
3.	https://www.ics.uci.edu/~yuzok/software-architecture.html
4.	http://www.nptel.ac.in/syllabus/106104027/
5.	https://www.coursera.org/learn/software-architecture
	SEE Scheme: There will be 8 questions of 20 marks each in the question paper divided into 3 Units as per the syllabi & contact hours and the student will have to answer 5 full questions, selecting 2 full questions from Unit - I & Unit – II and 1 full question from Unit – III .

SOFTWARE TESTING				
Course Code:		CS3314-1	Course Type	PEC
Teaching Hours/Week (L: T: P):		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives: This Course will enable students to:				
1.	Explain the concept of testing and the testing lifecycle.			
2.	Use the testing frameworks, process and test management to generate the test plans.			
3.	Generate the test plans for a business.			
4.	Illustrate the use of automation in testing.			
5.	Perform defect management and data management.			
UNIT-I				
INTRODUCTION TO TESTING – WHY AND WHAT:				15 Hours
Why is testing necessary? What is testing? Role of Tester, Testing and Quality, Overview of STLC SOFTWARE TESTING LIFE CYCLE – V MODEL: SDLC vs STLC, different stages in STLC, document templates generated in different phases of STLC, different levels of testing, different types of testing: Functional Testing, API Testing, Usability Testing, Exploratory Testing, Ad-hoc Testing. Static Testing:Static techniques, reviews, walkthroughs. BASICS OF TEST DESIGN TECHNIQUES: Various test categories, test design techniques for different categories of tests. Designing test cases using MS-Excel.				
UNIT-II				
TEST MANAGEMENT:				15 Hours
Documenting test plan and test case, effort estimation, configuration management, project progress management. Use of Testopia for test case documentation and test management. DEFECT MANAGEMENT: Test Execution, logging defects, defect lifecycle, fixing / closing defects. Use of Bugzilla for logging and tracing defects. TEST DATA MANAGEMENT: Test Data Management –Overview, Why Test Data Management, Test Data Types, Need for Test Data Setup, Test Data Setup Stages, Test data management Challenges. Creating sample test data using MS-Excel.				
UNIT-III				
BASICS OF AUTOMATION TESTING:				10 Hours
Introduction to automation testing, why automation, what to automate, tools available for automation testing. BASICS OF AUTOMATION TESTING USING SELENIUM: Introduction to Selenium, using Selenium IDE for automation testing, using Selenium Web driver for automation testing, understanding TestNG framework with Selenium Web driver for automation testing, Introduction to Maven automation tool.				
Course Outcomes: Upon Completion of this course students will be able to:				
1.	Apply the knowledge of engineering to understand the various terms and techniques used in testing domain.			

2.	Identify the different phases of software testing life cycle and types of testing.
3.	Analyze test management and test data management processes
4.	Analyze defect management life cycle and use open source tool for defect management
5.	Design test case and formulate automation testing with demonstration of open source testing tool.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3314-1.1	2	2	1										2	1	
CS3314-1.2	2	2	1										2	2	
CS3314-1.3	2	3	1										2	2	
CS3314-1.4	2	3	2										2	2	
CS3314-1.5	2	2	2										2	2	

1: Low 2: Medium 3: High

Graduate Attributes (GA)

This course will map the following GA as per NBA:

1. Engineering Knowledge
2. Design / development of solutions
3. Conduct investigation of complex problems
4. Modern tool usage
5. The engineer and society
6. Ethics
7. Life-long Learning

TEXTBOOKS:

1. Rex Black , “Managing the Testing Process”, 2nd edition, John Wiley & Sons, 2001
2. Dorothy Graham, Erik van Veenendaal, Isabel Evans, Rex Black,” Foundations of software testing”, Cengage Learning EMEA, 2008.
3. Elfriede Dustin, “Implementing Automated Software Testing: How to Save Time and Lower Costs While Raising Quality”, Addison-Wesley Professional, 2009

REFERENCE BOOKS:

1. Paul C. Jorgensen,” Software Testing, A Craftsman’s Approach”, Third Edition, Auerbach Publications, 2008
2. Mauro Pezze, Michal Young,” Software Testing and Analysis –Process”, Principles and Techniques, Wiley India, 2009.

E Books / MOOCs/ NPTEL

1. <https://www.softwaretestinghelp.com/selenium-tutorial-1/>
2. <http://softwaretestingfundamentals.com/software-testing-methods/>
3. https://www.tutorialspoint.com/software_testing/software_testing_tutorial.pdf
4. <http://docs.seleniumhq.org/docs/>
5. <http://www.seleniumhq.org/download/>
6. <http://nptel.ac.in/courses/106105150/>
7. <https://freevideolectures.com/course/3625/testing-with-selenium>

Professional Elective Courses (Systems, Networks & Security Stream)

SYSTEM MODELLING & SIMULATION			
Course Code:	CS2221-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives: This Course will enable students to:			
1.	Describe the appropriateness of the simulation, its application, types of simulation model, steps in simulation study and general principles in simulation and concepts in discrete-event simulation.		
2.	Describe the generation of random numbers and pseudo-random numbers and apply techniques for generating random numbers.		
3.	Illustrate and apply the techniques of random variate generation, Accept- Rejection techniques, input modelling on relevant exercise problems.		
4.	Explain the verification, validation, and calibration of simulation models.		
5.	Describe the high-level computer simulation, CPU simulation and memory simulation.		
UNIT-I			
INTRODUCTION TO SIMULATION:			15 Hours
When Simulation is the Appropriate Tool; When Simulation Is Not Appropriate; Advantages and Disadvantages of Simulation; Areas of Application; Systems and System Environment; Components of a System; Discrete and Continuous Systems; Model of a System; Types of Models; Discrete-Event System Simulation; Steps in a Simulation Study. General Principles: Concepts in Discrete-Event Simulation: The Event-Scheduling / Time-Advance Algorithm, World Views, Manual simulation Using Event Scheduling. (Text Book-1: Chapter 1, Chapter 3: 3.1)			
UNIT-II			
RANDOM-NUMBER GENERATION:			15 Hours
Properties of Random Numbers; Generation of Pseudo-Random Numbers; Techniques for Generating Random Numbers; Tests for Random Numbers. Random-Variate Generation: Inverse Transform technique: Exponential Distribution, Uniform Distribution, Discrete Distributions; Acceptance-Rejection Technique: Poisson Distribution. Input Modeling: Data Collection; Identifying the distribution with Data; Parameter Estimation; Goodness of Fit Tests; Selecting Input Models without Data; Multivariate and Time-Series Input Models. (Text Book-1: Chapter 7, Chapter 8: 8.1.1,8.1.2,8.1.7,8.2.1,Chapter 9:9.1,9.2,9.3,9.4,9.6,9.7)			
UNIT-III			
VERIFICATION AND VALIDATION OF SIMULATION MODELS:			10 Hours
Model Building, Verification and Validation; Verification of Simulation Models; Calibration and Validation of Models. Simulation of Computer Systems: Introduction; Simulation Tools; Model Input; High-Level Computer-System Simulation; CPU Simulation; Memory Simulation. (Text Book-1: Chapter 10, 14)			

Course Outcomes: At the end of the course the student will be able to:

1.	Describe the appropriateness of the simulation, its application, types of simulation model, steps in simulation study and general principles in simulation and concepts in discrete-event simulation.
2.	Describe the generation of random numbers and pseudo-random numbers and apply techniques for generating random numbers
3.	Illustrate and apply the techniques of random variate generation, Accept- Rejection techniques, input modelling on relevant exercise problems.
4.	Illustrate the verification, validation and calibration of simulation models.
5.	Illustrate the high-level computer simulation, CPU simulation and memory simulation.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS2221-1.1	2	3											3	1	
CS2221-1.2	3	3											3	1	
CS2221-1.3	3	3											3	1	
CS2221-1.4	3	3											3	1	
CS2221-1.5	3	3											3	1	

1: Substantial (High) 2: Moderate (Medium) 3: Poor (Low)

TEXTBOOKS:

1.	Jerry Banks, John S. Carson, Barry L. Nelson, David M. Nicol, “Discrete-Event System Simulation”, Third Edition, Prentice-Hall India, 2000
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REFERENCE BOOKS:

1.	Averill M. Law, W. David Kelton, “Simulation Modeling and Analysis” ,Third Edition, McGrawHill, 2000.
2.	Geoffrey Gordon, “System Simulation”, Second Edition, Prentice-Hall India, 1978.

E Books / MOOCs/ NPTEL

1.	https://ptolemy.berkeley.edu/books/Systems/PtolemyII_DigitalV1_02.pdf
2.	https://epdf.tips/system-modelling-and-simulation.html
3.	https://www.coursera.org/learn/modeling-simulation-natural-processes
4.	https://swayam.gov.in

SEE Scheme:

There will be 8 questions of 20 marks each in the question paper divided into 3 Units as per the syllabi & contact hours and the student will have to answer 5 full questions, selecting 2 full questions from Unit - I & Unit – II and 1 full question from Unit – III.

ADHOC WIRELESS NETWORKS				
Course Code:		CS3221-1	Course Type	PEC
Teaching Hours/Week (L: T: P):		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives:				
1.	Distinguish the characteristics of ad hoc Wireless networks with other Wireless networks.			
2.	Identify Ad-Hoc Wireless networks, issues, classification of MAC Protocols.			
3.	Describe and distinguish different types of ad hoc Routing Protocols, TCPover Ad hoc Protocol and a brief introduction to security issues in ad hocWireless networks.			
UNIT-I				
Review of Wireless Networks:				15 Hours
IEEE Wireless Standard, Basic 802.11 MAC layer mechanisms, CSMA/CA mechanisms and other MAC layer functionalities.				
Adhoc Networks: Introduction, Issues in AdHoc wireless networks, Adhoc wireless internet.				
MAC Protocols for Ad hoc wireless Networks: Introduction, Issues in designing a MAC Protocol for Adhoc wireless Networks,Design goals of a MAC protocol for Ad hoc wirelessNetworks.				
Classification of MAC Protocols: Contention based protocols: MACAW, FAMA busy tone protocols, receiver initiated protocol: MARCH.Contention based protocols withreservation mechanisms: DPRMA, HRMA, FPRP. Contention-based MAC protocols with scheduling mechanism:DPS&MA.				
Routing protocols for Adhoc wireless Networks: Introduction, Issues in designing a routing Protocol for Adhoc wireless Networks, Classification of routingProtocols.				
UNIT-II				
Table drive routing protocol:				15 Hours
DSDV, WRP, CGSR. On-demand routing protocol: DSR, AODV, LAR, FORP.				
Hybrid routing protocol: CEDAR, ZRP. Hierarchical routing protocols: FSR. Metrics used by power aware routing protocols.				
Transport layer protocols for Ad hoc wireless Networks: Introduction, IssuesindesigningatransportlayerProtocolforAdhocwirelessNetworks, Design goals of a transport layer protocol for Ad hoc wireless Networks, Classification of transport layer solutions, TCP over Ad hoc wireless Networks: TCP-F, TCP with ELFN, TCP-BuS, ATCP, Split TCP. Other transport layer protocols for Adhoc wireless Networks:ACTP,ATP.				
UNIT-III				
Security in wireless Ad hoc wireless Networks:				10 Hours
Network Security requirements, Issues & Challenges in security provisioning, Network security attacks, Key Management, Secure routing in Ad hoc wireless Networks: SAR, SEAD, Security-Aware AODV.				
Quality of service in Ad hoc wireless Networks: Introduction, Issues & challenges in providing QoS in Ad hoc wireless Networks, Classification of QoS solutions, MAC layer solutions, network layer solutions.				
Course Outcomes: At the end of the course student will be able to				
1.	Define the MAC layer functionalities of wirelessnetworks			
2.	DefinetheworkingofmajorMAClayerprotocolsforadhocwireless networks			
3.	ClassifyanddistinguishNetworklayerprotocolsforadhocwireless networks.			
4.	Identify the issues with TCP/IP Transport layer protocols with wireless networks and examine few solutions provided by ad hoc transport layerprotocols.			
5.	Identify security and QoS issues and challenges with ad hoc wireless networks.			

What is Malware Analysis, Malware Analysis Techniques, Basic Static Analysis, Basic Dynamic Analysis, General Rules for Malware Analysis. Lab – Debugging Malicious Binaries	10 Hours
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Course Outcomes: At the end of the course student will be able to:

1.	To understand different types of Malwares.
2.	To understand the approach required to capture Malwares.
3.	To write snort rules to detect malicious traffic.
4.	To perform memory forensics.
5.	To understand how to mitigate Malware threats.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CB3222-1.1	3	3											2	3	
CB3222-1.2		2	2										2	2	
CB3222-1.3	2	3											3	3	
CB3222-1.4	2	2	2										2	3	
CB3222-1.5	2	3											2	3	

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software 1st Edition -by Michael Sikorski
2.	Snort Intrusion Detection and Prevention Toolkit Kindle Edition
3.	by Brian Caswell (Author), Jay Beale (Author), Andrew Baker (Author)
4.	Malware Analysis and Detection Engineering: A Comprehensive Approach to Detect and Analyze Modern Malware 1st ed. Edition,

REFERENCE BOOKS:

1.	Cybersecurity Threats, Malware Trends, and Strategies: Learn to mitigate exploits, malware, phishing, and other social engineering attacks 1st Edition, by Tim Rains (Author).
2.	Learning Malware Analysis: Explore the concepts, tools, and techniques to analyze and investigate Windows malware, 29 June 2018 by Monnappa K A (Author).

E Books / MOOCs/ NPTEL

1.	https://www.coursera.org/learn/malware-analysis-and-assembly
2.	https://www.udemy.com/course/expert-malware-analysis-and-reverse-engineering/

BLOCKCHAIN TECHNOLOGY

Course Code:	CS3321-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50

Teaching Department: Information Science & Engineering (Cyber Security)

Prerequisites:

1. Cryptography Techniques
2. Data Structure and algorithms
3. Introduction to Programming

Course Objectives:

This Course will enable students to:																
1.	To provide conceptual knowledge of Block chain technology used to innovate and improve business processes															
2.	To provide theoretical and practical solution of Block chain technology															
3.	Identify the current and future trends in Block chain technology															
UNIT-I																
INTRODUCTION:														15 Hours		
Overview of Block chain, Public Ledgers, Bitcoin, Smart Contracts, Block in a Block chain, Transactions, Distributed Consensus, Public v/s Private Block chain, Understanding crypto currency to Block chain Permissioned model of Block chain, overview of security aspects of Block chain. (R1, R2, R3, R4) BITCOIN AND BLOCK CHAIN: Creation of coins, Payments and double spending, Bitcoin scripts, Bitcoin P2P network, transaction in bitcoin network, Block mining, Block propagation and Block relay. (R5,R6,R7,R8) WORKING WITH CONSENSUS IN BITCOIN: Distributed consensus in open environment, Consensus in Bitcoin network, Proof of Work (PoW)- basic introduction hash cash PoW, Bitcoin PoW, attacks on PoW, monopoly problem, proof of stake, proof of burn, and proof of elapsed time, the life of bitcoin miner, mining difficulty, mining pool. (R9,R10,R11,R12)																
UNIT-II																
PERMISSIONED BLOCK CHAIN:														15 Hours		
Permission model and use cases, design issues for permissioned Block chains, execute contracts, state machine replication, overview of consensus models for permissioned Block chain – distributed consensus in closed environment, Paxos, RAFT consensus, Byzantine general problem, Byzantine fault tolerant system, Lamport-Shostak Pease, BFT algorithm, BFT over asynchronous system. (R13,R14,R15,R16,R17) ENTERPRISE APPLICATION OF BLOCK CHAIN: Cross border payment, Know Your Customer (KYC), Food Security, Mortgage over Block chain, Block chain enabled trade, We trade-Trade finance network, supply chain financing, identity on Block chain.(R18, R19,R20,R21)																
UNIT-III																
BLOCK CHAIN APPLICATION DEVELOPMENT:														10 Hours		
Hyperledger fabric – architecture, fabric details and Channels, Fabric – Membership and Identity Management, Identities and Policies, Membership and Access Control, writing smart contract using Ethereum. (R24,R25,R26,R27,R28)																
Course Outcomes: At the end of the course student will be able to																
1.	Explain the Blockchain Technology and its application in Crypto currency															
2.	Make use of the bitcoin protocols for mining and select the optimal solution															
3.	Identify the types of Blockchain protocols for permissioned and permission less environment															
4.	Apply the blockchain protocols for enterprise applications															
5.	Analyze the Blockchain based solutions and write smart contracts using solidity in Hyperledger Fabric and Ethereum Framework															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS3321-1.1		3	3								1		1		1	
CS3321-1.2		2	3	3							1		1		2	
CS3321-1.3		2	3	3							1		1		2	
CS3321-1.4		2	2	3							1		1		2	
CS3321-1.5		2	2	3							1		1		2	
(L/1 = Low 30%-49%, M/2 = Medium 50%-69%, H/3 = High >70%)																
REFERENCE BOOKS:																

1.	Melanie Swan “Block Chain: Blue Print for a New Economy”, O’Reilly, 2015.
2.	Josh Thompsons, “The Block Chain for a Beginners – Guide To Block chain technology and Leveraging Block Chain Programming”.
3.	Daniel Dreschers “, Block Chain Basics”, A press; 1st edition 2017.
4.	Anhul Kaushik, “Block Chain and Crypto Currencies”, Khanna Publishing House, New Delhi
5.	Imran Bashir, “Mastering Block Chain: Distributed Ledger Technology and Decentralization, Smart Contracts Explained” Packt Publishing.
6.	sh Modi, “Solidity Progamming Essentials: A Beginners Guide to build Smart Contracts for Ethereum and Block Chain” Packt Publishing.
7.	Salman Baset, Luc Desrosiers, Nitin Gaur, Petr Navotny, Antony O’Dowd, Venkatraman Ramakrishna, “Hands-On Block Chain with Hyperledger: Building Decentralized Application with Hyperledger Fabric and Composer”, Import 2018.

E Books / MOOCs/ NPTEL

1.	E-Resources R1: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=5&lesson=9 R2: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=5&lesson=10 R3: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=5&lesson=11 R4: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=5&lesson=13 R5: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=6&lesson=14 R6: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=6&lesson=15 R7: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=6&lesson=16 R8: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=6&lesson=17 R9: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=6&lesson=18 R10: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=7&lesson=19 R11: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=7&lesson=20 R12: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=7&lesson=21 R13: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=7&lesson=22 R14: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=7&lesson=23 R15: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=8&lesson=24 R16: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=8&lesson=25 R17: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=8&lesson=26 R18: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=31&lesson=45 R19: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=31&lesson=46 R20: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=31&lesson=47 R21: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=31&lesson=49 R22: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=50&lesson=55 R23: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=32&lesson=57 R24: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=29&lesson=35 R25: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=29&lesson=36 R26: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=29&lesson=37 R27: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=32&lesson=60 R28: https://onlinecourses.nptel.ac.in/noc19_cs63/unit?unit=34&lesson=72
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SECURITY AND RESILIENCE																
Course Code:					CB3222-1			Course Type				PEC				
Teaching Hours/Week (L: T: P):					3:0:0			Credits				03				
Total Teaching Hours					40+0+0			CIE + SEE Marks				50+50				
Teaching Department: Computer Science & Engineering (Cyber Security)																
Course Objectives:																
1.	To understand the difference between IT and OT security.															
2.	To understand basic principles of Effective, Efficient and Cyber Resilient Organizations and Operations															
3.	To understand Industrial Control system (ICS) Security framework.															
4.	To understand Cyber Resilience best practices in Manufacturing sector.															
5.	To gain practical hands-on understanding of Cyber resilience on Cyber Range.															
UNIT-I																
Effective, Efficient and Cyber Resilient Organizations and Operations; An Insight into Multi-domain Command and Control Systems: Issues and Challenges; Cloud Technologies for Building a System of Data Centers for Defense and Security.													15 Hours			
UNIT-II																
Cyber Situational Awareness in Critical Infrastructure Organizations, Cybersecurity in Next Generation Energy Grids: Challenges and Opportunities for Blockchain and AI Technologies, A New Approach to Assess the Risk of Cyber Intrusion Attacks Over Drones Using Intuitionistic Fuzzy Estimations, Cyber Resilience Using Self-Discrepancy Theory.													15 Hours			
UNIT-III																
Insider Threats to IT Security of Critical Infrastructures; Empirical Study on Cyber Range Capabilities,Interactions and Learning Features.													10 Hours			
Course Outcomes: At the end of the course student will be able to:																
1.	To understand how under cyber resilience in various sectors.															
2.	To analyze Cyber Resilience plan of large manufacturing companies.															
3.	To understand how latest advancement in technologies are changing the security landscape.															
4.	To gain hands on experience working with OT attacks.															
5.	To understand key components of Cyber Resiliency Framework.															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CB3222-1.1		3	3											2	3	
CB3222-1.2			2	2										2	2	
CB3222-1.3		2	3											3	3	
CB3222-1.4		2	2	2										2	3	
CB3222-1.5		2	3											2	3	
1: Low 2: Medium 3: High																
TEXTBOOKS:																

1.	Digital Transformation, Cyber Security and Resilience of Modern Societies: 84 (Studies in BigData) by Todor Tagarev (Editor), Krassimir T. Atanasov (Editor), Vyacheslav Kharchenko (Editor), Janusz Kacprzyk (Editor)
2.	The Security of Critical Infrastructures: Risk, Resilience and Defense: 288 (International Series in Operations Research & Management Science) by Marcus Matthias Keupp (Editor)
REFERENCE BOOKS:	
1.	Hacking Exposed Industrial Control Systems: ICS and SCADA Security Secrets & Solutions - September 2016 by Clint Bodungen, Bryan Singer, Aaron Shbeeb, Kyle Wilhoit, Stephen Hilt.
2.	Industrial Cybersecurity: Efficiently secure critical infrastructure systems, 18 October 2017 by Pascal Ackerman.
E Books / MOOCs/ NPTEL	
1.	https://www.coursera.org/learn/security-management-and-governance
2.	https://www.udemy.com/course/cyber-security-in-9-steps/

SOFTWARE DESIGN			
Course Code:	CS3324-1	Course Type	PEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
The aim of this course is to teach fundamental patterns in software. This includes some of the most popular architectural patterns, more than two dozen design patterns from creational, structural and behavioural categories.			
UNIT-I			
SOLID Design Principles and Design Patterns			15 Hours
Introduction to Design Principles, Single Responsibility Principle, Open/Closed Principle, Liskov Substitution Principle, Interface Segregation Principle, Dependency Inversion Principle, Introduction to Design Patterns, Factory Design Pattern, Builder Design Pattern, Prototype Design Pattern, Singleton Design Pattern, Adapter Design Pattern, Bridge Design Pattern, Composite Design Pattern, Facade Design Pattern, Proxy Design Pattern, Decorator Design Pattern, Chain of Responsibility Design Pattern, Observer Design Pattern, Mediator Design Pattern, Case Study, Abstract Factory Design Pattern, Front Controller Design Pattern, Command Design Pattern, Iterator Design Pattern, Strategy Design Pattern.			
UNIT-II			
Hands-on with NoSQL - MongoDB>			15 Hours
Why NoSQL?, BASE property of NoSQL Database, Types of NoSQL Databases and Use Cases, Getting Familiar With The Coding Console, Getting Familiar with MongoDB, Setting Up MongoDB, Basic MongoDB Commands, CRUD Operations, Getting the Data, Comparison Operators, Logical Operators, Aggregation Pipeline in MongoDB, Aggregation Operations, Setup the Project Structure in IntelliJ, Connection Set Up, CRUD Operations in JAVA API, Advanced Querying in JAVA API, What is Indexing?, Types of Indexes, Indexing - Best Practices, Replication, Replica Set Architecture, Sharding, Shard Key, Embedded Documents and References, Demo On Creating References and Embedded Documents, Modeling Relationships: One-to-One, Modeling Relationships: One-to-Many, Modeling Relationships: Many-to-Many			
UNIT-III			
System Design and Design Patterns			10 Hours
Need for System Design, Characteristics of Distributed Systems, Load Balancing, Caching, Data Partitioning, Indexing, Redundancy and Replication, SQL and NoSQL, CAP Theorem, Types of Client-Server			

Communication, Distributed System Architecture, Requirement Collection, Scale Estimation, System Interface Definition, Defining Data Model, High-Level Design, Detailed Design, Resolving Bottlenecks, Designing Autocomplete, Designing Twitter Timeline, Designing BookMyShow, Practice: Designing Pastebin, Practice: Designing API Rate Limiter, Practice: Designing Dropbox, Solution for Designing Pastebin, Solution for Designing API Rate Limiter, Solution for Designing Dropbox.

Design patterns - Usability & Advantages, Creational Design Patterns - Factory Method, Creational Design Patterns - Builder, Creational Design Patterns - Singleton, Structural Design Patterns - Adapter and Decorator, Structural Design Patterns - Facade and Proxy, Behavioral Patterns - Command design and Iterator, Behavioral Patterns - Observer and Strategy, Design Patterns recap, An Example through Observer Pattern, Subject-Object Model, Java Observer Pattern, Differences between Observer Pattern and Java SDK

Course Outcomes: At the end of the course student will be able to

- | | |
|----|--|
| 1. | Understand and Implement design patterns in Python |
| 2. | Implement some of the design patterns in Java |
| 3. | Use multiple design patterns in software development |
| 4. | Apply the knowledge in the project work |

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3324-1.1	2	2						2				1		2	
CS3324-1.2	2	3						2				1		2	
CS3324-1.3	2	2						2				1		2	
CS3324-1.4	2	2						2				1		2	

1: Low 2: Medium 3: High

TEXTBOOKS:

- | | |
|----|--|
| 1. | Eric Freeman, Elisabeth Freeman, Kathy Sierra, Bert Bates Head First Design Patterns, O'Reilly |
| 2. | Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides Design Patterns: Elements of Reusable Object oriented Software Addison-Wesley |

Professional Elective Courses (Intelligent Systems & Analytics Stream)

INTRODUCTION TO DATA SCIENCE																			
Course Code:					CS1231-1			Course Type				PEC							
Teaching Hours/Week (L: T: P):					3:0:0			Credits				03							
Total Teaching Hours					40+0+0			CIE + SEE Marks				50+50							
Teaching Department: Computer Science & Engineering (Cyber Security)																			
UNIT-I																			
INTRODUCTION TO DATA ANALYSIS USING EXCEL														15 Hours					
Introduction to Data Analysis using Excel, Types of Data Analysis, Data Mining, Business Intelligence, Statistical Analysis, Predictive Analytics, Text Analytics Data Analysis with Excel, Data Analysis with Excel and Advanced Data Analysis with Excel.																			
Data Analysis Process: Data Requirements Specification, Data Collection, Data Processing, Data Cleaning, Data Analysis, Communication. Data Cleaning – Text Functions, Dates and Times: With Text Functions, Containing Date Values, Containing Time Values. Conditional Formatting, Sorting and Filtering, Subtotals with Ranges.																			
UNIT-II																			
DATA QUICK ANALYSIS														15 Hours					
Data Quick Analysis: Understanding Lookup Functions, PivotTables, Data Visualization, Data Validation, Financial Analysis.																			
Data Consolidation, Cleaning Data with Text Functions, Extracting Data Values from Text, Extracting Data Values with Convert Text to Columns Wizard, What-If Analysis, Importing Data into Excel, Data Model, Exploring Data with PivotTable, Exploring Data with PowerPivot, Exploring Data with Power View, Aesthetic Power View Reports, Key Performance Indicators (KPIs)Preparing Data for Consolidation																			
UNIT-III																			
R PROGRAMMING BASICS														10 Hours					
R Programming Basics: Overview of R programming, Environment setup with R Studio, R Commands, Variables and Data Types, Control Structures, Array, Matrix, Vectors, Factors, Functions, R packages. Data Visualization using R: Reading and getting data into R (External Data): Using CSV files, XML files, Web Data, JSON files, Excel files. Working with R Charts and Graphs: Histograms, Boxplots, Bar Charts, Line Graphs, Scatterplots, Pie Charts.																			
Course Outcomes: At the end of the course student will be able to																			
1.		Understand the types of data analysis and carry out the data analysis process.																	
2.		Practice out quick data analysis, extracting data values from text.																	
3.		Demonstrate the export of data to excel, PivotTable etc.																	
4.		Describe R basics, Variables and Data Types, Control Structures, Array, Matrix, Vectors, Factors, Functions, R packages and Develop Simple programs using R.																	
5.		Understand the R Data Visualization concept and apply R programming in plotting various Charts and Graphs.																	
Course Outcomes Mapping with Program Outcomes & PSO																			
		Program Outcomes→			1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
		↓ Course Outcomes															1	2	3
		CS1231-1.1			3	1	3	1	2				2			2	3	3	2
		CS1231-1.2			3	1	3	1	2				2			2	3	3	2
		CS1231-1.3			3	1	3	1	2				2			2	3	3	2
		CS1231-1.4			3	1	3	1	3				2			2	3	3	2
		CS1231-1.5			3	1	3	2	3				2			2	3	3	2
1: Low 2: Medium 3: High																			
TEXTBOOKS:																			
1.		Microsoft Excel 2019 Data Analysis and Business Modelling (Business Skills) 6th Edition ISBN-13: 978-1509305889.ISBN-10: 1509305882																	

2.	Tilman M. Davies, “The Book of R: A First Course in Programming and Statistics”, No Starch Press; 1st edition ,2016.
REFERENCE BOOKS:	
1.	Microsoft Excel 2019 Formulas and Functions (Business Skills) 1st Edition ISBN-13: 978-1509306190 ISBN-10: 1509306196 Paul McFedries
2.	Collect, Combine, and Transform Data Using Power Query in Excel and Power BI (Business Skills) 1st Edition ISBN-13: 978-1509307951 ISBN-10: 1509307958 Gil Raviv (Author)
3.	Microsoft Power BI Quick Start Guide: Bring your data to life through data modeling, visualization, digital storytelling, and more, 2nd Edition 2nd ed. Edition by Devin Knight (Author), Mitchell Pearson (Author), Bradley Schacht (Author), Erin Ostrowsky (Author) ISBN-13: 978-1800561571 ISBN-10: 1800561571
4.	Andrie de Vries and Joris Meys. “R for Dummies”, 2nd Edition, John Wiley & Sons; 2nd edition, 2015.
5.	Hadley Wickham, Garrett Golemund, “R for data Science: Import, Tidy, Transform, Visualize, And Model Data”, O'Reilly; 1st edition, 2017.
6.	Andrew Oleksy , “Data Science with R: A Step By Step Guide With Visual Illustrations & Examples”.
E Books / MOOCs/ NPTEL	
1	Excel Skills for Data Analytics and Visualization Specialization https://www.coursera.org/specializations/excel-data-analytics-visualization
2	IBM Data Analytics with Excel and R Professional Certificate https://www.coursera.org/professional-certificates/ibm-data-analyst-r-excel
3	Introduction to Data Analysis Using Excel https://www.coursera.org/learn/excel-data-analysis

ARTIFICIAL INTELLIGENCE				
Course Code:		CS2231-1	Course Type	PEC
Teaching Hours/Week (L: T: P):		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives:				
1.	Understand the history of artificial intelligence(AI) and its foundation.			
2.	Learn principles of AI in problem solving, Inference, perception, knowledge representation.			
3.	Explain various applications of AI techniques in intelligent agents, artificial neural networks and other machine learning models.			
4.	Developing simple applications in an AI language, expert system shell or datamining tool.			
5.	Find current scope and limitations, and societal implications of AI.			
UNIT-I				
INTRODUCTION: INTELLIGENT AGENTS				15Hours

What is AI? Foundation of AI, State of Art, Agents of Environment, Structure of agents. (Textbook-1: Chapter 1: 1.1 to 1.4 and 2.1 to 2.4)
PROBLEM SOLVING:
 Problem solving agents, Example Problems, Searching for solutions, Uniformed and Informed search strategies, Heuristic Functions
 (Textbook-1: Chapter 3: 3.1, 3.6),

UNIT-II

UNCERTIAN KNOWLEDGE AND REASONING

15Hours

Acting under uncertainty, Basic Probability Notation, Inference using full joint distributions, BayesRule and its use.
 (Textbook-1:Chapter13:13.1,13.5)

PROBABILISTICREASONINGOVERTIME:

Time uncertainty, TemporalModels, Hidden Markov Models.(Textbook-1:Chapter 15:15.1,15.3),

UNIT-III

REINFORCEMENT LEARNING:

10 Hours

Introduction, Passive Reinforcement Learning, Active Reinforcement Learning, Generalization in Reinforcement Learning, Applications of Reinforcement Learning.
 (Textbook-1: Chapter 21: 21.1 to 21.6)

Course Outcomes: At the end of the course student will be able to

1.	Explain the fundamental understanding of the history of artificial intelligence (AI)and its foundation.
2.	Interpret the basic principles of AI in solutions that require problem solving, Inference, perception, knowledge representation, and learning.
3.	Describe the awareness and a fundamental understanding of various applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.
4.	Identifyandexplaintheproficiencydevelopingapplicationsinan'AIlanguage',expert system shell or datamining tool.
5.	Determine an ability to share in discussions of AI, its current scope and limitations, and societal implications.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS2231-1.1	1	2											1	3	
CS2231-1.2	2	3											1	3	
CS2231-1.3	2	3											3	3	
CS2231-1.4	1	2	3		3								3	2	
CS2231-1.5	1	2											3	3	

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Stuart Russel and Peter Norvig, "Artificial Intelligence A Modern Approach", Pearson 3rd Edition, 2016.

REFERENCE BOOKS:

1. DANWPATTERSON, "IntroductiontoArtificialIntelligenceandExpertSystems", PEARSON, 1stedition2015.
2. Tom.M.Mitchel, "MachineLearning", McGrawHigher Ed, 1stedition2013.
3. Elaine Rich, "Artificial Intelligence", Mc Graw Hill 3rd Edition, 2017.
4. Er. Rajiv Chopra, "Artificial Intelligence – A practical approach", Chand publication, 1st edition 2012

COMPUTER VISION																
Course Code:					CS3231-1			Course Type				PEC				
Teaching Hours/Week (L: T: P):					3:0:0			Credits				03				
Total Teaching Hours					40+0+0			CIE + SEE Marks				50+50				
Teaching Department: Computer Science & Engineering (Cyber Security)																
Prerequisites: Linear Algebra																
Course Objectives:																
This Course will enable students to:																
1.		Understand the goal and scope of computer vision.														
2.		Learn the basics of image and image formation in computers.														
3.		Learn about camera projections and viewing in computer vision														
4.		Learn the basics of image segmentation and feature tracking														
5.		Understand and learn the principles of object detection and recognition in computer vision.														
UNIT-I																
Introduction: Introduction to Computer Vision:															15 Hours	
Goal, areas, Human Vision, Segmentation, Perception, Semantic information, Special effects, Modelling, Applications; Linear Algebra: Vectors & matrices, Transformation matrices, Matrix inverse, Matrix rank, SVD. Pixels, Features, and Cameras: Pixels and Filters: Images as functions, Linear Systems (filters), Convolution & Correlation. Edge detection: Simple, Canny, RANSAC; Feature detector: Local invariant, Harris, DOG, SIFT;																
UNIT-II																
Camera and Camera model:															15 Hours	
Pinhole Cameras, Cameras & lenses, Projection matrix, Intrinsic parameters, Extrinsic parameters; Stereo Vision: Epipolar geometry, Parallel images, Images rectification, Solving correspondence problem, Active Stereo Vision System. Regions of Images, and Segmentation: Basic Concepts of Segmentation: Gestalt theory; Agglomerative, K-means & Mean-shift Clustering; Optical flow, Feature tracking, Applications;																
UNIT-III																
Advanced Image Parsing Topic and Applications:															10 Hours	
Binary, Image Matting; Figure-ground Segmentation Using Clustering Algorithms. Recognizing Faces and Objects: Basic Concepts in Recognition & its pipeline, Nearest Neighbour Match; PCA and Eigenfaces; Tracking Millions of People: Detection, Tracklet Generation & Association.																
Course Outcomes: At the end of the course student will be able to																
1.		Outline Application of image processing														
2.		Explain edge detection and feature description techniques.														
3.		Summarize camera projections														
4.		Apply image segmentation and feature tracking to video streams														
5.		Apply the techniques of recognizing faces and objects to photo frames.														
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS3231-1.1		1	3							1	1		1	1	1	
CS3231-1.2		1	3							1	1		1	1	1	
CS3231-1.3		2	2							1	1		1	1	2	

Secondary)																
CS3231-1.4		2	2							1	1		1	2	2	
CS3231-1.5		1	3							1	1		1	3	2	
(L/1 = Low 30%-49%, M/2 = Medium 50%-69%, H/3 = High >70%)																
TEXTBOOKS:																
1.	Computer Vision: Algorithms and Applications, Richard Szeliski, Microsoft Research, Electronic draft (2010).															
2.	Multiple View Geometry in Computer Vision, Hartley & Zisserman, Cambridge University Press; 2 edition (2004).															
3.	Computer Vision: A Modern Approach, David A. Forsyth& Jean Ponce, Prentice Hall; 2 edition (2011).															
REFERENCE BOOKS:																
1.	Machine vision, Jain, Ramesh and Rangachar Kasturi and Brian G. Schunck; McGraw-Hill, Edition-1995.															
2.	Introductory computer vision and image processing, Low, Adrian; McGraw-Hill, Edition-1991.															
3.	Digital image processing, Gonzalez, Rafael C. and Richard E. Woods; Addison- Wesley, Edition: 3rd, Year:1998.															
E Books / MOOCs/ NPTEL																
1.	Youtube channel: First Principles of Computer Vision															
2.	Coursera : First Principles of Computer Vision specialization															
3.	Nptel course: https://nptel.ac.in/courses/106105216 (12 weeks)															
	<u>SEE Question Paper Pattern:</u> There will be 8 questions of 20 marks each in the question paper divided into 3 Units as per the syllabi & contact hours and the student will have to answer 5 full questions, selecting 2 full questions from Unit - I & Unit – II and 1 full question from Unit – III.															

DATA MINING			
Course Code:	CS3232-1	Course Type	PEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
UNIT-I			
INTRODUCTION TO DATA MINING – Why Data Mining? What is Data Mining? What kind of data can be mined? What kinds of patterns can be mined? Issues in data mining. GETTING TO KNOW YOUR DATA: Data Objects and Attribute Types, Basic Statistical Descriptions of Data (2.2.1,2.2.2), Measuring Data Similarity and Dissimilarity. DATA PREPROCESSING: Data Pre-processing: An Overview, Data Cleaning, Data integration, Data Reduction (3.4.1,3.4.6,3.4.7,3.4.8), Data transformation and Discretization (3.5.1,3.5.2,3.5.3). MINING FREQUENT PATTERNS: Associations and Correlations - Basic Concepts, Frequent Item-set Mining Methods (6.2.1, 6.2.2, 6.2.4) – Apriori Algorithm, Generating Association rules from frequent item-sets. A Pattern growth approach for mining frequent item-sets, Which Patterns are interesting? (6.3.1, 6.3.2)			15 Hours
UNIT-II			
CLASSIFICATION: BASIC CONCEPTS: Basic Concepts, Decision tree induction			15 Hours

(8.2.1,8.2.2), Bayes Classification methods (8.3.1,8.3.2)													Security)			
Rule based classification (8.4.1,8.4.2,8.4.3- Rule Induction Using a Sequential Covering Algorithm), Model evaluation and selection (8.5.1).																
UNIT-III																
CLUSTER ANALYSIS: Basic concepts and methods- Cluster Analysis (10.1.1, 10.1.2,10.1.3), Partitioning methods (10.2.1,10.2.2), Hierarchical methods (10.3.1,10.3.2), Evaluation of clustering.													10 Hours			
Course Outcomes: At the end of the course student will be able to																
1.	Explain the functionalities, interesting patterns and kind of data for data mining & the need of pre-processing of data.															
2.	To identify frequent patterns and develop the association's rules.															
3.	Apply classification algorithms like Decision tree and Bayesian classification to classify the data.															
4.	Experiment with rule-based classification method to classify the data.															
5.	Apply partitional clustering and Hierarchical clustering methods to cluster the data and outline the evaluation of clustering methods.															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS3232-1.1		2	2								1		1			
CS3232-1.2		2	2		2						1		1		2	
CS3232-1.3		2	2		2						1		1		2	
CS3232-1.4		2	2		3						1		1		2	
CS3232-1.5		3	3		2						1		1		3	
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.	Jiawei Han and MichelineKamber, “Data Mining – Concepts and Techniques”, Third Edition, Morgan Kaufmann Publishers.															
REFERENCE BOOKS:																
1.	M. H. Dunham, “Data Mining: Introductory and Advanced Topics”, Pearson Education. 2001.															
2.	D. Hand, H. Mannila and P. Smyth, “Principles of Data Mining”, Prentice-Hall. 2001.															
3.	I. H. Witten and E. Frank, “Data Mining: Practical Machine Learning Tools and Techniques”, Morgan Kaufmann. 2000.															

DEEP LEARNING				
Course Code:		CS3233-1	Course Type	PEC
Teaching Hours/Week (L: T: P):		3:0:0	Credits	03
Total Teaching Hours		40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)				
Course Objectives:				
1.	Explain the importance and basics of neural network and its training for deep learning.			
2.	Outline the structure and training of convolutional neural network.			
3.	Develop a working convolutional neural network.			
UNIT-I				
Introduction				15 Hours
Biological neural network vs Artificial neural networks, the perceptron architecture, simple neural network examples, classification of neural network.				
Biological neural network vs Artificial neural networks, the perceptron architecture, simple neural network examples, classification of neural network.				
Neural networks basics: Feed-forward; Multi-Layer Perceptron, linear vs nonlinear networks, activation functions, shallow vs deep neural network.				
Network Training - 1: Training process, Loss functions: regression (MSE, MSLE), binary classification loss (binary cross entropy, hinge loss), multiclass classification loss (multi-class cross entropy); cost function, maximum likelihood-based cost.				
UNIT-II				
Network Training – 2				15 Hours
Learning via backpropagation, recursive chain rule, computational graph; Vanishing and exploding gradient; Optimization Algorithms: learning rate decay, momentum-based learning, SGD, AdaGrad, RMSProp, Adam; Bias-variance trade-off; Practical issues in NN training: Overfitting - regularization (dropout, L1, L2), data augmentation, early stopping; Network and training parameters and hyperparameters.				
Convolutional Neural Networks - 1:				
Introduction, basic structure of a convolutional network, padding, strides, pooling (max, average), fully connected layer, Interleaving between layers, local response normalization, hierarchical feature engineering;				
UNIT-III				
Title: Convolutional Neural Networks - 2:				10 Hours
Basics of convolutional autoencoder, Design of a CNN for classification using keras; VGG-19 architecture, ResNet-50, Transfer learning; Applications of convolutional networks: content-based image retrieval, object localization, object detection, natural language and sequence learning, video classification, Introduction to Generative Adversarial Networks and its applications.				
Course Outcomes: At the end of the course student will be able to				
1.	Demonstrate the importance and basics of neural network for deep learning.			
2.	Explain the training process and approaches to evaluate the training error.			
3.	Explain how the network learns, learning optimization approaches and regulate practical issues in			

	the network design.
4.	Explain the concepts of CNN, its working and understand neural network designs
5.	Use standard CNN architecture, develop DCNN and apply them to solve real world problems.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS3233-1.1	3	2	1										1	3	
CS3233-1.2	2												2	2	
CS3233-1.3	1	3											2	2	
CS3233-1.4	2	3											2	2	
CS3233-1.5	1		1										1	1	

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Charu C. Aggarwal, "Neural Networks and Deep Learning", Springer, 2018.
2.	Ian Goodfellow and YoshuaBengio and Aaron Courville, "Deep Learning", MIT press, 2016.

REFERENCE BOOKS:

1.	Duda, R.O., Hart, P.E., and Stork, D.G., "Pattern Classification", Wiley-Interscience, 2nd Ed. 2001
2.	Theodoridis, S and Koutroumbas, K, "Pattern Recognition", 4th Ed, Academic Press, 2008
3	Bishop, C. M., "Neural Networks for Pattern Recognition", Oxford University Press, 1995
4	Russell, S. and Norvig, N, "Artificial Intelligence: A Modern Approach", Prentice Hall Series in Artificial Intelligence, 2003
5	Hastie, T., Tibshirani, R., and Friedman, J., "The Elements of Statistical Learning", Springer, 2001

SOCIAL AND WEB ANALYTICS

Course Code:	CS3234-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50

Teaching Department: Computer Science & Engineering (Cyber Security)

Course Objectives:

This course will enable students to:-

1.	Understand social media, web and social media analytics, and their potential impact.
2.	Determine how to Leverage social media for better services and Understand usability metrics, web and social media metrics
3.	Use various data sources and collect data relating to the metrics and key performance indicators
4.	Identify key performance indicators for a given goal, identify data relating to the metrics and key performance indicators
5.	Use ready-made web analytics tools (Google Analytics) and be able to understand a statistical programming language (R), also use its graphical development environment (Deduce) for data exploration and analysis.

UNIT-I

Introduction to web and social analytics: **15 Hours**

Overview of web & social media (Web sites, web apps, mobile apps and social media), Impact of social media on business, Social media environment, , How to leverage social media for better services, Usability, user experience, customer experience, customer sentiments, web marketing, conversion rates, ROI, brand reputation, competitive advantages.Need of using analytics, Web analytics technical requirements., current analytics platforms, Open Source

vs licensed platform, choosing right specifications & optimal solution, Web analytics and a Web analytics 2.0 framework (clickstream, multiple outcomes Relevant Data And its Collection using statistical Programming language R.:Data (Structured data, unstructured data, metadata, Big Data and Linked Data), Participating with people centric approach, Data analysis basics (types of data, metrics and data, descriptive statistics, comparing, Basic overview of R:R-Data Types, R- Decision Making, R-Loops, R-functions, R-Strings, Arrays, R-Lists, R-Data Frame, R-CSV Files, R- Pie Charts, R-Bar charts, R-Barplots. Basic Text Mining in R and word cloud.																
UNIT-II																
Kpi/Metrics:													15 Hours			
Understand the discipline of social analytics, Aligning social objectives with business goals, Identify common social business objectives, developing KPIs; Standard vs Critical metrics. PULSE metrics (Page views, Uptime, Latency, Seven-day active users) on business and technical Issues, HEART metrics (Happiness, Engagement, Adoption, Retention, and Task success) on user behaviour issues; Bounce rate, exit rate, conversion rate, engagement, strategically aligned KPIs, Measuring Macro & micro conversions, On-site web analytics, off-site web analytics, the goal- signal-metric process. Case study on Ready-made tools for Web and social media analytics (Key Google Analytics metrics, dashboard, social reports, Tableau Public and KNIME																
Mining Twitter:																
Exploring Trending Topics, Discovering What People Are Talking About, and More: Why Is Twitter All the Rage?, Exploring Twitter’s API, Fundamental Twitter Terminology, Creating a Twitter API Connection, Exploring Trending Topics, Searching for Tweets, Analyzing the 140 Character, Extracting Tweet Entities, Analyzing Tweets and Tweet Entities with Frequency Analysis, Computing the Lexical Diversity of Tweets, Examining Patterns in Retweets, Visualizing Frequency Data with Histograms.																
Mining Facebook:																
Analyzing Fan Pages, Examining Friendships, and More: Overview, Exploring Facebook’s Social Graph API, Understanding the Social Graph API, Understanding the Open Graph Protocol, Analyzing Social Graph Connections, Analyzing Facebook Pages, Examining Friendships.																
UNIT-III																
Data Mining in Social Media :													10 Hours			
Introduction, Data Mining in a Nutshell, Social Media, Motivations for Data Mining in Social Media, Data Mining Methods for Social Media, Data Representation, Data Mining - A Process, Social Networking Sites: Illustrative Examples, The Blogosphere: Illustrative Examples, Related Efforts, Ethnography and Netnography, Event Maps																
Text Mining in Social Networks																
Introduction, Keyword Search, Query Semantics and Answer Ranking, Keyword search over XML and relational data, Keyword search over graph data, Classification Algorithms, Clustering Algorithms, Transfer Learning in Heterogeneous Networks																
Course Outcomes: At the end of the course student will be able to																
1.	Understand social media, web and social media analytics, and their potential impact.															
2.	Identify and explain ready-made web analytics tools (Google Analytics) and able to understand a statistical programming language (R).															
3.	Identify key performance indicators for a given goal, identify data relating to the metrics and key performance indicators.															
4.	Determine how twitter mining can be done for better services.															
5.	Explain text mining and data mining in social networks.															
Course Outcomes Mapping with Program Outcomes & PSO																
↓ Course Outcomes	Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
														1	2	3
	CS3234-1.1	2	2											2		
	CS3234-1.2	3	2	3											3	
	CS3234-1.3	2	3											1		
	CS3234-1.4	1	2												2	
	CS3234-1.5	2	2												2	

1: Low 2: Medium 3: High	
TEXTBOOKS:	
1.	Matthew A. Russell,” Mining of Social web, O’Reilly”, Second Edition ,ISBN-13: 978- 1449367619, 2013,
2.	Charu C Agarwal, “Social Network Data Analytics”, Springer; October 2014.
REFERENCE BOOKS:	
1.	Hand, Mannila, and Smyth,”Principles of Data Mining”, Cambridge, MA: MIT Press, ISBN: 026208290X, 2001.
2.	Avinash Kaushik, “Web Analytics 2.0: The Art of Online Accountability and Science of Customer Centricity”, John Wiley & Sons; Pap/Cdr Edition, 2009.
3.	Tom Tullis, Bill Albert, “Measuring the User Experience: Collecting, Analyzing, and Presenting Usability Metrics”, First Edition ,Morgan Kaufmann ,2008.
4.	Jim Sterne, Social Media Metrics: “How to Measure and Optimize Your Marketing Investment”, John Wiley & Sons ,2010.
5.	Brian Clifton, “Advanced Web Metrics with Google Analytics”, Third Edition, John Wiley & Sons ,2012.
E Books / MOOCs/ NPTEL	
1.	http://www.webpages.uidaho.edu/~stevel/504/Mining-the-Social-Web-2nd-Edition.pdf
2.	http://dbmanagement.info/Books/MIX/Computer_Science_Mit_Press_Principles_Of_Data_Mining_Big_Data.pdf http://nptel.ac.in/courses/106106146/21#watch/ https://www.coursera.org/learn/social-media-data-analytics

BIO INFORMATICS																
Course Code:				CS1331-1				Course Type				PEC				
Teaching Hours/Week (L:T:P):				3:0:0				Credits				03				
Total Teaching Hours				40+0+0				CIE + SEE Marks				50+50				
Teaching Department: Computer Science & Engineering (Cyber Security)																
UNIT-I																
Bioinformatics and Computational Biology														15 Hours		
Bioinformatics and Computational Biology, Nature & Scope of Bioinformatics. The central dogma of molecular biology and bio-sequences associated with it, RNA classification –coding and non coding RNA- mRNA, tRNA, miRNA and sRNA, RNAi. DNA and RNA structure – Nucleic Acid structure and function, Genetic Code, Genes and Evolution Importance of databases - Biological databases-primary sequence databases, Composite sequence databases- Secondary databases- nucleic acid sequence databases - Protein sequence data bases - structure databases, Types of databases, Data retrieval tools – Entrez																
UNIT-II																
Sequence alignment														15 Hours		
Sequence alignment – local/global, pairwise sequence alignment, scoring methods. Needleman and Wunsch algorithm, global and local alignments. Multiple sequence alignment. Scoring matrices: basic concept of a scoring matrix, Matrices for nucleic acid and proteins sequences, PAM and BLOSUM series, principles based on which these matrices are derived. Differences between distance & similarity matrix. Introduction, Advantages, Phylogenetic Trees, Tree topologies, Methods for phylogenetic analysis- Distance Matrix methods, Character based methods. HMM (Hidden Markov Model): Introduction to HMM, Forward algorithm, Viterbi algorithm, applications in Bioinformatics																
UNIT-III																
Introduction to Gene expression														10 Hours		
General introduction to Gene expression in prokaryotes and eukaryotes Protein and RNA structure Prediction: Predicting RNA secondary structure - Nussinov Algorithm, Energy minimization methods - Zuker Algorithm. Amino Acids, Polypeptide Composition, Protein Structures, Algorithm for protein folding, Structure prediction																
Course Outcomes: At the end of the course student will be able to																
1.	Understand the concepts of bioinformatics															
2.	Identify different types of biological sequence															
3.	Analyse multiple sequences and find conserved regions															
4.	Understand RNA and Protein folding															
5.	Analyze the algorithm for protein folding and structure prediction															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS1331-1.1		3	1											2		
CS1331-1.2		3	1				1	1						2		
CS1331-1.3		3	2	3	2	2								3	1	
CS1331-1.4		3	1	1			1	1						1		
CS1331-1.5		3	2	3	1	1	1	1						3	1	
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.	S C Rastogi, N Mendiratta and P Rastogi, " Bioinformatics: Methods and Applications", New Delhi, 2015.															

2.	D E Krane and M L Raymer, Fundamental Concepts of Bioinformatics, 2006.
REFERENCE BOOKS:	
1.	Andreas D.Baxeavanis, B F Francis Ouellette, "Bioinformatics - A Practical Guide to the Analysis of Genes and Proteins", Third Edition,
2.	Neil C Jones and Pavel A Pevzner, An Introduction to Bioinformatics Algorithms, MIT press, 2004.

BIG DATA ANALYTICS			
Course Code	CS2331-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
1.	Study and comprehend in depth the fundamental issues behind the Big Data problem.		
2.	Understand various Big Data technologies and different NoSQL databases. Learn MongoDB NoSQL database.		
3.	Understand various Big Data technologies and Hadoop Components such as HDFS,MapReduce. Learn MapReduce Programming		
4.	Determine various techniques for analyzing the data such as Spark,Pig and Hive.		
UNIT-I			
Introduction to Big Data			15 Hours
Introduction to Big Data: Types of digital Data, Characteristics of Data, Evolution of Big Data, Definition of Big Data, Challenges with Big Data, What Is Big Data? Why Big data? Traditional BI versus Big data.Big Data Analytics: What is Big Data Analytics? Why this sudden Hype around Big Data analytics? Data Science, Terminologies used in Big Data environments			
Introduction to NoSQL: Where it is used, Types of NoSQL databases, Why NoSQL, Advantages of NoSQL,			
Introduction to MongoDB: What is MongoDB? Why MongoDB? Using JSON, Creating or generating a unique key, Data types in MongoDB, MongoDB Query Language: Insert method, Save method, Update method, Remove method, Find method, Dealing with Null values, Count, Limit, Sort, Skip.			
UNIT-II			
Introduction to Hadoop			15 Hours
Introduction to Hadoop : Introducing Hadoop, need of Hadoop, limitations of RDBMS, RDBMS versus Hadoop, Distributed Computing Challenges, History of Hadoop , Hadoop Overview, Use Case of Hadoop, Hadoop Distributors, HDFS (Hadoop Distributed File System) , Processing Data with Hadoop, Managing Resources and Applications with Hadoop YARN (Yet another Resource Negotiator).			
Writing Hadoop MapReduce Programs: Understanding the basics of MapReduce, Introducing Hadoop MapReduce,Understanding the different Java concepts used in Hadoop programming, Writing a Hadoop MapReduce example, Understanding several possible MapReduce definitions to solve business problems.			
SPARK: Spark applications, Jobs, stages and Tasks, Resilient Distributed Datasets(RDD), Anatomy of SPARK Job Run; SPARK on YARN			
UNIT-III			
			10 Hours
Hadoop Ecosystem : Understanding Hadoop subprojects: Mahout, Apache HBase, Hive, Pig, Apache Sqoop, Apache Zookeeper, Apache Solr, Ambari.			
HBase: What is HBase? Storage Mechanism in HBase, Features of HBase, HBase and RDBMS, HBase and HDFS.			
Introduction to Pig: What is Pig? Pig on Hadoop, Pig Philosophy, Pig Latin overview; Pig Data Types; Running Modes of Pig; Execution Modes of PIG, Relational operators, EVAL function, Complex data types.			
Introduction to Hive: What is Hive? Architecture; HIVE Data Types; HIVE File Format; Hive Query Language(HQL).			
Course Outcomes: At the end of the course student will be able to			
1.	Outline the theory of big data, and explain applications of big data.		
2.	Get the idea of NoSQL databases, different types of NoSQL datastores.		

3.	Analyse the technological foundations for Big data with hadoop and design of hadoop distributed file system.
4.	Understand the concept of MapReduce programmig and Spark workflow.
5.	Understand the need of Big Data Analytics and Analyze Hadoop Ecosystem

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
CS2331-1.1	3	1	3		1							2	2	3	
CS2331-1.2	2	1	3		2							2		3	
CS2331-1.3	2	2	3		3							3		3	
CS2331-1.4	3	1	2		2							2	2	3	
CS2331-1.5	3	2	3		3							2		3	

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Seema Acharya, Subhashini Chellappan, "Big Data Analytics", 1st Edition, Wiley, 2015.
2. Vignesh Prajapati, "Big Data Analytics with R and Hadoop", Packet Publishing 2013.
3. Tom White, Hadoop: The Definitive Guide, 4th Edition, O'Reilley, 2012.

REFERENCE BOOKS:

1. V1. Boris lublinsky, Kevin t. Smith, Alexey Yakubovich, "Professional Hadoop Solutions", Wiley, ISBN: 9788126551071, 2015.
2. Chris Eaton, Dirk deroosetal. , "Understanding Big data ", McGraw Hill, 2012.
3. E. Capriolo, D. Wampler, and J. Rutherglen, Programming Hive, O'Reilley, 2012.
4. Lars George, HBase: The Definitive Guide, O'Reilley, 2011.
5. Alan Gates, Programming Pig, O'Reilley, 2011

E Books / MOOCs/ NPTEL

1. <https://www.upgrad.com/big-data-analytics->
2. <https://www.coursera.org/courses?query=big%20data%20analytics.>
3. <https://www.edx.org/micromasters/big-data>

SOFT COMPUTING																
Course Code:				CS2332-1				Course Type				PEC				
Teaching Hours/Week (L:T:P):				3:0:0				Credits				03				
Total Teaching Hours				40+0+0				CIE + SEE Marks				50+50				
Teaching Department: Computer Science & Engineering (Cyber Security)																
UNIT-I																
Introduction To Soft Computing:														15 Hours		
Evolution of Computing, Soft and Hard Computing, Soft Computing characteristics, Constituents and Applications, AI Definitions and Intelligent systems architecture.																
Genetic Algorithms:																
Introduction to Genetic Algorithms (GA) – Conceptual GA algorithm, Reproduction operators Mutation and cross over, Applications of GA, Learning Definitions, strategies, Machine Learning Approach, applications and Architecture of learning agent																
UNIT-II																
Neural Networks :														15 Hours		
Introduction to Neural Networks, Applications, Structure and function of Biological Neuron, ANN introduction, Perceptron, Multi-layer feed forward Networks with Back propagation																
Fuzzy Logic:																
Fuzzy Sets,Operationson Fuzzy Sets,MembershipFunctions,FuzzyRules, Models,Fuzzy Reasoning and Fuzzy InferenceSystems.																
UNIT-III																
Decision Making And Expert Systems:														10 Hours		
Single person, multi person, Multi criteria and Multi stage decision making, Expert system features, architecture and applications																
Course Outcomes: At the end of the course student will be able to																
1.		Explain the constitutes and applications Soft Computing														
2.		Perform reproduce operations like mutations and crossover														
3.		Demonstrate the concepts of neural networks.														
4.		Explain the fuzzy member ship functions and models														
5.		Apply decision making strategies to real world examples														
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS2332-1.1		2	1	1									1	2	2	
CS2332-1.2		1	2	2		2	1		1				2	2	2	
CS2332-1.3		1	1	1									1	2	2	
CS2332-1.4		2	1	2									2	2	2	
CS2332-1.5		1	2	1		2	1		1				1	2	2	
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.		Jyh-ShingRogerJang,Chuen-TsaiSun,EijiMizutani,“Neuro-Fuzzy and Soft Computing”, Prentice-Hall of India,2003.														
2.		George J. Klir and Bo Yuan, “Fuzzy Sets and Fuzzy Logic-Theory and Applications”,Prentice Hall,1995.														
3.		James A. Freeman and David M.Skapura, “Neural Networks Algorithms, Applications, and Programming Techniques”, Pearson Edition.,2003.														
4.		Simon Haylion “Neural Networks”, Prentice-Hall of India,2003.														
REFERENCE BOOKS:																

1.	Mitchell Melanie, "An Introduction to Genetic Algorithm", Prentice Hall, 1998.
2.	David E. Goldberg, "Genetic Algorithms in Search, Optimization and Machine Learning", Addison Wesley, 1997.
3.	S. N. Sivanandam, S. Sumathi and S. N. Deepa, "Introduction to Fuzzy Logic using MATLAB", Springer, 2007.
4.	S. N. Sivanandam, S. N. Deepa, "Introduction to Genetic Algorithms", Springer, 2007.
5.	Jacek M. Zurada, "Introduction to Artificial Neural Systems", PWS Publishers, 1992.

HUMAN COMPUTER INTERACTION			
Course Code:	CS3331-1	Course Type	PEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
1.	Understand basics of HCI and different HCI models.		
2.	Analyze the research methods and the guidelines to be followed in designing HCI.		
3.	Analyze the task modeling in HCI systems.		
4.	Understand the cognitive architecture of HCI.		
5.	Apply HCI systems to various real time problems		
UNIT-I			
INTRODUCTION: Course Learning Objectives and overview, Historical evolution of the field. INTERACTIVE SYSTEM DESIGN: concept of usability -definition and elaboration, HCI and software engineering, GUI design and aesthetics, prototyping techniques. MODEL-BASED DESIGN AND EVALUATION: Introduction to different types of models, GOMS family of models(KLMandCMN-GOMS),Fitt’slaw and Hick-Hyman’s law, Model based design case studies.			15 Hours
UNIT-II			
Guidelines in HCI: Shneiderman’s eight golden rules, Norman’s seven principles, Norman’s model of interaction, Nielsen’s ten heuristics with example of its use, Heuristic evaluation, Contextual inquiry, Cognitive walk-through. Empirical research methods in HCI: Introduction (motivation, issues, research question formulation techniques),Experiment design and data analysis(with explanation of one-way ANOVA). Task modeling and analysis: Introduction to formalism in dialog design, design using FSM (finite state machines),State charts and(classical) Petri Nets in dialog design			15 Hours
UNIT-III			
Cognitive architecture: Introduction to CA, CA types, relevance of CA in IS design ,Model Human Processor (MHP). Design - Case Studies: Case Study 1- Multi- Key press Hindi Text Input MethodonaMobilePhone,CaseStudy2-GUIDesignforamobilephone based Matrimonial application. Case Study 3 - Employment Information System for unorganised construction workers on a MobilePhone.			10 Hours
Course Outcomes: At the end of the course student will be able to			
1.	Understand basics of HCI and different HCI models.		
2.	Analyze the research methods and the guidelines to be followed in designing HCI.		
3.	Analyze the task modeling in HCI systems.		
4.	Understand the cognitive architecture of HCI.		

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5.	Apply HCI systems to various real time problems															
Course Outcomes Mapping with Program Outcomes & PSO																
	Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
	↓ Course Outcomes													1	2	3
	CS3331-1.1	1	1	1	1									1	1	
	CS3331-1.2	3	3	3	1									1	1	
	CS3331-1.3	2	2	1	1									2	1	
	CS3331-1.4	1	1	2	1									1	1	
	CS3331-1.5	3	3	3	3	2								3	2	
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.	Jennifer Preece, Helen Sharp and Yvonne Rogers, Interaction design: Beyond Human-Computer Interaction, 4th edition Helen Sharp, John Wiley and Sons, 2015, ISBN:978-1-119-02075-2															
2.	Ben Shneiderman and Catherine Plaisant, Designing the User Interface: Strategies for Effective Human-ComputerInteraction.6th Edition. Pearson.2017															

NATURAL LANGUAGE PROCESSING			
Course Code:	CS3332-1	Course Type	PEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
1.	Explain the importance of NLP and breaking of words.		
2.	Outline the syntax, semantics, and pragmatics in speech language		
3.	Describe the models for different applications of NLP.		
UNIT-I			
Knowledge in speech and language processing; Ambiguity; Models and algorithms; Regular expressions - Basic Regular Expression Pattern, Disjunction, Grouping, and Precedence, A Simple Example, A More Complex Example, Advanced Operators(2.1.1 - 2.1.5), Using an FSA to Recognize Sheep-talk, Formal Languages, Another Example, Non-Deterministic FSAs, Using an NFSA to Accept String(2.2.1 - 2.2.5); Words and Transducers - Inflectional Morphology, Derivational Morphology, Cliticization, Non-concatenative Morphology, Agreement(3.1.1 - 3.1.5); Finite-state morphological parsing(3.2); Detecting and correcting spelling errors, Minimum edit distance(3.10 - 3.11), N-Grams - Counting words in corpora, Simple(un-smoothed) n-grams(4.1 - 4.2); Part – of - Speech Tagging - English word classes, tagsets for English(5.1 - 5.2), Hidden Markov Models - Markov chains, The Hidden Markov Model(6.1 - 6.2). (Refer Text Book 1)			15 Hours
UNIT-II			

Syntactic Parsing: Grammars and syntax structure, A top down parser, Depth first strategy vs Breadth first strategy, Bottom up chart parser, Efficiency considerations, Transition Network Grammars, Top down chart parser. (Refer Text Book 2) Representing Meaning: Computational desiderata for representations, Meaning structure of language, Model theoretic semantics, First order logic. (17.1 – 17.4) Computational Semantics: Syntax driven semantic analysis, Semantic augmentations to context-free grammar rules, Quantifier scope ambiguity and under specification, Unification based approaches to semantic analysis. (18.1 – 18.4) (Refer Text Book 1)													15 Hours			
UNIT-III																
Applications: Information Extraction - Named entity recognition, Relation detection and classification, Temporal and event processing, Template filling (22.1-22.4), Question Answering and Summarization – Information retrieval, Factoid question answering, Summarization, Multi document summarization (23.1 – 23.4); Dialog and Conversational Agents - Properties of human conversations, Basic dialogue systems, VoiceXML (24.1-24.3) (Refer Text Book 1)													10 Hours			
Course Outcomes: At the end of the course student will be able to																
1.	Explain the understanding of core tasks in NLP.															
2.	Demonstrate the syntax, semantics, and pragmatics in speech language.															
3.	Implement and experiment the models for different applications of NLP.															
4.	Demonstrate understanding of state-of-the-art algorithms and techniques for text-based processing of natural language.															
5.	Demonstrate understanding of human languages and be familiar with the most mainstream descriptive and theoretical frameworks for handling their properties.															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS3332-1.1		3	3											3	3	
CS3332-1.2		3	3	2										3	3	
CS3332-1.3		3	2											3	3	
CS3332-1.4		3	3	2										3	3	
CS3332-1.5		3	2											3	3	
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.	Jurafsky, D. and J. H. Martin, “Speech and language processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition”, Second Edition, Prentice Hall, 2008.															
2.	Allen, James, “Natural Language Understanding”, Second Edition, Benjamin/Cumming, 1995.															
REFERENCE BOOKS:																
1.	Steven Bird, S., Klein, E., Loper, E, “Natural Language Processing with Python- Analyzing Text with the Natural Language Toolkit”, O'Reilly Media, 2010.															
2.	Grant S Ingersoll, Thomas S. Morton, and Andrew L. Farris,”Taming text: how to ` find, organize, and manipulate it” Manning Publications Co., 2013.															
3.	Feldman Ronen, and James Sanger,”The text mining handbook: advanced approaches in															

	analyzing unstructured data”, Cambridge university press, 2007.
4.	Christopher D Manning, and Hinrich Schütze,” Foundations of statistical natural language processing”, MIT press, 1999.
E Books / MOOCs/ NPTEL	
1.	http://www.allitebooks.in/mastering-natural-language-processing-python/
2.	https://www.experfy.com/training/courses/natural-language-processing-for-retail
3.	https://www.class-central.com/course/coursera-natural-language-processing-9603
4.	Graduate Attributes (GA) This course will map the following GA as per NBA: 1. Engineering Knowledge 2. Design / development of solutions 3. Conduct investigation of complex problems 4. The engineer and society 5. Environment and sustainability 6. Ethics 7. Individual and team work 8. Communication 9. Life-long Learning

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VIRTUAL REALITY																
Course Code:				CS3333-1				Course Type				PEC				
Teaching Hours/Week (L: T: P):				3:0:0				Credits				03				
Total Teaching Hours				40+0+0				CIE + SEE Marks				50+50				
Teaching Department: Computer Science & Engineering (Cyber Security)																
UNIT-I																
INTRODUCTION TO VIRTUAL REALITY														15 Hours		
Virtual Reality & Virtual Environment : Introduction – Computer graphics – Real time computer graphics Flight Simulation – Virtual environments –requirement – benefits of virtual reality- Historical development of VR : Introduction – Scientific Landmark -3D Computer Graphics :Introduction – The Virtual world space – positioning the virtual observer – the perspective projection – human vision – stereo perspective projection – 3D clipping – Color theory – Simple 3D modeling Illumination models – Reflection models – Shading algorithms- Radiosity – Hidden Surface Removal – Realism-Stereographic image. (T1)																
GEOMETRIC MODELLING																
Geometric Modeling: Introduction – From 2D to 3D – 3D space curves – 3D boundary representation - Geometrical Transformations: Introduction – Frames of reference – Modeling transformations – Instances – Picking – Flying – Scaling the VE – Collision detection - A Generic VR system: Introduction – The virtual environment – the Computer environment – VR Technology – Model of interaction. (T1)																
VR Systems.																
UNIT-II																
VIRTUAL ENVIRONMENT														15 Hours		
Animating the Virtual Environment: Introduction – The dynamics of numbers – Linear and Non-linear interpolation - The animation of objects – linear and non- linear translation - shape & object inbetweening – free from deformation – particle system- Physical Simulation: Introduction – Objects falling in a gravitational field– Rotating wheels – Elastic collisions – projectiles – simple pendulum – springs – Flight dynamics of an aircraft.(T1)																
UNIT-III																
VR HARDWARES & SOFTWARES														10 Hours		
Human factors: Introduction – the eye - the ear- the somatic senses - VR Hardware: Introduction – sensor hardware – Head-coupled displays –Acoustic hardware – Integrated VR systems-VR Software: Introduction – Modeling virtual world –Physical simulation- VR toolkits – Introduction to VRML. (T1)																
VR APPLICATION																
Virtual Reality Applications: Introduction – Engineering – Entertainment – Science – Training – The Future: Introduction – Virtual environments – modes of interaction. (T1)																
Course Outcomes: At the end of the course student will be able to																
1.	Explain the various aspects of virtual reality and 3D Modeling Illumination models.															
2.	Demonstrate Geometric modeling Concepts.															
3.	Understand the concepts of linear and Non-linear interpolation and translation that are used to animate the virtual Environment.															
4.	Demonstrate the physical simulations in Gravitational field.															
5.	Understand the required VR Hardware and its applications.															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
CS3333-1.1		3		2							1			2		
CS3333-1.2		3	3	2							1			2	2	
CS3333-1.3		3									1			1		
CS3333-1.4		3									1		1	2		
CS3333-1.5		3				3			3		1		1	1		
1: Low 2: Medium 3: High																
TEXTBOOKS:																

1.	John Vince, “Virtual Reality Systems “, Pearson Education Asia, 2007.
REFERENCE BOOKS:	
1.	Adams, “Visualizations of Virtual Reality”, Tata McGraw Hill, 2000.
2.	Grigore C. Burdea, Philippe Coiffet , “Virtual Reality Technology”, Wiley Interscience, 2nd Edition, 2006.
3.	William R. Sherman, Alan B. Craig, “Understanding Virtual Reality: Interface, Application, and Design”, Morgan Kaufmann, 2008.
E Books / MOOCs/ NPTEL	
1.	www.vresources.org .
2.	www.vrac.iastate.edu .
3.	www.w3.org/MarkUp/VRML .

Ability Enhancement Courses

Section 1

INNOVATION AND DESIGN THINKING			
Course Code:	ME1654-1	Course Type	AEC
Teaching Hours/Week (L: T: P):	1:0:0	Credits	01
Total Teaching Hours	15+0+0	CIE + SEE Marks	50+50
Prerequisite	---		
Teaching Department: Mechanical Engineering			
Course Objectives:			
1.	To explain the concept of design thinking for product and service development		
2.	To explain the fundamental concept of innovation and design thinking		
3.	To discuss the methods of implementing design thinking in the real world.		
	Note: Teaching-Learning Process (General Instructions)		
	These are sample Strategies; which teachers can use to accelerate the attainment of the various course outcomes.		
	- Lecturer method (L) does not mean only the traditional lecture method, but a different type of teaching method may be adopted to develop the outcomes. - Show Video/animation films to explain concepts. - Encourage collaborative (Group Learning) Learning in the class. - Ask at least three HOTS (Higher-order Thinking) questions in the class, which promotes critical thinking. - Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develops thinking skills such as the ability to evaluate, generalize, and analyze information rather than simply recall it. - Topics will be introduced in multiple representations. - Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them. - Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.		
UNIT-I			
Design Thinking			03 Hours
Understanding Design Thinking:			
Shared model in team-based design – Theory and practice in Design thinking – Explore the presentation.			
Tools for Design Thinking:			
Real-Time design interaction capture and analysis – Empathy for design			
Teaching-Learning Process:			
Introduction about the design thinking: Chalk and Talk method			
Theory and practice through presentation			
Case studies on design thinking for real-time interaction and analysis			
UNIT-II			
Design Thinking for Strategic Innovations			05 Hours
Design Thinking in IT			
Design Thinking to Business Process modeling – Scenario-based Prototyping			
Design Thinking for Strategic Innovations			
Growth – Storytelling representation – Strategic Foresight - Change – Sense Making – Maintenance - Relevance – Value redefinition - Extreme Competition – experience design - Standardization – Humanization - Creative Culture – Rapid prototyping, Strategy and Organization – Business Model design.			
Teaching-Learning Process			
Case studies on design thinking and business acceptance of the design			

Business model examples of successful designs

UNIT-III

Design Thinking Workshop

07 Hours

Design Thinking Workshop Empathize, Design, Ideate, Prototype and Test

Teaching-Learning Process

Presentation by the students on the success of Live project on design thinking in a group of 4 students

Course Outcomes: At the end of the course student will be able to

1.	Explain various design process procedure
2.	Generate and develop design ideas through different techniques
3.	Explain the significance of Design Thinking to Understand products

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
ME1654-1.1	2		2												
ME1654-1.2							2	2							
ME1654-1.3										3	3				

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	John.R.Karsnitz, Stephen O’Brien and John P. Hutchinson, “Engineering Design”, Cengage learning (International edition) Second Edition, 2013.
2.	Roger Martin, "The Design of Business: Why Design Thinking is the Next Competitive Advantage", Harvard Business Press , 2009.
3.	Hasso Plattner, Christoph Meinel and Larry Leifer (eds), "Design Thinking: Understand – Improve– Apply", Springer, 2011.
4.	Idris Mootee, "Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School”, John Wiley & Sons 2013.
5.	Yousef Haik and Tamer M.Shahin, “Engineering Design Process”, Cengage Learning, Second Edition, 2011.
6.	Jeanne Liedtka, Andrew King and Kevin Bennett, "Solving Problems with Design Thinking - Ten Stories of What Works", Columbia Business School Publishing, Sep 2013.

E Books / MOOCs/ NPTEL

1.	www.tutor2u.net/business/presentations/. /productlifecycle/default.html
2.	https://docs.oracle.com/cd/E11108_02/otn/pdf/. /E11087_01.pdf
3.	www.bizfilings.com › Home › Marketing › Product Developmen
4.	https://www.mindtools.com/brainstm.html
5.	https://www.quicksprout.com/. /how-to-reverse-engineer-your-competit
6.	www.vertabelo.com/blog/documentation/reverse-engineering
7.	https://support.microsoft.com/en-us/kb/273814
8.	https://support.google.com/docs/answer/179740?hl=en
9.	https://www.youtube.com/watch?v=2mjSDiBaUIM
10.	thevirtualinstructor.com/foreshortening.html
11.	https://dschool.stanford.edu/.../designresources/.../ModeGuideBOOTCAMP2010L.pdf
12.	https://dschool.stanford.edu/use-our-methods/ 6. https://www.interactiondesign.org/literature/article/5-stages-in-the-design-thinking-process 7.
13.	http://www.creativityatwork.com/design-thinking-strategy-for-innovation/ 49 8.
14.	https://www.nngroup.com/articles/design-thinking/ 9.
15.	https://designthinkingforeducators.com/design-thinking/ 10.

BACK END DEVELOPMENT																
Course Code:				CS3003-1				Course Type				AEC				
Teaching Hours/Week (L: T: P):				1:0:2				Credits				02				
Total Teaching Hours				15+0+26				CIE + SEE Marks				50+50				
Teaching Department: Computer Science & Engineering (Cyber Security)																
UNIT-I																
														15 Hours		
Unit Testing, TDD and Refactoring :Introduction: Unit Testing, What Is a Unit and What Do We Unit Test?, Unit Test Cases, JUnit and Its Configurations, Assertions in Unit Testing, Unit Testing for Library Class, Mocking in Unit Testing, Mocking the Tests in Library class, Introduction: Test Driven Development and Refactoring, Characteristics of TDD, Requirements of a Good Test Case, Developing the Library Class the TDD way, Refactoring, Need of Refactoring and What to Refactor, Refactoring the Library Class																
UNIT-II																
MVC Architecture :Client-Server Architecture, Networking Terminologies, Hypertext Transfer Protocol (HTTP), Hypertext Markup Language (HTML), Understanding web applications, Introduction to dynamic website creation, Your first web application, Web application using JSP, Introduction to Model-View-Controller architecture, Spring MVC and Springboot, Hello World using Spring MVC, Technical Blog Application, Presentation Logic, Dependency injection, Login Feature, Registration, Create Blog Post, Assessment 1 - Run the application, Assessment 2 - Display images, Assessment 3 - Registration, Assessment 4 - Login, Assessment 5 - Image Details, Spring Introduction, Design Patterns, SpringBoot Basics, Servers, Servlets, JSP and Archiving in Spring, Spring MVC & SpringBoot, Test																
UNIT-III																
Databases and ORMs, Agile and Extreme Programming :Relational Database, Structured Query Language(SQL), Data Retrieval with SQL, Order by Clause, Aggregate Functions, Group by' and 'Having' Clause, Object Relational Mapping(ORM), Nested Queries and Inner Join, Introduction to JDBC API, Implementation of JDBC API, Introduction to ORM, JPA Configuration, JPA implementation, Repository Class, Create Post Implementation, Edit Post, Delete Post, User Registration, User Login, HTTP Session, Mapping User to a Post, Assigning categories to a post, Image Hoster Project. Requirements Engineering, Software Development: The Remaining Steps, Why SDLC Models?, Waterfall Model, Rational Unified Process, Agile and Extreme Programming, Industry Demonstration, Introduction to Agile, Agile Manifesto: Principles, What Is Extreme Programming?, Values and Practices of Extreme Programming, Advantages of Extreme Programming, User Stories, Comprehensive User Stories and Backlog Documents, Estimation and Planning, Planning Poker, Creating User Stories, Theme and Epic, Backlog Document, Setting Priorities, Release Planning: Planning Poker, Product Backlog With Estimates.																
Course Outcomes: At the end of the course student will be able to																
1.		Explain unit testing and test driven development with its characteristics.														
2.		Describe refactoring and refactoring library classes.														
3.		Discuss MVC, Explain how MVC architecture role in MVC spring and springboot.														
4.		Apply the knowledge of HTTP in dynamic web page creation using JSP.														
5.		Explain and understand Databases and ORMs, Agile and Extreme Programming in backend development.														
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes		1	2	3	4	5	6	7	8	9	10	11	12	PSO		
Course Outcomes														1	2	3
CS3003-1.1		2	1	3		2								1		3
CS3003-1.2		2	1	2										1		3
CS3003-1.3		3	2	2		3								1		3
CS3003-1.4		3	3	3		3								1		3

CS3003-1.5	2	2	2		2							1		3
1: Low 2: Medium 3: High														
TEXTBOOKS:														
1.Kathy Sierra, Bert Bates: Head First Java , 2 nd Edition.														

RESEARCH METHODOLOGY	
Course Code:	HU1010-1
Teaching Hours/Week (L: T: P):	2:0:0
Total Teaching Hours:	30+0+0
Course Type:	AEC
Credits:	02
CIE + SEE Marks:	50+50
Teaching Department: Respective Department	
Course Objectives:	
1.	Understand Research Formulation and Design
2.	Inculcate the ability to collect Data and its analysis
3.	Enhance knowledge of Soft Computing
4.	Comprehend Research Ethics and the art of publishing
5.	Develop Interpretative Skills and write reports
UNIT-I	
Research Formulation and Design	6 Hours
Motivation and Objectives – Research methods vis-a-vis Methodology. Types of research – Descriptive vis-a-vis Analytical, Applied vis-a-vis Fundamental, Quantitative vis-a-vis Qualitative, Conceptual vis-a-vis Empirical, concept of applied and basic research process, Criteria of good research. Defining and formulating the research problem, Selecting the problem, Importance of Literature Review, Literature Review - Primary and Secondary sources, reviews, monograph, patents, research databases, Web as a source, Critical literature review, Identifying gap areas from Literature Review, Development of working hypothesis.	
Data Collection and Analysis	6 Hours
Accepts of method validation, observation and collection of data, methods of data collection, sampling methods, data processing and analysis strategies and tools, data analysis with statically package (Sigma STAT, SPSS for student t-test, ANOVA, etc.), hypothesis testing.	
UNIT-II	
Soft Computing	6 Hours
Computer and its role in research, Use of statistical software SPSS, GRETL in research. Introduction to evolutionary algorithms - Fundamentals of Genetic algorithms, Simulated Annealing, Neural Network based optimization, Optimization of fuzzy systems.	
Research Ethics and Scholarly Publishing	6 Hours
Ethics-ethical issues, ethical committees (human & animal); IPR- intellectual property rights and patent law, commercialization, copy right, royalty, trade related aspects of intellectual property rights (TRIPS); scholarly publishing- IMRAD concept and design of research paper, citation and acknowledgement, plagiarism, reproducibility, and accountability	
UNIT-III	
Interpretation and Report Writing	3 Hours
Meaning of Interpretation, Technique of Interpretation, Precaution in Interpretation, Significance of Report Writing, Different Steps in Writing Report, Layout of the Research Report, Types of Reports, Oral Presentation, Mechanics of Writing a Research Report, Precautions for Writing Research Reports	
Course Outcomes: At the end of the course student will be able to	
1.	Formulate and design the research problem.
2.	Interpret and Analyze the Data for research.

3.	Identify and interpret the Data with Soft Computing.
4.	Apply research ethics and develop the art of publishing.
5.	Integrate interpretative skills and write reports.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes													1	2
HU1010-1.1	-	-	-	-	-	-	-	-	-	2	-	3	-	-
HU1010-1.2	-	-	-	-	-	-	-	-	-	3	-	3	-	-
HU1010-1.3	-	-	-	-	-	-	-	-	-	3	-	3	-	-
HU1010-1.4	-	-	-	-	-	-	-	-	2	2	-	2	-	-
HU1010-1.5	-	-	-	-	-	-	-	-	1	2	-	2	-	-

1: Low 2: Medium 3: High
REFERENCES:

1.	Garg, B.L., Karadia, R., Agarwal, F., & Agarwal, “An introduction to Research Methodology”, RBBSA Publishers, 2002.
2.	Wadehra, B.L., “Law relating to patents, trademarks, copyright designs and geographical indications” Universal Law Publishing, 2000.
3.	Kothari, C.R., “Research Methodology: Methods and Techniques”, New Age International, 1990.
4.	Trochim, W.M.K. “Research Methods: the concise knowledge base”, Atomic Dog Publishing, 2005.
5.	Sinha, S.C., & Dhiman, A.K., “Research Methodology”, EssEss Publications. (2 volumes), 2002.
6.	Satarkar, S.V., “Intellectual property rights and copyright”, EssEss Publications, 2000.
7.	Coley, S.M., & Scheinberg, C.A. “Proposal Writing”, Sage Publications, 1990.
8.	Day, R.A. “How to Write and Publish a Scientific Paper”, Cambridge University Press, 1992.
9.	Anthony, M., Graziano, A.M., & Raulin, M.L., “Research Methods: A Process of Inquiry”, Allyn and Bacon, 2009.

SOCIAL CONNECT AND RESPONSIBILITY

Course Code:	HU1007-1	Course Type:	AEC
Teaching Hours/Week (L: T: P):	1:0:0	Credits:	01
Total Teaching Hours:	15+0+0	CIE + SEE Marks:	50+50

Teaching Department: Respective Department
Course Objectives:

1.	Understand Rural Society
2.	Acquire the knowledge about Rural Economy
3.	Know the working of rural administration
4.	Familiarize the different rural schemes of Governance

UNIT-I

Appreciation of Rural Society	3 Hours
Rural Society, Caste and Gender relations, Rural values, Nature and Resources, Rural infrastructure.	
Understanding Rural Economy & Livelihood	3 Hours
Agriculture, Farming, Landownership, Water Management, Animal Husbandry, Non-Farm Livelihoods And Artisans, Rural Entrepreneurs.	

UNIT-II

Rural Institutions	3 Hours
Traditional Rural Organizations, Self-help Groups, Panchayat Raj Institutions - Gram Sabha, Gram Panchayat, Standing Committees	
Rural Development Programmes	3 Hours
History of Rural Development in India, Current National Programmes - Sarva Shiksha Abhiyaan, Beti Bachao – Beti Padhao, Ayushman Bharath, Swachh Bharath, PM Awaas Yojana, Skill India, Decentralised Planning, NRLM, MNREGA	

Security

UNIT-III													
Corporate Social Responsibility (CSR)												3 Hours	
Global Guidelines on CSR, Growing Importance of CSR, CSR in India													
Course Outcomes: At the end of the course student will be able to													
1.	Comprehend Rural Society and its Economy												
2.	Identify the working of Rural Administration and different rural schemes												
3.	Grasp the working of Corporate Social Responsibility												
Course Outcomes Mapping with Program Outcomes & PSO													

EMPLOYABILITY SKILL DEVELOPMENT			
Course Code:	UM1003-1	Course Type	AEC
Teaching Hours/Week (L: T: P):	1:0:0	Credits	01
Total Teaching Hours	15+0+0	CIE + SEE Marks	50+00
Teaching Department: Computer Science & Engineering (Cyber Security)			
Course Objectives:			
1.	To explain the students the necessity of clearing the aptitude tests irrespective of the written test is for jobs or higher education.		
2.	To assess the readiness of the students to appear for the aptitude test and assisting them to better it if already ready, else train them.		
3.	To evaluate the understanding of the students in answering quantitative multiple-choice questions and guide them to improve it.		
4.	To evaluate the preparedness of the students to answer the analytical and logical questions.		
5.	To evaluate the quality of the students with regard to their professional language grammar, vocabulary and communication skills.		
UNIT-I			
Quantitative			06 Hours
Numbers (Odd, even, H.C.F & L.C.M, Square roots & cube roots, Average, Percentage), Ratios & Proportions, Partnership, Time & work, Pipes & Cistern, Speed, Problems on trains, Problems on boats & streams, Allegation & Mixtures.			
UNIT-II			
Analytical/ Logical			06 Hours
Numerical logic (next number in series, odd man out), Coded language, Syllogism, Direction (N-E-W-S), Seating arrangement, Blood relations, Statement & Conclusion			

Section

UNIT-III																
Verbal															03 Hours	
Vocabulary (root words, prefix, suffix, synonyms, antonyms), One word substitution, Idiom/phrases, Sentence completion, Active & Passive voice, Direct and indirect speech.																
Course Outcomes: At the end of the course student will be able to																
1.	Answer the quantitative multiple-choice questions.															
2.	Analyse the analytical and logical questions.															
3.	Improve the professional language grammar, vocabulary and communication skills.															
4.	Clear the aptitude tests of any employer or higher educational institution.															
5.	Advance in the chosen field of interest by appending aptitude skills with the technical skills															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
UM1003-1.1		3	3							2	2	1				
UM1003-1.2		3	3							2	2	1				
UM1003-1.3		3	3	2						2	2	1				
UM1003-1.4		3	3	2						2	2	1				
UM1003-1.5		3	3	2						2	2	1				
1: Low 2: Medium 3: High																
TEXTBOOKS:																
1.	Aggarwal R.S, “Quantitative Aptitude for Competitive Examinations”, S Chand Publishing.															
2.	Aggarwal R.S, “A modern approach to verbal and non-verbal reasoning”, S Chand Publishing.															
REFERENCE BOOKS:																
1.	Bharath Patodi and Aditya Choudhary, “Verbal Ability & Comprehension”, Disha Publication, Second edition, 2015.															
2.	Shakuntala Devi, “Joy of numbers”, Orient Black Swan.															
3.	Shakuntala Devi, “More puzzles to puzzle you”, Orient Black Swan.															
E Books / MOOCs/ NPTEL																
1.	https://www.indiabix.com															
2.	https://www.faceprep.in															

LIFE SKILLS FOR ENGINEERS			
Course Code:	HU1008-1	Course Type:	AEC
Teaching Hours/Week (L: T: P):	1:0:0	Credits:	01
Total Teaching Hours:	15+0+0	CIE + SEE Marks:	50+50
Teaching Department: Respective Department			
Course Objectives:			
1.	Understand Time Management, Managing Information Overload, Coping with Peer pressure and Stress Management		
2.	Familiarize the Science behind Personal Health Management and Addictions.		
3.	Appreciate the importance of cultivating good hobbies, need for forming good habits and discarding bad habits and holding difficult conversations during crises		
4.	Comprehend the importance of Creative Thinking, Continuous and Lifelong Learning, Collaboration and Team Work.		
5.	Equip them to excel in real work environment proactively		
UNIT-I			
Introduction to Life Skills			3 Hours
Meaning and Importance of Life Skills, Competitive Job market, Fast paced changes in Technology, Proliferation of Electronic Gadgets and harmful online content.			
Time Management			
Introduction to Time Management, Impulsive Behaviour vis-a-vis goal Directive Behaviour, Time log, Information Overload and coping with Information & Communication Technology (ICT) Revolution; Proliferation of Electronic Media; Exponential growth in online content; Impact of Information Overload on human brain			
Science behind Personal Health Management			3 Hours
Ignorance in Society on health issues, World Health Organization (WHO) - Definition of Health, Human Evolution, Importance of physical work for human body & mind, Dangers of sedentary lifestyle, Germ diseases versus Lifestyle diseases, Integrating physical exercise into daily life			
Science behind Addictions			
Addiction - Meaning, Neurology and Hormonal basics of Addictive Behaviour, How addictions are formed; Harmful effects of addictions on Physical and Mental Health, Recognizing addictions in oneself, Coming out of addictions			
UNIT-II			
Need for cultivating good hobbies			3 Hours
Need for Hobbies in maintaining Work-Life Balance; how hobbies help in maintaining good physical and mental health, Various Hobbies			
Habits			
Difference between hobbies & habits, Cultivating good habits & discarding bad habits: Role of habits for a successful life, How habits form; Analyzing one's own habits; Recognizing useless & harmful habits, Cultivating & Sustaining useful habits			
Peer pressure and How to cope with it			3 Hours
Human being as a Social Animal, Physical Pain & Social Pain; Awareness of Harmful Social Pressure, Role of Prefrontal Cortex in Judgement and Decision Making, why teenagers are vulnerable to peer pressure, strategies to overcome harmful peer pressure.			
Stress Management			
Stress, Types of Stress, Fight & Flight Response of Humans; Harmful effects of chronic stress; Symptoms of Poor Coping Skills of Stress, Stress & Psychiatric problems, Easy coping strategies for stress			
UNIT-III			
Continuous & Lifelong Learning			3 Hours
Accelerated change in Technology Landscape, Shorter Life Cycles of Technologies, Need for Continuous Learning of other skills.			
Team Working Skills & Collaboration			
Team Work – Meaning, Skills and Relevance, Importance of Collaboration to succeed in one's own career, How to be a good team member			
Course Outcomes: At the end of the course student will be able to			
1.	Apply the concept of Time Management, cope with Information Overload and withstand harmful peer pressure		
2.	Comprehend the need to stay away from addictions by realizing the biological basis behind these concepts		
3.	Develop good hobbies to maintain ideal work-life balance		
4.	Develop the aptitude for finding creative solutions to problems and realize the importance of continuous and lifelong learning		

5.

Demonstrate positive and progressive abilities

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes													1	2
HU1008-1.1	-	-	-	-	-	-	-	-	-	2	1	3	-	-
HU1008-1.2	-	-	-	-	-	-	-	-	-	3	2	3	-	-
HU1008-1.3	-	-	-	-	-	-	-	-	-	3	1	3	-	-
HU1008-1.4	-	-	-	-	-	-	-	-	2	2	1	2	-	-
HU1008-1.5	-	-	-	-	-	-	-	-	1	2	1	2	-	-

1: Low 2: Medium 3: High

REFERENCES:

1.	Lieberman, D.E., “The Story of the Human Body”, Pantheon Books, 2013.
2.	Ratey, J.J., “Spark. Little Brown Spark”, 2013.
3.	De Bono, E., “Creative Thinking”, Penguin UK, 2016.
4.	Pachter, B., “The Power of Positive Confrontation”, Da Capo Lifelong Books, 1999.
5.	Duhigg, C., “The Power of Habit”, Random House Trade Paperbacks, 2012.
6.	Sharma, S., & Mishra, B., “Communication Skills for Engineers and Scientists”, PHI Learning, 2009.
7.	Tracy, B., “Time Management”, AMACOM, 2014.

Humanities & Management Courses

ENHANCING SELF-COMPETENCE																
Course Code:				HU2001-1			Course Type			HSMC						
Teaching Hours/Week (L: T: P):				2:0:0			Credits			02						
Total Teaching Hours				25+0+0			CIE + SEE Marks			50+50						
Prerequisite				---												
Teaching Department: Humanities																
Course Objectives:																
1.	Introspect and learn about oneself.															
2.	Develop professional writing skills.															
3.	Acquaint with the various social behaviour and etiquette.															
4.	Apply the techniques of fundamental communication skills.															
5.	Develop necessary techniques for formal presentations.															
UNIT-I																
Personality Traits										09 Hours						
Types & Kinds of personality, Ways to Identify Self (SWOT Analysis, Johari Window), Concepts of Self-Management and Self-Motivation																
Effective Communication Skills																
One-way and Two-way Communication, Interpersonal & Social Skills																
UNIT-II																
Social Behaviour and Cultural Etiquette										09 Hours						
Time Management, Personal Grooming, Making Small Talk, Customs & Manners																
Professional Presentation Techniques																
Formal Presentation, Sensitivity towards multi-cultural workspaces																
UNIT-III																
Job-Related Communication										08 Hours						
Resume & Cover Letter, Formal E-mails, Framing Requests, Greetings, Salutations, Close																
Course Outcomes: At the end of the course student will be able to																
1.	Understand the importance of human conduct.															
2.	Demonstrate knowledge of theory and competence in office communication.															
3.	Develop and assess various types of communication.															
4.	Be Familiar with the current practices of social behaviour.															
5.	Prepare and deliver presentation appropriate for the workplace.															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
HU2001-1.1			1				2	2		3						
HU2001-1.2									3	2	1		1			
HU2001-1.3				2			2	2	2				2			
HU2001-1.4			3							2	3	2				
HU2001-1.5		2	2		1					2						
1: Low 2: Medium 3: High																
REFERENCE BOOKS:																
1.	R R Gaur, R Sangal, G P Bagaria, “Human Values and Professional Ethics”, Excel Books, New Delhi, 2010.															
2.	Ronald B Adler and Jeanne Marquardt Elmhurst, "Communicating at Work – Principles and Practices															

	for Business and the Professions", 6th Edition, McGraw Hill College.
3.	Stephen R. Covey, "The 7 Habits of Highly Effective People", Simon & Schuster, 1994.
4.	Sarvesh Gulati, "Corporate grooming and Etiquette", Rupa Publications India Pvt. Ltd., 2010.
5.	Fred. Luthans, "Organizational Behaviour", McGraw Hill International.
6.	Tom Rath, "Strengths Finder 2.0", Gallup Press, 2007.
7.	M Ashraf Rizvi, "Effective Technical Communication", Tata McGraw- Hill, 2005.
8.	Stephen P. Robbins, "Organizational Behaviour", Prentice Hall.
9.	Dale Carnegie, "How to Win Friends and Influence People", Gallery Books, 2016.

BALAKE KANNADA (COMMUNICATION IN KANNADA)				
Course Code	HU1003-1	Course Type	MNC	
Teaching Hours/Week (L: T: P):	1:0:0	Credits	0	
Total Teaching Hours	13+0+0	CIE + SEE Marks	50+0	
Prerequisite				
Teaching Department: Humanities				
Course Objectives:				
1.	The course will enable the students to cognize Kannada and communicate in basic Kannada language.			
UNIT – I				
Basic Kannada Grammar			05 Hours	
- Personal Pronouns, Possessive Forms, Interrogative words - Possessive forms of nouns, Dubitive question and Relative nouns - Qualitative, Quantitative and Colour Adjectives, Numerals - Predictive Forms, Locative Case - Dative Cases, and Numerals - Ordinal numerals and Plural markers - Defective / Negative Verbs and Colour Adjectives - Permission, Commands, encouraging and Urging words (Imperative words and sentences) - Accusative Cases and Potential Forms used in General Communication - Helping Verbs “iru and iralla”, Corresponding Future and Negation Verbs - Comparative, Relationship, Identification and Negation Words - Different types of forms of Tense, Time and Verbs - Formation of Past, Future and Present Tense Sentences with Verb Forms - Karnataka State and General Information about the State - Kannada Language and Literature - Do's and Don'ts in Learning a Language				
UNIT – II				
Kannada Language Script Part – 1				05 Hours
UNIT – III				
Kannada Vocabulary List & Kannada Words in Conversation				03 Hours

Course Outcomes: At the end of the course student will be able to

1.	Understand the parts of speech of Kannada
2.	Know the script in Kannada
3.	Able to Converse daily usages in Kannada
4.	Enrich Basic Kannada Vocabulary
5.	Have knowledge about Karnataka and its culture

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
HU1003-1.1								3			1	1			
HU1003-1.2								2			1	1			
HU1003-1.3			2				1	2			1	1			
HU1003-1.4								1							
HU1003-1.5			1					3			1	1			

1: Low 2: Medium 3: High

REFERENCE MATERIALS:

1.	English –Kannada Rapidex Dictionary of Spoken Words, S N Raju, Bengaluru
2.	English Kannada Standard Dictionary, D K Bharadwaj, Sankeshwar Printers Pvt Ltd, Bengaluru
3.	ಮಾತಾಡುವಕನ್ನಡ, ಕನ್ನಡಸಾಹಿತ್ಯಪರಿಷತ್, ಬೆಂಗಳೂರು (೨೦೧೬).
4.	ಸಂಕ್ಷಿಪ್ತಕನ್ನಡನಿಗಂಟು (ಪರಿಷ್ಕೃತ), ಕನ್ನಡಸಾಹಿತ್ಯಪರಿಷತ್, ಬೆಂಗಳೂರು.
5.	ಆಡಳಿತಪದಕೋಶ, ಕನ್ನಡಅಭಿವೃದ್ಧಿಪ್ರಾಧಿಕಾರ, ಬೆಂಗಳೂರು.
6.	ಕನ್ನಡಭಾಷಾಕೈಪಿಡಿ, ಸಂಗಮೇಶ್ವರವದತ್ತಿಮಠ, ರೂಪರಶ್ಮಿಪ್ರಕಾಶನ, ಗುಲ್ಬರ್ಗ, ೧೯೯೫.
7.	ಡಿ.ಎನ್. ಶಂಕರ್ಭಟ್, ಕನ್ನಡವಾಕ್ಯಗಳಒಳರಚನೆ, ೨೦೦೬, ಭಾಷಾಪ್ರಕಾಶನ, ಮೈಸೂರು.
8.	ಕಾನೂನುಪದಕೋಶ (ಪರಿಷ್ಕೃತ) ಕನ್ನಡ- ಇಂಗ್ಲಿಷ್, ಕನ್ನಡಮತ್ತುಸಂಸ್ಕೃತಿನಿರ್ದೇಶನಾಲಯ, ಬೆಂಗಳೂರು.

ESSENCE OF INDIAN CULTURE															
Course Code:					HU1005-1					Course Type:			HEC		
Teaching Hours/Week (L: T: P):					1:0:0					Credits:			01		
Total Teaching Hours:					15+0+0					CIE + SEE Marks:			50+00		
Teaching Department: Respective Department															
Course Objectives:															
1.	To facilitate students with the concepts of Indian Culture and to make them understand the roots of knowledge system.														
2.	To acquaint students with Indian Culture and inculcate an ability to analyze it.														
3.	To apply various approaches for the enhancement of living ideals based on Indian traditional knowledge.														
UNIT-I															
Introduction to Traditional Knowledge													6 Hours		
Define traditional knowledge, nature and characteristics, scope and importance, kinds of traditional knowledge, Indigenous Knowledge and its characteristics, Traditional Knowledge vis-a-vis Indigenous Knowledge, Traditional Knowledge vis-a-vis Western Knowledge															
UNIT-II															
Significance of Traditional Knowledge													6 Hours		
Value of Traditional Knowledge in global economy, Role of Government in harnessing Traditional Knowledge, Traditional medicine system, Traditional Knowledge in agriculture. food and healthcare.															
UNIT-III															
Holistic Healthcare for Human Well-being													3 Hours		
Definition of Ayurveda, Ayurveda for Life, Health and Well-being, Introduction to principles of Ayurvedic healing and Astanga Ayurveda.															
Course Outcomes: At the end of the course student will be able to															
1.	Identify the concept of Traditional Knowledge and its importance.														
2.	Explain the need for and importance of protecting Traditional Knowledge.														
3.	Illustrate the various enactments related to Traditional Knowledge.														
4.	Familiarize the importance of Holistic Healthcare.														
Course Outcomes Mapping with Program Outcomes & PSO															
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes														1	2
HU1005-1.1		-	-	-	-	-	-	-	-	-	2	2	3	-	-
HU1005-1.2		-	-	-	-	-	-	-	-	-	3	2	3	-	-
HU1005-1.3		-	-	-	-	-	-	-	-	-	3	2	3	-	-
HU1005-1.4		-	-	-	-	-	-	-	-	2	2	2	2	-	-
HU1005-1.5		-	-	-	-	-	-	-	-	1	2	2	2	-	-
1: Low 2: Medium 3: High															
REFERENCES:															
1.Jha, A., "Traditional Knowledge System in India", Atlantic Publishers, 2002.															
2.Kapoor, K., &Danino, M., "Knowledge Traditions and Practices of India", 2012.															
3.Kapil Kapoor, Michel Danino, "Knowledge Traditions and Practices of India", Medknow Publications and															

Media.

4. Jha, R.N., "Science of Consciousness Psychotherapy and Yoga Practices", Delhi: VidyanidhiPrakashan, 2015.

5. TEDx Talks. (2015, February 6). Unleashing the Power of Traditional Medicine | Dr. Arvind Singh [Video file]. Retrieved from <https://www.youtube.com/watch?v=LZP1StpYEPM>

DEVELOPING PROFESSIONAL SKILLS			
Course Code	HU2002-1	Course Type	HSMC
Teaching Hours/Week (L: T: P: S)	1:0:0:0	Credits	01
Total Teaching Hours	15	CIE + SEE Marks	50+50
Pre-requisite	HU1001-1 (Technical English)		
Teaching Department: Humanities			
Course Objectives:			
1.	Developing an understanding of one's own strengths, weaknesses, values, and beliefs through exercises and assessments		
2.	Enhancing verbal and non-verbal communication skills to express thoughts and feelings clearly, listen actively, and resolve conflicts constructively		
3.	Building professional online presence through personal branding		
4.	Learning techniques for setting achievable goals, prioritizing tasks, and managing time effectively to enhance productivity and personal fulfilment		
UNIT - I			
Understanding Personality			02 Hours
- Introduction to 16 types of personalities (Myers & Briggs) and their significance in professional growth - MBTI Self-assessment exercise to identify individual Personality Type and areas for development			
Campus Placement: First Impressions			04 Hours
- Comprehensive overview of campus placement opportunities and challenges - Crafting tailored resumes, cover letters, and portfolios for different job applications - Self-Introductions: Styles and strategies			
UNIT - II			
Personal Branding and Professional Image			02 Hours
- Understanding the importance of personal branding in career advancement - Developing a strong online presence through LinkedIn, and GitHub			
Voicing Perspectives in Interviews and Group Discussions			04 Hours
- Exploring various interview formats - Addressing common interview challenges and handling difficult questions with poise			

- Group Discussions

UNIT - III

Goal Setting

- Work Ethics and Professionalism
- Setting SMART (Specific, Measurable, Achievable, Relevant, Time-bound) goals for ongoing skill development and career advancement

02 Hours

Wrap-Up

- Participant feedback collection to assess the effectiveness of the modules

01 Hours

Course Outcomes: At the end of the course, students will be able to

1.

Demonstrate an increased awareness of their strengths, weaknesses, values, and beliefs and set a goal for their life

2.

Effectively express their thoughts and feelings verbally and non-verbally, demonstrate active listening skills, and employ constructive conflict resolution techniques in interpersonal interactions in interviews, GD and online platforms

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes													1	2
HU2002-1.1	-	1	-	-	-	2	2	-	3	-	-	-	-	-
HU2002-1.2	-	-	-	-	-	-	-	3	2	1	-	1	-	-

1: Low 2: Medium 3: High

REFERENCE BOOKS:

1.

Adler, R. B., & Elmhurst, J. M. "Communicating at Work: Principles and Practices for Business and the Professions", Edition number). McGraw Hill College.

2.

Covey, S. R. "The 7 Habits of Highly Effective People", Great Britain: Simon & Schuster, 2013.

3.

Gulati, S. "Corporate Grooming and Etiquette", New Delhi: Rupa Publications India Pvt. Ltd, 2010.

4.

Luthans, F. "Organizational Behavior". McGraw Hill International, 1998.

5.

Rath, T. "Strengths Finder 2.0", New York: Gallup Press, 2020.

6.

Rizvi, M. A. "Effective Technical Communication", New Delhi: Tata McGraw-Hill, 2007.

7.

Robbins, S. P. "Organizational Behavior", New Delhi: Prentice Hall, 2016.

8.

Carnegie, D. "How to Win Friends and Influence People", New York: Gallery Books, 2019.

9.

Gaur, R. R., Sangal, R., & Bagaria, G. P. "Human Values and Professional Ethics", New Delhi, India: Excel Books, 2010.

INNOVATION AND DESIGN THINKING				
Course Code	ME1654-2	Course Type	AEC	
Teaching Hours/Week (L: T: P: S)	1:0:0:0	Credits	01	
Total Teaching Hours	15	CIE + SEE Marks	50+50	
Teaching Department: Mechanical Engineering				
Course Objectives:				
1.	To explain the fundamental concept of innovation and design thinking			
2.	To discuss the methods of implementing design thinking in the real world.			
	Note: Teaching-Learning Process (General Instructions) These are sample Strategies; which teachers can use to accelerate the attainment of the various course outcomes. 1. Lecturer method (L) does not mean only the traditional lecture method, but a different type of teaching method may be adopted to develop the outcomes. 2. Show Video/animation films to explain concepts 3. Encourage collaborative (Group Learning) Learning in the class 4. Ask at least three HOTS (Higher-order Thinking) questions in the class, which promotes critical thinking 5. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develops thinking skills such as the ability to evaluate, generalize, and analyze information rather than simply recall it. 6. Topics will be introduced in multiple representations. 7. Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them. 8. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.			
List of Modules				
1.	Process of Design Understanding Design thinking Shared model in team-based design – Theory and practice in Design thinking – Explore the presentation Tools for Design Thinking Real-Time design interaction capture and analysis – Empathy for design Teaching-Learning Process Introduction about the design thinking: Chalk and Talk method Theory and practice through presentation Case studies on design thinking for real-time interaction and analysis			
2.	Design Thinking in IT Design Thinking to Business Process modeling – Scenario-based Prototyping DT For strategic innovations Growth – Storytelling representation – Strategic Foresight - Change – Sense Making – Maintenance - Relevance – Value redefinition - Extreme Competition – experience			

	design - Standardization – Humanization - Creative Culture – Rapid prototyping, Strategy and Organization – Business Model design. Teaching-Learning Process Case studies on design thinking and business acceptance of the design Business model examples of successful designs
3.	Design thinking workshop Design Thinking Workshop Empathize, Design, Ideate, Prototype and Test Teaching-Learning Process Presentation by the students on the success of Live project on design thinking in a group of 4 students

Course Outcomes: Upon the successful completion of the course, students will be able to:

- | | |
|-----------|---|
| 1. | Generate and develop design ideas through a different techniques |
| 2. | Identify the significance of Design Thinking to Understand products |

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes													1	2	3
ME1654-1.1	-	-	-	-	-	-	2	2	-	-	-	-	1	1	1
ME1654-1.2	-	-	-	-	-	-	-	-	-	3	3	-	1	1	1

1: Low 2: Medium 3: High

REFERENCE BOOKS:

- John.R.Karsnitz, Stephen O'Brien and John P. Hutchinson, "Engineering Design", Cengage learning (International edition) Second Edition, 2013.
- Roger Martin, "The Design of Business: Why Design Thinking is the Next Competitive Advantage", Harvard Business Press, 2009.
- Hasso Plattner, Christoph Meinel and Larry Leifer (eds), "Design Thinking: Understand – Improve– Apply", Springer, 2011
- Idris Mootee, "Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School", John Wiley & Sons 2013.
- Yousef Haik and Tamer M. Shahin, "Engineering Design Process", Cengage Learning, Second Edition, 2011.
- Jeanne Liedtka, Andrew King, Kevin Bennett, "Solving Problems with Design Thinking - Ten Stories of What Works", Columbia Business School Publishing Hardcover, 2013.

E Resources

- www.tutor2u.net/business/presentations/. /productlifecycle/default.html
- https://docs.oracle.com/cd/E11108_02/otn/pdf/. /E11087_01.pdf
- www.bizfilings.com > Home > Marketing > Product Developmen
- <https://www.mindtools.com/brainstm.html>
- <https://www.quicksprout.com/. /how-to-reverse-engineer-your-competit>
- www.vertabelo.com/blog/documentation/reverse-engineering
<https://support.microsoft.com/en-us/kb/273814>
- <https://support.google.com/docs/answer/179740?hl=en>
- <https://www.youtube.com/watch?v=2mjSDIBaUIM>
thevirtualinstructor.com/foreshortening.html

	https://dschool.stanford.edu/.../designresources/.../ModeGuideBOOTCAMP2010L.pdf https://dschool.stanford.edu/use-our-methods/ 6. https://www.interactiondesign.org/literature/article/5-stages-in-the-design-thinking-process 7. http://www.creativityatwork.com/design-thinking-strategy-for-innovation/ 49 8. https://www.nngroup.com/articles/design-thinking/ 9. https://designthinkingforeducators.com/design-thinking/ 10. www.designthinkingformobility.org/wp-content/.../10/NapkinPitch_Worksheet.pdf
9.	Activity Based Learning (Suggested Activities in Class)/ Practical Based learning · http://dschool.stanford.edu/dgift/

UNIVERSAL HUMAN VALUES			
Course Code	HU1004-1	Course Type	HSMC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	1+0+0	CIE + SEE Marks	50+50
Teaching Department: Respective Department			
Course Objectives			
1.	Enable students to appreciate values, skills, and behaviour with an appropriate understanding of 'Self' to attain sustained happiness and prosperity with right aspirations of life.		
2.	Develop a holistic perspective among the students towards physical needs and prosperity of life.		
3.	Develop a holistic approach and understand the importance of co-existence and living in harmony ensuring mutually fulfilling interaction with the society and nature.		
UNIT - I			
Need, Basic Guidelines, Content and Process for Value Education			6 Hours
Self-Exploration; 'Natural Acceptance' and Experiential Validation; Continuous Happiness and Prosperity; Right understanding, Relationship and Physical Facility; Understanding Happiness and Prosperity - living in harmony at various levels.			
UNIT - II			
Understanding Harmony in the Human Being, Family and Society			6 Hours
Understanding human being as a co-existence of the sentient 'I' and the material 'Body'; the needs of Self ('I') and 'Body'; the Body as an instrument; Holistic perspective of Physical needs and Prosperity; Visualizing a universal harmonious order in society- Undivided Society, Universal Order- from family to world family			
UNIT - III			
Whole existence as Coexistence: Implications of the above Holistic Understanding of Harmony and Professional Ethics			3 Hours
Understanding the harmony in the Nature and Existence; Existence as Co-existence, Holistic perception of harmony at all levels of existence; Natural acceptance of human values,			

Professional Ethics

Course Outcomes: At the end of the course student will be able to

1.	Have a better self-exploration and understanding with a capacity to identify the priorities of life with focus on human values having a holistic perspective of physical needs.
2.	Understand and practice living in harmony, co-existence, and natural acceptance and also to exhibit Professional Ethics in the workplace

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes													1	2
HU1004-1.1	-	-	2	-	-	-	-	3	-	-	2	2	-	-
HU1004-1.2	-	-	1	-	-	-	-	3	-	-	2	2	-	-

1: Low 2: Medium 3: High

REFERENCE BOOKS:

1.	Gaur, R. R., Sangal, R., & Bagaria, G. P., "Human Values and Professional Ethics", New Delhi, India: Excel Books, 2010.
2.	Nagaraj, A. "Jeevan Vidya: Ek Parichaya", Amarkantak: Jeevan Vidya Prakashan, 1999.
3.	Tripathi, A.N. "Human Values", New Delhi: New Age Intl. Publishers, 2004.
4.	Leonard, A. "The Story of Stuff", 2007.
5.	Gandhi, M. K. "The Story of My Experiments with Truth", 1927.
6.	Schumacher, E. F. "Small is Beautiful: Economics as if People Mattered", 1973
7.	Andrews, C. "Slow is Beautiful: New Visions of Community, Leisure, and Joie de Vivre", 2006.
8.	Kumarappa, J.C. "Economy of Permanence", 2018.
9.	Sunderlal, P. "Bharat Mein Angreji Raj", 2021.
10.	Dharampal. "Rediscovering India, 2003.
11.	Gandhi, M. "Indian Home Rule", 2017.
12.	Azad, M. A. K. "India Wins Freedom", 1989.
13.	Rolland, R. "Vivekananda", 2017.
14.	Rolland, R. "Gandhi", 2002.

INTRODUCTION TO IPR																																																																																																																								
Course Code:					HU1006-1					Course Type:					HSMC																																																																																																									
Teaching Hours/Week (L: T: P):					1:0:0					Credits:					01																																																																																																									
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Teaching Department: Respective Department																																																																																																																								
Course Objectives:																																																																																																																								
1.	Enhancing the learning system through innovation and creative thinking skills for effective business process.																																																																																																																							
2.	Acquaint with special challenges of starting new ventures.																																																																																																																							
3.	Facilitate Entrepreneurial skills in recognizing opportunities for competitive advantages.																																																																																																																							
4.	Provide insights of financial aspects in planning and executing a business plan.																																																																																																																							
5.	Ascertain the role of IPR to protect innovations and intangible assets.																																																																																																																							
UNIT-I																																																																																																																								
Intellectual Property Rights (IPR) 6 Hours																																																																																																																								
Introduction to IPR: Business Perspective, IPR in India – Genesis and Development, International Context, Concept of IP Management, Uses in marketing																																																																																																																								
UNIT-II																																																																																																																								
Types of Intellectual Property6 Hours																																																																																																																								
Patent - Procedure, Licensing and Assignment, Infringement and Penalty, Trademark, Example of Trademarks - Domain name, Geographical Indications, Copyright, Industrial Designs, Class Discussion - Major Court Cases regarding violation of Patents																																																																																																																								
UNIT-III																																																																																																																								
Basic Tenets of Information Technology Act, 20003 Hours																																																																																																																								
IT Act – Introduction, E-Commerce and Legal Provisions, E- Governance, Digital signature and Electronic Signature, Cybercrimes																																																																																																																								
Course Outcomes: At the end of the course student will be able to																																																																																																																								
1.	Comprehend Innovation, its process and sources.																																																																																																																							
2.	Apply the process of building an innovative organization.																																																																																																																							
3.	Recognize the characteristics of different types of Entrepreneurships																																																																																																																							
4.	Formulate a business plan based on a business idea in Technology.																																																																																																																							
5.	Interpret basic tenets of Information Technology Act, 2000.																																																																																																																							
Course Outcomes Mapping with Program Outcomes & PSO																																																																																																																								
<table><tr><td>Program Outcomes→</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td colspan="2">PSO↓</td></tr><tr><td>↓ Course Outcomes</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>1</td><td>2</td></tr><tr><td>HU1006-1.1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>2</td><td>-</td><td>3</td><td>-</td><td>-</td></tr><tr><td>HU1006-1.2</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td><td>-</td><td>3</td><td>-</td><td>-</td></tr><tr><td>HU1006-1.3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td><td>-</td><td>3</td><td>-</td><td>-</td></tr><tr><td>HU1006-1.4</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>2</td><td>2</td><td>-</td><td>2</td><td>-</td><td>-</td></tr><tr><td>HU1006-1.5</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>1</td><td>2</td><td>-</td><td>2</td><td>-</td><td>-</td></tr></table>																Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		↓ Course Outcomes													1	2	HU1006-1.1	-	-	-	-	-	-	-	-	-	2	-	3	-	-	HU1006-1.2	-	-	-	-	-	-	-	-	-	3	-	3	-	-	HU1006-1.3	-	-	-	-	-	-	-	-	-	3	-	3	-	-	HU1006-1.4	-	-	-	-	-	-	-	-	2	2	-	2	-	-	HU1006-1.5	-	-	-	-	-	-	-	-	1	2	-	2	-	-
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓																																																																																																											
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REFERENCES:																																																																																																																								
1.	Tidd, J., & Bessant, J., "Managing Innovation: Integrating Technological, Market and																																																																																																																							

	Organizational Change", Wiley, 2021.
2.	Case Study Materials: To be distributed for Class Discussion
3.	Reddy, G. B., "Intellectual Property Rights and the Law", Gogia Law Agency, 2012.
4.	Wadehra, B. L., "Law relating to Intellectual Property", Universal Law Publishing Co., 2011.
5.	Narayanan, P., "IPR", Eastern Law House Private Ltd, 2017.

MANAGEMENT & ENTREPRENEURSHIP			
Course Code:	MG1003-1	Course Type	HSMC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	39	CIE + SEE Marks	50+50
Teaching Department: Any			
Course Objectives:			
1.	To introduce the field of management, task of the manager, importance of planning and types of planning, staff recruitment and selection process.		
2.	To discuss the ways in which work is allocation, structure of organizations, modes of communication and need of coordination between the manager and staff		
3.	To explain the role and importance of the entrepreneur and their functions in economic development and the concepts of entrepreneurship.		
4.	To discuss the importance of Small Scale Industries and methods for generating new business ideas and business opportunities		
5.	To introduce the concepts of financial concepts in enterprises.		
UNIT-I			
Management:			03 Hours
Definition, Importance – Nature and Characteristics of Management, Management Functions, Roles of Manager, Levels of Management, Managerial Skills, Management & Administration, Management as a Science, Art & Profession.			
Planning:			03 Hours
Nature, Importance and Purpose of Planning, Types of Plans, Steps in Planning, Limitations of Planning, Decision Making – Meaning, Types of Decisions- Steps in Decision Making.			
Organizing and Staffing			04 Hours
Meaning, Nature and Characteristics of Organization – Process of Organization, Principles of Organization, Departmentalization, Committees – meaning, Types of Committees, Centralization Versus Decentralization of Authority and Responsibility, Span of Control (Definition only), Nature and Importance of Staffing, Process of Selection and Recruitment.			
Directing and Controlling			04 Hours
Meaning and Nature of Directing-Leadership Styles, Motivation Theories Communication – Meaning and Importance, Coordination- Meaning and Importance, Techniques of Coordination. Controlling – Meaning, Steps in Controlling			
UNIT-II			
Social Responsibilities of Business:			03 Hours

Meaning of Social Responsibility, Social Responsibilities of Business towards Different Groups, Social Audit, Business Ethics, and Corporate Governance.														
Entrepreneurship													05 Hours	
Definition of Entrepreneur, Importance of Entrepreneurship, concepts of Entrepreneurship, Characteristics of successful Entrepreneur, Classification of Entrepreneurs, Intrapreneur – An Emerging Class, Comparison between Entrepreneur and Intrapreneur, Myths of Entrepreneurship, Entrepreneurial Development models, Entrepreneurial development cycle, Problems faced by Entrepreneurs and capacity building for Entrepreneurship.														
Modern Small Business Enterprises													05 Hours	
Role of Small Scale Industries, Concepts and definitions of SSI Enterprises, Government policy and development of the Small Scale sector in India, Growth and Performance of Small Scale Industries in India, Sickness in SSI sector, Problems for Small Scale Industries, Impact of Globalization on SSI, Impact of WTO/GATT on SSIs, Ancillary Industry and Tiny Industry (Definition only).														
Institutional Support for Business Enterprises													02 Hours	
Introduction, Policies & Schemes of Central–Level Institutions, State-Level Institutions														
UNIT-III														
Finance Management in enterprises													10 Hours	
Introduction, functions, Accounting and Bookkeeping, Financial Statements, Working Capital Management, Break even Analysis, Financial ratio Analysis.														
Course Outcomes: At the end of the course student will be able to														
1.	Describe the field of management, the task of the manager, planning, and steps in decision making.													
2.	Discuss the structure of the organization, importance of staffing, leadership styles, modes of communication, techniques of coordination, and importance of managerial control in the business.													
3.	Describe the concepts of entrepreneurship and a businessman’s social responsibilities towards different groups.													
4.	Develop an understanding of the role of SSI’s in the development of country and state/central level institutions/agencies supporting business enterprises.													
5.	Apply the concepts of financial management for effective use in enterprises													
Course Outcomes Mapping with Program Outcomes & PSO														
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes													1	2
MG1003-1.1	3	-	-	-	-	-	-	2	2	-	3	-	-	1
MG1003-1.2	3	-	-	-	-	-	-	2	2	-	3	-	-	2
MG1003-1.3	3	-	-	-	-	-	-	2	2	-	3	-	-	2
MG1003-1.4	3	-	-	-	-	-	-	2	2	-	3	-	-	2
MG1003-1.5	3	-	-	-	-	-	-	2	2	-	3	-	-	2
1: Low 2: Medium 3: High														
TEXTBOOKS:														
1.P. C. Tripathi, P. N. Reddy, “Principles of Management”, McGraw Hill, 6 th Edition, 2017.														
2.Poornima M. Charanthimath, “Entrepreneurship Development and Small Business Enterprises”, Pearson 2 nd Edition, 2014.														
3.W.D Stevenson, “Elements of Power System Analysis”, 4 th edition, TMH, 2001.														
REFERENCE BOOKS:														
1.Vasant Desai, “Dynamics of Entrepreneurial Development and Management”, Himalaya Publishing House, 2007.														
2.Harold Koontz, Heinz, Weihrich, “Essentials of Management: An International, Innovation and Leadership perspective”, McGraw Hill, 10 th Edition, 2016.														

FINANCIAL MANAGEMENT																
Course Code:				MG1002-1				Course Type				HSMC				
Teaching Hours/Week (L: T: P):				3:0:0				Credits				03				
Total Teaching Hours				39+0+0				CIE + SEE Marks				50+50				
Teaching Department: Any																
Course Objectives:																
1.	Develop basic financial management knowledge essential to make a managerial career in professional life.															
2.	Impart some of the crucial and basic skills required to work in the area of budgeting, investment and financial decision making.															
3.	Enable in making a right decisions on selection of projects for investment.															
4.	Understand the basics of finance and financial markets, project evaluation and selection.															
UNIT-I																
Time Value of Money														15 Hours		
Financial Management: Concepts and Meaning – Introduction to Finance; Objectives of Financial Management; Profit Maximization; EVA; Changing Role of Financial Managers. Time Value of Money: Techniques and Applications of Compounding and Discounting.																
UNIT-II																
Capital Budgeting and Working Capital														15 Hours		
Capital Budgeting (Investment Evaluation Techniques): Payback Period Method; Present Worth Method; Annual Worth Method; Profitability index method; Estimation of IRR. Cost of Capital: Sources of various Types of Capital; Cost of Debenture Capital; Cost of Preferential Capital; Cost of Term Loans; Cost of Equity Capital. Working Capital: Factors influencing Working Capital Requirements.																
UNIT-III																
Inventory Management and Break Even Analysis														9 Hours		
Inventory Management: Techniques of Inventory Management and Control – EOQ, ABC Analysis, Just-in-Time (JIT) System Break Even Analysis: Estimation of Break-Even Point and Values.																
Course Outcomes: At the end of the course student will be able to																
1.	Describe the basic financial management skills required for a professional.															
2.	Explain techniques and applications of compounding and discounting and calculate compounded/discounted amount for the given proposal.															
3.	Evaluate the given investment option by capital budgeting techniques.															
4.	Describe the basics of cost of capital and working capital. Determine the cost of capital for the given investment option.															
5.	Describe the basics of inventory management and calculate the economic order quantity and reorder point for the given conditions. Calculate breakeven point for the given manufacturing setup.															
Course Outcomes Mapping with Program Outcomes & PSO																
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
↓ Course Outcomes														1	2	3
MG1002-1.1		3	-	-	-	-	-	-	-	1	1	-	1	-	-	-

MG1002-1.2	1	3	-	-	-	-	-	-	1	1	-	1	-	-	-
MG1002-1.3	2	3	-	-	-	-	-	-	1	1	-	1	-	-	-
MG1002-1.4	2	3	-	-	-	-	-	-	1	1	-	1	-	-	-
MG1002-1.5	1	3	-	-	-	-	-	-	1	1	-	1	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	M Y Khan, P K Jain , “Financial Management – Text, Problems & Cases”, 7th Edition, 2015; McGraw Hill Education (India) Pvt. Ltd, New Delhi.
2.	I M Pandey, "Financial Management", 11th Edition, 2015; Vikas Publishing House Pvt. Ltd. (UP) India.
3.	James L. Riggs, David D. Bedworth and Sabah U. Randhawa, “Engineering Economics”, 4th Edition, Tata McGraw Hill Edition.

REFERENCE BOOKS:

1.	Prasanna Chandra, “Financial Management”, 6th Edition, 2004; Tata McGraw Hill Publishing Company Ltd, New Delhi.
2.	S. D. Sharma, “Operation Research” , Kedar Nath Ram Nath Publishers, 2015.

INDIAN KNOWLEDGE SYSTEMS

Course Code:	HU1009-1	Course Type:	HEC
Teaching Hours/Week (L: T: P):	1:0:0	Credits:	01
Total Teaching Hours:	15+0+0	CIE + SEE Marks:	50+50

Teaching Department: Respective Department

Course Objectives:

1.	Enhance knowledge about the History of Ancient India and Rich Culture of the country
2.	Gain an introduction to ancient Indian Engineering Technology and Architecture
3.	Familiarize Indian indigenous wisdom in Modern scientific paradigm
4.	Understanding the Scientific Value of the Traditional Knowledge of our country
5.	Comprehend and compare the Ancient and Current Knowledge Systems

UNIT-I

Indian History	6 Hours
History - Land, Environment, and people in Ancient India; Ancient Education System, Takṣaśilā and Nālandā University, Hunting to Agriculture; Introduction to Vedas and Upanishads; Great Indian Epics; Indian Festivals	

UNIT-II

Engineering, Technology, and Architecture	6 Hours
Pre-Harappan and Sindhu Valley Civilization, Laboratory and Apparatus, Juices, Dyes, Paints and Cements, Glass and Pottery, Metallurgy, Engineering Science and Technology in the Vedic Age and Post-Vedic Records, Iron Pillar of Delhi, Rakhigarhi, Mehrgarh, Sindhu Valley Civilization, Marine Technology	

UNIT-III

Science, Astronomy, and Mathematics	3 Hours
Concept of Matter, Life and Universe, Gravity, Sage Agastya’s Model of Battery, Velocity of Light, Vimāna: Aeronautics, Vedic Cosmology and Modern Concepts, History and Culture of Astronomy, Sun, Earth, Moon, Eclipses, Rotation of Earth, Concepts of Zero and Pi, Number System, Pythagoras Theorem and Vedic Mathematics.	

Course Outcomes: At the end of the course student will be able to

1.	Understand the relevance of studying history
2.	Comprehend the origin of Vedas and epics
3.	Realize the scientific value of the Traditional Knowledge of India
4.	Converting the Bhāratiya wisdom into the applied aspect of the modern scientific paradigm
5.	Preserve and disseminate Indian Knowledge Systems in Research and Societal applications

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓	
↓ Course Outcomes													1	2
HU1009-1.1	-	-	-	-	-	-	-	-	-	-	2	3	-	-
HU1009-1.2	-	-	-	-	-	-	-	-	-	-	3	3	-	-
HU1009-1.3	-	-	-	-	-	-	-	-	-	-	2	3	-	-
HU1009-1.4	-	-	-	-	-	-	-	-	-	-	2	2	-	-
HU1009-1.5	-	-	-	-	-	-	-	-	-	-	2	2	-	-

1: Low 2: Medium 3: High
REFERENCES:

1.	Tripati, R.S., “History of Ancient India”, Motilal Banarsidass, 1942.
2.	Mahajan, V.D.. “Ancient India”, S. Chand and Company, 1985.
3.	Ramasubramanian, K., & Srinivas, M.D., “Development of Calculus in India”, 2010.
4.	Ramasubramanian, K., Srinivas, M.D., & Sriram, M.S., “The Traditional Indian Planetary Model and its Revision by NilakanthaSomayaji”, 2011.
5.	Srinivas, M.D., “Proofs in Indian Mathematics”, Hindustan Book Agency, 2005.
6.	Srinivas, M.D., “The Algorithmic Approach of Indian Mathematics”, 2015.
7.	Srinivas, M.D. “Indian Tradition of Science: An Introductory Overview”, 2016.
8.	Rahika, M., & Balasubramanian, A.V., “Ayurvedic Principles of Food and Nutrition”, Part 1. Lok Swasthya Parampara Samvardhan Samithi, 1990.

Vocational Education Course

PYTHON PROGRAMMING WITH DATA SCIENCE																
Course Code: CS1551-1								Course Type: PCC Lab								
Teaching Hours/Week (L: T: P): 0:0:2								Credits: 01								
Total Teaching Hours: 0+0+26								CIE + SEE Marks: 50+50								
Prerequisite				-----												
Teaching Department: Computer Science & Engineering (Cyber Security)																
Course Objectives:																
1.	Execute the basic operations of python programming.															
2.	Explain the elementary programming constructs and file operations and use it in Python programming.															
3.	Describe the concepts like strings, conversion of strings to numbers, lists, tuples, and dictionaries and apply these in python programming.															
4.	Illustrate the functions, recursive functions and object-oriented programming concepts in Python.															
5.	Write data handling program in python.															
List of Experiments																
	The concept of data types; immutable variables; Conditions, Boolean logic, logical operators; ranges;															
	control statements: if-else, loops (for, while); text files: reading/writing text and numbers from/to a file; creating and reading a formatted file (csv or tab-separated).															
	String operations, slicing a string; strings and number system: converting strings to numbers and vice versa.															
	Lists, tuples, and dictionaries: Basic list operators, replacing, inserting, removing an element; searching and sorting lists															
	Dictionary literals, adding and removing keys, accessing and replacing values; traversing dictionaries.															
	Python Functions: Arguments and return values; formal vs actual arguments. Named arguments.															
	Object Oriented Concepts: Classes, objects, attributes and methods, inheritance, types of inheritance.															
	Introduction to NumPy - The Basics of NumPy Arrays, Computation on NumPy Arrays: Universal Functions,															
	Aggregations: Min, Max, Computation on Arrays: Broadcasting, Boolean Logic, Fancy Indexing, Sorting Arrays, Structured Data: NumPy's Structured Arrays															
	Data Manipulation with Pandas- Introducing Pandas Objects, Data Indexing and Selection, Operating on Data in Pandas, Handling Missing Data, Hierarchical Indexing,															
	Combining Datasets: Concat and Append, Combining Datasets: Merge and Join, Data Manipulation with Pandas- Aggregation and Grouping, Pivot Tables, Vectorized String Operations.															
	Visualization with Matplotlib - General Matplotlib Tips, Simple Line Plots, Simple Scatter Plots, Pie chart, Bar chart.															
Course Outcomes: At the end of the course student will be able to																
1.	Execute basic python functionalities. Demonstrate the file handling and change of the permission according to the user's requirement.															
2.	Execute and comprehend the data handling operations.															
3.	Apply the knowledge of basic program constructs and file operations of python to develop the solutions for engineering problems. Implement programs using a suitable modern tool.															
4.	Illustrate the usage of strings, conversion of strings to numbers, lists, tuples and dictionaries to develop data handling programs in Python.															
5.	Apply the knowledge of functions and data handling libraries of Python to analyzethe problem and develop solutions.															
Course Outcomes Mapping with Program Outcomes & PSO																
	Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓		
	↓ Course Outcomes													1	2	3
	CS1551-1.1	2	3							1	1		1		3	
	CS1551-1.2	2	3							1	1		1		3	

CS1551-1.3	1	2	3		1				1	1		1		3	
CS1551-1.4	1	2	3		1				1	1		1		3	
CS1551-1.5	1	2	3		1				1	1		1		3	

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Sumitaba Das, "UNIX-Concepts and Applications", Fourth Edition, Tata McGraw Hill, 2006. (Chapters 1.1,1.2,2,3,4,5,6,7,8,9,11,12,13,14,19,21).
2. Kenneth A. Lambert, "The Fundamentals of Python: First Programs", Cengage Learning, 2011.
3. Magnus Lie Hetland, "Beginning Python from Novice to Professional", Second Edition, Apress, 2009.
4. Mark Summerfield, "Programming in Python 3 - A Complete Introduction to the Python Language", Second Edition, Addison-Wesley, 2009.
5. Y. Daniel Liang, "Introduction to Programming Using Python", Pearson, ISBN: 978-0-13-274718-9, 2013.

REFERENCE BOOKS:

1. Chun, J Wesley, "Core Python Programming", 2nd Edition, Pearson, 2007 Reprint 2010.
2. David Beazley and Brian K. Jones, Shroff, "Python Cookbook", Third Edition, Publishers & Distributors Pvt. Ltd., ISBN : 978-93-5110-140-6, 2013.
3. Mark Lutz, "Learning Python" Fifth Edition, 2013
4. Mark Lutz, "Programming Python (English)", Fourth Edition, 2011.
5. David Sale, "Testing Python", Wiley India (P) Ltd., ISBN : 978-81-265-5277-1, 2014
6. Behrouz A. Forouzan and Richard F. Gilberg, "UNIX and Shell Programming", Thomson 2005. (Chapters Appendix H,9).

E Resources

1. http://www.davekuhlman.org/python_book_01.pdf
2. [http://slav0nic.org.ua/static/books/python/OReilly%20%20Core%20Python%20Program ming.pdf](http://slav0nic.org.ua/static/books/python/OReilly%20%20Core%20Python%20Program%20ming.pdf)
3. <http://www.freebookcentre.net/UnixCategory/Free-Unix-Books-Download.html>
4. <http://nptel.ac.in/courses/106105166/26>
5. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016/lecture-slides-code/>

Value added Course:

INTRODUCTION TO DEVOPS	
Course Code:	CS3602-1
Course Type:	PCC Lab
Teaching Hours/Week (L: T: P):	0:0:2
Credits:	01
Total Teaching Hours:	0+0+26
CIE + SEE Marks:	50+50
Teaching Department: Computer Science & Engineering	
List of Experiments	
1.	Overview of DevOps
2.	Version Control with Git • Git Installation • Commonly used commands in Git • Working with Remote repository
3.	Git, Jenkins & Maven Integration • Execute branching and merging operations • Perform various Git commands Understand Maven Architecture and dependencies
4.	Continuous Integration using Jenkins • Create pipeline view using DevCompile and QATest • Adding Slave node in Jenkins

	Build Pipeline project using Groovy script
5.	Containerization using Docker • Understand the Docker Architecture • Perform Various actions using Docker CLI • Bind container ports to the Machine ports • Run containers in different modes Write and build a Dockerfile to create a Docker Image
6.	Orchestration using Kubernetes
7.	Provisioning
8.	DevOps on Cloud

University Core Courses (UCC)

INTERNSHIP-II																		
Course Code:						UC2001-1			Course Type:				UCC					
Teaching Hours/Week (L: T: P):						-			Credits:				08					
Total Teaching Hours:						-			CIE + SEE Marks:				50+50					
Course Objectives:																		
1.		This course is meant to provide students an avenue to understand the work environment, ethics and practices in an industry/organization and take up assignments/jobs in the future.																
Course Outcomes: At the end of the course student will be able to																		
1.		Analyse and develop technical solutions for a specific problem that is assigned to them.																
2.		Communicate ideas that are developed through brainstorming, presentation and prepare a report.																
3.		Understand and inculcate industry practices in their professional career.																
Course Outcomes Mapping with Program Outcomes & PSO																		
		Program Outcomes→														PSO↓		
		↓ Course Outcomes												1		2 3		
		UC2001-1.1		3	2	-	-	1	1	-	-	2	3	1	-	1	1	1
		UC2001-1.2		3	2	-	-	1	1	-	-	2	3	1	-	1	1	1
		UC2001-1.3		3	2	-	-	1	1	-	-	2	3	1	-	1	1	1
1: Low 2: Medium 3: High																		

1: Low 2: Medium 3: High

MAJOR PROJECT			
Course Code:	UC3001-1& UC3002-1	Course Type:	UCC
Teaching Hours/Week (L: T: P):	24	Credits:	2+8
Total Teaching Hours:	-	CIE + SEE Marks:	(100+0) + 100+100
Course Objectives:			
1.	To perform effective literature survey, identification of research problem / project idea.		
2.	To develop skills of planning to execute the project		
3.	To assess the needs and necessity of a project.		
4.	To learn time management and documentation.		
5.	To expose the students to research aspects like literature review, executing experiments and analysis of results.		
6.	To expose the students to research aspects like literature review, executing experiments and analysis of results.		
A group of students (not more than 4) is assigned to a guide/projectsupervisor. The students must do a thorough literature review and come out with aproject plan. They are expected submit a project proposal (not more than 10 pages)including project idea, protocols, designs (if any), expected outcome, majorrequirements, and approximate budget. They shall present the same in a proposalseminar in front of the panel of internal examiners (involving guide) and shall get theirproposal approved. The presentation must involve projected timeline of the projectexecution.			
Assessment Details (both CIE and SEE)			
CIE procedure: Shall involve project proposal, proposal seminar, continuous evaluation of theproject progress by Guide and HOD. Monthly progress is evaluated.			
Semester End Examination:			

SEE procedure:

- i) Single discipline: Contribution to the project and the performance of each group member shall be assessed individually in semester end examination (SEE) conducted at the department.
- ii) Interdisciplinary: Contribution to the project and the performance of each group member shall be assessed individually in semester end examination (SEE) conducted separately at the departments to which the student/s belong to.

SCHEME OF EVALUATION:

Project demonstration, Viva voce

Total marks: 100 Marks

The distribution of marks shall be proportioned based on the type of the project and it is based on fulfilling the following requisites.

The evaluation of students is proposed to be done by internal faculty with active involvement of industrial personnel. The evaluation may be based on following criteria:

- Punctuality and Attendance “ Interpersonal relations
- Sense of Responsibility
- Clarity of concepts, principles and procedures
- Self-expression/communication skills
- Report Writing Skills
- Creativity/conceiving new and unusual ideas
- Problem-solving skills

At the end of the project work course students are required to submit a working model of the equipment they have designed and developed or if it is a theoretical or experimental work, they are expected to study a detailed analysis and findings from their work.

Course Outcomes: At the end of the course student will be able to

1.	Use various methods or sources for finding literature and analyze data for relevance and appropriateness to the research project undertaken.
2.	Identify and propose suitable methods of analysis and/or design or develop appropriate experiments to address the specific research objectives.
3.	Apply suitable standardized method/s for experimental design.
4.	Analyze and interpret the research findings and compare with reported results to arrive at suitable conclusions.
5.	Adopt appropriate documentation protocol to organize research findings, learn good laboratory practices and work in a team.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes → ↓ Course Outcomes													PSO ↓		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
UC3001-1/UC3002-1.1	-	1	-	-	2	2	3	1	-	-	-	1	1	2	2
UC3001-1/UC3002-1.2	-	1	2	1	1	-	1	2	1	-	1	1	1	2	2
UC3001-1/UC3002-1.3	-	1	2	2	1	-	1	1	1	1	1	1	1	2	2
UC3001-1/UC3002-1.4	1	3	2	2	1	2	2	3	3	3	3	2	1	2	2
UC3001-1/UC3002-1.5	-	1	1	-	1	2	2	3	3	3	3	1	1	2	2

1: Low 2: Medium 3: High

Open Elective Courses

LIST OF OPEN ELECTIVE COURSES

Sl No.	Department	Course Codes	Open Elective Courses
1	BT	BT1501-1	Bio Fuel Engineering
2	BT	BT1502-1	Solid Waste Management
3	CS	CS2501-1	Fundamentals of AI and ML
4	CS	CS2502-1	Introduction to Data Structures
5	CV	CV2501-1	Disaster Management
6	CV	CV2502-1	Environmental Hygiene, Sanitation and Waste Management
7	CV	CV2503-1	Environmental Impact Assessment
8	CV	CV2504-1	Introduction to Geoinformatics
9	CY	CY2501-1	Corrosion Science (Only for CV and ME)
10	CY	CY2502-1	Natural Products Chemistry (Only For BT)
11	EC	EC1501-1	Artificial Neural Network Systems
12	EC	EC1502-1	Introduction to MATLAB Programming: A Hands-on Approach (only for CV and BT)
13	EC	EC1503-1	Robotics
14	EC	EC2501-1	Consumer Electronics
15	EC	EC2502-1	PCB Design and Fabrication
16	EC	EC2503-1	Space Technology and Applications
17	EE	EE2501-1	Battery Management System
18	EE	EE2502-1	Biomedical Instrumentation
19	EE	EE2503-1	Electric Vehicle Technology
20	EE	EE2504-1	Fundamentals of PLC and its applications
21	EE	EE2505-1	Motors and Motor Control Circuits
22	EE	EE2506-1	Non-Conventional Energy sources
23	HU	HU1501-1	Elements of Yoga
24	HU	HU1502-1	Intellectual Property Rights
25	HU	HU1503-1	Introduction to German Language
26	HU	HU1504-1	Introduction to Japanese Language
27	HU	HU1505-1	National Cadet Corps: Organization, Functions & Capabilities
28	HU	HU1506-1	Overview of Indian Culture
29	HU	HU1507-1	Philosophy
30	HU	HU1508-1	Principles of Physical Education
31	HU	HU1509-1	Indian Culture – Dance *
32	HU	HU1510-1	Indian Culture – Music *
33	HU	HU1511-1	Engineering Ethics *
34	HU	HU1512-1	Art of Communication and Interpersonal Skills*
35	HU	HU2501-1	Common sense and Critical Thinking
36	HU	HU2502-1	Linguistics & Language Technology
37	IS	IS2501-1	Introduction to Cyber Security (except EC, EE, AM, AD, CC, CS, IS)
38	IS	IS2502-1	Python Application Programming
39	IS	IS2503-1	Software Engineering Practices
40	IS	IS2504-1	Web technologies
41	MA	MA1501-1	Graph Theory (for BT, CV, EC, EE, ME and RI)
42	MA	MA1502-1	Number Theory
43	MA	MA3501-1	Linear Algebra (for BT, CV, EE, ME and RI)
44	ME	ME1501-1	Automotive Engineering
45	ME	ME1502-1	Industrial Pollution Control
46	ME	ME1503-1	Sustainable Development Goals
47	ME	ME1504-1	Technology Innovation
48	MG	MG1501-1	Human Resource Management
49	MG	MG1502-1	Management Accounting and Control Systems
50	MG	MG1503-1	Operations and Quality Management
51	MG	MG1504-1	Organizational Behaviour
52	MG	MG1505-1	Taxation for Engineers
53	MG	MG1506-1	Working Capital Management
54	PH	PH2501-1	Nanotechnology
55	PH	PH2502-1	Optoelectronic Devices (EC, EE, CSE, ISE, AM and CC branches)

56	RI	RI2501-1	Autonomous Mobile Robots
57	RI	RI2502-1	Medical Robotics (for all except AI)
58	RI	RI2503-1	PLC Control of Hydraulic and Pneumatic Circuits (for all except AI)

*** For students admitted under Twinning Program**

BIOFUEL ENGINEERING			
Course Code:	BT1501-1	Course Type:	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits:	03
Total Teaching Hours:	40+0+0	CIE + SEE Marks:	50+50
Teaching Department: Biotechnology			
Course Objectives:			
1.To learn the fundamental concepts of biofuels, types of biofuels, their production technologies.			
2.To learn the concepts of feedstock utilization and energy conversion technologies.			
UNIT-I			
Liquid Biofuels			15 Hours
Description and classification of Biofuels; Primary biomass: Plant Materials-Woody biomass, Lignocellulosic and agroindustrial by-products, starchy and sugary crops. Secondary biomass: Waste residues and co-products- wood residues, animal waste, municipal solid waste. Biomass production for fuel – algal cultures, yeasts (Lipid and carbohydrate). Production of biodiesel: Sources of Oils – edible and non-edible; Esterification and Transesterification. Free fatty acids; saponification; Single step and two step biodiesel production. Catalysts for biodiesel production – homogeneous (alkali/acidic) and heterogeneous; Lipase mediated process. General procedure of biodiesel production and purification Quality Control Aspects: GC analysis of biodiesel, fuel property measurements, ASTM (D-6751) and Indian standards (IS15607). Algal Biodiesel production. Production of Bioethanol: Bioethanol production using Sugar; Starch and Lignocellulosic feedstocks; Pretreatment of lignocellulosic feed stock			
UNIT-II			
Biohydrogen and Microbial Fuel Cells			15 Hours

Enzymes involved in H₂ Production; Photobiological H₂ Production: Biophotolysis and Photo fermentation; H₂ Production by Fermentation: Biochemical Pathway, Batch Fermentation, Factors affecting H₂ production, Carbon sources, Detection and Quantification of H₂. Reactors for biohydrogen production.

Microbial Fuel cells: Biochemical Basis; Fuel Cell Design: Anode & Cathode Compartment, Microbial Cultures, Redox Mediators, Exchange Membrane, Power Density; MFC Performance Methods: Substrate & Biomass Measurements, Basic Power Calculations, MFC Performance: Power Density, Single vs Two-Chamber Designs, Wastewater Treatment Effectiveness; Advances in MFC.

UNIT-III

Recovery of Biological Conversion Products

10 Hours

Bio gasification of municipal solid waste: Anaerobic processing; Types of digesters, Biogas plant in India.

Thermochemical processing: Planning an incineration facility, Incineration technologies: Mass burning system; Refuse derived fuel (RDF) system; modular incineration; Fluidized bed incineration; energy recovery; Fuel production through biomass incineration, Pyrolysis and gasification, hydrothermal processing.

Course Outcomes: At the end of the course student will be able to

1. Mark the significance of biofuels and raw materials and Identify suitable feedstock for production of biofuels.
2. Illustrate the production of liquid biofuels from various feed stocks.
3. Demonstrate production of biohydrogen using microbial sources.
4. Extend the concepts of microbial fuel cells towards development of specific application.
5. Understand and apply the concepts of biochemical processing to harvest energy from waste products/streams.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
BT1501-1.1	-	2	-	-	-	-	-	-	1	-	-	-
BT1501-1.2	-	2	-	-	-	-	-	-	1	-	-	-
BT1501-1.3	-	2	-	-	-	-	-	-	1	-	-	-
BT1501-1.4	-	2	-	-	-	-	-	-	1	-	-	-
BT1501-1.5	-	2	-	-	-	-	-	-	1	-	-	-

1: Low 2: Medium 3: High

REFERENCE BOOKS:

1. Drapcho, C. M., Nhuan, N. P. and Walker, T. H. , "Biofuels Engineering Process Technology", Mc Graw Hill Publishers, New York, 2008.
2. Jonathan R.M, Biofuels, "Methods and Protocols (Methods in Molecular Biology Series)", Humana Press, New York, 2009.
3. Olsson L. (Ed.), "Biofuels (Advances in Biochemical Engineering/Biotechnology Series)", Springer-Verlag Publishers, Berlin, 2007.

4.	Glazer, A. and Nikaido, H., "Microbial Biotechnology – Fundamentals of Applied Microbiology", 2 Ed., Cambridge University Press, 2007.
5.	Godfrey Boyle (Ed). "Renewable Energy- Power for sustainable future", 3 rd Ed. Oxford. 2012.
6.	Ramachandran, T. V., "Management of municipal solid waste", Environmental Engineering Series. Teri Press, 2016.

SOLID WASTE MANAGEMENT				
Course Code:	BT1502-1	Course Type:	OEC	
Teaching Hours/Week (L: T: P):	3:0:0	Credits:	03	
Total Teaching Hours:	40+0+0	CIE + SEE Marks:	50+50	
Teaching Department: Biotechnology				
Course Objectives:				
1.	To learn types of solid wastes, collection, treatment and disposal methods.			
2.	To understand various processing techniques and regulations of treatment and disposal.			
UNIT-I				
Introduction to Solid Wastes and its Segregation & Transportation				15 Hours
Solid waste – Definition, Sources of waste, Classification of Solid waste, Characteristics of Solid Waste (Physical, Chemical, Biological), Solid waste problems – impact on environment and health. Concept of waste reduction, recycling and reuse. Waste collection and segregation: Solid waste generation, Onsite handling and segregation of wastes at source, Collection and storage of municipal solid wastes, Equipment used and manpower required in collection, Collection systems and routes. Transportation: Transfer stations: types, location, maintenance, Methods and means of transportation.				
UNIT-II				
Processing Techniques, Recovery of Resources and Waste Disposal				15 Hours
Processing Techniques: Unit operations for separations and processing, mechanical and thermal volume reduction, Incineration of solid wastes – process and types of incinerators (liquid injection, rotary kiln and fluid bed), Biological processing – composting, vermicomposting, biomethanation, fermentation, Drying and dewatering of wastes. Recovery of Resources: Heat recovery in incineration process, energy recovery and conversion of products from biological processes. Dumping of solid wastes, Landfills – Types, site selection, preliminary design, operation, case study, Advantages and disadvantages of landfills, Leachate and landfill gases: Collection and treatment, Landfill disposal for hazardous wastes, biomedical waste.				

Security

UNIT-III																																																																																																								
Solid Waste Management Rules and Planning Issues												10 Hours																																																																																												
Legislative trends and impacts: Major legislations, Government agencies. Municipal Solid Waste Management Act (1999), Hazardous Wastes (Handling and Management) Rules, Biomedical Waste (Handling and Management) Rule (1998), e-Waste (Management and Handling) Rule 2011. Planning and developing a site for solid waste management, Site Remediation: Assessment and Inspection, Remedial techniques, Siting guidelines.																																																																																																								
Course Outcomes: At the end of the course student will be able to																																																																																																								
1.	Identify the sources, classification and characteristics of solid wastes																																																																																																							
2.	Develop insight into the collection, transfer, and transport of solid waste.																																																																																																							
3.	Apply waste processing techniques and recovery of resources from the waste.																																																																																																							
4.	Select the alternatives of solid waste disposals and its impacts.																																																																																																							
5.	Acquire knowledge about solid and hazardous waste management legislative rules.																																																																																																							
Course Outcomes Mapping with Program Outcomes																																																																																																								
<table><tr><td>Program Outcomes→</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>↓ Course Outcomes</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>BT1502-1.1</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>1</td><td>-</td><td>-</td><td>-</td></tr><tr><td>BT1502-1.2</td><td>1</td><td>1</td><td>-</td><td>-</td><td>-</td><td>1</td><td>1</td><td>-</td><td>1</td><td>-</td><td>-</td><td>-</td></tr><tr><td>BT1502-1.3</td><td>-</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>1</td><td>-</td><td>-</td><td>-</td></tr><tr><td>BT1502-1.4</td><td>-</td><td>2</td><td>-</td><td>-</td><td>-</td><td>1</td><td>1</td><td>-</td><td>1</td><td>-</td><td>-</td><td>-</td></tr><tr><td>BT1502-1.5</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>1</td><td>-</td><td>-</td><td>1</td></tr></table>														Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	↓ Course Outcomes													BT1502-1.1	1	-	-	-	-	-	-	-	1	-	-	-	BT1502-1.2	1	1	-	-	-	1	1	-	1	-	-	-	BT1502-1.3	-	2	-	-	-	-	-	-	1	-	-	-	BT1502-1.4	-	2	-	-	-	1	1	-	1	-	-	-	BT1502-1.5	1	-	-	-	-	-	-	-	1	-	-	1
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12																																																																																												
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BT1502-1.2	1	1	-	-	-	1	1	-	1	-	-	-																																																																																												
BT1502-1.3	-	2	-	-	-	-	-	-	1	-	-	-																																																																																												
BT1502-1.4	-	2	-	-	-	1	1	-	1	-	-	-																																																																																												
BT1502-1.5	1	-	-	-	-	-	-	-	1	-	-	1																																																																																												
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REFERENCE BOOKS:																																																																																																								
1.	Tchobanaglou, G., Theisen, H. and Vigil, S. A. "Integrated Solid Waste Management", McGraw – Hill. 1993.																																																																																																							
2.	Tchobanoglous, G., Thiesen, H., Ellasen, "Solid Waste Engineering Principles and Management", McGraw – Hill, 1997.																																																																																																							
3.	Landrefh, R. E. and Sundaresan, B. B. "Solid Waste Management in Developing Countries", Indian National Scientific Documentation Centre. New Delhi, 2000.																																																																																																							

FUNDAMENTALS OF AI AND ML

Course Code:	CS2501-1	Course Type:	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits:	03
Total Teaching Hours:	40+0+0	CIE + SEE Marks:	50
Prerequisite	CS1002-1		

Teaching Department: Computer Science & Engineering

Course Objectives:

1.	Analyze the most fundamental knowledge to the students so that they can understand what the AI is.
2.	Gain a historical perspective of AI and its foundations
3.	Investigate applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.
4.	Experience AI development tools such as an 'AI language', expert system shell, and/or data mining tool.
5.	Explore the current scope, potential, limitations, and implications of intelligent systems.

UNIT-I

Introduction **15 Hours**

What is AI? Foundation of AI, Early History of AI, The Middle Ages and Dark Ages of AI, Renaissance, Future of AI. Intelligence of AI
 AI An Impossible Task, Animal Intelligence, Brain Size And Performance, Sensing And Movement, Subjective Intelligence, Iq Tests. Comparative Intelligence,
 Chapter No 1: Introduction and Intelligence (Page No 11-37)

UNIT-II

Classical Artificial Intelligence **15 Hours**

Introduction, Expert Systems, Conflict Resolution, Multiple Rules, Forward Chaining, Backward Chaining, Problems With Expert Systems, Fuzzy Logic, Fuzzification, Fuzzy Rules, Defuzzification, Fuzzy Expert System, Problem Solving. Chapter No 2: Classical AI (Page No 38-45)

UNIT-III

Foundations of Machine Learning **10 Hours**

What is machine learning? Applications of Machine learning, Understand Data, Types of machine learning: Supervised, Unsupervised, Reinforcement Learning, Theory of learning: feasibility of learning, error and noise, training versus testing, theory of generalization, bias and variance, learning curve,.

Course Outcomes: At the end of the course student will be able to

1.	Explain the fundamental understanding of the history of artificial intelligence (AI) and its foundation
2.	Interpret the basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.
3.	Describe the awareness and a fundamental understanding of various applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models
4.	Identify and explain the proficiency developing applications in an 'AI language', expert system shell, or data mining tool.
5.	Explain the fundamental concept and importance of machine learning.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
CS2501-1.1	3	3	-	-	-	-	-	-	-	-	-	-
CS2501-1.2	3	3	-	-	-	-	-	-	-	-	-	-
CS2501-1.3	3	3	-	-	-	-	-	-	-	-	-	-
CS2501-1.4	3	3	2	-	-	-	-	-	-	-	-	-
CS2501-1.5	3	3	2	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High	
TEXTBOOKS:	
1.	Kevin Warwick, "Artificial Intelligence the basics", Typeset in Bembo by Wearset Ltd, Boldon, Tyne and Wear, Library of Congress Cataloging in Publication Data Warwick, K. ISBN: 978-0-415-56482-3 (hbk).
REFERENCE BOOKS:	
1.	Stuart Russel and Peter Norvig, "Artificial Intelligence A Modern Approach", Pearson 3 rd Edition, 2016.
2.	Dan W Patterson, "Introduction to Artificial Intelligence and Expert Systems", Pearson, 1st edition 2015.
3.	Elaine Rich, "Artificial Intelligence", Mc Graw Hill 3rd Edition, 2017.
E Books / MOOCs/ NPTEL	
1.	Practical Artificial Intelligence Programming With Java, Third Edition, Mark Watson
2.	Artificial Intelligence - http://www.nptelvideos.in/2012/11/artificial-intelligence.html
3.	http://nptel.ac.in/courses/106105077/
4.	https://www.udemy.com/artificial-intelligence
5.	https://www.edx.org/course/artificial-intelligence-ai-columbia-csimm-101x-4

INTRODUCTION TO DATA STRUCTURES

Course Code:	CS2502-1	Course Type:	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits:	03
Total Teaching Hours:	40+0+0	CIE + SEE Marks:	50
Prerequisite	CS1001-1		

Teaching Department: Computer Science & Engineering

Course Objectives:

1.	Outline the concepts of data structures, types, operations, structures, pointers
2.	Implement linear data structures stacks, queues and usage of stacks in various applications.
3.	Implement the operations of singly linked lists
4.	Identify and differentiate different types of binary trees and binary search trees data structures
5.	Illustrate and classify threaded binary trees.

UNIT-I

Introduction	15 Hours
Data Structure, Classification (Primitive and non-primitive), data structure operations, Arrays, Pointers and structures, Dynamic Memory Allocation Functions,	
Linear Data Structures – Stacks	
Introduction and Definition, Representation of stack: Array and structure representation of stacks, Operations on stacks,	
Applications of Stack	
Conversion of Expressions, Evaluation of expressions, Recursion: Implementation, Simulating Recursion, examples on Recursion.	

UNIT-II

Linear Data Structures – Queues	15 Hours
Introduction and Definition Representation of Queue: Array and Structure, representation of Queue, Various queue structures: ordinary queue, circular Queue	
Linear Data Structures - Linked Lists	
Definition and concepts singly linked List: Representation of link list in memory, Operations on singly Linked List, Circular Linked List, Doubly Linked List: Representation and Operations, Circular doubly Link list: Representation and Operations.	

UNIT-III
10 Hours
Nonlinear Data Structures- Tree Data Structures

Basic Terminologies, Binary Trees: Properties, Representation of Binary Tree: Linear representation, Linked representation, Operations on Binary Tree: Insertion, traversals. Introduction to Binary Search Tree

Course Outcomes: At the end of the course student will be able to

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|----|--|
| 1. | Acquire the fundamental knowledge of various types of data structures and pointers. |
| 2. | Apply the fundamental programming knowledge of data structures to design stack and use them for solving problems. |
| 3. | Apply the fundamental programming knowledge of data structures to design queues and use them for solving problems. |
| 4. | Design various functions for implementation of linked list. |
| 5. | Implement and apply the concept of binary trees and binary search tree data structure. |

Course Outcomes Mapping with Program Outcomes

Program Outcomes→ ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12
CS2502-1.1	-	-	-	-	-	-	-	-	-	-	-	-
CS2502-1.2	3	1	2	-	-	-	-	1	-	-	-	1
CS2502-1.3	3	2	2	-	-	-	-	1	-	-	-	1
CS2502-1.4	3	2	-	-	-	-	-	1	-	-	-	1
CS2502-1.5	-	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High
TEXTBOOKS:

- | | |
|----|---|
| 1. | Aaron M. Tenenbaum, YedidyahLangsam& Moshe J. Augenstein, "Data Structures using C", Pearson Education/PHI, 2009. |
| 2. | Ellis Horowitz and Sartaj Sahni, "Fundamentals of Data Structures in C", 2nd edition, Universities Press, 2014. |

REFERENCE BOOKS:

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|----|--|
| 1. | Seymour Lipschutz, "Data Structures, Schaum's Outlines", Revised 1st edition, McGraw Hill, 2014. |
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E Books / MOOCs/ NPTEL

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|----|--|
| 1. | Data Structures Using C, ISRD Group, Tata McGraw Hill, 2006. |
| 2. | Data Structures Using C, Reema Thareja, 2nd edition, Oxford University Press, 2014 |
| 3. | Introduction to Data Structures by edx , URL: https://www.edx.org/course/ |
| 4. | Data structures by Berkley, URL: https://people.eecs.berkeley |
| 5. | Advance Data Structures by MIT OCW , URL: https://www.mooclab.club/ |
| 6. | Data Structure by Harvard Extension School, URL: http://www.extension.harvard . |

DISASTER MANAGEMENT

Course Code:	CV2501-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	CV1002-1		

Teaching Department: Civil Engineering

Course Objectives:

1.	Understand difference between Disaster, Hazard, Vulnerability, and Risk.
2.	Know the Types, Trends, Causes, Consequences and Control of Disasters
3.	Apprehend Disaster Management Cycle and Framework.
4.	Know the Disaster Management in India
5.	Appreciate Applications of Science and Technology for Disaster Management.

UNIT-I

Understanding Disasters	04 Hours
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Understanding the Concepts and definitions of Disaster, Hazard, Vulnerability, Risk, Capacity – Disaster and Development, and disaster management.

Types, Trends, Causes, Consequences and Control of Disasters	10 Hours
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Geological Disasters (earthquakes, landslides, tsunami, mining); Hydro-Meteorological Disasters (floods, cyclones, lightning, thunder-storms, hail storms, avalanches, droughts, cold and heat waves) Biological Disasters (epidemics, pest attacks, forest fire); Technological Disasters (chemical, industrial, radiological, nuclear) and Manmade Disasters (building collapse, rural and urban fire, road and rail accidents, nuclear, radiological, chemicals and biological disasters) Global Disaster Trends – Emerging Risks of Disasters – Climate Change and Urban Disasters

UNIT-II

Disaster Management Cycle and Framework	10 Hours
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Disaster Management Cycle and Framework: Disaster Management Cycle – Paradigm Shift in Disaster Management Pre-Disaster – Risk Assessment and Analysis, Risk Mapping, zonation and Micro zonation, Prevention and Mitigation of Disasters, Early Warning System; Preparedness, Capacity Development; Awareness During Disaster – Evacuation – Disaster Communication – Search and Rescue – Emergency Operation Centre – Incident Command System – Relief and Rehabilitation – Post-disaster – Damage and Needs Assessment, Restoration of Critical Infrastructure – Early Recovery – Reconstruction and Redevelopment; IDNDR, Yokohama Strategy, Hyogo Framework of Action.

Disaster Management in India	06 Hours
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Disaster Management in India: Disaster Profile of India – Mega Disasters of India and Lessons Learnt, Disaster Management Act 2005 – Institutional and Financial Mechanism National Policy on Disaster Management, National Guidelines and Plans on Disaster Management; Role of Government (local, state and national), Non-Government and Inter-Governmental Agencies.

UNIT-III

Applications of Science and Technology for Disaster Management	06 Hours
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Geo-informatics in Disaster Management (RS, GIS, GPS and RS) Disaster Communication System (Early Warning and Its Dissemination) Land Use Planning and Development Regulations Disaster Safe Designs and Constructions Structural and Non Structural Mitigation of Disasters S&T Institutions for Disaster Management in India

Case Studies	04 Hours
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Study of Recent Disasters (at local, state and national level), Preparation of Disaster Risk Management Plan of an Area or Sector, Role of Engineers in Disaster Management

Course Outcomes: At the end of the course student will be able to

1.	Explain Concepts, Types, Trends, Causes of Disasters
2.	Describe Consequences and Control of Disasters
3.	Explain Disaster Management Cycle and Framework
4.	Explain the lesson learnt from the disasters in India and discuss the financial mechanism, roles and responsibilities of Non-Government and Inter-Governmental Agencies for Disaster management
5.	Describe the Applications of Science and Technology recent disasters, role of engineers for Disaster Management and prepare a report of Disaster Risk Management Plan.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
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↓ Course Outcomes													
CV2501-1.1	-	-	-	-	-	-	3	2	-	-	-	1	2
CV2501-1.2	-	-	-	-	-	-	3	2	-	-	-	1	2
CV2501-1.3	-	-	-	-	-	-	3	2	-	-	-	1	2
CV2501-1.4	-	-	-	-	-	-	3	2	-	-	-	1	2
CV2501-1.5	-	-	-	-	-	-	3	2	-	-	-	1	2

1: Low 2: Medium 3: High

TEXTBOOKS:

- Noble, L. , "Introduction to environmental impact assessment. A Guide to Principles and Practice", 2nd edition, Oxford University Press, Don Mills, Ontario, 2010.
- Larry W. Canter, "Environmental Impact Assessment", McGraw Hill Inc. Singapore, 1996.

REFERENCE BOOKS:

- Morris and Therivel, "Methods of Environmental Impact Assessment", 3rd edition. New York, NY: Routledge, 2009.
- Hanna, K. S., "Environmental impact assessment", Practice and Participation. 2nd edition. Oxford, University Press, Don Mills, Ontario, 2009.

E Books / MOOCs/ NPTEL

- <http://nptel.ac.in/courses/120108004/>
- <http://nptel.ac.in/courses/120108004/module3/lecture3.pdf>

ENVIRONMENTAL HYGIENE, SANITATION AND WASTE MANAGEMENT

Course Code:	CV2502-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	CV1002-1		

Teaching Department: Civil Engineering

Course Objectives:

1.	Creation of awareness among student's health issues and Swachh Bharath mission and the consequent responsibilities.
2.	To understand the culture cleanliness, engineering applications in creation of ODF (Open defecation free) concept, Importance of legal & cultural issues related to Environmental Hygiene.
3.	To know the importance of sanitation, gender sensitive sanitation issues & use of engineering technology in construction of toilets.
4.	To know the importance of waste management system, wastewater audit and waste water treatment process.
5.	To study the role of student in Swachh Bharata Abhiyan, solid and waste water treatment process.

UNIT-I

Prospective: Environmental Hygiene (EH), Sanitation, Solid Waste and Wastewater

06 Hours

Introduction- Swachh Bharath Mission (SBM)-Mission Objectives-Duration- Components .Environmental Hygiene-Benefits-Sanitation-Waste Management. Work opportunities in Environmental Hygiene, Sanitation and Waste Management. Participatory Learning for Environmental Hygiene, Sanitation and Waste Management.

Sociology of environmental hygiene management, solid waste and waste water and impacts

08 Hours

Open Defecation-Habits & attitude towards waste-Goals of SBA. Community Consciousness and Engagement on Sanitation Aspects, Roles & Responsibilities, Job Charts, Frequency, Schedules and Timelines in Swachhata Management, Culture of Cleanliness (Swachh Bharat Abhiyan), Behaviour Change Communication, Role of Habits and Attitudes in Environmental Hygiene Management, Waste and Wastewater Disposal; Change Management.

UNIT-II

Infrastructure for Sanitation

08 Hours

Containment-Preparation of toilets –Toilet Types Evaluation of Construction and Maintenance of Community, Public, Institutional and Individual Sanitation Infrastructure Toilets-Proportion and Number of toilets, Gender Sensitive Sanitation

Facilities, Ramps for Differently Aabled, Types – Indian and Western. Faecal Sludge treatment - Single / Twin pit, Eco San, Septic Tank and Formal Sewerage.																																																																																																							
Solid Waste Management												08 Hours																																																																																											
Swachh Survekshan- Solid Waste management- Steps- Waste Audit-Classification Methods of Solid Waste Disposal and Management-Composting-Different types of composting- Waste Minimization-Waste Management.																																																																																																							
UNIT-III																																																																																																							
Waste & Wastewater Audit												06 Hours																																																																																											
Waste Audit -Environmental Impact Assessment, Waste Characterization, Quantity Determination, Primary Collection Methods, Secondary Transportation. Wastewater Audit-Water Budget, Types of Wastewater, Survey of Distribution Network and Feasibility of Various Wastewater Treatment Methods.																																																																																																							
Swachh Bharath Mission and Inclusivity												04 Hours																																																																																											
Swachh Bharath Mission in rural & Urban Context-Gender Issues in sanitation. Role of women in Sanitation.																																																																																																							
Course Outcomes: At the end of the course student will be able to																																																																																																							
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Course Outcomes Mapping with Program Outcomes																																																																																																							
<table><tr><td>Program Outcomes→</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>↓ Course Outcomes</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CV2502-1.1</td><td>1</td><td>1</td><td>-</td><td>-</td><td>-</td><td>2</td><td>3</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CV2502-1.2</td><td>1</td><td>1</td><td>-</td><td>-</td><td>-</td><td>2</td><td>3</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CV2502-1.3</td><td>1</td><td>1</td><td>-</td><td>-</td><td>-</td><td>2</td><td>3</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CV2502-1.4</td><td>1</td><td>1</td><td>-</td><td>-</td><td>-</td><td>2</td><td>3</td><td>2</td><td>-</td><td>3</td><td>-</td><td>-</td></tr><tr><td>CV2502-1.5</td><td>1</td><td>1</td><td>-</td><td>3</td><td>-</td><td>2</td><td>3</td><td>2</td><td>-</td><td>-</td><td>-</td><td>3</td></tr></table>													Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	↓ Course Outcomes													CV2502-1.1	1	1	-	-	-	2	3	2	-	-	-	-	CV2502-1.2	1	1	-	-	-	2	3	2	-	-	-	-	CV2502-1.3	1	1	-	-	-	2	3	2	-	-	-	-	CV2502-1.4	1	1	-	-	-	2	3	2	-	3	-	-	CV2502-1.5	1	1	-	3	-	2	3	2	-	-	-	3
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CV2502-1.5	1	1	-	3	-	2	3	2	-	-	-	3																																																																																											
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TEXTBOOKS:																																																																																																							
1.	Joanne E. Drinan and Frank Spellman, "Water and Wastewater Treatment: A Guide for the Non-engineering Professional".																																																																																																						
2.	M. S. Bhatt and Asherefilliyan, "Solid Waste Management: An Indian Perspective".																																																																																																						
3.	Jagbir Singh, "Solid Waste Management: Present and Future Challenges".																																																																																																						
4.	M. S. Bhatt, "Solid Waste Management: An Indian Perspective".																																																																																																						
5.	T. V. Ramachandra, "Management of Municipal Solid Waste".																																																																																																						
6.	Syed R. Qasim, "Wastewater Treatment Plants: Planning, Design and Operation".																																																																																																						
REFERENCE BOOKS:																																																																																																							
1.	Swachhbharatmission.gov.in/																																																																																																						
2.	https://www.india.gov.in/swachh-bharat-mission-gramin-portal																																																																																																						
3.	https://www.swachhsurvekshan2018.org/																																																																																																						
4.	https://zerowasteurope.eu/																																																																																																						
5.	www.zerowasteindia.in/																																																																																																						
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1.	http://www.un.org/waterforlifedecade/pdf/award_south_africa_eng_for_web.pdf																																																																																																						
2.	http://www.sulabhinternational.org																																																																																																						
3.	http://swachhbharatmission.gov.in/sbmcms/writereaddata/images/pdf/Guidelines/Complete-set-guidelines.pdf																																																																																																						

ENVIRONMENTAL IMPACT ASSESSMENT

Course Code:	CV2503-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	CV1002-1		

Teaching Department: Civil Engineering

Course Objectives:

1.	Identify the need to assess and evaluate the impact of projects on environment.
2.	Explain major principles of environmental impact assessment.
3.	Understand the different steps within environmental impact assessment.
4.	Appreciate the importance of EIA for sustainable development and a healthy environment.

UNIT-I

Evolution of EIA

16 Hours

Concepts of EIA, EIA methodologies (Adhoc, Network Analysis, Checklists, Map overlays, Matrix method), Screening and scoping, Rapid EIA and Comprehensive EIA, General Framework for Environmental Impact Assessment, EIA Specialized areas like environmental health impact assessment, Environmental risk analysis.

UNIT-II

14 Hours

Baseline data study, Prediction, and assessment of impacts on physical, biological, and socio-economic environment, Legislative and environmental clearance procedures in India, Public participation, Resettlement, and rehabilitation.

UNIT-III

10 Hours

Fault free analysis, Consequence Analysis, Introduction to Environmental Management Systems, Environmental management plan-Post project monitoring Environmental Audit: Cost Benefit Analysis, Life cycle Assessment. Case studies on project, regional and sectoral EIA.

Course Outcomes: At the end of the course student will be able to

1.	Understand phenomena of impacts and know the impact quantification of various projects in the environment.
2.	Liaise with and list the importance of stakeholders in the EIA process.
3.	Know the role of public in EIA studies.
4.	Overview and assess risks posing threats to the environment.
5.	Assess different case studies/examples of EIA in practice.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
CV2503-1.1	1	1	-	-	-	2	3	2	-	-	-	-
CV2503-1.2	1	1	-	-	-	2	3	2	-	-	-	-
CV2503-1.3	1	1	-	-	-	2	3	2	-	-	-	-
CV2503-1.4	1	1	-	-	-	2	3	2	-	3	-	-
CV2503-1.5	1	1	-	3	-	2	3	2	-	-	-	3

1: Low 2: Medium 3: High

TEXTBOOKS:

- Noble, L., "Introduction to environmental impact assessment. A Guide to Principles and Practice", 2nd edition, Oxford University Press, Don Mills, Ontario, 2010.

2.	Larry W. Canter, "Environmental Impact Assessment", McGraw Hill Inc. Singapore, 1996.
REFERENCE BOOKS:	
1.	Morris and Therivel, "Methods of Environmental Impact Assessment", 3rd edition. New York, NY: Routledge, 2009.
2.	Hanna, K. S., "Environmental impact assessment. Practice and Participation". 2nd edition. Oxford, University Press, Don Mills, Ontario, 2009.
E Books / MOOCs/ NPTEL	
1.	http://nptel.ac.in/courses/120108004/
2.	http://nptel.ac.in/courses/120108004/module3/lecture3.pdf

INTRODUCTION TO GEOINFORMATICS			
Course Code:	CV2504-1	Course Type	OEC
Teaching Hours/Week (L:T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	CV1001-1, CV1002-1		
Teaching Department: Civil Engineering			
Course Objectives:			
1.	Explain the basic principles of Geoinformatics comprising Remote sensing, Photogrammetry, GPS, GNSS & GIS.		
2.	Explain the stages and techniques of photogrammetry, aerial photo interpretation, visual & digital image processing, enhancement and interpretation.		
3.	Explain and Appraise GIS - its components, data structures, process and operation, Map and its projections, components, preparation and overlays		
4.	Explain the GIS functionality and Appraise the significance of GEOINFORMATICS (Photogrammetry, RS, GPS, GNSS & GIS) in real world applications.		
UNIT-I			
			16 Hours
Remote sensing and its Principles: Physics of remote sensing, EM spectrum, Blackbody concept, atmospheric windows, spectral response of common earth features. Platforms & Sensors: Ground based, Air borne and Space borne platforms, Active and Passive Sensors, Photographic sensors, scanners, radiometers, RADAR and thermal infrared, hyper spectral remote sensing, Indian satellites and sensors: capabilities, data products Photogrammetry: Basic principles of Aerial photography and Photogrammetry, Flight procedures, Aerial Photo Interpretation and Analysis techniques. Satellite Image Interpretation and Analysis techniques: Visual & Digital Image interpretation, Interpretation elements, False Colour Composites (FCC).			
UNIT-II			
			15 Hours
Digital Image Processing and Analysis: Digital image formats, pre-processing and processing (DIP), image restoration/enhancement procedures, information extraction, pattern recognition concepts, post processing procedures. Geographic Information System -concept and spatial models: Fundamentals of GIS, spatial and non-spatial data, vector and raster GIS, GIS Hardware and software, georeferencing, digitization, thematic maps, Overlay Analysis, Operation of GIS, Co-ordinate systems and map projections, Map scale, data display and cartography.			
UNIT-III			
			09 Hours
Geoinformatics and Virtual GIS: Modern Surveying and Geoinformatics, GPS & GNSS, GIS Functionality: Introduction, data acquisition, preliminary data processing, data storage and retrieval, spatial search and analysis, graphics and interaction, Virtual GIS and Real world applications.			
Course Outcomes: At the end of the course student will be able to			
1.	Define and explain the principles of Remote Sensing and list various types of platforms, sensors & resolutions in RS with a special reference to Indian satellites and data products.		

2.	Explain Photogrammetry, its basic principles, elements of photo interpretation, Visual & Digital Image interpretation techniques
3.	Explain different stages involved in Digital Image Processing, various image enhancement techniques, list and classify the digital image formats and the extracted information for various purposes.
4.	Explain and Appraise GIS - its components, data structures, process and operation, Map and its projections, components, preparation and Overlays.
5.	Explain the GIS functionality and appraise the significance of GEOINFORMATICS (Photogrammetry, RS, GPS, GNSS & GIS) and Virtual GIS in real world applications.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
CV2504-1.1	2	2	-	-	-	2	-	-	-	-	-	-
CV2504-1.2	2	2	-	-	-	2	1	-	-	-	-	-
CV2504-1.3	2	2	-	-	-	2	1	-	-	-	-	-
CV2504-1.4	2	2	-	-	-	2	1	-	-	-	-	-
CV2504-1.5	2	2	-	-	-	2	1	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Anji Reddy, M, "Text Book of Remote Sensing and Geographical Information Systems", Fourth Edition, BS Publication, Hyderabad, 2012.
2.	Bhatta, Basudeva, "Remote Sensing and GIS", 2nd edition, Oxford University Press, N. Delhi, 2011.
3.	Lillesand, T.M., Kiefer, R.W and Chipman, J. W., "Remote sensing and Image Interpretations", 7th edition, John Wiley and sons, New Delhi, 2015.

REFERENCE BOOKS:

1.	Anji Reddy, M. and Hari Shankar, Y., "Digital Image Processing", BS Pub., Hyd, 2006.
2.	Bernhardsen, Tor, "Geographic Information Systems", 3rd Ed., Wiley India, Delhi, 2002.
3.	Canada Centre for Remote Sensing, Fundamentals of Remote sensing-Tutorial, 2011.
4.	Chang, Kang-tsung, "Introduction to Geographic Information Systems", 4th Ed., Tata McGraw Hill Publishing Company Limited, New Delhi, 2008.
5.	Korte, George B., "The GIS Book", Onword Press, Thomson Learning Inc., USA, 2001.
6.	Kumar, S., "Basics of Remote sensing and GIS", Laxmi Publications (P) Ltd., Delhi, 2008.
7.	Longler, Paul A., Goodchild, Michael F., Maguire, David J., Rhind. David W., "Geographic Information Systems and Science", John Wiley & Sons Ltd., ESRI Press, 2004.
8.	Sabins, F. L., "Remote Sensing: Principles and Interpretation" 3rd edn. WH Freeman and Company, New York, 1997.

E Books / MOOCs/ NPTEL

1.	https://www.youtube.com/user/edusat2004
2.	https://eclass.iirs.gov.in/login

CORROSION SCIENCE

Course Code:	CY2501-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	CY1001-1		
Teaching Department: Chemistry			
Course Objectives:			
1.	To provide fundamental understanding aspects of electrochemistry and material science related to corrosion. To understand the types of corrosion attacking on the metal and its preventions.		
2.	To impart knowledge on corrosion science and its applications to the engineering materials.		
3.	To identify practice for the prevention and remediation of the corrosion. To provide methodologies for measuring the corrosion performance of materials.		

UNIT-I																																																																														
Fundamentals of Corrosion												09 Hours																																																																		
Definition, cost of corrosion, Corrosion Damage and consequences, Classification of corrosion, Electrochemical Aspects of corrosion, Electrochemical reactions, Different Environmental aspects, polarization and passivity, Corrosion Rate Expression, Determination. Standard electrode potential, EMF and Galvanic series, Potential-pH (Roubaix Diagram).																																																																														
Forms of Corrosion												08 Hours																																																																		
Galvanic corrosion, Crevices corrosion, Filiform corrosion, Pitting corrosion, Uniform corrosion and Atmospheric corrosion, Inter granular corrosion, Selective leaching, Erosion corrosion, Cavitation damage, Stress corrosion , Impingement attack, Inlet tube corrosion, Corrosion fatigue, Hydrogen blistering, Hydrogen embrittlement.																																																																														
UNIT-II																																																																														
Corrosion at Elevated Temperature												08 Hours																																																																		
High temperature materials, Metal oxides, Pilling bed worth rule, oxide defect structure, Hot corrosion, Corrosion of mineral acids-corrosion of steel, stainless steel, Cu and Al.																																																																														
Corrosion Testing												07 Hours																																																																		
Weight loss method, Tafel extrapolation test, linear polarization test and AC impedance method.																																																																														
UNIT-III																																																																														
Corrosion Prevention Methods												08 Hours																																																																		
Materials Selections, Design, Change of the environments: Atmospheric corrosion, Control of atmospheric corrosion, Changing medium, Inhibitors, Cathodic and Anodic protection, Protective coatings.																																																																														
Course Outcomes: At the end of the course student will be able to																																																																														
<table><tr><td>1.</td><td>Explain the fundamentals of difference in electrode potential across an interface in particular a metal/ electrolyte and the relationship between rates of electrochemical reactions and the potential drop across interfaces.</td></tr><tr><td>2.</td><td>Analyze the causes and mechanisms of various types of corrosion including uniform, galvanic, crevice, pitting, inter granular and various modes of environmentally cracking. Acquire knowledge of influence of a materials composition, the effect of an electrolytes composition on the corrosion of metals and microstructure on its corrosion performance.</td></tr><tr><td>3.</td><td>Identify the materials that will exhibit adequate corrosion resistance in a particular environment and remedial action that will reduce corrosion to a acceptable level. Explain the concepts of different measuring techniques of corrosion.</td></tr></table>														1.	Explain the fundamentals of difference in electrode potential across an interface in particular a metal/ electrolyte and the relationship between rates of electrochemical reactions and the potential drop across interfaces.	2.	Analyze the causes and mechanisms of various types of corrosion including uniform, galvanic, crevice, pitting, inter granular and various modes of environmentally cracking. Acquire knowledge of influence of a materials composition, the effect of an electrolytes composition on the corrosion of metals and microstructure on its corrosion performance.	3.	Identify the materials that will exhibit adequate corrosion resistance in a particular environment and remedial action that will reduce corrosion to a acceptable level. Explain the concepts of different measuring techniques of corrosion.																																																											
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1: Low 2: Medium 3: High																																																																														
TEXTBOOKS:																																																																														
1.	Mars G Fontana, "Corrosion Engineering", 3 rd Edition, Tata McGraw-Hill Edition.																																																																													
REFERENCE BOOKS:																																																																														
1.	Chamberlian and K. Trethway, "Corrosion", Longman scientific and technical, John Wiley and Sons.																																																																													

NATURAL PRODUCTS CHEMISTRY

Course Code:	CY2502-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	CY1001-1		

Teaching Department: Chemistry

Course Objectives:

1.	Identify the structure of terpenoids and their biosynthesis. Elucidate the structure of β -carotene, haemoglobin and chlorophyll.
2.	Understand the chemistry underlying steroids and sex hormones. Get introduced to the different types of prostaglandins as well as theory and chemistry behind natural dyes.
3.	Gain knowledge on general methods of structural determination of some of the important alkaloids.

UNIT-I

Terpenoids & Carotenoids	08 Hours
Introduction and classification, isoprene rules, general methods of determination of structure of terpenoids. Structure elucidation of the following terpenoids-geraniol, α -pinene, camphene and farnesol. Biosynthesis of terpenoids. Introduction and classification of carotenes. Structural elucidation of β -carotene.	
Porphyrins	07 Hours
Introduction to porphyrins, structure and degradation products of haemoglobin and chlorophyll.	

UNIT-II

Steroids	08 Hours
Introduction, Dile's hydrogenation. Chemistry of cholesterol, Blanc's rule, Barbier-Wielman degradation, Oppenauer oxidation. Constitution of bile acids. Sex hormones: Chemistry of oestrone, progesterone, androsterone and testosterone.	
Prostaglandins & Natural Dyes	08 Hours
Introduction, nomenclature, classification, and biological role of prostaglandins. Structure elucidation of PGE_1 , Biosynthesis of PGE_2 and $\text{PGF}_{2\alpha}$. Introduction, Witt's theory of colour, methods of dyeing, chemical constitution of alizarin.	

UNIT-III

Alkaloids	09 Hours
Definition, Classification and isolation of alkaloids. General methods of structural determination of alkaloids. Detailed study of structure elucidation of the following alkaloids- papaverine, cinchonine and nicotine.	

Course Outcomes: At the end of the course student will be able to

1	Elucidate the structure of terpenoids like geraniol, α -pinene, camphene and farnesol. Explain the structural chemistry of carotenoids and porphyrins.
2	State the basic reactions governing steroids and sex hormones. Explain the biological role and structure of prostaglandins and state the methods employed for dyeing.
3	Apply the general methods of structural determination to elucidate the structure of alkaloids like papaverine, cinchonine and nicotine.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
CY2502-1.1	3	3	-	-	-	1	1	-	-	-	-	-
CY2502-1.2	3	3	-	-	-	1	1	-	-	-	-	-
CY2502-1.3	3	3	-	-	-	1	1	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Agarwal, "Organic Chemistry of Natural Products", Vol.-I & Vol.-II, O.P. Goel Publishing House, 2014.
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REFERENCE BOOKS:	
1.	K. Nakanishi, T. Goso, S. Ito, S. Natori and S. Nozoe, "Natural Products Chemistry", Vol. I & II, Academic Press, Ny, 1974.
2.	Gurudeep R. Chatwal, "Organic Chemistry of Natural Products", Vol. I & II, Himalaya Publishing House, 2013.
3.	G.A. Swal, "An Introduction to Alkaloids", Backwell Scientific Publications, 1967.
4.	Hand book of naturally occurring Compounds, Vol. II, terpenes, T.K. Davon, A.I. Scott, Academic Press, Ny, 1974.

ARTIFICIAL NEURAL NETWORK SYSTEMS																																																																																																							
Course Code:				EC1501-1			Course Type			OEC																																																																																													
Teaching Hours/Week (L: T: P)				3:0:0			Credits			03																																																																																													
Total Teaching Hours				40+0+0			CIE + SEE Marks			50+50																																																																																													
Teaching Department: Electronics & Communication Engineering																																																																																																							
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Introduction to Artificial Neural networks											16 Hours																																																																																												
Introduction, Basic building blocks: network architecture, setting the weights, activation functions, ANN terminologies: weights, activation functions, bias, threshold, McCulloch-Pitts Neuron Model, Learning Rules																																																																																																							
UNIT-II																																																																																																							
Single Layer Perceptron Classifiers											15 Hours																																																																																												
Classification Model, Features, and Decision Regions, Discriminant Functions, Linear Machine and Minimum Distance Classification, Nonparametric Training Concept, Training and Classification Using the Discrete Perceptron: Algorithm and Example, Single-Layer Continuous Perceptron Networks for Linearly Separable Classifications, Multicategory Single-Layer Perceptron Networks																																																																																																							
UNIT-III																																																																																																							
Single-Layer Feedback Networks											09 Hours																																																																																												
Basic Concepts of Dynamical Systems, Mathematical Foundations of Discrete-Time Hopfield Networks, Mathematical Foundations of Gradient-Type Hopfield Networks. Transient Response of Continuous-Time Networks, Relaxation Modeling in Single-Layer Feedback Networks																																																																																																							
Course Outcomes: At the end of the course student will be able to																																																																																																							
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EC1501-1.5	3	-	-	-	-	-	-	-	-	-	-	-																																																																																											

1: Low 2: Medium 3: High	
TEXTBOOKS:	
1	S. N. Sivanandam, S. Sumathi, S. N. Deepa, "Introduction to Neural Networks Using MATLAB 6.0", Tata McGraw-Hill Education, 2006
2	Jacek M. Zurada "Introduction to Artificial Neural Systems", 1st Edition, St. Paul West Publishers-USA, 1992.
3	Michael A Neilsen, "Neural Networks and Deep Learning", Determination Press, 2015

INTRODUCTION TO MATLAB PROGRAMMING: A HANDS-ON APPROACH			
Course Code:	EC1502-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	2:0:2	Credits	03
Total Teaching Hours	27+0+26	CIE + SEE Marks	50+50
Teaching Department: Electronics & Communication Engineering Offered to Civil & BT			
Course Objectives:			
1.	To demonstrate basic understanding of MATLAB programming		
2.	To use and write functions		
3.	To use MATLAB programming for image processing		
Unit-I			27 Hours
Introduction to MATLAB: Starting MATLAB and familiarization with its user interface, syntax and semantics, ways in which MATLAB provides help, create plots in MATLAB. Matrices and Operators: defining matrices, manipulation of matrices, extract parts of them and combine them to form new matrices, use of operators to add, subtract, multiply, and divide matrices, and we will learn that there are several different types of multiplication and division. Functions: creating reusable functions, how the environment inside a function is separated from the outside via a well-defined interface through which it communicates with that outside world, define a function to allow input to it when it initiates its execution. Programmer's Toolbox: polymorphism and how MATLAB exploits it to change a function's behavior on the basis of the number and type of its inputs, random number generator, how to get input from the keyboard, how to print to the Command Window, and how to plot graphs in a Figure window, how to find programming errors with the help of the debugger, how to print to the Command Window, and how to plot graphs in a Figure window, how to find programming errors with the help of the debugger. Selection Statement and Loops: how to use the if-statement, how to use relational operators and logical operators, how to write polymorphic functions and how to make functions resistant to error, the for-loop and the while-loop, how the break-statement works, nested loops, logical indexing and implicit loops. Data Types: character arrays and how the characters in them are encoded as numbers, string and datetime datatype, how to produce heterogeneous collections of data via structs and cells. File Input/Output: reading and writing files, how to create, read from, and write into MAT-files, Excel files, text files, and binary files, how to navigate among folders with MATLAB commands. Image Processing using MATLAB: pre-processing – conversion of color image to gray scale image, decomposition of color images to single color component image, histogram of image, thresholding			
List of Experiments			
1	Starting MATLAB and familiarization with its user interface, syntax and semantics, ways in which MATLAB provides help, create plots in MATLAB.		
2	Defining matrices, manipulation of matrices, extract parts of them and combine them to form new matrices, use of operators to add, subtract, multiply, and divide matrices, and we will learn that there are several different types of multiplication and division.		
3	creating reusable functions, how the environment inside a function is separated from the outside via a well-defined interface through which it communicates with that outside world, define a function to allow input to it when it initiates its execution.		

4	Polymorphism and how MATLAB exploits it to change a function's behavior on the basis of the number and type of its inputs, random number generator, how to get input from the keyboard, how to print to the Command Window
5	How to plot graphs in a Figure window, how to find programming errors with the help of the debugger, how to print to the Command Window, and how to plot graphs in a Figure window, how to find programming errors with the help of the debugger.
6	How to use the if-statement, how to use relational operators and logical operators, how to write polymorphic functions and how to make functions resistant to error.
7	The for-loop and the while-loop, how the break-statement works, nested loops, logical indexing and implicit loops.
8	Character arrays and how the characters in them are encoded as numbers, string and datetime datatype, how to produce heterogeneous collections of data via structs and cells.
9	Reading and writing files, how to create, read from, and write into MAT-files, Excel files, text files, and binary files, how to navigate among folders with MATLAB commands.
10	Reading an image, saving, basic manipulation of images, arithmetic operations
11	Pre-processing – conversion of color image to gray scale image, decomposition of color images to single color component image.
12	Histogram processing.
13	Thresholding operation.

Course Outcomes: At the end of the course student will be able to

1.	Use matrices and operators in MATLAB programming
2.	Use and write functions; use MATLAB toolbox
3.	Use toolbox and selection statement in MATLAB programming
4.	Write MATLAB programs using loops and summarize data types
5.	Summarize file input/output methods using MATLAB commands and apply pre-processing and thresholding operations on images

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
EC1502-1.1	1	-	-	-	3	-	-	-	-	-	-	-
EC1502-1.2	1	-	-	-	3	-	-	-	-	-	-	-
EC1502-1.3	1	-	-	-	3	-	-	-	-	-	-	-
EC1502-1.4	1	-	-	-	3	-	-	-	-	-	-	-
EC1502-1.5	1	-	-	-	3	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Stormy Attaway, "Matlab: A Practical Introduction to Programming and Problem Solving", Second Edition, Butterworth-Heinemann, 2011
2. Fitzpatrick and Ledeczi, "Computer Programming with MATLAB", eBook, 2013
3. Rafael C. Gonzalez, Richard E. Woods and Steven L. Eddins, Digital Image Processing using MATLAB, first edition, Dorling Kindersley Pvt Ltd, 2006.

REFERENCE BOOKS:

1. Duane C. Hanselman, Bruce L. Littlefield, "Mastering MATLAB", first edition, Pearson, 2011

E Books / MOOCs/ NPTEL

1. <https://nptel.ac.in/courses/103/106/103106118/>
2. <https://www.coursera.org/learn/matlab>

ROBOTICS

Course Code:	EC1503-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50

Teaching Department: Electronics & Communication Engineering

Course Objectives:

1.	Understand Anatomy of a robot.
2.	Analyse the robot motion using translation and rotational matrix.
3.	Discuss Robot trajectory planning and robot control.
4.	Categorise the various sensors used in robotics
5.	Understand the robot programming.

UNIT-I

Introduction	16 Hours
Definition, anatomy of robot, classification configurations, robot links and joints, robot specifications, resolution accuracy and repeatability, simple numerical problems, robot drive systems, hydraulic, pneumatic and electric drive systems, wrist and its motions, end effectors, types of end effectors, mechanical & Non-mechanical grippers, methods of constraining parts in grippers.	

Motion analysis	
Direct kinematics and inverse kinematics, 3D homogeneous transformations, rotation, translation and displacement matrix, composite rotation matrix, rotation matrix about an arbitrary axis.	

UNIT-II

Control and trajectory planning	15 Hours
Trajectory planning, definition, steps in trajectory planning, joint space techniques, use of a p-degree polynomial as interpolation function, cubic polynomial trajectories, linear function with parabolic blends, joint space verses, simple numerical problems on joint space trajectory planning.	

Sensors	
Classification, Types- Contact & Non-Contact sensors.	

Machine Vision	
Machine vision, functions of machine vision system, sensing and digitizing, imaging devices, analog to digital signal conversion, quantization and encoding, simple numerical problems, image storage, image processing and analysis, image data reduction, segmentation, feature extraction, object recognition, robotic machine vision applications, inspection, identification, visual surveying and navigation.	

UNIT-III

Programming	09 Hours
Introduction to robot programming, robot cell layout, work cell control and interlocks, manual programming, lead through and walkthrough programming, off-line programming, robot programming languages, examples	

Course Outcomes: At the end of the course student will be able to

1.	Explain the working principle, various performance parameters of robots and identify the types of robots employed in industry.
2.	Discuss the concept of direct and inverse kinematics. Determine the position and orientation of End-Effector subjected to transformations. Demonstrate the applications of Denavit-Hartenberg (DH) method for different robot configurations.
3.	Determine the technique of trajectory planning, control schemes for robot joints and understand the types of the sensors used in robotics.
4.	Apply engineering knowledge in robot visual surveying and navigation.
5.	Analyze and formulate different types of robot cell layouts and use modern tools to write robot programs for different tasks.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
EC1503-1.1	3	2	2	1	-	-	-	-	-	-	-	1
EC1503-1.2	3	3	2	2	-	-	-	-	3	3	-	1

Security

	EC1503-1.3	3	2	2	2	-	-	-	-	3	3	-	1
	EC1503-1.4	3	2	2	1	-	-	-	-	-	-	-	1
	EC1503-1.5	3	3	3	2	2	-	-	-	-	-	-	1

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	R. K. Mittal and I. J. Nagrath, “Robotics and Control”, Tata-McGraw-Hill Publications, 2007.
2.	Mikell P. Groover, Mitchel Weiss, Roger N. Nagel and Nicholas G. Odrey, “Industrial Robotics”, McGraw-Hill Publications, International Edition, 2008

REFERENCE BOOKS:

1.	Fu K. S., Gonzelez R. C., Lee C. S. G., "Robotics: Control, Sensing, Vision, Intelligence," , McGraw Hill Book Co., International edition, 2008.
2.	YoremKoren, "Robotics for Engineers", McGraw-Hill Publication, International edition, 1987.
3.	Craig, J. J., "Introduction to Robotics: Mechanics and Control", 3rd Edition, Pearson PrenticeHall Publications, 2005.
4.	Schilling R. J., "Fundamentals of Robotics, Analysis and Control", Prentice-Hall Publications, Eastern Economy edition, 2007.
5.	AppuKuttan K. K., "Robotics", I.K. International Publications, First Edition, 2007.
6.	James G. Keramas, "Robot Technology Fundamentals", Cengage Learning, 1999.
7.	Richard K. Barnhart, Stephen B. Hottman, Douglas M. Marshall, Eric Shappee, "Introduction to Unmanned Aircraft Systems", CRC Press, 2012.
8.	Ghosh, "Control in Robotics and Automation", Allied Publishers.
9.	Deb, "Robotics Technology", Wiley India.

E Books / MOOCs/ NPTEL

1.	https://nptel.ac.in/courses/112105249
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CONSUMER ELECTRONICS			
Course Code:	EC2501-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	EC1001-1		
Teaching Department: Electronics & Communication Engineering			
Course Objectives:			
1.	To provide basic knowledge on sound and transducers		
2.	To provide basic knowledge on different display units and camera		
3.	To understand the recording process and storage mechanism		
4.	To provide basic knowledge on communication and broadcasting		
5.	To understand the working of various electronic gadgets		
UNIT-I			
Sound & Vision	15 Hours		
Sound: Definition and properties of sound, Transducers: Micro Phone – characteristics and types, and Loud Speakers – characteristics and types, Enclosures and baffles, mono-stereo, audio amplifiers-characteristics, Synthesizers.			
Vision: Displays-LED, LCD, PLASMA, Camera: basic principle, CCTV Camera.			
UNIT-II			
Recording, Playback, Communication & Broadcasting Systems	15 Hours		
Recording and Playback: Audio recording methods-magnetic recording, optical recording, digital recording, erasing methods, optical discs- recording and playback, Film projector, Theatre Sound, HiFi system.			
Communications And Broadcasting: Modulation: AM, FM PCM, Radio transmitters, Radio receivers - Tuned radio frequency receiver and Superhetrodyne receiver. Fiber optics, Radio and TV broadcasting. Cellular communication: digital cellular phone, establishing a call.			
UNIT-III			

Security

Other Electronic Systems	10 Hours																																																																																											
Fax machine, Xerox machine, electronic Calculator, Microwave ovens, Washing Machines, A/C and refrigeration, ATM, Auto Electronics, Industrial Electronics and Robotics, Electronics in health / Medicine.																																																																																												
Course Outcomes: At the end of the course student will be able to																																																																																												
1.	Recall basics of sound and transducers.																																																																																											
2.	Understand the working principles of display units and CCTV camera.																																																																																											
3.	Explain basic working of Recording, storage devices																																																																																											
4.	Explain basics of communication and broadcasting																																																																																											
5.	Recall basic working of commonly used electronic gadgets																																																																																											
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1.	Anand, “Consumer Electronics”, Khanna publications, 2011.																																																																																											
2.	Bali S. P., “Consumer Electronics”, Pearson Education, 2005.																																																																																											
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1.	Gulati R. R. "Modern Television Engineering", Wiley Eastern.																																																																																											

PCB DESIGN AND FABRICATION				
Course Code		EC2502-1	Course Type	OEC
Teaching Hours/Week (L: T: P):		1:0:4	Credits	03
Total Teaching Hours		15+0+52	CIE + SEE Marks	50+50
Prerequisite		EC1001-1		
Teaching Department: Electronics & Communication Engineering				
Course Objectives:				
1.	To enable students to gain knowledge of Schematic Design techniques & PCB design techniques			
2.	To expose students to complete PCB Design & manufacturing process			
Unit-I				
Circuit Schematic			05 Hours	
Introduction to Kicad schematic design tool, features, node connections, labeling, creating new component.				
Unit-II				
PCB Layout:			05 Hours	
Introduction to Kicad layout editor, features, layer selections, manual and auto routing in Kicad, verification of footprint, creating footprint for a given component.				
Unit-III				
PCB Fabrication			05 Hours	
Generating and verifying the PCB Gerber file, preparing artwork for a single side PCB fabrication, preparing PCB artwork for double side PCB, Etching process, tin plating, legend printing, green masking and through hole plating				
List of Experiments				

1	Exploring the Kicad Schematic and layout tool												
2	Developing a schematic circuit for microphone preamplifier												
3	Designing a single side PCB layout for microphone preamplifier												
4	Developing a schematic circuit for a microcontroller development board												
5	Designing a double side PCB layout for a microcontroller development board												
6	Choosing a new sensor/display module and building a schematic circuit for the user level application												
7	Building a layout using single or double side PCB for the sensor/display module												
8	Preparing the film for the bottom copper, solder mask and top silk (legend) to fabricate a single side PCB using chemical process												
9	Preparing the film for the top copper, bottom copper, top solder mask, bottom solder mask and legend to fabricate double side PCB using chemical process												
10	PCB routing, etching, cutting and drilling using CNC machine												
Course Outcomes: At the end of the course student will be able to													
1.	Draw schematic circuit and create PCB layout for single or multilayer PCB												
2.	Fabricate single and double-layer PCB												
Course Outcomes Mapping with Program Outcomes													
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TEXTBOOKS:													
1.	Peter Dalmaris, “Kicad Like a Pro”, Tech Exploration.												
REFERENCE BOOKS:													
1.	Peter Dalmaris, “Kicad Like a Pro”, Tech Exploration.												
2.	David L. Jones, “PCB Design Tutorials”, Alternate zone, 2004.												
E Books / MOOCs/ NPTEL													
1.	www.alternatezone.com												

SPACE TECHNOLOGY AND APPLICATIONS			
Course Code:	EC2503-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	EC1001-1		
Teaching Department: Electronics & Communication Engineering			
Course Objectives:			
1.	Understand the general laws governing satellite orbits and its parameters.		
2.	Discuss effect of space environment on satellite signal propagation.		
3.	Illustrate various segments employed in satellite and ground station.		
4.	Calculate the uplink / downlink subsystem characteristics.		
5.	know the effects on the EM waves in propagation through space.		
6.	Explain the satellite launch in the space and their applications in remote sensing.		
7.	Discuss the different communication systems used for satellite access.		
8.	Summarise Advanced space systems for mobile communication, VSAT, GPS.		
UNIT-I			

Security

Satellite Technology	15 Hours																																																																																																						
Satellite communications: Introduction, Kepler’s laws, definitions, orbital element, apogee and perigee heights, orbit perturbations, inclined orbits. Space environment: Earth’s Atmosphere, Ionosphere and Meteorological effects on space systems, propagation of signal, Transmission losses in space environment. Satellite Technology: Space segment, Ground segment, Quality and Reliability, Satellite Communication systems.																																																																																																							
UNIT-II																																																																																																							
Space Applications	15 Hours																																																																																																						
Launch Vehicles: Working, stages, Fuel, payload protection, Navigation, guidance and control, Reliability, launching into outer space and launch bases. Types of launch vehicles. Space Applications: Digital DBS TV, DBS-TV System Design, Master Control Station and Uplink Antennas. Introduction, Radio and Satellite Navigation, Remote Sensing: Introduction to Remote Sensing, Concepts and Applications of satellite Remote sensing.																																																																																																							
UNIT-III																																																																																																							
Advanced Space Systems	10 Hours																																																																																																						
Satellite Access: Introduction, Single Access, Pre-assigned FDMA, Demand-Assigned FDMA, Spade system. Advanced space systems: Satellite mobile services, VSAT, Radarsat, orbital communication. Global Positioning Satellite System (GPS). Course Outcomes: At the end of the course student will be able to																																																																																																							
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Security

BATTERY MANAGEMENT SYSTEM			
Course Code:	EE2501-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	EE1001-1		
Teaching Department: Electrical & Electronics Engineering			
Course Objectives:			
1	To familiarize various concepts of BMS		
2	To understand functional blocks of BMS		
3	To study design steps of BMS		
4	To introduce hardware implementation of BMS		
UNIT-I			
Battery System	08 Hours		
Introduction, Cells, Batteries, and Packs, Resistance, Li-Ion Cells, Formats, Chemistry, Safety, Safe Operating Area, Efficiency, Aging, Modeling, Unequal Voltages in Series Strings, Li-Ion BMSs, BMS Definition, Li-Ion BMS Functions, Custom Versus Off-the-Shelf, Li-Ion Batteries, SOC, DOD, and Capacity, Balance and Balancing, SOH			
BMS Options	07 Hours		
Functionality, CCCV Chargers, Regulators, Meters, Monitors, Balancers, Protectors, Functionality Comparison, Technology, Simple (Analog), Sophisticated (Digital), Technology Comparison, Topology, Centralized, Modular Master-Slave, Distributed, Topology Comparison			
UNIT-II			
BMS Functions	07 Hours		
Measurement, Voltage, Temperature, Current, Management, Protection, Thermal Management, Balancing, Redistribution, Distributed Charging, Evaluation, State of Charge and Depth of Discharge, Capacity, Resistance, State of Health (SOH), External Communications, Dedicated Analog Wire, Dedicated Digital Wire, Data Link, Logging and Telemetry, Off-the-Shelf BMSs, Cell Manufacturers' BMSs, Comparison			
Custom BMS Design	08 Hours		
Using BMS ASICs , BMS ASIC Comparison, Analog BMS Design, Analog Regulator, Analog Monitor, Analog Balancer, Analog Protector, Ready-Made, Digital BMS Designs, ATMEL's BMS Processor, Elithion's BMS Chip Set, National Semiconductors' Complete BMS, Peter Perkin's Open Source BMS, Texas Instruments' bq29330/bq20z90, Texas Instruments' bq78PL114/bq76PL102, Custom Digital BMS Design, Voltage and Temperature Measurement, Current Measurement, Evaluation, Communications, Optimization, Switching, Logging, Cell Interface, Non-distributed, Distributed, Distributed Charging			
UNIT-III			
Deploying a BMS	10 Hours		
Installing, Battery Pack Design, BMS Connections to Pack, BMS Connections to System, Configuring, Cell Configuration, Pack Configuration, System Configuration, Testing, Troubleshooting, Grounding, Shielding, Filtering, Wire Routing			
Course Outcomes: At the end of the course student will be able to			
1	Identify process to implement BMS		
2	Describe various communication protocol involved in BMS		
3	Illustrate functionality of BMS		

Security

4	Apply concepts of BMS using application specific IC																																																																																																							
5	Analyse the hardware implementation aspects of BMS																																																																																																							
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1	Rui Xiong, "Battery Management Algorithm for Electric Vehicles", Springer 2019.																																																																																																							
2	Nicolae Tudoroiu, "Battery Management Systems of Electric and Hybrid Electric Vehicles", MDPI 2021																																																																																																							

BIOMEDICAL INSTRUMENTATION			
Course Code:	EE2502-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	EC1001-1		
Teaching Department: Electrical & Electronics Engineering			
Course Objectives:			
1.	The course is designed to give the basic concepts of Instrumentation involved in medical field and human physiology.		
2.	To introduce an fundamental of transducers as applicable to physiology		
3.	To explore the human body parameter measurements setups		
4.	To make the students understand the basic concepts of forensic techniques.		
5.	To give basic ideas about Electrophysiological measurements, medical imaging		
UNIT-I			
Physiology and transducers			08 Hours
Cell and its structure, Resting and Action Potential, Nervous system: Functional organization of the nervous system, Structure of nervous system, neurons, synapse, transmitters and neural communication, Cardiovascular system, respiratory system, Basic components of a biomedical system, Transducers, selection criteria, Piezo-electric, ultrasonic transducers, Temperature measurements, Fiber optic sensors.			
Electro – Physiological measurements			09 Hours
Electrodes: Limb electrodes, floating electrodes, pre-gelled disposable electrodes, Micro, needle and surface electrodes, Amplifiers: Preamplifiers, differential amplifiers, chopper amplifiers, Isolation			

amplifier. ECG, EEG, EMG, ERG, Lead systems and recording methods, Typical waveforms. Electrical safety in medical environment: shock hazards, leakage current-Instruments for checking safety parameters of biomedical equipment.

UNIT-II

Non-electrical parameter measurements

08 Hours

Measurement of blood pressure, Cardiac output, Heart rate, Heart sound Pulmonary function measurements, spirometer, Photo Plethysmography, Body Plethysmography, Blood Gas analyzers : pH of blood, measurement of blood pCO₂, pO₂, finger-tip oximeter, ESR, GSR measurements

Medical Imaging

07 Hours

Radiographic and fluoroscopic techniques, X rays, Computer tomography, Mammography, MRI, fMRI, Ultrasonography, Endoscopy, Thermography, Different types of biotelemetry systems and patient monitoring

UNIT-III

Assisting and therapeutic equipments:

08 Hours

Pacemakers, Defibrillators, Ventilators, Nerve and muscle stimulators, Diathermy, Heart Lung machine, Audio meters, Dialyzers, Lithotripsy

Course Outcomes: At the end of the course student will be able to

1	Understand the physiology of biomedical system
2	Measure biomedical and physiological information
3	Discuss the application of Electronics in diagnostics and therapeutic area.
4	Analyze the images and do a prediction using image processing.
5	Understand the different equipment's used for various measurements of physiology

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
EE2502-1.1	3	3	-	2	1	1	-	-	-	-	-	-
EE2502-1.2	2	2	2	2	-	-	-	-	-	-	-	-
EE2502-1.3	3	2	2	1	2	1	-	-	-	-	-	-
EE2502-1.4	2	3	-	-	1	-	-	-	-	-	1	-
EE2502-1.5	3	3	-	-	2	-	-	-	-	-	2	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Leslie Cromwell, Fred J. Weibell, Erich A. Pfeiffer, "Bio-Medical Instrumentation and Measurements", II edition, Pearson Education, 2002.
2. R. S. Khandpur, "Handbook of Bio-Medical instrumentation", Tata McGraw Hill Publishing Co Ltd., 2003.
3. J. Webster, "Medical Instrumentation", John Wiley & Sons, 1995.
4. L. A. Geddes and L. E. Baker, "Principles of Applied Bio-Medical Instrumentation", John Wiley & Sons, 1975.
5. David. Cooney and Michel Deckker, "Bio- Medical Engineering Principles", INC.

REFERENCE BOOKS:

1. David Cooney, "Bio-Medical Engineering Principles", 2015, 1st Edition, Marcel Deckker Pub Co., New York.

ELECTRIC VEHICLE TECHNOLOGY			
Course Code:	EE2503-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	EE1001-1		
Teaching Department: Electrical & Electronics Engineering			
Course Objectives:			
1	To Understand the fundamental laws and vehicle mechanics.		
2	To Understand working of Electric Vehicles and recent trends.		
3	Ability to analyze different power converter topology used for electric vehicle application		
4	Ability to develop the electric propulsion unit and its control for application of electric vehicles		
UNIT-I			
Vehicle Mechanics			07 Hours
Roadway Fundamentals, Laws of Motion, Vehicle Kinetics, Dynamics of Vehicle Motion, Propulsion Power, Force-Velocity Characteristics, Maximum Gradability, Velocity and Acceleration, Constant FTR, Level Road, Velocity Profile, Distance Traversed, Tractive Power, Energy Required, Nonconstant FTR, General Acceleration, Propulsion System Design..			
Electric and Hybrid Electric Vehicles			07 Hours
Configuration of Electric Vehicles, Performance of Electric Vehicles, Traction motor characteristics, Tractive effort and Transmission requirement, Vehicle performance, Tractive effort in normal driving, Energy consumption Concept of Hybrid Electric Drive Trains, Architecture of Hybrid Electric Drive Trains, Series Hybrid Electric Drive Trains, Parallel hybrid electric drive train).			
UNIT-II			
Energy storage for EV and HEV			08 Hours
Energy storage requirements, Battery parameters, Types of Batteries, Modelling of Battery, Fuel Cell basic principle and operation, Types of Fuel Cells, PEMFC and its operation, Modelling of PEMFC, Supercapacitors.			
Electric Propulsion			08 Hours
EV consideration, DC motor drives and speed control, Induction motor drives, Permanent Magnet Motor Drives, Switch Reluctance Motor Drive for Electric Vehicles, Configuration and control of Drives.			
UNIT-III			
Design of Electric and Hybrid Electric Vehicles			10 Hours
Series Hybrid Electric Drive Train Design: Operating patterns, control strategies, Sizing of major components, power rating of traction motor, power rating of engine/generator, design of PPS Parallel Hybrid Electric Drive Train Design: Control strategies of parallel hybrid drive train, design of engine power capacity, design of electric motor drive capacity, transmission design, energy storage design.			
Course Outcomes: At the end of the course student will be able to			
1	Explain the roadway fundamentals, laws of motion, vehicle mechanics and propulsion system design		
2	Explain the working of electric vehicles and hybrid electric vehicles in recent trends.		
3	Model batteries, Fuel cells, PEMFC and super capacitors.		
4	Analyze DC and AC drive topologies used for electric vehicle application.		

Security

5	Develop the electric propulsion unit and its control for application of electric vehicles.																																																																																																							
Course Outcomes Mapping with Program Outcomes																																																																																																								
<table><tr><td>Program Outcomes→</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>↓ Course Outcomes</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>EE2503-1.1</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>EE2503-1.2</td><td>1</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>EE2503-1.3</td><td>1</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>EE2503-1.4</td><td>1</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>EE2503-1.5</td><td>1</td><td>2</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td><td>-</td></tr></table>														Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	↓ Course Outcomes													EE2503-1.1	2	3	-	-	-	-	-	-	-	-	-	-	EE2503-1.2	1	2	3	-	-	-	-	-	-	-	-	-	EE2503-1.3	1	2	3	-	-	-	-	-	-	-	-	-	EE2503-1.4	1	2	3	-	-	-	-	-	-	-	-	-	EE2503-1.5	1	2	2	-	-	-	-	-	-	-	3	-
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12																																																																																												
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1: Low 2: Medium 3: High																																																																																																								
TEXTBOOKS:																																																																																																								
1	Iqbal Husain, "Electric and Hybrid Vehicles: Design Fundamentals", CRC Press, 2003.																																																																																																							
2	M. Ehsani, Y. Gao, S.Gay and Ali Emadi, "Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design", CRC Press, 2005.																																																																																																							
REFERENCE BOOKS:																																																																																																								
1	Sheldon S. Williamson, "Energy Management Strategies for Electric and Plug-in Hybrid Electric Vehicles", Springer, 2013.																																																																																																							
2	C.C. Chan and K.T. Chau, "Electric Vehicle Technology", OXFORD University, 2001																																																																																																							
3	Chris Mi, M. Abul Masrur, David Wenzhong Gao, "Hybrid Electric Vehicles Principles And Applications with Practical Perspectives", Wiley Publication, 2001																																																																																																							
E Books / MOOCs/ NPTEL																																																																																																								
1.	Introduction to Mechanics Coursera																																																																																																							
2.	Electric Vehicles - Part 1 - Course (npTEL.ac.in)																																																																																																							
3.	NPTEL: Electrical Engineering - Introduction to Hybrid and Electric Vehicles																																																																																																							
4.	Hybrid Vehicles (edX) MOOC List (mooc-list.com)																																																																																																							
5.	Electric Cars: Technology My MOOC (my-mooc.com)																																																																																																							

FUNDAMENTALS OF PLC AND ITS APPLICATIONS			
Course Code:	EE2504-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	EC1001-1		
Teaching Department: Electrical & Electronics Engineering			
Course Objectives:			
1.	To understand main parts and their functions, basic sequence of operation of PLC.		
2.	To study the different programming languages and fundamental wiring diagrams.		

3.	To explain the functions of PLC counter instructions, applying combinations of counters and timers to control systems.
4.	To explain the basic operation of PLC closed-loop control system, various forms of mechanical sequencers and their operations
5.	To discuss the operation of various processes, structures of control systems and the method of communication between different industrial processes
UNIT-I	
Programmable Logic Controllers	02 Hours
Introduction, Parts of a PLC, Principles of Operation, PLC Size and Application.	
PLC Hardware Components	05Hours
The I/O Section, Discrete I/O Modules, Analog I/O Modules, Special I/O Modules, I/O Specifications, The Central Processing Unit (CPU), Memory Design, Memory Types, Programming Terminal Devices, Human Machine Interface (HMIs).	
Basic Programming Language	05Hours
Ladder diagrams, Ladder conventions, Logic functions with timing diagram, latching, multiple outputs, entering programs, Functional blocks, Program examples, instruction list, branch codes, programming examples, Sequential functions charts, branching and convergence, actions, Structured Text, conditional and iteration statements	
Developing Fundamental PLC Wiring Diagrams and Ladder Logic Programs	03Hours
Electromagnetic Control Relays, Contactors, Motor Starters, Manually Operated Switches, Mechanically Operated Switches, Sensors, Output Control Devices, Seal-In Circuits, Latching Relays, Converting Relay Schematics into PLC Ladder Programs, Writing a Ladder Logic Program Directly from a Narrative Description.	
UNIT-II	
Programming Timers	02 Hours
Introduction, Necessity of Energy Storage and Methods of Energy Storage (Classification and brief description using block diagram representation)	
Programming Counters	04 Hours
Counter Instructions, Up-Counter, Down-Counter, Cascading Counters, Incremental Encoder-Counter Applications, Combining Counter and Timer Functions.	
Program Control Instructions	05 Hours
Master Control Reset Instruction, Jump Instruction, Subroutine Functions, Immediate Input and Immediate Output Instructions, Forcing External I/O Addresses, Safety Circuitry, Selectable Timed Interrupt, Fault Routine, Temporary End Instruction, Suspend Instruction.	
Data Manipulation Instructions	02 Hours
Data Manipulation, Data Transfer Operations, Data Compare Instructions, Data Manipulation Programs, Numerical Data I/O Interfaces, Closed-Loop Control.	
Math Instructions	02 Hours
Math Instructions, Addition Instruction, Subtraction Instruction, Multiplication Instruction, Division Instruction, Other Word-Level Math Instructions, File Arithmetic Operations	
UNIT-III	
Sequencer and Shift Register Instructions	05 Hours
Mechanical Sequencers, Sequencer Instructions, Sequencer Programs, Bit Shift Registers, Word Shift Operations.	
Process Control, Network Systems, and SCADA	05 Hours
Types of Processes, Structure of Control Systems, On/Off Control, PID Control, Motion Control, Data Communications, Supervisory Control and Data Acquisition (SCADA).	

Course Outcomes: At the end of the course student will be able to

1.	Identify main parts, functions of PLC and describe basic circuitry for I/O modules to select PLC for desired application
2.	Apply suitable logic using various programming languages to achieve specific control mechanism for a given application
3.	Identify timer/counter resources of a PLC to design control logic for interfaced device.
4.	Interpret data manipulation and math instructions as they apply to a PLC program
5.	Develop programs that use shift registers and explain functions of control elements of a closed loop control system

Course Outcomes Mapping with Program Outcomes

Program Outcomes→ ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12
EE2504-1.1	3	-	-	-	-	-	-	-	-	-	-	-
EE2504-1.2	1	3	-	-	-	-	-	-	-	-	-	-
EE2504-1.3	1	2	3	-	-	-	-	-	-	-	-	-
EE2504-1.4	1	2	3	-	-	-	-	-	-	-	-	-
EE2504-1.5	1	2	3	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Frank Petruzella, "Programming Logic Controllers", Fifth Edition.
2.	W Bolton, "Programmable Logic controllers", 6th edition, Elsevier- newness, 2015.

REFERENCE BOOKS:

1.	John W Webb, Ronald A Reis, "Programmable logic controllers - principles and applications", 5th edition, 2nd impression, Pearson education, 2009
2.	L. A Bryan, E. A Bryan, "Programmable Controller Theory and Implementations", 2nd edition, 2003
3.	S. P. Sukhumi, J. K. Nayak, "Solar Energy: Principles Collection and Storage", 3rd edition, McGraw-Hill Education (India) , 2009.

E Books / MOOCs/ NPTEL

	https://library.automationdirect.com/category/product/programmable-control/
2.	https://www.coursera.org/lecture/intelligent-machining/programmable-logic-controllers-plc-fGz3r
3.	https://www.udemy.com/course/plc-programming-from-scratch/

MOTORS AND MOTOR CONTROL CIRCUITS													
Course Code:	EE2505-1	Course Type	OEC										
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03										
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50										
Prerequisite	EE1001-1												
Teaching Department: Electrical & Electronics Engineering													
Course Objectives:													
<table><tr><td>1.</td><td>Study architecture of induction motor and synchronous motor</td></tr><tr><td>2.</td><td>Understanding control of AC motor</td></tr><tr><td>3.</td><td>Study principle of operation of different dc motors</td></tr><tr><td>4.</td><td>Understand the different types of control techniques</td></tr><tr><td>5.</td><td>Study different sensors and their role in control of a motor</td></tr></table>				1.	Study architecture of induction motor and synchronous motor	2.	Understanding control of AC motor	3.	Study principle of operation of different dc motors	4.	Understand the different types of control techniques	5.	Study different sensors and their role in control of a motor
1.	Study architecture of induction motor and synchronous motor												
2.	Understanding control of AC motor												
3.	Study principle of operation of different dc motors												
4.	Understand the different types of control techniques												
5.	Study different sensors and their role in control of a motor												
UNIT-I													
AC Motor Designs	08 Hours												
Introduction, Three phase AC motor architecture, Torque speed curve, wound rotor, Synchronous motors Single phase AC motors, split phase motor, capacitor start and shaded pole motors, Universal and gear motors, AC Motor Specifications, Specifying an AC motor for an application.													
AC Motor Control:	07 Hours												
AC motor Enclosures, AC motor control components, Manual motor starting systems, Direct On Line Starter, semi-automatic star delta starter, fully automatic star delta starter, control circuit for sequence operation of two motors													
UNIT-II													
DC Motors	07 Hours												
DC motor principle of operation, Brushed DC motors, shunt, series and compound wound motors, Brushless DC motors, driving a brushless DC motor, Commutation, Specifying a DC motor													
DC Motor Control and Stepper Motors	08 Hours												
Stepper motor principles of operation, Illustrative example of a stepper motor drive, stepper motor specification and operation, commercial stepper motor drive chips and packages, Direction Controller- H Bridge, Speed Controller: Pulse Width Modulation (PWM), Armature Controller: Variable resistance, DC vs.AC motors													
UNIT-III													
Sensors	10 Hours												
Unipolar Hall Effect Switches, Omnipolar Hall Effect Switches, Latched Hall Effect Switches, Current Sensors: Shunt resistor, Current-sensing transformer, Hall effect current sensor, Speed/position sensors: Quadrature encoder, Hall effect tachometer, Back EMF/Sensorless control method, BLDC motor control with Hall sensor, Block diagram approach of BLDC Fan and Motor Control													
Course Outcomes: At the end of the course student will be able to													
<table><tr><td>1.</td><td>Demonstrate an understanding of the general principles of AC Motor.</td></tr></table>				1.	Demonstrate an understanding of the general principles of AC Motor.								
1.	Demonstrate an understanding of the general principles of AC Motor.												

2.	Understand the basic principles of AC motor controls which includes starters, contactors, and control relays
3.	Demonstrate an understanding of the general principles of DC Motor.
4.	Understand the basic principles of DC motor controls which includes starters, contactors, and control relays
5.	Set up sensors in order to give feedback to a control circuit

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
EE2505-1.1	3	-	-	-	-	-	-	-	-	-	-	-
EE2505-1.2	2	3	3	-	-	2	-	-	-	-	-	-
EE2505-1.3	3	-	-	-	-	-	-	-	-	-	-	-
EE2505-1.4	2	3	3	-	-	2	-	-	-	-	-	-
EE2505-1.5	2	3	3	-	-	2	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	S. K. Bhattacharya Birjindersingh, "Control of electrical machines", New Age International.
2.	Gary J. Rockis& Glen A. Mazura, "Electrical Motor Controls", 5th Edition, ISBN number is 9780826912268

REFERENCE BOOKS:

1.	Stephen L. Herman, "Industrial Motor Control", Delmar Publishers, Inc., latest Edition.
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E Books / MOOCs/ NPTEL

	https://www.coursera.org/learn/motors-circuits-design
2.	http://ww1.microchip.com/downloads/en/appnotes/00894a.pdf

NON-CONVENTIONAL ENERGY SOURCES

Course Code:	EE2506-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Prerequisite	EE1001-1		

Teaching Department: Electrical & Electronics Engineering

Course Objectives:

1.	To understand the principle of extraction of energy from conventional, nonconventional sources
2.	To understand the working principle and applications of solar based thermal, electrical and PV systems.
3.	To justify the usage of energy storage techniques and understand the process of design and implement wind based energy conversion systems.

4.	To understand the process of design and implement biomass based energy conversion systems	
UNIT-I		
Energy Sources		03 Hours
Introduction, Importance of Energy Consumption as Measure of Prosperity, Per Capita Energy Consumption, Classification of Energy Resources, Conventional Energy Resources- Availability and their Limitations, Non-Conventional Energy Resources- Classification, Advantages, Limitations, Comparison of Conventional and Non-Conventional Energy Resources, World Energy Scenario, Indian Energy Scenario		
Solar Energy Basics		05 Hours
Introduction, Solar Constant, Basic Sun-Earth Angles – definitions and their representation, Solar Radiation Geometry (numerical problems), Estimation of Solar Radiation of Horizontal and Tilted Surfaces (numerical problems), Measurement of Solar Radiation Data – Pyranometer and Pyrhelimeter		
Solar Thermal Systems		04Hours
Principle of Conversion of Solar Radiation into Heat, Solar Water Heaters (Flat Plate Collectors), Solar Cookers – Box type, Concentrating dish type, Solar driers, Solar Still, Solar Furnaces, Solar Green House.		
Solar Electric Systems		04Hours
Solar Thermal Electric Power Generation, Solar Pond and Concentrating Solar Collector(Parabolic Trough, Parabolic Dish, Central Tower Collector), Advantages and Disadvantages; Solar Photovoltaic – Solar Cell fundamentals, characteristics, classification, construction of module, panel and array. Solar PV Systems- stand-alone and grid connected, Applications- Street lighting, Domestic lighting and Solar Water pumping systems.		
UNIT-II		
Energy Storage		04 Hours
Introduction, Necessity of Energy Storage and Methods of Energy Storage (Classification and brief description using block diagram representation)		
Wind Energy		04 Hours
Introduction, Wind and its Properties, History of Wind Energy Wind Energy Scenario – World and India. Basic principles of WECS, Classification, Parts of a WECS, Derivation for Power in the wind, Electrical Power Output and Capacity Factor of WECS. Wind site selection consideration, Advantages and Disadvantages of WECS.		
Biomass Energy		06 Hours
Introduction, Photosynthesis process, Biomass fuels, Biomass conversion technologies, Urban waste to Energy Conversion, Biomass Gasification, Biomass to Ethanol Production, Biogas production from waste biomass, Factors affecting biogas generation, types of biogas plants- KVIC and Janata model, Biomass program in India		
UNIT-III		
Energy From Ocean		05 Hours
Tidal Energy – Principle of Tidal Power, Components of Tidal Power Plant, Classification of Tidal Power Plant, Estimation of Energy – Single basin and Double basin type TPP (no derivations, Simple numerical problems), Advantages and Limitation of TPP. Ocean Thermal Energy Conversion (OTEC): Principle of OTEC system, Methods of OTEC power generation – Open Cycle (Claude cycle), Closed Cycle (Anderson cycle), Hybrid cycle, Site-selection criteria, Biofouling, Advantages & Limitation of OTEC		
Emerging Technologies		05 Hours

Fuel Cell, Small Hydro Resources, Hydrogen Energy and Wave Energy (Principle of Energy generation using block diagrams, advantages and limitations)

Course Outcomes: At the end of the course student will be able to

1.	Describe non-conventional energy sources and solar radiation geometry to estimate and measure solar radiation.
2.	Apply the principle of solar radiation into heat to understand the operation of solar thermal and solar electric systems.
3.	Describe energy storage methods and wind–energy conversion systems to understand the factors influencing power generation.
4.	Review the biomass conversion technologies to design biomass-based energy systems.
5.	Describe tidal, ocean thermal and fuel cell energy conversion systems to understand emerging non-conventional energy technologies.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→ ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12
EE2506-1.1	2	3	-	-	-	1	2	1	-	-	-	-
EE2506-1.2	2	3	-	-	-	1	2	1	-	-	-	-
EE2506-1.3	2	3	-	-	-	1	2	1	-	-	-	-
EE2506-1.4	2	3	-	-	-	1	2	1	-	-	-	-
EE2506-1.5	2	3	-	-	-	1	2	1	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

- | | |
|----|---|
| 1. | Rai G. D., “Non-Conventional Sources of Energy”, 4th Edition, Khanna Publishers, New Delhi, 2007. |
|----|---|

REFERENCE BOOKS:

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|----|--|
| 1. | Mukherjee D. and Chakrabarti, S., “Fundamentals of Renewable Energy Systems”, New Age International Publishers, 2005. |
| 2. | Khan, B. H., “Non-Conventional Energy Resources”, TMH, New Delhi, 2006. |
| 3. | S. P. Sukhumi, J. K. Nayak “Solar Energy: Principles Collection and Storage”, 3rd edition, McGraw-Hill Education (India) , 2009. |

E Books / MOOCs/ NPTEL

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|----|---|
| 1. | https://nptel.ac.in/courses/108108078 |
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ELEMENTS OF YOGA																																									
Course Code:				HU1501-1			Course Type				OEC																														
Teaching Hours/Week (L:T:P):				3:0:0			Credits				03																														
Total Teaching Hours				40+0+0			CIE + SEE Marks				50+50																														
Teaching Department: Mechanical Engineering																																									
Course Objectives:																																									
<table><tr><td>1.</td><td>To give a brief history of the development of Yoga</td></tr><tr><td>2.</td><td>Identify names of different classical texts on Yoga</td></tr><tr><td>3.</td><td>To illustrate how Yoga is important for healthy living</td></tr><tr><td>4.</td><td>To explain the Asanas and other Yogic practices</td></tr><tr><td>5.</td><td>To explain, how Yoga practices can be applied for overall improvement</td></tr></table>														1.	To give a brief history of the development of Yoga	2.	Identify names of different classical texts on Yoga	3.	To illustrate how Yoga is important for healthy living	4.	To explain the Asanas and other Yogic practices	5.	To explain, how Yoga practices can be applied for overall improvement																		
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5.	To explain, how Yoga practices can be applied for overall improvement																																								
UNIT-I																																									
Yoga				09 Hours																																					
Meaning and initiation, definitions and basis of yoga, History and development, Astanga yoga, Streams of yoga. Yogic practices for healthy living. General guidelines for Yoga practices for the beginners: Asanas, Pranayama.																																									
Classification of Yoga and Yogic texts				07 Hours																																					
Yogasutra of Patanjali, Hatha yogic practices- Asanas, Pranayama, Dharana, Mudras and bandhas.																																									
UNIT-II																																									
Yoga and Health				06 Hours																																					
Concept of health and Diseases-Yogic concept of body – pancakosaviveka, Concept of disease according to Yoga Vasistha.																																									
				04 Hours																																					
Yogic concept of healthy living- rules & regulations, yogic diet, ahara, vihara. Yogic concept of holistic health.																																									
Applied Yoga for elementary education				04 Hours																																					
Personality development- physical level,mental level,emotional level. Specific guidelines and Yoga practices for - Concentration development,Memory development																																									
UNIT-III																																									
Yoga and physical development				05 Hours																																					
Mind-body, Meditation, Yogasanas and their types. Different Yoga practices and Benefits.																																									
				05 Hours																																					
Specific guidelines and Yoga practices for – Flexibility, Stamina, Endurance (Surya Namaskara)																																									
Course Outcomes: At the end of the course student will be able to																																									
<table><tr><td>1.</td><td>Understand a brief history of the development of Yoga</td></tr><tr><td>2.</td><td>Know important practices and principles of Yoga</td></tr><tr><td>3.</td><td>Explain how Yoga is important for healthy living</td></tr><tr><td>4.</td><td>Practice meditation to improvement of concentration etc.</td></tr><tr><td>5.</td><td>Have knowledge about specific guidelines of yoga practices</td></tr></table>														1.	Understand a brief history of the development of Yoga	2.	Know important practices and principles of Yoga	3.	Explain how Yoga is important for healthy living	4.	Practice meditation to improvement of concentration etc.	5.	Have knowledge about specific guidelines of yoga practices																		
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Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12																												
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	HU1501-1.1	-	-	-	-	-	1	-	-	1	-	-	1
	HU1501-1.2	-	-	-	-	-	1	-	-	1	-	-	3
	HU1501-1.3	-	-	-	-	-	2	-	-	1	-	-	3
	HU1501-1.4	-	-	-	-	-	3	-	-	2	-	-	3
	HU1501-1.5	-	-	-	-	-	2	-	-	2	-	-	3
1: Low 2: Medium 3: High													
TEXTBOOKS:													
	1.	B. K. S. Iyengar, “Light on Yoga: The Classic Guide to Yoga by the World’s Foremost Authority”, Thorsons publisher 2016.											
	2.	Makarand Madhukar Gore, “Anatomy and Physiology of Yogic Practices: Understanding of the Yogic Concepts and Physiological Mechanism of the Yogic Practices”, Motilal Banarsidass Publishers; 6 edition (2016).											
	3.	Swami Satyananda Saraswati, “Asana, Pranayama, Mudra and Bandha: 1”, Yoga Publications Trust.											
REFERENCE BOOKS:													
	1.	Ann Swanson, "Science of Yoga: Understand the Anatomy and Physiology to Perfect Your Practice".											
	2.	Dianne Bondy, "Yoga for Everyone : 50 Poses For Every Type of Body".											
E Books / MOOCs/ NPTEL													
	1.	https://onlinecourses.swayam2.ac.in/aic19_ed29/preview											
	2.	https://youtu.be/FMf3bPS5wDs											

INTELLECTUAL PROPERTY RIGHTS			
Course Code	HU1502-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Humanities			
Course Objectives:			
1.	Understand the creativity component in intellectual property, different types of legal protection of intellectual properties and other basic concepts of Intellectual property.		
2.	Analyze different types of protection for inventions, different types of agreements and treaties for Intellectual properties with an ability to examine patent types, specifications and patent search and database for 'prior art'.		
3.	Understand the basic procedure of drafting claims, apply for patents, other legal forms of intellectual property rights and also to examine the protocol involved in protection of inventions like patents.		
UNIT - I			
Introduction to Intellectual Property		08 Hours	
Invention and Creativity - Intellectual Property (IP) – Importance, Jurisprudential definition and concept of property, rights, duties and their correlation; History and evaluation of IPR – like Patents, Trademarks, Copyright & Related Rights, Industrial Design, Traditional Knowledge, Geographical Indications.			
Agreements and Treaties		08 Hours	
History - General Agreement on Trade and Tariff (GATT). Indian Position vis-a-vis WTO and Strategies; TRIPS Agreement; Madrid Agreement; Hague Agreement; WIPO Treaties; International convention relating to Intellectual Property - Establishment of WIPO - Mission and Activities – Budapest Treaty; PCT; Indian Patent Act 1970 & recent amendments – Patent (Amendment) Rules, 2017			
UNIT - II			

Basics of Patents and Concept of Prior Art		08 Hours											
Introduction to Patents; Types of patent applications: Ordinary, PCT, Conventional, Divisional and Patent of Addition; Specifications: Provisional and complete; Forms and fees Invention in the context of “prior art”; Patent databases; Searching International Databases; Country-wise patent searches (USPTO, EPO, WIPO, IPO, etc.)													
Patent filing procedures		08 Hours											
National & PCT filing procedure; Time frame and cost; Status of the patent applications filed; Structure of Patent document, Precautions while patenting – disclosure/non-disclosure; Financial assistance for patenting - introduction to existing schemes; Patent licensing and agreement; Patent infringement- meaning, scope, litigation, case studies.													
UNIT - III													
Case Studies		08 Hours											
Patents: Biological Cases - i) Basmati rice ii) Turmeric iii) Neem; Non-biological cases – (i) TVS V/S Hero, (ii) Samsung V/S Nokia – Copyright and related rights – Trade Marks – Trade secrets - Industrial design and Integrated circuits – Geographic indications – Protection against unfair competition; Technology transfer and license agreements (US anti-HIV drug license to Africa).													
Course Outcomes: At the end of the course student will be able to													
1.	Have a General understanding of the Intellectual Property Rights.												
2.	Have awareness of different forms of intellectual property rights, national and international IPR related legislations.												
3.	Have a general understanding about the provisions, privileges and limitations of intellectual property right holders with an understanding of the legal aspects (civil or criminal) of the use of intellectual property rights.												
4.	Acquire Knowledge of National and International Trade Agreements and Agencies functioning in relation to intellectual property rights												
5.	Be aware and have a general understanding of patenting procedures and licensing.												
Course Outcomes Mapping with Program Outcomes													
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes													
HU1502-1.1		-	3	3	2	-	3	-	-	2	2	-	3
HU1502-1.2		2	2	3	-	-	3	-	3	1	1	2	2
HU1502-1.3		2	-	-	2	-	3	-	-	2	2	2	3
HU1502-1.4		-	-	1	1	-	3	-	-	1	2	-	3
HU1502-1.5		3	2	1	-	-	3	-	-	3	1	-	2
1: Low 2: Medium 3: High													
REFERENCE MATERIALS:													
1.	BAREACT, "Indian Patent Act 1970 Acts & Rules", Universal Law Publishing Co. Pvt. Ltd., 2007.												
2.	Kankanala C., "Genetic Patent Law & Strategy", 1st Edition, Manupatra Information Solution Pvt. Ltd., 2007.												
3.	Subbaram N.R., "Handbook of Indian Patent Law and Practice", S. Viswanathan (Printers and Publishers) Pvt. Ltd., 1998.												
4.	Eli Whitney, United States Patent Number: 72X, Cotton Gin, March 14, 1794.												
5.	Intellectual Property Today: Volume 8, No. 5, May 2001.												
6.	M B Rao, "WTO and International Trade", Vikas Publishing House Pvt. Ltd.												
7.	Correa, Carlos M. "Intellectual property rights, the WTO and developing countries: the TRIPS agreement and policy options", Zed Books, New York 2000.												
8.	Wadehra, B. L. "Law relating to patents, trademarks, copyright designs & geographical indications", 2 ed. Universal Law Publishing 2000.												
9.	Sinha, Prabhas Chandra, "Encyclopedia of Intellectual Property Rights", 3 Vols. Eastern Book Corporation, 2006.												
10.	Rachna Singh Puri and Arvind Vishwanathan, “Practical Approach to Intellectual Property Rights”; I. K. International Publishing House Pvt. Ltd.												

E-RESOURCES:	
1.	http://www.w3.org/IPR/
2.	http://www.wipo.int/portal/index.html.en
3.	http://www.ipr.co.uk/IP_conventions/patent_cooperation_treaty.html
4.	www.patentoffice.nic.in
5.	www.iprlawindia.org/

INTRODUCTION TO GERMAN LANGUAGE			
Course Code	HU1503-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Mechanical			
Course Objectives:			
1.	Distinguish - definite and indefinite articles, declension of singular and plural nouns by adding certain endings to them to differentiate between subjects, objects and indirect objects and construct sentences of simple day to day usage.		
2.	Differentiate between nomnative and akkusative cases with transitive and intransitive verbs, and negation with Kein/e/er		
3.	Differentiate use of dative object besides the subject for some specific verbs and Apply the grammar principles of use of personal pronoun as a substitute for noun as per the case, number and gender of the noun.		
4.	Differentiate preposition forms when used exclusively in akkusative or Dative forms or on combination of the two cases		
5.	Differentiate conjugation of verbs in present, present-perfect and past participle tenses, separable and inseparable verbs, application of conjugation of modal verbs and position of modal verb in a sentence.		
UNIT – I			
	15 Hours		
Introduction: Mein Name ist (saying who you are, greeting people and saying goodbye, asking people where they come from and where they live. Language point: I and you), Lesen der politischen Karte der Welt, Nationalitaeten und Spachen, Die Uhrzeit (The time) telling time and talking about daily routine, Tage der Woche, die Monate, die vier Jahreszeiten, die Jahre Mir geht es gut: Asking people how they are, saying how you are, saying which cities and counries people come from, Language points: verb endings), Wie schreibt man das (how do you write that?) Counting from 1-100 and above, alphabet, spelling our names and words, talking about us and them. Language points: Yes-no questions Artikel (Articles): As in English, there are definite (der/die/das) and indefinite (ein/eine) articles: the { der/die/das; a/an { ein/eine Die vierFälle (The four cases): Nominativ, Akkusativ, Dativ, Genitiv(Not in level A-1) Deklination des bestimmtenArtikels der/die/das Deklination des unbestimmtenArtikelsein/eine (Deklination/Declension: the variation of the form of a noun, pronoun, or adjective, by which its grammatical case, number, and gender are identified) Deklination von Substantiven (Declension of nouns) (Singular and Plural) (German nouns are declined by attaching certain endings to them, according to case, number and gender. This helps to differentiate between subjects, objects and indirect objects). Nominativ und Akkusativ(nominative and accusative cases) The verb determines the case of the noun. Some verbs only go with the nominative, others only with the accusative (or the dative). Thus, German verbs are either transitive or			

	intransitive. (Nominative and accusative cases) Intransitive Verben (intransitive verbs) Transitive Verben (transitive verbs) Negation „kein/e/er “(negation with „kein/e/er “) (Singular und Plural) The negation of the indefinite article (ein/eine/ein) is kein/keine/kein. For this, you just have to put a „k“ at the beginning of the declined form of ein/eine/ein. Peter sieht ein Haus. ◇ Negation ◇ Peter sieht kein Haus. (Peter sees a house. ◇ negation ◇ Peter does not see a house.) (With examples, writing and hearing exercises, and German to English Glossary as applicable)
UNIT – II	
	14 Hours
	Dativ (the dative) (You are already familiar with verbs which require a direct accusative object in addition to the subject, which is in the nominative case. But there also some verbs which require a dative object besides the subject. To identify the dative object you ask “(To) whom?”) Der Plural (the plural) There are many different forms of the plural in the German language. Principally, the gender and the ending of the noun determine the plural form. Then, you either attach a plural ending to the noun, change a vowel, or keep the noun as it is in the singular. Das Personalpronomen (the personal pronoun) The personal pronoun is a substitute for a noun. Its forms are determined by the case, number and gender of the noun which is to be replaced. Die Formen des Personalpronomen im Nominativ (The nominative forms of the personal pronoun): Präpositionen (prepositions) German prepositions are followed by an object, either in the accusative or the dative case. Some prepositions always take an accusative object, others always a dative object. But there are also prepositions which can be followed by both. In this case, the question “Where(to)?” (◇ accusative) or “Where?” (◇ dative) determines the case of the object. Präpositionen mit Akkusativ und Dativ (Prepositions with accusative and dative) 1. Präpositionen mit Akkusativ (prepositions with accusative) 2. Präpositionen mit Dativ (prepositions with dative) 3. Präpositionen mit Akkusativ oder Dativ (prepositions with accusative or dative) (With examples, writing and hearing exercises, and German to English Glossary as applicable)
UNIT - III	
	11 Hours
	Konjugation von Verben im Präsens (Conjugation of verbs in present tense) Verbs are conjugated by attaching certain endings, depending on the person and number of the subject.

Trennbare und untrennbare Verben

(separable and inseparable verbs)

Verbs with prefixes are distinguished between separable and inseparable verbs.

The prefix of an inseparable verb must never be separated from the stem. Here the stress is on the stem: be-kommen. The prefix of a separable verb gets separated from the stem when the verb is conjugated.

In the infinitive, the stress is on the prefix: an-kommen

1. Trennbare Verben (separable verbs)

2. Untrennbare Verben (inseparable verbs)

Konjugation von Verben im Perfekt

(Conjugation of verbs in present perfect)

The present perfect (Perfekt) describes something which happened in the past and is especially used in spoken German. It is formed with the present tense form of „haben“ or

„sein“ and the past participle of the main verb.

1. Die Bildung des Partizips

(the formation of the past participle)

2. Die Bildung des Perfekts mit „haben“ und „sein“

(the formation of the present perfect with „haben“ and „sein“)

Modalverben (modal verbs)

A modal verb is rarely used as a main verb; instead, it usually modifies the main verb. While the main verb remains in the infinitive, the modal verb is conjugated.

In German, there are 7 modal verbs:

können (can/be able), dürfen (may/be allowed), wollen (want),

müssen (must/have to), sollen (shall), mögen (to like), möchten (wish/would like)

1. Konjugation der Modalverben

(Conjugation of the modal verbs)

2. Stellung des Modalverbs im Satz

(Position of the modal verb within a sentence)

(With examples, writing and hearing exercises, and German to English Glossary as applicable)

Course Outcomes: At the end of the course student will be able to

1.	Distinguish - definite and indefinite articles, declension of singular and plural nouns by adding certain endings to them to differentiate between subjects, objects and indirect objects and construct sentences of simple day to day usage.
2.	Differentiate between nominative and accusative cases with transitive and intransitive verbs, and negation with Kein/e/er
3.	Differentiate use of dative object besides the subject for some specific verbs and Apply the grammar principles of use of personal pronoun as a substitute for noun as per the case, number and gender of the noun.
4.	Differentiate preposition forms when used exclusively in accusative or Dative forms or on combination of the two cases
5.	Differentiate conjugation of verbs in present, present-perfect and past participle tenses, separable and inseparable verbs, application of conjugation of modal verbs and position of modal verb in a sentence.

Course Outcomes Mapping with Program Outcomes

Program Outcomes →	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
HU1503-1.1	-	-	-	-	-	3	-	-	2	1	-	1
HU1503-1.2	-	-	-	-	-	3	-	-	2	1	-	1
HU1503-1.3	-	-	-	-	-	3	-	-	2	1	-	1
HU1503-1.4	-	-	-	-	-	3	-	-	2	1	-	1

Security

HU1503-1.5		-	-	-	-	-	3	-	-	2	1	-	1
1: Low 2: Medium 3: High													
TEXT BOOKS:													
1.	Ulrich Haessermann, Georg Dietrich, Christianne C. Guenther, Diethelm Kaminski, Ulrike Woods and Hugo Zenker, Sprachkurs Deutsch Neusaffung 1, Unterrichtswerk fuer Erwachsene, Verlag Moritz Diesterweg, Universitaetsdruckerei H. Stuertz AG Wuerzburg, 1989.												
2.	Paul Coggle and Heiner Schenke, Teach Yourself German (a complete course in understanding, speaking and writing), Teach Yourself Books, Hodden& Stoughton Educational, UK, 2001												
3.	Langenscheidt German In 30 Days: Book + Cd Paperback, www.amazon.in, – 1 September 2011												
REFERENCE MATERIALS:													
1.	Deutsche Sprachlehre für Ausländer.												
2.	ThemenAktuell (Text and workbook).												
3.	Deutsch als Fremdsprache 1A.												
4.	Tangram Aktuell 1A/1B (Text and workbook).												
5.	Wherever required the Videos/Audios are also played in the class room sessions												
E-RESOURCES:													
1.	https://onlinecourses.nptel.ac.in/noc21_hs30/preview NPTEL-Swayam, German-I by Prof. Milind Brahme IIT Madras												
2.	https://www.traingerman.com/en/ powered by Sprachinstitut TREFFPUNKT Online												

INTRODUCTION TO JAPANESE LANGUAGE			
Course Code	HU1504-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0+0	CIE + SEE Marks	50+50
Teaching Department:			
Course Objectives:			
1.	Have basic spoken communication skills		
2.	Write Simple Sentences		
3.	Listen and comprehend basic Japanese spoken Japanese		
4.	Read and understand basic Japanese characters including Kanji		
UNIT - I			
(Lessons 1-6)		15 Hours	
Grammar – Introduction, Alphabets, Accents, Noun, Pronoun, Present Tense, Past tense			
Vocabulary – Numbers, Days, week days, months, Seasons, Nature, Dialogs and Video Clips			
UNIT - II			
(Lessons 7-13)		14 Hours	
Communication skills – Time, Adjective, Seasons, Conversation, Q&A, Hobby, 5-W/1-H, Entering School/Company, Body Parts, Colours, Features etc.			
UNIT - III			
(Lessons 14-20)		11 Hours	
Japanese Counting System, Birth/Death, Dialogs (Going to Party, Restaurant), My day, Success/Failure, Kanji Characters, and sentence making, Video Clips			
Course Outcomes: At the end of the course student will be able to			

1.	Understand Simple words, expressions and sentences, spoken slowly and distinctly
2.	Speak slowly and distinctly to comprehend
3.	Read and Understand common words and sentences
4.	Ask Basic questions and speak in simple sentences
5.	Write Hiragana/Katakana and Kanji (120) characters.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
HU1504-1.1	-	-	-	-	-	3	-	-	2	1	-	1
HU1504-1.2	-	-	-	-	-	3	-	-	2	1	-	1
HU1504-1.3	-	-	-	-	-	3	-	-	2	1	-	1
HU1504-1.4	-	-	-	-	-	3	-	-	2	1	-	1
HU1504-1.5	-	-	-	-	-	3	-	-	2	1	-	1

1: Low 2: Medium 3: High

NATIONAL CADET CORPS: ORGANIZATION, FUNCTIONS AND CAPABILITIES

Course Code	HU1505-1	Course Type	OEC
Teaching Hours/Week (L:T:P)	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50

Teaching Department: Chemistry

Course Objectives:

1.	To create evolved youth, who will be equipped to contribute in the development of the nation.
2.	To train students so as to achieve their physical and mental endurance. To acquire body language of smart soldier and to inculcate the sense of authority by commanding the troop under him/her.
3.	To inculcate spirit of adventure, undertake adventure activities, to hone leadership qualities and risk-taking abilities.
4.	To understand and develop life skills, soft skills and to improve emotional quotient of the student.
5.	To impart basic military training, to develop awareness about the defense forces and expose learners to military ethos / values

UNIT - I

NCC: Aims, Objectives and Organization
07 Hours

NCC General, Aims, Objectives and Organization of NCC. Duties of NCC Cadets, NCC Camps: Types and Conduct. National Integration: Importance and Necessity, Unity in Diversity.

Personality Development
07 Hours

Self-Awareness, Empathy, Critical and Creative Thinking, Decision Making and Problem Solving. Communication Skills, Coping with stress and emotions. Leadership: Traits, Indicators, motivation, moral values, Honor Code. Social Service and Community Development.

UNIT - II

Naval Communication and Seamanship
08 Hours

Naval Communication: Introduction, Semaphore, Navigation: Navigation of Ships- Basic requirements, Chart work. Seamanship: Introduction to Anchor work, Rigging Capsule, Boat work- Parts of Boat, Boat pulling instructions, Whaler sailing instructions. Ship Modeling.																																																																																									
Disaster management and environmental awareness										08 Hours																																																																															
Disaster Management- Organization, Types of Disasters, Essential Services, Assistance, Civil Defence organization. Adventure Activities. Dos and Don'ts, Fire services and Firefighting, Environmental Awareness and Conservation.																																																																																									
UNIT - III																																																																																									
Naval Orientation										10 Hours																																																																															
Naval Orientation- Armed Forces and Navy Capsule, EEZ Maritime Security & ICG. Border & Coastal Areas: Security setup and Boarder/Coastal management in the area. Naval Orientation: Modes of Entry- IN, ICG, Merchant Navy. Border and Coastal areas: Security Challenges & role of cadets in Border management																																																																																									
Course Outcomes: At the end of the course student will be able to																																																																																									
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1: Low 2: Medium 3: High																																																																																									
REFERENCE BOOKS:																																																																																									
<table><tr><td>1.</td><td>R.K. Gupta, "Cadets Handbook", Ramesh Publishing House, New Delhi.</td></tr></table>															1.	R.K. Gupta, "Cadets Handbook", Ramesh Publishing House, New Delhi.																																																																									
1.	R.K. Gupta, "Cadets Handbook", Ramesh Publishing House, New Delhi.																																																																																								

OVERVIEW OF INDIAN CULTURE

Course Code	HU1506-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50

Teaching Department: Humanities

Course Objectives:

1.	To understand the relevance of Culture in Human Life, dynamism of Indian Culture and Arts through ages.
2.	To understand the local culture and its vibrancies.
3.	To develop awareness about Indian Society, Culture and Arts under Western rule.
4.	To comprehend different dimension and aspects of the Indian culture and arts.
5.	To appreciate cultural performances in India.

UNIT - I

Knowing Culture	08 Hours
What is Culture, Different aspects of Culture, Cultural expression, Importance of Culture	
Influence of Culture	07 Hours
Relationship of Culture with: Language, Religion and History, Gender	

UNIT - II

Media and Culture	07 Hours
Role of News Papers, Indian Cinema, Music, Advertisements	
Languages, Literature and Culture	07 Hours
Role of Sanskrit, Vedas, Upanishads, Ramayana and Mahabharata, Puranas, other Sanskrit Literature, Buddhist and Jain Literature, Dravidian Languages and Literature, North Indian Languages and Literature, Subaltern Literature	

UNIT - III

Arts and Culture	07 Hours
Indian Theatre and Performing Arts, Ritual performances, and Tuluva cultural and ritual performances.	
(Self-study Component)	04 Hours
Contribution of Indian History to Culture	

Ancient India – Persian and Macedonian invasions and its impact on Indian Culture, Development of Culture and Arts during the Mauryan Empire (Ashoka), the Guptas, the South Indian Dynasties – the Cholas, Nalanda as a Centre of Learning.

Medieval India – Life of People under Delhi Sultanate, Rise of Islam and Sufism, Political Scene of India, Bhakti Movement, Folk Arts, Rise of Modern Indian Languages.

Modern India – British Ruling and its impact on Indian Culture, Social and Religious Reforms, Indian National Movement and Achievement of Independence.

Course Outcomes: At the end of the course student will be able to

1.	Examine how the culture has a very important role in human life and growth of human civilization and have a general awareness on historical perspective of growth of Indian Culture and Arts.
2.	Appreciate their own local culture from an academic perspective.
3.	Know about the impact of Western Rule in India and Indian Struggle for Freedom and also its impact on Indian Culture and Arts and able to appreciate and the role of language in connecting people, growth of culture and arts beyond the barriers of religion and ages.
4.	Take interest in learning these forms of arts, and also appreciate and preserve them for the future generations feeling proud of Indian Culture, Arts and Architecture.
5.	Appreciate art performances in India which will enable them to get exposed to an artistic sphere, which eventually help them to be creative and imaginative.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												

HU1506-1.1	-	1	-	-	-	3	-	3	3	1	-	3
HU1506-1.2	-	-	-	2	-	3	-	2	3	3	-	3
HU1506-1.3	-	-	-	-	-	3	-	1	-	-	-	1
HU1506-1.4	-	-	-	-	-	3	-	2	1	2	-	3
HU1506-1.5	-	-	-	-	-	3	-	3	3	3	-	2

1: Low 2: Medium 3: High

PHILOSOPHY			
Course Code	HU1507-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Teaching Department: Visiting			
Course Objectives:			
1.	To provide a new understanding based on which one can move to overcome the current problems, both at the individual level as well as at the societal level.		
2.	To introduce an orientation course for humanities courses in general and for philosophy courses in particular.		
3.	To relate philosophy to literature, culture, society and lived experience.		
4.	To train students in already available philosophical systems.		
5.	To bridge the gap between theory and practice.		
UNIT - I			
Knowledge (Vidya) and Ignorance (Avidya)		14 Hours	
Upanishads Six systems orthodox and Heterodox schools of Indian philosophy Greek philosophy			
Origin of the universe			
NasidiyaSukta: “Who really knows?” Brhadaranyaka Upanishad; Chandogya Upanishad: Non-Self, real and unreal Taithriya Upanishad: SikshaValli Plato’s Symposium: Lack as the source if desire and knowledge. Socratic method of knowledge as discovery Language: word as root of knowledge (Bhartrahari’sVakyapadiyam) Fourteen Knowledge basis as a source of Vidya: Four Vedas, six auxiliary sciences (vedangas); Purana, Nyaya, Mimamsa and Dharma Sastras.			
UNIT - II			
Knowledge as Power		16 Hours	
Francis Bacon. Knowledge as both power and self- realization in Bhagavad Gita.			
Knowledge as Oppression			
M. Foucault. Discrimination between Ram and Satyam in Indian Philosophy.			
Knowledge as Invention			
Modern definition of creativity; scientific activity in the claim that science invents new things at least through technology.			
UNIT - III			
		10 Hours	
Knowledge about the self, transcendental self; knowledge about society, polity and nature Knowledge about moral an ethics codes.			

Course Outcomes: At the end of the course student will be able to

1.	To provide a new understanding based on which one can move to overcome the current problems, both at the individual level as well as at the societal level.
2.	To introduce an orientation course for humanities courses in general and for philosophy courses in particular.
3.	To relate philosophy to literature, culture, society and lived experience.
4.	To train students in already available philosophical systems.
5.	To bridge the gap between theory and practice.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→ ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12
HU1507-1.1	-	-	-	-	-	3	-	-	2	1	-	1
HU1507-1.2	-	-	-	-	-	3	-	-	2	1	-	1
HU1507-1.3	-	-	-	-	-	3	-	-	2	1	-	1
HU1507-1.4	-	-	-	-	-	3	-	-	2	1	-	1
HU1507-1.5	-	-	-	-	-	3	-	-	2	1	-	1

1: Low 2: Medium 3: High

REFERENCE MATERIALS:

1.	Copleston, Frederick, "History of Philosophy", Vol. 1. Great Britain: Continuum.
2.	Hiriyanna, M. , "Outlines of Indian Philosophy", Motilal Banarsidass Publishers; Fifth Reprint edition, 2009.
3.	Sathaye, Avinash, "Translation of NasadiyaSukta".
4.	Raju, P. T. "Structural Depths of Indian Thought", Albany: State University of New York Press.
5.	Plato, Symposium, Hamilton Press

PRINCIPLES OF PHYSICAL EDUCATION

Course Code	HU1508-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50

Teaching Department: Physical Education

Course Objectives:

1.	Express understanding of constitution of sports organizations
2.	Demonstrate considerate familiarity of various food practices
3.	Grasp understanding of first aid and physical education
4.	Awareness on the importance of exercise
5.	Leadership skills and the rules of different sports

UNIT - I

15 Hours

History of Physical Education - Olympic games, Modern Olympic games, Olympic Ideals & Objectives, Olympic Symbols, Olympic Flag, Olympic Emblem, Olympic Motto, Olympic Flame, Asian games

Security

International Olympic Committee (IOC), Indian Olympic Association (IOA) Sports awards - Eligibility, Objectives & Criteria Yoga - Meaning and Importance World Health organization (WHO)													
UNIT - II													
14 Hours													
Concept of Health - Meaning of Health, Health Definition, Factors Affecting Health, Qualities of Healthy Person. Health Hazards of College Students, Physical Fitness and Exercises. Food and Nutrition - Food & Nutrition Defined, Nutrients and their Functions - i) Proteins ii) Carbohydrates iii) Fats iv) Vitamins Balanced Diet & Malnutrition Health Education - Meaning of Health Education, Health Education Defined, Scope of Health Education, Importance of Health Education. Posture - Concept of Posture, Correct Postures, Common Postural Defects First Aid - First Aid Defined, Need and importance of First Aid, The Requisites of First Aid, Scope of First Aid, Qualities of a First Aider, Fundamental Principles to be followed and the Duties to be performed by the First Aider, First Aid in Different Cases. Physical Education - Concept of Physical Education, Physical Education Defined, Importance of Physical Education, Scope of Physical Education, Aims and Objectives of Physical Education. Teaching Aid in Physical Education Competition - Introduction, Types of competition, Knock out, League or Round Robin Tournament.													
UNIT - III													
11 Hours													
Training in Sports – Meaning, Principles, Warming Up & Limbering Down Importance of Anatomy and Physiology in Physical Education, Oxygen Debt and Second wind Leadership and Supervision – Leadership, Qualities of a good leader in Physical Education, Types of Leadership in Physical Education - 1. Teacher Leadership 2. Student Leadership. Measurement & specification of various playing fields – Cricket, Volley Ball, Basket Ball, Badminton, Ball Badminton, Foot Ball, Hand Ball & their basic playing skills.													
Course Outcomes: At the end of the course student will be able to													
1.	Demonstrate knowledge of structure of the world sports organizations												
2.	Display understanding of different type of food and nutrition for a healthy diet												
3.	Comprehend awareness of first aid and physical education												
4.	Elucidate about training and the importance of Physical Education												
5.	Aware of leadership skills and the knowledge of various sports												
Course Outcomes Mapping with Program Outcomes													
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	
↓ Course Outcomes													
HU1508-1.1	-	-	-	-	-	3	-	-	2	1	-	1	
HU1508-1.2	-	-	-	-	-	3	-	-	2	1	-	1	
HU1508-1.3	-	-	-	-	-	3	-	-	2	1	-	1	
HU1508-1.4	-	-	-	-	-	3	-	-	2	1	-	1	
HU1508-1.5	-	-	-	-	-	3	-	-	2	1	-	1	
1: Low 2: Medium 3: High													

LINGUISTICS & LANGUAGE TECHNOLOGY

Course Code	HU2501-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Pre-requisite	HU1001-1 (Technical English)		

Teaching Department: Humanities

Course Objectives:

1.	Introspect about the consciousness in one's language
2.	Learn pronunciation and how the process helps to communicate effectively.
3.	Build contextual speech and writing with the pedagogy in sentence structure.
4.	Improve skill of applying language to enunciate words.
5.	Progress on the speech aspects by understanding the acquisition of Second Language.

UNIT – I

Introduction to Linguistics	08 Hours
Broad understanding of Linguistics, Language and characteristic features, Scientific Language, Levels of Linguistic Analysis (Phonetics, Phonology, Morphology, Syntax and Semantics); Approach to Linguistics (Traditional, Structural and Cognitive).	

Phonology and Morphology	08 Hours
Perspectives in Linguistics, Phonemes, Allophones, Phonemic Analysis, Morphology and Morphemes, Word building process, Morphological Analysis.	

UNIT - II

Syntax	16 Hours
Constituent structure (Simple Sentence, Noun Phrase, Verb Phrase, Prepositional Phrase, Adjective Phrase, Adverb Phrase, Structure Rules), Tree Diagrams, Case	

UNIT - III

Sociolinguistics & Psycholinguistics, Artificial Intelligence	08 Hours
Notion of Language Variety, Languages in Contact, Language and Mind, Error Analysis.	

Course Outcomes: At the end of the course student will be able to

1.	Understand the importance of language and its facets.
2.	Demonstrate knowledge of sounds and competence in process of word building.
3.	Evolve to reason the constituent parts of a sentence.
4.	Understand the techniques of how 'meaning' is applied.
5.	Analyze errors in day-to-day-conversations and how language is related to society.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
HU2501-1.1	-	1	-	-	1	1	-	-	1	-	-	2
HU2501-1.2	-	-	2	-	-	-	-	-	2	2	-	-
HU2501-1.3	2	3	-	3	-	-	-	-	3	2	-	-
HU2501-1.4	-	-	-	-	2	-	-	-	1	2	-	-
HU2501-1.5	-	2	-	-	-	2	1	-	-	-	-	1

1: Low 2: Medium 3: High

REFERENCE MATERIALS:

1.	Akmaijan, A, R. A. Dimers and R. M. Harnish. "Linguistics: An Introduction to Language and Communication". London: MIT Press, 1979.
2.	Chomsky, Noam. "Language in Mind". New York: Harcourt Brace Jovanovich, 1968.
3.	Fabb, Nigel. "Sentence Structure". London: Routledge, 1994.
4.	Hockett, C. "A Course in Modern Linguistics". New York: Macmillan, 1955.
5.	O'Grady, W., O. M. Dobrovolsky and M. Aronoff. "Contemporary Linguistics: An Introduction". New York: St. Martin's Press, 1991.
6.	Pride, J. B. and J. Holmes. "Sociolinguistics". Harmondsworth: Penguin, 1972.
7.	Richards, J. C. "Error Analysis: Perspectives in Second Language Acquisition". London: Longman, 1974.
8.	Salkie, R. "The Chomsky Update: Linguistics and Politics". London: Unwin Hyman Ltd., 1990.
9.	Sinclair, J. M. C. H. and R. M. Coulthard. "Towards an Analysis of Discourse". Oxford: OUP, 1975.
10.	Thomas, Linda. "Beginning Syntax". Oxford: Blackwell, 1993.
11.	Verma, S. K. and N. Krishnaswamy. "Modern Linguistics: An Introduction". New Delhi: OUP, 1989.
12.	Wekker, Herman and Liliane Haegeman. "A Modern Course in English Syntax". Kent: Croom Helm, 1985.

PROFESSIONAL & COGNITIVE COMMUNIQUÉ

Course Code	HU2502-1	Course Type	OEC
Teaching Hours/Week (L:T:P):	3:0:0	Credits	03
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50
Pre-requisite	HU1001-1 (Technical English)		

Teaching Department: Humanities	
Course Objectives:	
1.	To Problematize Commonsense & Apply Critical thinking skills
2.	Comprehend etiquettes and manners in different situations
3.	Be gender sensitive in both offline and online behavior
4.	Exhibit better comprehension of the social implications of human body
5.	Understand the importance of reading and writing skills

UNIT - I	
Common sense and Emotional Intelligence	15 Hours
Common sense, Commonsensical Consensus, Critical thinking, Unsettling commonsensical Consensus, Role of language in Common sense and Critical Thinking;Nature & Functions of Emotional Intelligence, Emotions, Intelligence and Creativity, Growth of Emotional Intelligence	
Etiquettes & Workplace	
Etiquette, Workplace Etiquettes, Workplace Readiness Skills, Significance of Cross-Cultural Understanding; Cultural Sensitivity, Impact of social media in Workplace	
UNIT - II	
Social Networking Sites and its Impacts	15 Hours

Security

Emergence of social media, Impact on Gender and Self Representation, Regulatory and Liberatory aspects of social media, Offline Norms & Online Behaviour													
Gender and Body													
Gender & Sex, Genderization, Homogeneity and Heterosexuality, Gender Expressions, Gender Schooling, Representations of Body, Objectification, Gender Perspectives of Body, Different Ways of Seeing the Body, Discipline & Coercion, ISA & RSA													
UNIT - III													
Writing				10 Hours									
Types of Writing, Note Taking Methods, Plagiarism													
Reading													
Styles of Reading, Types of Reading, Scanning, Skimming													
Course Outcomes: At the end of the course student will be able to													
1.	Problematize Commonsense & Apply Critical thinking skills												
2.	Comprehend etiquettes and manners in different situations												
3.	Be gender sensitive in both offline and online behavior												
4.	Exhibit better comprehension of the social implications of human body												
5.	Understand the importance of reading and writing skills												
Course Outcomes Mapping with Program Outcomes													
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes													
HU2502-1.1		-	3	-	-	-	-	-	-	3	3	-	3
HU2502-1.2		-	2	-	-	-	-	-	3	2	3	-	2
HU2502-1.3		-	3	-	-	-	-	-	-	2	2	-	3
HU2502-1.4		-	3	-	-	-	-	-	-	2	2	-	3
HU2502-1.5		-	2	-	-	-	-	-	-	3	3	-	2
1: Low 2: Medium 3: High													
REFERENCE MATERIALS:													
1.		Geetha.V. Gender. Kolkatta: Web Impressions, 2009.											
2.		Bailey, Jane, et al. "Negotiating with Gender Stereotypes On Social Networking Sites: From "Bicycle Face" to Facebook." Journal of Communication Enquiry 37.2 (2013): 91-112.											
3.		Barry, Peter. "Beginning Theory". New Delhi: Viva Books, 2010.											
4.		Berger, John. "Ways of Seeing". London: Penguin Books, 1977.											
5.		Cranny-Francis, Anny, et al. "Gender Studies: Terms and Debates". New York: Palgrave Macmillan, 2003.											
6.		Gauntlett, David. "Media, Gender and Identity: An Introduction". London: Routledge, 2008											
7.		Pilcher, Jane, and Imelda Whelehan. "50 Key Concepts in Gender Studies". London: Sage, 2004. Print.											
8.		Jeanne, Haraway Donna. Simians, Cyborgs, and Women. London: Free Association Books, 1991. Web.											
9.		Koskela, Hille. "Webcams, TV Shows and Mobile Phones: Empowering Exhibitionism." Surveillance & Society 2.3 (2004): 199-215.Web.											
E-RESOURCES:													
1.		http://www.cyberpsychology.eu/view.php?cisloclanku=2009061501/ >.											
2.		http://www.surveillance-and-society.org/articles2(2)/webcams.pdf											
3.		http://eprints.rclis.org/19790/ >.											

INTRODUCTION TO CYBER SECURITY			
Course Code:	IS2501-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	IS1651-1		
Teaching Department: Information Science & Engineering			
Course Objectives:			
1.	Define the area of cybercrime and forensics and to understand the security threat		
2.	Explain the motive and causes for cybercrime, detection, and handling.		
3.	Investigate Areas affected by cybercrime.		
4.	Illustrate tools used in cyber forensic		
UNIT-I			
Introduction to Cyber Security		15 Hours	
Concepts of Cyber Security, Formal Methods of Security Validation, CIA framework-Confidentiality, Integrity and Authenticity, Threat modelling, Domains of cyber security, Security attacks, Security services, Security Mechanisms, Fundamental security design principles, Types of Cyber Threat.			
UNIT-II			
Tools and methods used in Cybercrime		14 Hours	
Introduction, Proxy Servers and Anonymizers, Intruders and Hackers, Insider threats, Cybercrimes. Network Threats: Active/ Passive – Interference – Interception –Impersonation – Worms –Virus – Spam’s – Ad ware - Spy ware – Trojans and covert channels –Backdoors – Bots – IP, Spoofing - ARP spoofing - Session Hijacking, Introduction to Phishing, Identity Theft (ID Theft).			
UNIT-III			
Understanding Computer Forensics		11 Hours	
Introduction, Digital Forensics Science, The Need for Computer Forensics, Cyberforensics and Digital Evidence, Forensics Analysis of E-Mail, Digital Forensics Life Cycle, Chain of Custody Concept, Network Forensics, Approaching a Computer Forensics Investigation, Setting up a Computer Forensics Laboratory: Understanding the Requirements, Computer Forensics and Steganography, Relevance of the OSI 7 Layer Model to Computer Forensics, Forensics and Social Networking Sites: The Security/Privacy Threats, Computer Forensics from Compliance Perspective, Challenges in Computer Forensics, Special Tools and Techniques, Forensics Auditing, Antiforensics.			
Course Outcomes: At the end of the course student will be able to			
1.	Comprehend the Cybercrime and its origin		
2.	Analyse Security Threat Management and understand the security elements.		
3.	Apply tools and methods used in Cyber crimes		
4.	Analyse Phishing and ID Theft		
5.	Comprehend Digital Forensics		
Course Outcomes Mapping with Program Outcomes			

Program Outcomes→ ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12
IS2501-1.1	2	-	-	-	-	1	-	3	-	-	-	-
IS2501-1.2	-	3	-	1	-	2	-	-	2	-	-	-
IS2501-1.3	-	3	2	-	-	-	-	-	-	-	-	-
IS2501-1.4	2	-	-	-	-	2	-	-	-	-	-	-
IS2501-1.5	-	-	-	-	-	-	-	3	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	William Stallings, "Cryptography and Network Security: Principles and Practice", Pearson Education, 2006.
2.	Swiderski, Frank and Syndex, "Threat Modeling", Microsoft Press, 2004.
3.	SunitBelapure and Nina Godbole, "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", Wiley India Pvt Ltd, ISBN: 978-81-265-21791, Publish Date 2013.

REFERENCE BOOKS:

1.	Thomas J. Mowbray, "Cybersecurity: Managing Systems, Conducting Testing, and Investigating Intrusions", John Wiley & Sons, Inc, ISBN: 978 -1-118 -84965 -1, 2014.
2.	James Graham, Ryan Olson, Rick Howard, "Cyber Security Essentials", CRC Press, 15-Dec 2010. Anti- Hacker Tool Kit (Indian Edition) by Mike Shema, Publication Mc Graw-Hill.
3.	Santosh B. J., K. V. S. S. S. S. Sairam, Shubham Kumar, Chandu Jagan Sekhar M, "Information and Cyber Security", Scientific International Publishing House, ISBN- 978-93-5625-694-1.

PYTHON APPLICATION PROGRAMMING

Course Code:	IS2502-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	CS1002-1		

Teaching Department: Information Science & Engineering

Course Objectives:

1.	Construct Python programs using data types and looping.
2.	Design object-oriented Python programs using classes and objects.
3.	Design useful stand-alone and CGI applications in

UNIT-I

Functions, Classes and OOP	15 Hours
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Functions: Design with functions: hiding redundancy, complexity; arguments and return values; formal vs actual arguments, named arguments. Program structure and design. Recursive functions

Classes and OOP: Classes, objects, attributes and methods; defining classes; design with classes, data modelling; persistent storage of objects, inheritance, polymorphism, operator overloading (_eq_, _str_, etc); abstract classes; exception handling, try block

UNIT-II

Lists, Tuples, and Dictionaries

14 Hours

Lists, tuples, and dictionaries: Basic list operators, replacing, inserting, removing an element; searching and sorting lists; dictionary literals, adding and removing keys, accessing, and replacing values; traversing dictionaries.

File Handling: Reading From Text Files, Writing to Text Files, Working with Excel Sheets ,CSV, PDF, Word,

UNIT-III

Essential Python Libraries

11 Hours

Working with SciPy, Numpy, Matplotlib, Pandas.

Graphical user interfaces: event-driven programming paradigm; creating simple GUI; buttons, labels, entry fields, dialogs; widget attributes - sizes, fonts, colors layouts, nested frames Simple CGI form.

Course Outcomes: At the end of the course student will be able to

1.	Demonstrate the basics of Python programming like data types and looping
2.	Apply the basic data structures in solving the problems
3.	Experiment with usage of functions in a given problem
4.	Develop Objects by creating classes and apply object-oriented features
5.	Develop applications in Python using File Programming & User Interface

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
IS2502-1.1	2	-	-	-	2	-	-	-	-	-	-	3
IS2502-1.2	2	-	-	-	2	-	-	-	-	-	-	3
IS2502-1.3	2	-	-	-	2	-	-	-	-	-	1	3
IS2502-1.4	-	-	-	-	-	-	-	-	-	-	-	-
IS2502-1.5	-	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Kenneth A. Lambert, "The Fundamentals of Python: First Programs", 2011, Cengage Learning, ISBN: 978-1111822705.

SOFTWARE ENGINEERING PRACTICES			
Course Code:	IS2503-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	CS1002-1		
Teaching Department: Information Science & Engineering			
Course Objectives:			
1.	Outline software engineering principles and activities involved in building large software programs.		
2.	Explain the importance of architectural decisions in designing the software.		
3.	Describe the process of Agile project development.		
4.	Recognize the importance of software testing and describe the intricacies involved in software evolution.		
5.	Identify several project planning and estimation techniques and explain the importance of software quality.		
UNIT-I			
Introduction	15 Hours		
Need for Software Engineering, Professional Software Development, Software Engineering Ethics, Case Studies.			
Software Processes			
Models: Waterfall Model, Incremental Model and Spiral Model; Process activities			
Requirements Engineering			
Functional and non-functional requirements, Requirements engineering processes, Requirements Elicitation and Analysis, Requirements specification, Software requirements document, Requirements validation & management.			
UNIT-II			
System Models	15 Hours		
Context models, Interaction models, Structural models, Behavioral models.			
T Architectural Design			
Architectural design decisions. Architectural Views and patterns, Application architectures.			
Design and implementation			
Object oriented Design using UML.			
Agile Software Development			
Agile methods, Plan-driven and agile development, Extreme Programming, Agile project management.			
UNIT-III			
Project Management	10 Hours		
Risk management, Teamwork.			
Project Planning			
Software pricing, Plan-driven development, Project Scheduling.			
Quality Management			
Software quality, Reviews and inspections, Software measurement and metrics, Software standards.			
Course Outcomes: At the end of the course student will be able to			

1.	Recognise the basics of software system, component, process and Software Requirement Specification to meet desired needs within realistic constraints and outline the professional and ethical responsibility
2.	Describe the waterfall, incremental and iterative models and architectural design in implementing the software
3.	Make use of the techniques, skills, modern engineering design tools and agile methods necessary for engineering practice.
4.	Describe the methods for maintaining software system.
5.	Discuss project planning and management and illustrate the quality of software Products

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
IS2503-1.1	-	3	1	-	-	-	-	2	-	-	-	-
IS2503-1.2	1	3	1	-	-	-	-	-	-	-	-	-
IS2503-1.3	1	1	3	-	-	-	-	-	-	-	-	-
IS2503-1.4	1	3	2	-	-	-	-	-	-	-	-	-
IS2503-1.5	1	2	2	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Ian Sommerville, “Software Engineering”, 9th Edition, Pearson Education, 2012.
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REFERENCE BOOKS:

1.	Roger S. Pressman: “Software Engineering-A Practitioners approach”, 7th Edition, Tata McGraw Hill, 2017.
2.	Pankaj Jalote: “An Integrated Approach to Software Engineering”, Wiley, India, 2010.

E Books / MOOCs/ NPTEL

1.	http://agilemanifesto.org/
2.	http://www.jamesshore.com/Agile-Book/
3.	https://www.mooc-list.com/course/uml-class-diagrams-software-engineering-edx
4.	https://www.mooc-list.com/course/enterprise-software-lifecycle-management-edx

WEB TECHNOLOGIES

Course Code:	IS2504-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	CS1002-1		
Teaching Department: Information Science & Engineering			
Course Objectives:			
1.	Illustrate the Semantic Structure of HTML and CSS		
2.	Compose forms and tables using HTML and CSS		
3.	Design Client-Side programs using JavaScript and Server-Side programs using PHP		
4.	Illustrate the Database connectivity using PHP		

Security

5.	Examine JavaScript frameworks such as jQuery																																																																																												
UNIT-I																																																																																													
Introduction to HTML	15 Hours																																																																																												
HTML tags and simple HTML forms, web site structure, HTML table, Need for CSS, introduction to CSS, basic syntax and structure, using CSS, background images, colours and properties, manipulating texts, using fonts, borders and boxes, margins, padding lists, positioning using CSS, Selectors, The Cascade: How Styles Interact, The Box Model, CSS Text Styling.																																																																																													
UNIT-II																																																																																													
Client side Scripting	15 Hours																																																																																												
Introduction to JavaScript: JavaScript language – declaring variables, scope of variables functions, event handlers (on click, on submit etc.), Document Object Model, Form validations. Introduction to PHP: Declaring variables, data types, arrays, strings, operations, expressions, control structures, functions, Reading data from web form controls like Text Boxes, radio buttons, lists etc.,																																																																																													
UNIT-III																																																																																													
PHP Databases	10 Hours																																																																																												
Basic command with PHP examples, Connection to server, creating database, selecting a database, listing database, listing table names creating a table, inserting data, altering tables, queries, deleting database, deleting data and tables, File Handling in PHP, PHP Arrays and Superglobals, Arrays, \$_GET and \$_POST Superglobal Arrays, jQuery Introduction: What is jQuery, Adding jQuery in to your web pages, jQuery Syntax, jQuery Selectors, jQuery Events.																																																																																													
Course Outcomes: At the end of the course student will be able to																																																																																													
1.	Adapt HTML and CSS syntax and semantics to build web pages																																																																																												
2.	Construct and visually format tables and forms using HTML and CSS.																																																																																												
3.	Experiment with the usage of Event handling and Form validation using JavaScript.																																																																																												
4.	Understand the principles of object-oriented development using PHP and Database concepts.																																																																																												
5.	Inspect JavaScript frameworks like jQuery which facilitates developers to focus on core features.																																																																																												
Course Outcomes Mapping with Program Outcomes																																																																																													
<table><tr><td>Program Outcomes→</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>↓ Course Outcomes</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>IS2504-1.1</td><td>1</td><td>2</td><td>-</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>1</td></tr><tr><td>IS2504-1.2</td><td>1</td><td>-</td><td>-</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>1</td></tr><tr><td>IS2504-1.3</td><td>1</td><td>2</td><td>-</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>1</td></tr><tr><td>IS2504-1.4</td><td>1</td><td>2</td><td>-</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>1</td></tr><tr><td>IS2504-1.5</td><td>1</td><td>-</td><td>-</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>1</td></tr></table>			Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	↓ Course Outcomes													IS2504-1.1	1	2	-	2	-	-	-	-	-	-	-	1	IS2504-1.2	1	-	-	2	-	-	-	-	-	-	-	1	IS2504-1.3	1	2	-	2	3	-	-	-	-	-	-	1	IS2504-1.4	1	2	-	2	3	-	-	-	-	-	-	1	IS2504-1.5	1	-	-	2	3	-	-	-	-	-	-	1
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12																																																																																	
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IS2504-1.5	1	-	-	2	3	-	-	-	-	-	-	1																																																																																	
1: Low 2: Medium 3: High																																																																																													
TEXTBOOKS:																																																																																													
1.	Randy Connolly, Ricardo Hoar, "Fundamentals of Web Development", 1 st Edition, Pearson Education India. (ISBN:978-9332575271).																																																																																												
E Books / MOOCs/ NPTEL																																																																																													
1.	npTEL.ac.in/courses/106105084/11																																																																																												

GRAPH THEORY

Course Code:	MA1501-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50

Teaching Department: Mathematics

Course Objectives:

1.	Explain subgraphs, bipartite graphs, isomorphic graphs etc. Apply the concept of trees and its properties
2.	Distinguish between Hamilton and Eulerian graph. Distinguish between planar and nonplanar graphs and apply their properties to solve problems.
3.	Represent a graph in terms of adjacency matrix, incidence matrix etc. and vice-versa.
4.	Find the shortest path between two vertices in a graph. Find minimal spanning tree.

UNIT-I

Introduction to graphs

15 Hours

Graphs and Graph Models, digraphs, Konigsberg bridge problem. Special Types of Graphs: Subgraphs-spanning and induced subgraphs, complete graph, Bipartite Graphs. Isomorphism of graphs. Complement of a graph and its properties. Connectivity-point and line connectivity. Trees and its properties.

Euler and Hamilton graphs and their applications.

UNIT-II

Planar graphs

09 Hours

Euler's polyhedron formula, outer planar graphs, applications

Colorability

07 Hours

Chromatic number, five color theorem, chromatic polynomial, Applications of graph coloring.

Matrix representation of graphs

Adjacency matrix, incidence matrix, circuit matrix, cut set matrix, Path matrix.

UNIT-III

Network Flows

04 Hours

Max -flow and Min-cut Theorem(statement), problems.

Shortest paths in weighted graphs

Dijkstra's algorithm to find shortest paths.

Spanning trees

05 Hours

Algorithms to find a spanning tree, minimal spanning tree-Kruskal's & Prim's algorithm.

Course Outcomes: At the end of the course student will be able to

1.	Distinguish between bipartite and complete bipartite graphs, identify whether two graphs are isomorphic, find subgraphs of a graph etc.
2.	Distinguish between Eulerian and Hamiltonian graphs.
3.	Identify whether a graph is planar and to find the chromatic polynomial of a graph.
4.	Representing graphs interms of Matrices.
5.	Apply algorithmic methods to find the shortest path between two given vertices. Use a suitable algorithm to find a minimal spanning tree.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
MA1501-1.1	3	3	-	-	-	-	-	-	-	-	-	-
MA1501-1.2	2	1	-	-	-	-	-	-	-	-	-	-
MA1501-1.3	2	3	-	-	-	-	-	-	-	-	-	-

	MA1501-1.4	3	2	-	-	-	-	-	-	-	-	-	-
	MA1501-1.5	3	2	-	-	-	-	-	-	-	-	-	-
1: Low 2: Medium 3: High													
TEXTBOOKS:													
	1.	F. Harary, “Graph theory”, Narosa Publishing House, 1988.											
	2.	Narsing Deo, “Graph Theory with applications to Engg. and Comp. Sciences”, PHI,1974.											
	3.	Kenneth H. Rosen, “Discrete Mathematics and its applications”, Tata McGraw Hill, V Edition-2003.											
REFERENCE BOOKS:													
	1.	D. B. West, “Introduction to Graph Theory”, PHI,2001.											
	2.	Chartrand and Zhang, “First Course in Graph Theory”, 2012											
E Books / MOOCs/ NPTEL													
	1.	http://diestel-graph-theory.com .											
	2.	https://nptel.ac.in/courses/111106102											

NUMBER THEORY			
Course Code:	MA1502-1	Course Type	OEC
Teaching Hours/Week (L: T: P) :	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Mathematics			
Course Objectives:			
1.	Understand the divisibility of integers, study of prime numbers and basic properties of congruences.		
2.	Study Fermat's little theorem and understand Euler's function.		
3.	Study the existence of primitive roots and quadratic residues.		
4.	Study the cryptographic applications in number theory.		
UNIT-I			
Divisibility and the theory of congruences	15 Hours		
Division algorithm, Euclid's algorithm for the greatest common divisor. Linear Diophantine equations. Prime numbers, fundamental theorem of arithmetic. Basic properties of congruences, Linear congruences and Chinese remainder theorem.			
UNIT-II			
	07 Hours		
Fermat's theorem, Wilson's theorem, Euler's Phi function, Euler's theorem.			
Primitive roots and Quadratic congruences	08 Hours		
Order of an integer modulo n, primitive roots for primes, Euler's criterion, Legendre symbol and its properties.			
UNIT-III			
Cryptography	10 Hours		
Introduction to public key cryptography, RSA cryptosystem, an application of primitive roots to cryptography.			
Course Outcomes: At the end of the course student will be able to			
1.	Use divisibility and Greatest common divisor in Euclidean algorithm. Solve Diophantine equations. Identify prime factorization of an integers.		
2.	Understand the properties of congruences. Use Chinese remainder theorem to find solution of system of linear congruences		
3.	Use Fermat's Little Theorem and Wilson's Theorem. Use of Euler's Phi function.		

4.	Identify primitive roots of an integers. Apply Euler's criterion and Legendre symbols.
5.	Code and decode numbers in the RSA cryptosystem.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→ ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12
MA1502-1.1	2	3	-	-	-	-	-	-	-	-	-	-
MA1502-1.2	2	3	-	-	-	-	-	-	-	-	-	-
MA1502-1.3	2	3	-	-	-	-	-	-	-	-	-	-
MA1502-1.4	2	3	-	-	-	-	-	-	-	-	-	-
MA1502-1.5	2	3	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	D. Burton, "Elementary Number Theory", McGraw-Hill, 2005.
2.	Niven, H.S. Zuckerman & H.L. Montgomery, "Introduction to the Theory of Numbers", Wiley, 2000.

REFERENCE BOOKS:

1.	H. Davenport, "The Higher Arithmetic", Cambridge University Press, 2008.
2.	G. A. Jones & J. M. Jones, "Elementary Number Theory", Springer UTM, 2007.
3.	Thomas Koshy, "Elementary Number Theory with Applications", 2nd edition, Elsevier, 2007.
4.	William J. LeVeque, "Fundamentals of Number Theory".

E Books / MOOCs/ NPTEL

1.	http://refkol.ro/matek/mathbooks/ro.math.wikia.com%2520wiki%2520Fisiere pdf_incarcate/Elementary-Number-Theory.pdf
2.	https://nptel.ac.in/courses/111104138
3.	https://nptel.ac.in/courses/111103020

LINEAR ALGEBRA

Course Code:	MA3501-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	MA1001-1 and MA2009-1		

Teaching Department: Mathematics

Course Objectives:

1.	Understand the concepts of vectors, bases.
2.	Determine the kernel, range, rank, and nullity of a linear transformation and apply them suitably in their field of study.
3.	Find the canonical forms and appraise its importance in various fields.
4.	Make use of Gram-Schmidt process to produce an orthonormal basis.
5.	Learn the concepts of singular value decomposition and PCA.

UNIT-I

Vector spaces	08 Hours
Vector spaces, subspaces, bases and dimensions, coordinate vecotrs, null spaces and column spaces of the matrices.	
Linear Transformations	07 Hours
Linear transformations, rank-nullity theorem, algebra of linear transformations, change of basis, linear operators, linear functionals, transpose of a linear transformation.	

UNIT-II

Canonical Forms	08 Hours
Review of characteristic values, similarity of matrices, Cayley Hamilton theorem, annihilating polynomials, invariant subspaces, Jordan and rational canonical forms.	
Inner Product Spaces	07 Hours
Inner products; inner product spaces; orthogonal sets and projections; Gram-Schmidt process; QR-factorization, Least-squares problems.	

UNIT-III

Symmetric Matrices and Quadratic Forms	10 Hours
Diagonalization, quadratic forms, constrained optimization, singular value decomposition and principal component analysis. Applications to linear recurrence relations.	

Course Outcomes: At the end of the course student will be able to

1.	Interpret vectors in two and three-dimensional spaces both algebraically and geometrically.
2.	Analyze the concept of a linear transformation as a mapping from one vector space to another and be able to calculate its matrix representation with respect to standard and nonstandard bases.
3.	Understand the concepts of Jordan and rational canonical forms.
4.	Make use of Gram-Schmidt process to produce an orthonormal basis and also able to use least square approximation method to obtain the solution of ill conditioned system.
5.	Apply techniques of constrained optimization singular value decomposition and PCA for problems arising in various engineering fields.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
MA3501-1.1	3	2	-	-	-	-	-	-	-	-	-	-
MA3501-1.2	2	2	-	-	-	-	-	-	-	-	-	-
MA3501-1.3	3	1	-	-	-	-	-	-	-	-	-	-
MA3501-1.4	3	2	-	-	-	-	-	-	-	-	-	-
MA3501-1.5	3	2	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Kenneth Hoffman and Ray Kunze, "Linear Algebra," 2 nd edition, Pearson Education (Asia) Pte. Ltd, 2004.
2.	David C. Lay, "Linear Algebra and its Applications", 3 rd edition, Pearson Education (Asia) Pte. Ltd, 2005.
REFERENCE BOOKS:	
1.	M. Artin, "Algebra", Prentice Hall of India, 2004.
2.	Gilbert Strang, "Linear Algebra and its Applications", 4th edition, Thomson Learning Asia, 2003.
3.	Bernard Kolman and David R. Hill, "Introductory Linear Algebra with Applications", Pearson Education (Asia) Pte.Ltd, 7 th edition, 2003.
4.	Sheldon Axler, "Linear Algebra Done Right", Springer International Publication, Third Edition, 2015.

AUTOMOTIVE ENGINEERING

Course Code:	ME1501-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50

Teaching Department: Mechanical Engineering	
Course Objectives:	
1. Get an idea on the different components of an engine and its types with lubrication system. 2. Understand the fuel supply system and ignition systems used in automobiles. 3. Demonstrate the working of transmission system. 4. Explain the importance of suspension system, steering geometry and drives in automobiles 5. Know the concept of braking system, tyres and emission control.	

UNIT-I	
Engine Components and Cooling & Lubrication Systems	08 Hours
SI & CI engines, Cylinderarrangements and their relative merits, Liners, Piston, connectingrod, crankshaft, valves, valve actuating mechanisms, valve and port timing diagrams, Choiceof materials for different engine co mponents, engine positioning, cooling requirements,methods of cooling, thermostat valves, different lubrication arrangements,crankshaft/flywheel position sensor, accelerator pedal sensors, engine coolant watertemperature sensor.	
Fuel Supply Systems for SI and CI Engines	08 Hours
Fuel mixture requirements for SI engines, types of carburetors, simplecarburetor, multi point and single point fuel injection systems, CRDI, fuel transfer pumps: AC Mechanical Pump, SU Electrical Pumps, injectors, Fuel gauge sensor, Throttle position sensor, Mass air flow sensors. Ignition Systems : Battery Ignition systems, magneto Ignition system, Transistor assisted contacts. Electronic Ignition, Automatic Ignition advance systems, Lighting systems, Rain/Light sensors, starting device (Bendix drive) Pedagogy: Chalk and talk method, Power Point Presentation	

UNIT-II	
Power Trains	07 Hours
Clutches - Single plate, multiplate and centrifugal clutches. Gear box: Necessity for gear ratios in transmission, Constant m esh gear box, Synchromesh gear box, principle of automatic transmission, Vehicle Speed Sensors, calculation of gear ratios, Types of transmission systems. No numerical.	
Drive to Wheels	08 Hours
Propeller shaft, universal joints, Hotchkiss. and torque tube drives, differential, rear axle, steering geometry, camber, king pin inclination, included angle, castor, toe in & toe out, condition for exact steering, power steering, over steer, under steer & neutral steer, Steering angle sensors, numerical problems. Suspension and Springs: Requirements, leaf spring, coil spring, Torsion bar suspension systems, independent suspension for front Wheel, Air suspension system. Collective bargaining; Characteristics, Necessity, Forms Safety & Health; Industrial accidents, Safety Quality circle; Meaning, Structure Pedagogy: Chalk and talk method, Power Point Presentation	

UNIT-III	
Brakes	09 Hours
Types of brakes, mechanical, compressed air, vacuum and hydraulic braking systems, construction and working of master and wheel cvlinder. brake shoe arrangements. Disk brakes. Drum brakes.	

Tyres: Desirable tyre properties, Types of tyres.

Automotive Emission: Automotive exhaust emissions, sources and emission control method: EGR, SCR, Emission Standards, Exhaust sensors.

Electric Vehicles.

Pedagogy Chalk and talk method, Power Point Presentation

Course Outcomes: At the end of the course student will be able to

1.	Describe and demonstrate the layout of an automobile and components of an automobile engine. Explain cooling and lubrication systems.
2.	Explain and demonstrate the fuel supply and Ignition systems for SI and CI engines.
3.	Describe and demonstrate the transmission system
4.	Explain and demonstrate the components of drive to wheel and suspension system, calculate the parameters of steering geometry.
5.	Describe and demonstrate automotive braking system. Explain types and construction of tyres and wheels. Explain the significance of automotive emissions and its controlling methods

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
ME1501-1.1	3	1	-	-	-	1	-	-	3	1	-	1
ME1501-1.2	3	1	-	-	-	1	-	-	3	1	-	1
ME1501-1.3	3	1	1	-	-	1	-	-	3	1	-	1
ME1501-1.4	2	3	1	-	-	1	-	-	3	1	-	1
ME1501-1.5	3	1	1	-	-	1	1	1	3	1	-	1

1: Low 2: Medium 3: High

TEXTBOOKS:

1. S. Srinivasan, "Automotive Mechanics", Tata McGraw Hill, 2003.
2. Kirpal Singh, "Automobile Engineering", Vol I and II, 2013.
3. A. K. Babu, "Automotive Electrical and Electronics", Khanna Publishers, 2nd edition, 2016.

REFERENCE BOOKS:

1. R. B. Gupta, "Automobile Engineering", Satya Prakashan, 4th Edn., 1984.
2. Naran G, "Automobile Engineering", Khanna Publishers 2002

INDUSTRIAL POLLUTION CONTROL

Course Code:	ME1502-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50

Teaching Department: Mechanical Engineering

Course Objectives:

1.	Know the Consequences of pollution, relationship between man and environment over the last few decades, necessity of modern awareness on pollution and how carbon audit can help in developing a carbon strategy.
2.	Identify the Importance of Meteorology in pollution control and global warming, various types of plume dispersions and its effect; analyze various levels of plume height for different pollutants.
3.	Distinguish Particulates and fly ash separation techniques such as cyclone separator, electrostatic precipitator efficiency calculations etc.
4.	Illustrate Formation, measurement and control techniques for Smoke and gaseous pollutants.

5.	Summarize the Effects of water, soil, plastics and odor pollution their control techniques, Different Pollution Control Acts, Legal aspects of pollution control and how these acts can help in bringing down the pollution rate.																																																																																											
UNIT-I																																																																																												
Introduction to Pollution	08 Hours																																																																																											
Man and the environment, types of pollution and its consequences, Changing environmental management concept, sustainable industrial growth, carbon audit, Ill effects of various pollutants, permissible concentration levels & AQI.																																																																																												
Meteorology	08 Hours																																																																																											
Meteorology, Wind rose, Lapse rate, plume dispersion studies & Numerical problems. Pedagogy: Chalk and talk method, Power Point Presentation																																																																																												
UNIT-II																																																																																												
Separation techniques	08 Hours																																																																																											
Different types of Particulates, Need for Separation techniques, Sources of Particulates Matter Fly Ash Electrostatic precipitator (Problems) Theory of settling processes (Design Problems), Bag House fabric filter Cyclone separator Spray Tower Scrubbers & Venturi Scrubber																																																																																												
Smoke and gaseous pollutants:	08 Hours																																																																																											
Smoke- White, blue and black smoke, Sources of smoke, T,T,T-O Principle of smoke Measurement of stack smoke intensity using Ringlemann Chart and Smokescope&Bosch Smoke meter, Domestic and Industrial Incinerators-Design factors, Pollutant gaseous So ₂ , Co, UBHC, Nox their ill effects and & control methods. Pedagogy: Chalk and talk method, Power Point Presentation																																																																																												
UNIT-III																																																																																												
	08 Hours																																																																																											
Water, soil, noise, and odor pollution, their control methods, problems associated with nuclear reactors, Legal aspects of pollution control in India, brief details of Euro and BS standards Pedagogy: Chalk and talk method, Power Point Presentation																																																																																												
Course Outcomes: At the end of the course student will be able to																																																																																												
1.	Identify the various types of pollutants and distinguish between them with regards to Particulate matters and AQI.																																																																																											
2.	Outline the instruments for Meteorological measurements, distinguish types of plume dispersions and its effect; analyze the concentration of various gaseous pollutants from T-Z diagrams																																																																																											
3.	Explain the Particulates and fly ash separation techniques, compare and Interpret their efficiency																																																																																											
4.	Illustrate Formation, measurement and control techniques for Smoke and gaseous pollutants																																																																																											
5.	Identify Effects of water, soil, plastics and odor pollution on environmental Pollution and explain the Legal aspects of pollution control.																																																																																											
Course Outcomes Mapping with Program Outcomes																																																																																												
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1.	“Environmental Pollution Control Engineering”, Wiley Eastern Ltd.,																																																																																											
2.	Gilbert M Masters, “Introduction to Environmental Engineering & Science”, PHI,1995																																																																																											
3.	C. S Rao, “Environmental Pollution Control Engineering”, New Age Int.																																																																																											
REFERENCE BOOKS:																																																																																												
1.	Henry C. Perkins, “Air Pollution”, Mc-Graw Hill, 1974.																																																																																											
2.	W. L. Faith, “Air Pollution control”, John Wiley																																																																																											
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1.	http://nptel.ac.in/courses/105106119/36																																																																																											

SUSTAINABLE DEVELOPMENT GOALS

Course Code:	ME1503-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50

Teaching Department: Mechanical Engineering

Course Objectives:

1.	To provide the knowledge, skills, attitudes and values necessary to address sustainable development challenges
2.	Address the global challenges including poverty, inequality, climate change, environmental degradation, peace and justice.
3.	To learn more and take action.
4.	Addresses critical global challenges put forth by UN.
5.	Analyze how sustainable development can be achieved in practice.

UNIT-I

08 Hours

The origin, development and idea of the SDGs History and origins of the Sustainable Development Goals. What are the SDGs? What are their aims, methodology and perspectives? How are they related to the Millennium Development Goals?

SDGs and Society

08 Hours

Ensuring resilience and primary needs in society In-depth discussion and analysis of goals related to poverty, hunger, health & well-being and education

Pedagogy: Chalk and talk method, Power Point Presentation

UNIT-II

SDGs and Society

14 Hours

Strengthening Institutions for Sustainability In-depth discussion and analysis of goals related to gender equality, affordable and clean energy, sustainable cities & communities, and peace, justice & strong institutions

SDGs and the Economy: Shaping a Sustainable Economy In-depth discussion and analysis of goals related to work & economic growth, industry, innovation & infrastructure, inequalities, responsible production & consumption

Pedagogy: Chalk and talk method, Power Point Presentation

UNIT-III

SDGs and Biosphere

the

10 Hours

Development within Planetary Boundaries In-depth discussion and analysis of goals related to clean water, climate, life below water and life on land

Realizing the SDGs: Implementation through Global Partnerships In-depth discussion and analysis of SDG 17 which aims to implement the SDGs through partnerships, finance, technology and the development of coherence between policies.

Pedagogy: Chalk and talk method, Power Point Presentation

Course Outcomes: At the end of the course student will be able to

1.	Summarize the UN's Sustainable Development Goals and how their aims, methodology and perspectives.
2.	Analyze the major issues affecting sustainable development and how sustainable development can be achieved in practice.
3.	Identify and apply methods for assessing the achievement/possibilities of sustainable development in Nitte gram panchayath.
4.	Evaluate the implications of overuse of resources, population growth and economic growth. sustainability & Explore the challenges the society faces in making transition to renewable resource use.
5.	Create skills that will enable students to understand attitudes on individuals, society and their role regarding causes and solutions in the field of sustainable development.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
ME1503-1.1	1	2	1	1	1	3	3	1	1	1	-	2
ME1503-1.2	2	2	1	1	1	3	3	2	1	1	-	1

Security

ME1503-1.3	3	2	2	1	1	3	3	2	3	1	-	1
ME1503-1.4	3	2	3	1	1	3	3	2	1	1	-	1
ME1503-1.5	1	2	2	1	1	3	3	2	2	2	-	1

1: Low 2: Medium 3: High

TEXTBOOKS:	
1.	Sachs, Jeffrey D. "The age of sustainable development" Columbia University Press, 2015
2.	Gagnon, B., Leduc, R., and Savard, L., "Sustainable development in engineering: a review of principles and definition of a conceptual framework", Cahier de recherche / Working Paper 08-18, 2008.

REFERENCE BOOKS:

1.	Elliott, Jennifer, "An introduction to sustainable development", Routledge, 2012.
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E Books / MOOCs/ NPTEL

1.	https://www.un.org/sustainable-development/poverty/
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TECHNOLOGICAL INNOVATION			
Course Code:	ME1504-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Mechanical Engineering			
Course Objectives:			
1. Understand basics of operations management and Quality. 2. Define the concept of technological innovation. 3. Discuss Innovation management and the difference between Invention and Innovation. 4. Appreciate the importance of Innovation as a management process and Innovation management techniques. 5. Discuss the Innovation system, Understand the importance of Technology management and Transfer and basics of Technological Forecasting.			
UNIT-I			
Production and Operations Management and Introduction to Quality Concepts	04 Hours		
Production and Operations Management: Introduction - Functions within business organizations - the operation management function - Classification of production systems. Introduction to Quality Concepts: The Meaning of Quality and Quality Improvement - Key dimensions of Quality - Concept of cost of quality - Customers' perception of quality.			
Introduction to Technological Innovation	09 Hours		

Basic Concepts and Definitions: Technology - Technology Management – Invention – Creativity – Innovation - The Concept of Technological Innovation - Innovation Posture, Propensity and Performance - Innovation Measurement - Key factors linking creativity and innovation – Classifications of Innovations – Innovation Process.																																																																																																							
Startup Idea Pitching	03 Hours																																																																																																						
UNIT-II																																																																																																							
Introduction to Innovation Management and Innovation & Competitiveness	07 Hours																																																																																																						
Introduction to Innovation Management: Innovation Management Through Management of Knowledge and Education – Types of Learning - Difference Between Innovation and Invention - Types and Characteristics of Innovation. Innovation and Competitiveness: Case Study – Barriers for Innovation and Competitiveness																																																																																																							
Innovation as a Management Process	08 Hours																																																																																																						
Activities to enhance companies’ capacity for innovation – Management of Technological Innovation: Corporate Perspective, National Perspective, Theoretical Perspective and Individual Perspective - Challenges in Technological Innovation Management - Case Study in Technological Innovation Management - Innovation Management Techniques (IMTs).																																																																																																							
UNIT-III																																																																																																							
Innovation Systems and Technology Management & Transfer	04 Hours																																																																																																						
Innovation Systems: The Concept of Innovation Systems - Innovation Systems: Sectoral, Regional, National. Technology Management and Transfer: Technology Transfer - Impacts of MNCs in technology transfer																																																																																																							
Introduction to Technological Forecasting	05 Hours																																																																																																						
Introduction - Applications & Limitations of Technological Forecasting – Technology Forecasting Techniques – Exploratory Forecasting – Normative Forecasting – Delphi Technique – Problems of Technological Forecasting																																																																																																							
Course Outcomes: At the end of the course student will be able to																																																																																																							
1.	Define operations management and quality.																																																																																																						
2.	Describe technological innovation and its key features for business.																																																																																																						
3.	Discuss innovation management and the difference between invention and innovation.																																																																																																						
4.	Explain innovation as a management process, its management and perspectives. Understand Innovation management techniques.																																																																																																						
5.	Explain innovation systems, technology management transfer and basics of technological forecasting.																																																																																																						
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TEXTBOOKS:																																																																																																							
1.	Carayannis, Elias G., Samara, Elpida T., Bakouros, Yannis L., “Innovation and Entrepreneurship Theory, Policy and Practice”, Springer, 2015.																																																																																																						
REFERENCE BOOKS:																																																																																																							
1.	Dick Whittington, “Digital Innovation and Entrepreneurship”, Cambridge University Press, 2018.																																																																																																						

E Books / MOOCs/ NPTEL	
1.	https://krishi.icar.gov.in/jspui/bitstream/123456789/46063/1/21_Technological%20forecasting.pdf dtd 12/06/2022
2.	http://www.oiepec.eu/wp-content/uploads/2017/07/Introduction-to-Technology-Forecasting.pdf dtd 12/06/2022

HUMAN RESOURCE MANAGEMENT			
Course Code:	MG1501-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Mechanical Engineering			
Course Objectives:			
1.	To develop a meaningful understanding of HRM theory, functions and practices.		
2.	To understand concepts and skills recruitment.		
3.	To understand the concepts of training and development.		
4.	To deal with employees' grievances, safety and health types of organizations.		
5.	To understand the concepts of e-HRM.		
UNIT-I			
Human Resource Management & HRP			08 Hours
Introduction, meaning, nature, scope of HRM. Major functions of HRM, Personnel Management vs Human Resource Management, job design, job evaluation, job analysis, job specification, job enlargement, job enrichment. Role of HR Manager.HR Planning. Process HRP.			
Recruitment			08 Hours
Definition, Sources and Methods of Recruitment Selection: Definition and Process of Selection. Cost benefit analysis of selection. Placement: Meaning, Induction/Orientation, Internal Mobility, Transfer, Promotion, Demotion and Employee Separation. Performance Appraisal methods. Pedagogy: Chalk and talk method, Power Point Presentation			
UNIT-II			
Training and development			07 Hours
Training v/s development, stages in training, Training Methods, Executive Development, Methods and Development of Management Development, Career and Succession Planning.			
Compensation			08 Hours
Employee remuneration, rewards, Wage and Salary Administration, Bonus, fringe benefits. Internal Mobility, External Mobility, Trade union Act (Amendment) 2001. Employee Grievances: Employee Grievance procedure. Discipline procedure Collective bargaining; Characteristics, Necessity, Forms Safety & Health; Industrial accidents, Safety Quality circle; Meaning, Structure Pedagogy: Chalk and talk method, Power Point Presentation			
UNIT-III			
IHRM and e-HRM			09 Hours

Security

Managing IHRM. e-HR Activities, Global recruitment, selection, expatriates. Industrial conflict –Causes, Types, Prevention and Settlement. Aspects of e-HRM,e-Job design & Analysis, Ethical issues in employment Pedagogy: Chalk and talk method, Power Point Presentation																																																																																																							
Course Outcomes: At the end of the course student will be able to																																																																																																							
1.	Describe the basic concepts of HRM & HRP.																																																																																																						
2.	Elucidate the HRM functions of recruitment, selections, and appraisal.																																																																																																						
3.	Apply the training, development and compensation methods in HRD.																																																																																																						
4.	Identify the employee grievances to spell out the remedial measures.																																																																																																						
5.	Infer the concepts of e-HRM and I-HRM.																																																																																																						
Course Outcomes Mapping with Program Outcomes																																																																																																							
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							1.	P Courseba Rao, "Essentials of Human Resource Management & Industrial Relations", Third Revised Edition.																																																																																															
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							1.	John M. Ivancevich, "Human Resource Management", 10/e, McGraw Hill.																																																																																															
							2.	Flippo, "Human Resource Management".																																																																																															
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MANAGEMENT ACCOUNTING AND CONTROL SYSTEM			
Course Code:	MG1502-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Management			
Course Objectives:			
1.	Apply Cost Accounting concepts and techniques in the decision making process.		
2.	Make decisions such as pricing, special order pricing, make-or-buy and elimination of a part of the company or replacement of equipment.		
3.	Understand the relevance of different types of costs in the decision making process such as relevant costs, sunk costs or controllable costs.		
4.	Understand fundamental concepts in Financial, Cost & Management Accounting.		
5.	Develop analytical skills associated with the preparation and interpretation of Financial Statement		
UNIT-I			
Introduction to Cost and Management Accounting and Marginal Costing		07 Hours	
Cost Accounting – Meaning, Objectives and Scope, Management Accounting – Meaning, Objectives and Scope, Tools and Techniques of Management Accounting, Relationship of Cost Accounting, Financial Accounting, Management Accounting			

and Financial Management, Conflicts in Profit versus Value Maximization Principle, Role of Management Accountant in Decision Making.																																																																																																							
Marginal Costing							08 Hours																																																																																																
Meaning, Advantages, Limitations and Applications. Breakeven Analysis, Cost Volume Profit Analysis, P/V Ratio and its Significance, Margin of Safety, Absorption Costing: System of Profit Reporting and Stock Valuation, Difference between Marginal Costing and Absorption Costing, Income Measurement under Marginal Costing and Absorption Costing. (Practical Problems)																																																																																																							
UNIT II																																																																																																							
Standard Costing and Budgetary Control							07 Hours																																																																																																
Standard Costing – Definition, Significance and Applications, Various Types of Standards, Installation of Standard Costing System-for Material, Labour, and Overhead. Variance Analysis for Materials, Labour and Overheads, Accounting Treatment of Variances. Benchmarking for Setting of Standards, Variance Reporting to Management. (Practical Problems)																																																																																																							
Budgetary Control							08 Hours																																																																																																
Budget Concept, Manual, Fixed and Flexible Budgets, Preparation and Monitoring of Various Types of Budgets, Budgetary Control System- Advantages, Limitations and Installation. Zero Base Budgeting, Programme and Performance Budgeting. (Practical Problems)																																																																																																							
UNIT III																																																																																																							
Fund Flow and Cash Flow Statement							05 Hours																																																																																																
Fund Flow Statement Analysis – Definition, Features, Steps for Preparation of Fund Flow Statement.																																																																																																							
Cash Flow Statement Analysis							05 Hours																																																																																																
Classification, Preparation of Cash Flow Statement, Uses of Cash Flow statement, Difference between Cash Flow and Fund Flow Statement. (Practical Problems)																																																																																																							
Course Outcomes: At the end of the course student will be able to																																																																																																							
1.	Describe the Cost Accounting concepts and techniques in the decision making process.																																																																																																						
2.	Elucidate the Make decisions such as pricing, special order pricing, make-or-buy and elimination of a part of the company or replacement of equipment.																																																																																																						
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5.	Infer the analytical skills associated with the preparation and interpretation of Financial Statement																																																																																																						
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1.							M.Y. Khan and P.K. Jain. “Management Accounting”, McGraw-Hill Education																																																																																																
2.							Robert N. Anthony, “Management Accounting”, Richard Dirwin.																																																																																																
3.							I.M. Pandey , “Management Accounting”, Vikas Publishing House.																																																																																																
4.							Paresh shaw, “Management Accounting”, Oxford University Press.																																																																																																
5.							A. Murthy and S. Gurusamy , “Management Accounting”, McGraw Hill.																																																																																																
6.							NM Singhvi and Ruzbeh J. Bodhanwala, “Management Accounting”, PHI learning Pvt. Ltd.																																																																																																

OPERATIONS AND QUALITY MANAGEMENT			
Course Code:	MG1503-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Management			
Course Objectives:			
1.	Define production/operations management. Differentiate between Production and service system and types of production systems. Discuss continuous and intermittent production systems with their advantages and disadvantages. Discuss CRM and ERP systems.		
2.	Solve problems on fundamentals of statistics and normal distribution. Draw and Analyze variable process control chart and determine process capability.		
3.	Discuss Total Quality Management tools and methods. Calculate reliability of series and parallel systems using the information on failure rate and time.		
4.	Solve decision-making problems using break even analysis and decision tree methods. Apply the concepts of Design and System capacity. Solve problems on facility location using break even analysis and transportation method. Solve problems related to product and process layouts.		
5.	Use concepts of replacement theory to solve problems of replacing items that fail gradually and suddenly.		
UNIT-I			
Production and Operations Management			06 Hours
Introduction, Functions within business organizations, the operation management function, Classification of production systems, Productivity, factors affecting productivity, Introduction to Customer Relationship Management (CRM) and Enterprise Resource Planning (ERP).			
Philosophy of statistical process control and modeling process quality			11 Hours
Normal distribution tables, Finding the Z score, Central limit theorem, Chance and assignable causes of variation, Statistical Basis of the Control Charts (basic principles, choices of control limits, significance of control limits, warning limits) Control charts for variables: Control Charts for X-Bar and R- Charts, Type I and Type II errors, Simple Numerical Problems, Process capability: The foundation of process capability, Natural Tolerance limits, c_p – process capability index, c_{pk} , p_p – process performance index, summary of process measures. Numerical problems. Concept of Six sigma. Pedagogy: Chalk and talk method, Power Point Presentation			
UNIT II			
Quality Concepts and Reliability			06 Hours

Introduction to Quality Concepts: The Meaning of Quality and Quality Improvement, Key dimensions of Quality, Concept of cost of quality. Customers' perception of quality.
 TOTAL Quality Management: Definition, Principles of TQM, Gurus of TQM, Benefits of TQM.
 Managing Quality: Quality circles, Continuous Improvement- Juran's Trilogy, PDCA cycle, Kaizen, 7 QC tools.
 Introduction to reliability, Mean time to failure, Mean time between failures, Bath tub curve, Reliability of series and parallel systems, Numerical problems on the above topics.

Operations Management activities
12 Hours

Decision Making: The decision process, characteristics of operations decisions, use of models - decision making environments. Break even Analysis, Decision trees.

Capacity Planning: Importance of capacity decisions, defining and measuring capacity, determinants of effective capacity. Design, System an actual capacity. System efficiency and utilization. Determination of Equipment requirement for a single stage production processes. Numerical problems on the above.

Facilities location planning: Need for location decisions, nature of locations decisions, general procedure for making locations decisions, Use of Breakeven analysis and Transportation algorithms for making location decisions.

Facilities layout planning: Need for layout decisions. Minimizing material handling cost in process aout using Load distance analysis, Simple line balancing problems in product layout.

UNIT III
Replacement Theory
05 Hours

Replacement policy for equipment which deteriorates gradually. Replacement of items that fail suddenly.

Pedagogy: Chalk and talk method, Power Point

Course Outcomes: At the end of the course student will be able to

1.	Define production/operations management. Differentiate between Production and service system and types of production systems Discuss continuous and intermittent production systems with their advantages and disadvantages. Discuss CRM and ERP systems.
2.	Solve problems on fundamentals of statistics and normal distribution. Draw and Analyze variable process control charts and determine process capability.
3.	Discuss Total Quality Management tools and methods. Calculate reliability of series and parallel systems using the information on failure rate and time.
4.	Solve decision-making problems using break even analysis and decision tree methods. Apply the concepts of Design and System capacity. Solve problems on faculty location using break even analysis and transportation method. Solve problems related to product and process layouts.
5.	Use concepts of replacement theory to solve problems of replacing items that fail gradually and suddenly.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→ ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12
MG1503-1-1.1	2	1	-	-	-	-	-	-	-	-	2	-
MG1503-1-1.2	2	2	-	-	-	-	-	-	-	-	2	-
MG1503-1-1.3	1	1	-	-	-	-	-	-	-	-	2	-
MG1503-1-1.4	3	2	-	-	-	-	-	-	-	-	3	-
MG1503-1-1.5	1	1	-	-	-	-	-	-	-	-	1	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1	Joseph G Monks, "Production / Operations Management", McGraw Hill Books
2	William J Stevenson, "Production and Operations Management", Tata McGraw Hill, 8th Edition.
3	RC Gupta, "Statistical Quality Control", Khanna Publishers, New Delhi, 2005.
4	N.D. Vohra, "Quantitative Techniques in Management", Tata McGraw Hill, 2015

REFERENCE BOOKS:

1	E.L. Grant and R.S. Leavenworth, "Statistical Quality Control", 7th edition, McGraw- Hill publisher, 2004.
2	Prem Kumar Gupta, D S. Hira, "Operations Research", S Chand Publications, New Delhi, 2 nd edition 2008, Prentice Hall.

3	W S Messina, "Statistical Quality Control for Manufacturing Managers", Wiley & Sons, Inc. New York, 1987
4	Montgomery, Douglas, "Statistical Quality Control", 5th Edition, John Wiley & Sons, Inc. 2005, Hoboken, NJ.
5	Jerry Banks, "Principles of Quality Control", Wiley & Sons, Inc. New York.

ORGANIZATIONAL BEHAVIOUR

Course Code:	MG1504-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Management			
Course Objectives:			
1	Describe the Nature and Characteristics, Determinants and Approaches of Organizational Behaviour. Describe the concepts of Perception, Attitudes and values and their implications.		
2	Describe the concepts of learning and motivation along with their managerial implications.		
3	Describe the concepts of Leadership along with their managerial implications.		
4	Discuss the concepts of group dynamics and conflict management along with their implications.		
5	Discuss the concepts of Organization culture and change and conflict management along with their implications.		
UNIT-I			
	15 Hours		
Introduction: Conceptual Foundation of Organisational Behaviour; Nature and Characteristics; Determinants; Contributing Disciplines; Challenges and Opportunities for Organisational Behaviour, Models and Approaches of Organizational Behaviour, OB and Emotional Intelligence. Perception, Attitude, and Values: Nature, Process, Importance, Factors Influencing Perception; Attribution Theory of Perception; Issues Involved in Perception: Selective Perception, Halo Effect, Contrast Effect, Projection, Stereotyping; Concept of Pygmalion Effect; an overview of Emotions and feelings, Values, Beliefs and Attitudes with Managerial Implications. Learning: Concept; Theories of Learning: Conditioning, Social Learning, Managerial Implication of Learning Theories. Reinforcement. Motivation: Concept, Major Theories and Process of Motivation: Maslow’s Need-Hierarchy Theory; Herzberg’s Motivation-Hygiene Theory; McGregor’s Theory X and Theory Y; Goal- Setting Theory; ERG Theory; Vroom’s Expectancy Theory; Equity Theory; Managerial implications of Various Theories. Pedagogy: Chalk and talk method, Power Point Presentation, Case studies			
UNIT II			
	15 Hours		

Leadership: Concept and Functions; Style and Theories of Leadership: Traits, Behavioural and Situational/ Contingency Groups of Theories; Inspirational approaches to Leadership; Charismatic Leadership, Transformational Leadership, and Transactional Leadership, Contemporary Leadership Roles; Challenges to the Leadership Construct; Substitutes and Neutralizers to Leadership. Group Behaviour: Groups: Concept and Classification; Stages of Group Development; Group Structure; Roles and Norms; Premise and Issues; Group Decision-Making: Group vs Individual; Groupthink and Groups Shift; Group Decision Making Techniques and Process. Conflict Management: Concept; Causes; Types; Stages; Effects; Management of Conflicts. Pedagogy: Chalk and talk method, Power Point Presentation, Case studies																																																																																																								
UNIT III																																																																																																								
	10 Hours																																																																																																							
Organizational Culture: Concept; Dominant Culture; Strong vs Weak Cultures ; Creating and Sustaining Culture; Employees Learning of The Culture; Creating a Customer-Responsive Culture. Organizational Changes: Concept and Forces for Change; Managing Planned Changes; Resistance to Change; Approaches to Manage Organizational Change; Organizational Development; Culture-Boundedness of Managing the Change. Pedagogy: Chalk and talk method, Power Point Presentation, Case studies																																																																																																								
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1.	Robbins, SP Stephen P, Timothy Judge and Nehasika Vohra, "Organisational Behaviour", 12th or 16th edition, Pearson Education, 2011.																																																																																																							
2.	Fred Luthans, "Organisational Behaviour", 11th edition, Mc Graw Hill, 2009.																																																																																																							
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1.	W. Newstrom, John, "Organisational Behaviour", 10 th edition, Tata Mc Graw –Hill 2009.																																																																																																							
2.	Paul Heresy, Kenneth H. Blanchard, and Dewey E. Johnson, "Management of Organisational Behaviour", Leading Human Resources, 2008.																																																																																																							
3.	Dr S S Khanka, "Organisational Behaviour", S. Chand & Co, New Delhi, 2008.																																																																																																							
4.	Sanghi Seema, "Organisational Behaviour", Pearson, 2011.																																																																																																							

TAXATION FOR ENGINEERS

Course Code:	MG1505-1	Course Type	OEC
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Teaching Hours/Week (L: T: P):	3:0:0	Credits	03									
Total Teaching Hours	40	CIE + SEE Marks	50+50									
Teaching Department: Management												
Course Objectives:												
1	To make students understand the overview of Income Tax Law in India.											
2	To make students understand the basic concepts of income tax such as residential status, tax incidence.											
3	To make students understand the income tax provisions involved in determination of income from salary, House property, business and profession, capital gain and other sources.											
4	To help students understand the determination of tax liability Individual assessees.											
5	To make students understand the deductions u/s 80.											
UNIT-I												
Basic concepts and Explanation under various Heads of Income		15 Hours										
Basic concepts: Assessment Year, Previous Year, Person, Assessee, Income, Charges on Income, Gross Total Income, Capital and Revenue Receipts, Residential status, Connotation of income, Deemed to accrue or arise in India, Incidence of tax, Tax Planning, Tax Evasion, Tax Management. (Problems on Residential Status of Individual assessee)												
Explanation under various Heads of Income: Income from Salary (theory, basic and full-fledged problems on allowances, perquisites and retirement benefits)												
UNIT II												
Income under the head Profit and gains of Business or Professions and Income under Capital Gain		15 Hours										
Income under the head Profit and gains of Business or Professions and its computation - basis - Method of accounting - Scheme of business deductions/ allowance - Deemed profits - maintenance of books, (Problems on computation of Income from Business/ Profession of Individual assessee)												
Income under Capital Gain: Basis of charge, Transfer of capital asset, inclusion & exclusion from Capital Asset, Capital Gain, Computation of Capital Gains (theory & problems), Exemptions/deductions from capital gains												
UNIT III												
Income from House Property and Other Sources		10 Hours										
Income from House Property - Basic problems on House Property												
Income from Other Sources (theory only)												
Deductions under section 80C to 80U (No problems - Provisions only)												
Course Outcomes: At the end of the course student will be able to												
1.	Exhibit an understanding of the Income Tax Law in India.											
2.	Identify the nature of Incomes and their tax incidence.											
3.	Demonstrate how to determine the income from salary, house property, business and profession, capital gain.											
4.	Demonstrate the determination of tax liability of Individual assessees.											
5.	Exhibit a clear understanding of various provisions of deductions u/s 80.											
Course Outcomes Mapping with Program Outcomes												
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
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MG1505-1-1.2	2	-	-	-	-	1	-	-	1	-	2	1
MG1505-1-1.3	3	-	-	-	-	1	-	-	1	-	2	1
MG1505-1-1.4	3	-	-	-	-	1	-	-	1	-	2	1
MG1505-1-1.5	3	-	-	-	-	1	-	-	1	-	2	1
1: Low 2: Medium 3: High												
REFERENCE BOOKS:												

1.	Vinod Singhania, "Students Guide to Income Tax", Taxman Publications.
2.	Mehrotra & Goyal, "Direct Tax", Sahitya Bhavan.
3.	Lal & Vashisht, "Direct Tax", Pearson Ed. 28E.
4.	V S Datey, "Indirect Taxes", Taxman Publications.
5.	Vinod Singhania, "Direct Taxes", Taxman Publications.
6.	T N Manoharan, "Students Guide to Income Tax", Snow White.
7.	Kul Bushan, "How to deal with VAT", Pearson Education/PHI, 1/e.
8.	Mahesh Chandra & Shukla, "Income Tax Law & Practice", Pragathi Publications.
9.	Dr. Pillai, "VAT", Jaico Publications.

WORKING CAPITAL MANAGEMENT

Course Code:	MG1506-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Need of the Course: The course will enable the student to manage activities in the area of working capital in an enterprise and help the students to do advance study in the field of financial-management through detailed analysis of financial statements, liquidity crises, cash optimization, credit analysis etc. The student will learn how to apply sound techniques for managing inventory.			
Description of the Course: Every business needs adequate liquid resources in order to maintain day-to-day cash flow. It needs enough cash to pay wages and salaries as they fall due and to pay creditors if it is to keep its workforce and ensure its supplies. Maintaining adequate working capital is not just important in the short-term. Sufficient liquidity must be maintained in order to ensure the survival of the business in the long-term as well. Even a profitable business may fail if it doesn't have adequate cash flow to meet its liabilities as they fall due.			
Teaching Department: Management			
Course Objectives:			
1	Discuss the importance of working capital management.		
2	Evaluate working capital requirement.		
3	Assess the challenges faced in managing working capital in domestic and international operations.		
4	Plan for financing working capital requirement.		
UNIT-I			
Working Capital Decisions, Working Capital Management and Sources of Working Capital		15 Hours	
Working Capital Decisions: Meaning, Concepts, components Importance & types of working Capital.			

Working Capital Management: Meaning, objectives, Principles, Importance of adequate working capital & consequences of inadequate working capital, Dangers of excessive working capital, determinants of working capital - operating cycle and Cash cycle. Approaches to determine an appropriate financing mix, Estimation of working capital requirements (problems) important working capital ratios. Sources of Working Capital: Financing of long term working capital & short term working capital. Factoring - Meaning mechanism, Functions, types, merits & demerits.

UNIT II

Liquidity Management and Receivable Management	15 Hours
Liquidity Management: Cash Management - Meaning - Objectives of Cash Management - Nature of Cash - Motives of holding cash - Cash Management planning aspects - Cash Budgets (Problems), Cash Management control aspects - Concentration banking - Lock box system - Playing the float - Cash Management models - William J Baumol Model - Miller-Orr Model (Problems using these models) Receivable Management: Definition, Objectives, cost and benefits of receivable. Credit policy & its variables. Types of Credit policy & their merits & demerits, Factors influencing the size of investment in receivables. Control of receivables. Framing optimum credit policy & Average collection period (Problems)	

UNIT III

Inventory Management	10 Hours
Meaning of Inventory - Need/Purpose of holding inventory - Benefits of holding inventory - Risk and cost of holding inventory - Management of Inventory - Objectives of Inventory Management - Techniques of Inventory Management - Economic Order Quantity (EOQ) - Determination of Stock levels - ABC analysis - Just in Time (JIT).	

Course Outcomes: At the end of the course student will be able to

1.	Understand the meaning of working capital
2.	Realize the importance of management of working capital in an organization
3.	Learn about some key liquidity ratios used to understand more about a business' working capital position
4.	Understand various techniques used to manage working capital.
5.	Be aware of the techniques of cash, inventory and receivables management.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→ ↓ Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12
MG1506-1-1.1	2	-	-	-	-	1	-	-	-	1	2	1
MG1506-1-1.2	2	-	-	-	-	1	-	-	-	1	2	1
MG1506-1-1.3	2	-	-	-	-	1	-	-	-	1	2	1
MG1506-1-1.4	2	-	-	-	-	1	-	-	-	1	2	1
MG1506-1-1.5	2	-	-	-	-	1	-	-	-	1	2	1

1: Low 2: Medium 3: High

REFERENCE BOOKS:

1.	Sekhar Satya G.V., "Working Capital Management", 1/e; New Delhi: Wiley, 2014.
2.	Bhalla V. K., "Working Capital Management", 1/e; New Delhi: S. Chand Publishing, 2014.
3.	Sagner James S., "Working Capital Management, Applications and Cases", 1/e, New Delhi: Wiley, 2015.

ENGINEERING ECONOMICS & FINANCIAL MANAGEMENT

Course Code:	MG1507-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50

Teaching Department: Mechanical Engineering
Course Objectives:

- | | |
|---|---|
| 1 | Analyse the time value of money. |
| 2 | Evaluate the worth of creations, by comparing the alternatives visa, vis the cost (cost- benefit analysis). |
| 3 | Take decisions with the limited resources, the relevant course of action, with the help of suitable tools. |
| 4 | Determine the depreciated values of assets and also cost involved in each operation, a product should undergo with an aim to fix suitable selling price for the products. |
| 5 | Know the fundamentals of Financial Management. |

UNIT-I
Fundamental economic concepts
07 Hours

Consumer goods, Producer goods, Factors of production, Economy of organization, Demand theory, Law of demand, Exceptions to law of demand, Law of supply, Determinants of supply, Law of increasing returns and law of diminishing returns(No exercises)

Interest
07 Hours

Rate of interest, Determining rate of interest, Time value of money, Simple interest, Compound interest, Nominal and effective interest rate, Equivalence involving interest, Interest formulae [single payment, uniform series and arithmetic gradient only], problems using interest formulae [discrete compounding only].

UNIT II
Economic Analysis of Alternatives
09 Hours

Analysis based on: Present Worth [equal life and unequal life situations], Future Worth, Equivalent Annual Worth, Exercises. Analysis based on Rate of Return, Exercises.

Depreciation
04 Hours

Causes of depreciation, Depletion, Methods of depreciation [Straight line, Declining balance, Double declining balance] Exercises.

Estimating and Costing
03 Hours

Components of cost [Material cost, Labour cost, Overhead expenses, Prime cost, Factory cost, Total cost], Determination of selling price of a product, Exercises.

UNIT III
Financial management
05 Hours

Terminologies used in accounting, Journal and ledger, Profit and loss statement, Balance sheet, Understanding basic financial ratios, Simple exercises.

Working Capital Management
05 Hours

Factors influencing working capital requirement, determination of operating cycle and working capital.

Capital Budgeting: Risk analysis in Capital Budgeting
Course Outcomes: At the end of the course student will be able to

- | | |
|----|--|
| 1. | Explain the fundamental economic concepts. |
| 2. | Use simple interest and compound interest to determine compounded and discounted amount. |
| 3. | Compare the alternatives using Present Worth, Equivalent Annual Worth, Future Worth and IRR methods. |
| 4. | Calculate the depreciated amount of a given assets using Straight line, Declining balance, Double declining g balance method. Estimate the selling price of given product. |
| 5. | Prepare Balance Sheet & Profit and Loss account for given data of a firm. Estimate working capital. Explain capital budgeting. |

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
MG1507-1.1	3	1	-	-	-	1	-	-	1	1	-	1

Security

	MG1507-1.2	2	3	-	-	-	1	-	-	1	1	-	1
	MG1507-1.3	2	3	-	-	-	1	-	-	1	1	-	1
	MG1507-1.4	2	3	-	-	-	1	-	-	1	1	-	1
	MG1507-1.5	2	3	-	-	-	1	-	-	1	1	-	1
1: Low 2: Medium 3: High													
TEXTBOOKS:													
1.	Riggs J.L., “Engineering Economics”, 4th edition, Tata McGraw-Hill, 2004.												
2.	Banga and Sharma, “Mechanical Estimating and Costing”, 16 th edition, Khanna Publishers, 2012.												
3.	I M Pandey, “Financial Management”, Vikas Publishing House, 2002.												
REFERENCE BOOKS:													
1.	E Paul Degarmo, “Engineering Economy”, Macmillan Publishing, 2001.												
2.	Gerald J Thuesen & W J Fabrycky, “Engineering Economy”, Prentice Hall of India, 9th ed.												
3.	Tarachand, “Engineering Economics”, Nemchand& Bros, 1996.												
E Books / MOOCs/ NPTEL													
1.	http://nptel.ac.in/courses/112107209/												

NANOTECHNOLOGY

Course Code:	PH2501 -1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	PH1001 -1		

Teaching Department: PHYSICS

Course Objectives:

1.	To understand the basic scientific concepts of nanoscience, properties of nano materials, synthesis and fabrication of nano materials.
2.	To understand the various characterization techniques of nano materials.
3.	Study of carbon nano technology and its characterizations.
4.	To understand the applications of nano technology in various science, engineering and technology fields.

UNIT-I

Properties of Materials	07 Hours
Introduction: History of nano science, definition of nano meter, nanomaterials, nano technology. Classification of nano materials. Crystal symmetries, crystal directions, crystal planes, Band structure. Properties Of Materials: Mechanical properties, electrical properties, dielectric properties, thermal properties, magnetic properties, opto electronic properties. Effect of size reduction on properties, electronic structure of nano materials.	
Synthesis and Fabrication	08 Hours
Synthesis of bulk polycrystalline samples, growth of single crystals, Synthesis techniques for preparation of nano particle – Bottom Up Approach – sol gel synthesis, hydro thermal growth, thin film growth, PVD and CVD; Top Down Approach – Ball milling, micro fabrication, lithography, Requirements for realizing semiconductor nano structure, growth techniques for nano structures.	

UNIT-II

Characterization Techniques	15 Hours
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Security

X-Ray diffraction and Scherrer method, scanning electron microscopy, transmission electron microscopy (TEM), scanning probe microscopy (SEM), atomic force microscopy (AFM), piezoresponse microscopy, X-ray photoelectron spectroscopy, XANES and XAFS, angle resolved photoemission spectroscopy, diffuse reflectance spectra, photoluminescence spectra, UV-VIS-IR Spectrophotometers, Magnetic and electrical measurements and Infrared/ Raman, EPR and NMR													
UNIT-III													
Carbon Nano Technology				05 Hours									
Characterization of carbon allotropes, synthesis of diamond – nucleation of diamond, growth and morphology. Applications of nano crystalline diamond films, grapheme, and applications of carbon nano tubes.													
Applications of Nano Technology				05 Hours									
Applications in material science, biology and medicine, surface science, energy and environment. Applications of nano structured thin films, applications of quantum dots.													
Course Outcomes: At the end of the course student will be able to													
1.	Ability to choose the appropriate nano material to meet the requirement of a particular application.												
2.	Identify the essential concepts used in nanotechnology.												
3.	Identify the materials, properties, synthesis and fabrication of nanomaterials.												
4.	Understand the various characterization techniques of nano materials.												
5.	Applications of nanomaterials in various fields												
Course Outcomes Mapping with Program Outcomes													
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes													
PH2501-1.1		3	3	-	-	-	-	-	-	-	-	-	-
PH2501-1.2		3	3	-	-	-	-	-	-	-	-	-	-
PH2501-1.3		3	3	-	-	-	-	-	-	-	-	-	-
PH2501-1.4		3	3	-	-	-	-	-	-	-	-	-	-
PH2501-1.5		3	3	-	-	-	-	-	-	-	-	-	-
1: Low 2: Medium 3: High													
TEXTBOOKS:													
1.		M.S. Ramachandra Rao, Shubra Singh, "Nano science and nano technology", Wiley publishers.											
REFERENCE BOOKS:													
1.		Charles P. Poole, Jr. Frank J. Owens, "Introduction to Nano Technology", Wiley publishers.											
2.		Jermy J Ramsden, "Nanotechnology", Elsevier publishers.											
3.		A. K. Bandyopadhyay, "Nano Materials", New Age publishers.											
4.		T. Pradeep, "Nano Essentials", TMH.											
5.		M. A. Shah, "Nanotechnology the Science of Small", Wiley publishers.											
6.		Phani Kumar, "Principles of Nanotechnology", Scitech.											
E Books / MOOCs/ NPTEL													
1.		https://youtu.be/ebO38bbq0_4											
2.		https://youtu.be/0Mzlh7wkgMs											

OPTOELECTRONIC DEVICES

Course Code:	PH2502-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03

Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	PH1001 -1		

Teaching Department: PHYSICS
Course Objectives:

1.	To understand the basic principles of construction, working and applications of various optoelectronic devices.
2.	Study of sources of radiation like lasers and LED, their specific properties and hence their use for applications.
3.	Study of radiation detectors like semiconductor detector, diode as detector and photo multiplier.
4.	Understanding the fabrication and applications of optical fibers, optical modulators and waveguides for optical communication

UNIT-I
Optical processes in Semiconductor, Display devices & Optical fibers
15 Hours

Elements of optical phenomena in Semiconductors- fundamentals of Fermi-Dirac distribution, band structure, direct and indirect band gap semiconductors, generation-recombination mechanisms, absorption and emission processes.
 Display devices- cathode ray tube, liquid crystal display, charge coupled devices, plasma display.
 Optical fibers- types of fibers, modes of propagation, attenuation and losses, optical fiber communication system, advantages.

UNIT-II
Optical Sources and Detectors
15 Hours

Lasers- basic principles, optical resonator-types, modes and quality factor, practical lasers- Nd-YAG, CO₂, Excimer laser, Semiconductor laser- basic structure, laser action, heterojunction laser, quantum well laser, applications.
 Light emitting diode- electroluminescence in p-n junction, LED characteristics, efficiency and responsivity, Heterojunction LED, Surface-Emitting LED and Edge emitting LED.
 Photo detectors- photo conductor detector, junction photo diode, p-i-n photo diode, avalanche photo diode. Photo multiplier tube.

UNIT-III
Integrated Optics and Modulators
10 Hours

Modulation of light- Analog and digital modulation, Direct modulation - using LED and Semiconductor diode laser (SDL). External modulation - Electro-optic modulators (Pockels effect), Electro-absorption modulators. Acousto-optic modulation. Waveguides- device structure, waveguide devices – waveguide lenses, light bending devices, optical power dividers, directional couplers, waveguide polarizer, wavelength multiplexers and demultiplexers. Waveguide coupling. Optoelectronic integrated circuit

Course Outcomes: At the end of the course student will be able to

1.	Ability to choose the appropriate device to meet the requirement of a particular application.
2.	Making modifications to device structures by understanding the factors affecting their performance.
3.	Attempting better efficiency and utility through an understanding of the principles of performance.
4.	Use the technical knowledge acquired to troubleshoot and rectify devices and circuits.
5.	Explore the possibility of designing devices with better characteristics.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes												
PH2502-1.1	3	3	-	-	-	-	-	-	-	-	-	-
PH2502-1.2	3	3	-	-	-	-	-	-	-	-	-	-
PH2502-1.3	3	3	-	-	-	-	-	-	-	-	-	-
PH2502-1.4	3	3	-	-	-	-	-	-	-	-	-	-
PH2502-1.5	3	3	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	P.R.Sasikumar, "Photonics – an introduction", PHI Learning Pvt.Ltd.,New Delhi, 2012 edition.
2.	Pallab Bhattacharya, "Semiconductor Opto Electronic

	Devices", Prentice Hall of India Pvt., Ltd., New Delhi, 2006.
REFERENCE BOOKS:	
1.	J.Wilson and J.Haukes, "Opto electronics- an introduction", Prentice Hall of India, New Delhi.
2.	Jasprit Singh, "Opto electronics- an introduction to Materials and Devices", McGraw Hill international ed., 1998.
3.	A.Ghatak and Thyagarajan, "Introduction to opto electronics", New Age International Publication.
E Books / MOOCs/ NPTEL	
1.	http://nptel.ac.in/courses/115102026/

AUTONOMOUS MOBILE ROBOTS													
Course Code:	RI2501-1	Course Type	OEC										
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03										
Total Teaching Hours	40+0+0	CIE + SEE Marks	50+50										
Prerequisite	EC 1001-1, ME 1003-1												
Teaching Department: Robotics and Artificial Intelligence													
Course Objectives:													
<table><tr><td>1.</td><td>Explain different types of locomotion in mobile robots to obtain a required task.</td></tr><tr><td>2.</td><td>Understand the different types of kinematics and dynamics involved in a mobile robot.</td></tr><tr><td>3.</td><td>Study the different types of sensors used in an autonomous mobile robot.</td></tr><tr><td>4.</td><td>Understand the different types of algorithms to identify the position of the mobile robot.</td></tr><tr><td>5.</td><td>Understand the various algorithms for planning and navigation of the mobile robot.</td></tr></table>				1.	Explain different types of locomotion in mobile robots to obtain a required task.	2.	Understand the different types of kinematics and dynamics involved in a mobile robot.	3.	Study the different types of sensors used in an autonomous mobile robot.	4.	Understand the different types of algorithms to identify the position of the mobile robot.	5.	Understand the various algorithms for planning and navigation of the mobile robot.
1.	Explain different types of locomotion in mobile robots to obtain a required task.												
2.	Understand the different types of kinematics and dynamics involved in a mobile robot.												
3.	Study the different types of sensors used in an autonomous mobile robot.												
4.	Understand the different types of algorithms to identify the position of the mobile robot.												
5.	Understand the various algorithms for planning and navigation of the mobile robot.												
UNIT-I													
Robot locomotion	07 Hours												
Types of locomotion, hopping robots, legged robots, wheeled robots, stability, maneuverability, and controllability.													
Mobile robot kinematics and dynamics	09 Hours												
Forward and inverse kinematics, holonomic and nonholonomic constraints, kinematic models of simple car and legged robots, dynamics simulation of mobile robots.													
UNIT-II													
Perception	07 Hours												
Proprioceptive/Exteroceptive and passive/active sensors, performance measures of sensors, sensors for mobile robots like global positioning system (GPS), Doppler effect-based sensors, vision-based sensors, uncertainty in sensing, filtering.													
Localization	07 Hours												
Odometric position estimation, belief representation, probabilistic mapping, Markov localization, Bayesian localization, Kalman localization, and positioning beacon systems.													
UNIT-III													
Introduction to planning and navigation	10 Hours												
Path planning algorithms based on A-star, Dijkstra, Voronoi diagrams, probabilistic roadmaps (PRM), rapidly exploring random trees (RRT), Markov Decision Processes (MDP), stochastic dynamic programming (SDP).													
Course Outcomes: At the end of the course student will be able to													
<table><tr><td>1.</td><td>Explain different types of locomotion in mobile robots to obtain a required task.</td></tr></table>				1.	Explain different types of locomotion in mobile robots to obtain a required task.								
1.	Explain different types of locomotion in mobile robots to obtain a required task.												

Security

2.	Identify the different types of kinematics and dynamics involved in a mobile robot.	
3.	Apply the different types of sensors used in an autonomous mobile robot.	
4.	Apply the different types of algorithms to identify the position of the mobile robot.	
5.	Apply the various algorithms for planning and navigation of the mobile robot to reach the destination.	

Course Outcomes Mapping with Program Outcomes													
	Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12
	↓ Course Outcomes												
	RI2501-1.1	3	3	3	3	2	1	-	-	-	-	-	3
	RI2501-1.2	3	3	3	3	2	1	-	-	-	-	-	3
	RI2501-1.3	3	3	3	3	2	1	-	-	-	-	-	3
	RI2501-1.4	3	3	3	3	2	1	-	-	-	-	-	3
	RI2501-1.5	3	3	3	3	2	1	-	-	-	-	-	3

1: Low 2: Medium 3: High

TEXTBOOKS:	
1.	R. Siegwart, I. R. Nourbakhsh, “Introduction to Autonomous Mobile Robots”, The MIT Press, 2011.
2.	Peter Corke, "Robotics, Vision and Control: Fundamental Algorithms in MATLAB", Springer Tracts in Advanced Robotics, 2011.
3.	S. M. LaValle, “Planning Algorithms”, Cambridge University Press, 2006. (Available online http://planning.cs.uiuc.edu/)

REFERENCE BOOKS:	
1.	Thrun, S., Burgard, W., and Fox, D., "Probabilistic Robotics". MIT Press, Cambridge, MA, 2005.
2.	Melgar, E. R., Diez, C. C., "Arduino, and Kinect Projects: Design, Build, Blow Their Minds", 2012.
3.	H. Choset, K. M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, L. E. Kavraki, and S. Thrun, "Principles of Robot Motion: Theory, Algorithms, and Implementations", PHI Ltd., 2005.

E Books / MOOCs/ NPTEL	
1.	https://archive.nptel.ac.in/courses/112/106/112106298/
2.	https://www.edx.org/course/autonomous-mobile-robots

MEDICAL ROBOTICS			
(For All except AI)			
Course Code:	RI2502-1	Course Type	PEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	PH 1001-1, IS 1001-1, CY 1001-1		
Teaching Department: Robotics and Artificial Intelligence			
Course Objectives:			
1.	Understand the types of medical robots used in the field of healthcare.		
2.	Explain the various localization and tracking sensors		
3.	Understand the applications of surgical robots with the help of few case studies		
4.	Understand Rehabilitation of limbs and brain machine interface with the help of few case studies		
5.	Understand the design methodology of medical robots.		
UNIT-I			
Introduction	07 Hours		
Types of medical robots - Navigation - Motion Replication - Imaging - Rehabilitation and Prosthetics - State of art of robotics in the field of healthcare. Localization And Tracking			
Position sensors requirements	09 Hours		

Tracking - Mechanical linkages - Optical - Sound-based - Electromagnetic -Impedance-based - In-bore MRI tracking - Video matching - Fiber optic tracking													
UNIT-II													
Control Modes Radiosurgery			07 Hours										
Orthopedic Surgery - Urologic Surgery and Robotic Imaging - Cardiac Surgery – Neurosurgery – case studies.													
Rehabilitation			07 Hours										
Rehabilitation for Limbs - Brain-Machine Interfaces - Steerable Needles – case studies.													
UNIT-III													
Design of Medical Robots			10 Hours										
Characterization of gestures to the design of robots- Design methodologies- Technological choices - Security													
Course Outcomes: At the end of the course student will be able to													
1.	Describe the types of medical robots and the concepts of navigation and motion replication.												
2.	Describe about the sensors used for localization and tracking												
3.	Explain the applications of surgical robots												
4.	Explain the concepts in Rehabilitation of limbs and brain machine interface												
5.	Classify the types of assistive robots and analyze the design characteristics, methodology and technological choices for medical robots.												
Course Outcomes Mapping with Program Outcomes													
Program Outcomes→		1	2	3	4	5	6	7	8	9	10	11	12
↓ Course Outcomes													
RI2502-1.1		3	-	1	-	-	-	-	-	-	-	-	1
RI2502-1.2		3	-	1	-	-	-	-	-	-	-	-	1
RI2502-1.3		3	-	1	-	-	-	-	-	-	-	-	1
RI2502-1.4		3	-	1	-	-	-	-	-	-	-	-	1
RI2502-1.5		3	-	3	-	-	-	-	-	-	-	-	1
1: Low 2: Medium 3: High													
TEXTBOOKS:													
1.		Mark W. Spong, Seth Hutchinson, and M. Vidyasagar, "Robot Modeling and Control", Wiley Publishers, 2006.											
2.		Paula Gomes, "Medical robotics- Minimally, Invasive surgery", Woodhead, 2012.											
3.		Achim Schweikard, Floris Ernst, "Medical Robotics", Springer, 2015.											
REFERENCE BOOKS:													
1.		Jocelyne Troccaz, "Medical Robotics", Wiley-ISTE, 2012.											
2.		Vanja Bonzovic, "Medical Robotics", I-tech Education publishing Austria, 2008.											
3.		Daniel Faust, "Medical Robotics", Rosen Publishers, 2016.											
4.		Jocelyne Troccaz, "Medical Robotics", Wiley, 2013.											
E Books / MOOCs/ NPTEL													
1.		https://www.futurelearn.com/courses/medtech-ai-and-medical-robots											
2.		https://web.stanford.edu/class/me328/											

PLC CONTROL OF HYDRAULIC AND PNEUMATIC CIRCUITS			
(For All except AI)			
Course Code:	RI2503-1	Course Type	OEC
Teaching Hours/Week (L: T: P):	3:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	EE 1001-1, EC 1001-1		
Teaching Department: Robotics and Artificial Intelligence			
Course Objectives:			

1.	To understand the fundamentals of fluid power transmission systems																																																																																												
2.	To design various hydraulic system components.																																																																																												
3.	To design various pneumatic system components.																																																																																												
4.	Learn various types of hydraulic and pneumatic power circuits.																																																																																												
5.	Learn various types of applications in fluid power circuits using PLC.																																																																																												
UNIT-I																																																																																													
Fluid power systems and fundamentals		06 Hours																																																																																											
Introduction to fluid power, Advantages of fluid power, Application of fluid power system. Types of fluid power systems, General types of fluids - Properties of hydraulic fluids -Fluid power symbols. Basics of Hydraulics-Applications of Pascal's Law																																																																																													
Hydraulic system components		05 Hours																																																																																											
Sources of Hydraulic Power: Pumping theory - Pump classification - construction and working of pumps - Variable displacement pumps, pump performance. Actuators: Linear hydraulic actuators-Single acting and double acting cylinders, Rotary actuators - Fluid motors.																																																																																													
Control Components		04 Hours																																																																																											
Direction control valve - Valve terminology - Various center positions. Shuttle valve - check valve - pressure control valve - pressure reducing valve, sequence valve. Flow control valves - Fixed and adjustable Safety valves.																																																																																													
UNIT-II																																																																																													
Pneumatic system components		07 Hours																																																																																											
Pneumatic Components: Properties of air. Compressors. FRL Unit -Air control valves, Quick exhaust valves and pneumatic actuators- cylinders, air motors. Basics of low-cost automation																																																																																													
Fluidics & Pneumatic circuit design		08 Hours																																																																																											
Fluidics - Introduction to fluidic devices, simple circuits. Introduction to Electrohydraulic Pneumatic logic circuits, PLC applications in fluid power control, Sequential circuit design for simple applications using classic, cascade, logic with Karnaugh- Veitch Mapping and combinational circuit design methods.																																																																																													
UNIT-III																																																																																													
Fluid power circuits		10 Hours																																																																																											
Electrical control of pneumatic and hydraulic circuits-use of relays, timers, counters, Ladder diagram. Programmable logic control of Hydraulics Pneumatics circuits, PLC ladder diagram for various circuits, motion controllers, use of field busses in circuits. Electronic drive circuits for various Motors.																																																																																													
Course Outcomes: At the end of the course student will be able to																																																																																													
1.	Compare the basics of hydraulics to the performance of fluid power systems																																																																																												
2.	Explain the working principle of hydraulic systems including pumps and control components.																																																																																												
3.	Explain the working principle of pneumatic systems and their components.																																																																																												
4.	Design various types of Electrohydraulic and electro pneumatic circuits																																																																																												
5.	Design various types of applications in fluid power circuits using PLC.																																																																																												
Course Outcomes Mapping with Program Outcomes																																																																																													
<table><tr><td>Program Outcomes→</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr><tr><td>↓ Course Outcomes</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>RI2503-1.1</td><td>3</td><td>2</td><td>3</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td></tr><tr><td>RI2503-1.2</td><td>3</td><td>2</td><td>3</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td></tr><tr><td>RI2503-1.3</td><td>3</td><td>2</td><td>3</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td></tr><tr><td>RI2503-1.4</td><td>3</td><td>2</td><td>3</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td></tr><tr><td>RI2503-1.5</td><td>3</td><td>2</td><td>3</td><td>2</td><td>3</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td></tr></table>			Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	↓ Course Outcomes													RI2503-1.1	3	2	3	2	3	-	-	-	-	-	-	3	RI2503-1.2	3	2	3	2	3	-	-	-	-	-	-	3	RI2503-1.3	3	2	3	2	3	-	-	-	-	-	-	3	RI2503-1.4	3	2	3	2	3	-	-	-	-	-	-	3	RI2503-1.5	3	2	3	2	3	-	-	-	-	-	-	3
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12																																																																																	
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RI2503-1.3	3	2	3	2	3	-	-	-	-	-	-	3																																																																																	
RI2503-1.4	3	2	3	2	3	-	-	-	-	-	-	3																																																																																	
RI2503-1.5	3	2	3	2	3	-	-	-	-	-	-	3																																																																																	
1: Low 2: Medium 3: High																																																																																													
TEXTBOOKS:																																																																																													
1.	Majumdar S.R., "Pneumatic systems - Principles and maintenance", Tata McGraw Hill, 2008.																																																																																												
2.	Anthony Esposito, "Fluid Power with Applications", Pearson Education 2009.																																																																																												
REFERENCE BOOKS:																																																																																													
1.	Majumdar S.R., "Oil Hydraulics", Tata McGraw-Hill, 2000.																																																																																												

2.	Harry L. Stevart D. B, "Practical guide to fluid power", Taraoeala sons and Port Ltd.Broadey, 2010.
3.	Michael J, Prinches and Ashby J. G, "Power Hydraulics", Prentice Hall, 2011.
4.	Dudelyt, A. Pease and John T. Pippenger, "Basic Fluid Power", Prentice Hall, 2011.
E Books / MOOCs/ NPTEL	
1.	https://nptel.ac.in/courses/108/105/108105088/
2.	https://plc-coep.vlabs.ac.in/List%20of%20experiments.html?domain=Electrical%20Engineering
3.	http://vlabs.iitb.ac.in/vlabs-dev/vlab_bootcamp/bootcamp/COEP_KNOWLEDGE_SEEKERS/labs/exp1/theory.html